Rafik Hariri University
جامعة رفيق الحريري

Undergraduate Catalog

2026-2027





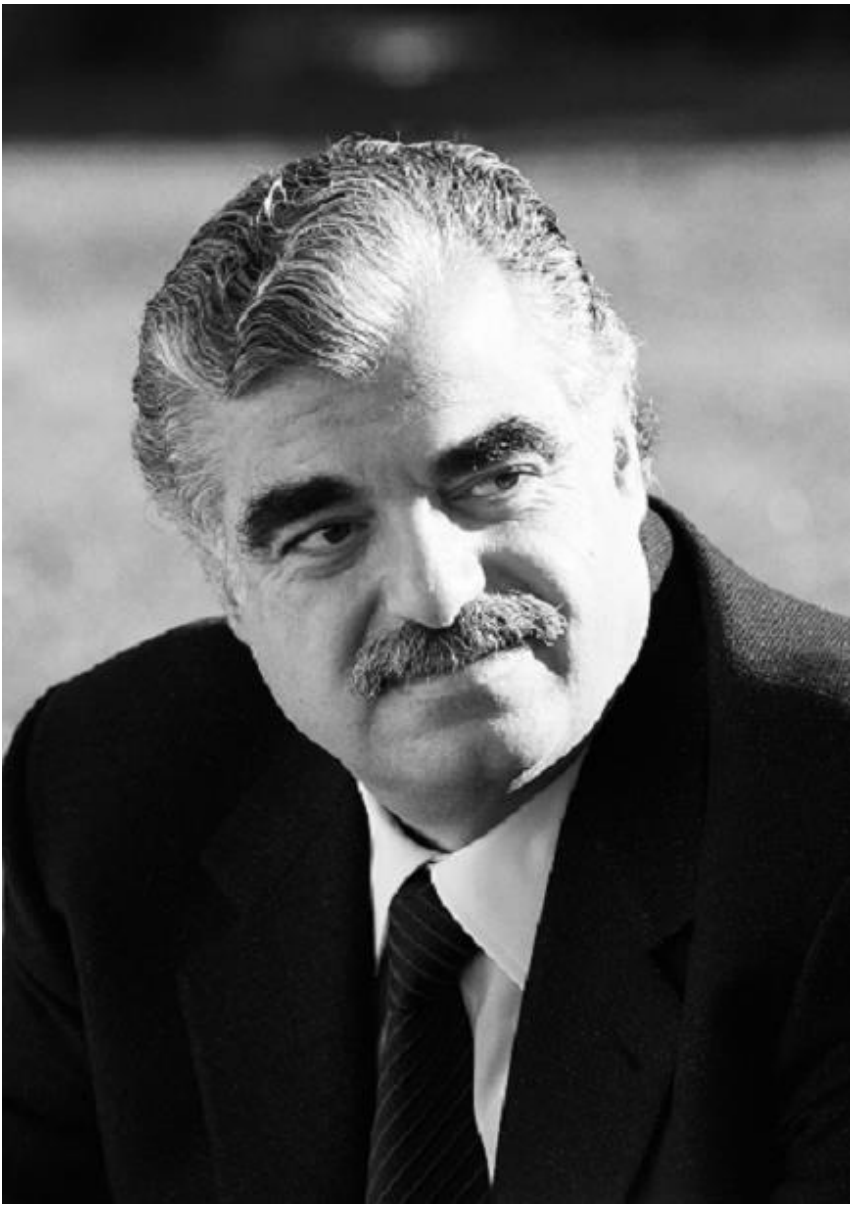


RHU for All,
RHU is Here to Stay

Rafik Hariri University
جامعة رفيق الحريري

UNDERGRADUATE CATALOG

2026-2027



"إن أملي هو فتح أبواب الجامعات والمعاهد العليا أمام المواهب والكفاءات المغمورة"

Notice

The information in this catalog applies to the academic year 2026-2027 as of September 1, 2026. The University reserves the right to make changes to the programs, course offerings, academic requirements, and teaching staff as it deems in the best interests of students, without prior notice.

This catalog conforms to applicable Lebanese laws and the rules and regulations of the Directorate of Higher Education (DoHE). In the event of a contradiction, the Lebanese laws and DoHE rules and regulations take precedence.

Student Responsibility for Catalog Information

Students are responsible for reading, understanding, and adhering to the information in this catalog. Failure to comply with the stated university, college, and program regulations will not exempt the students from the ramifications or penalties that could be incurred due to ignorance.

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You can also view this catalog on the RHU website at
https://www.rhu.edu.lb/academics/student-catalog.

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MESSAGE FROM THE PRESIDENT

As we celebrate our silver anniversary and lay the foundation of our golden anniversary and beyond, a new dawn is rising over Rafik Hariri University (RHU). This dawn brings new winds of change, designed to transform RHU into a stronger and more formidable institution.

The waves of change are at full speed and in constant motion. Soon, RHU will conclude its self-assessment journey and continue reinventing itself and its business model to become a sustainable, agile, and nimble institution.

Guided by a strong vision handed to us by our founder, Prime Minister Rafik Hariri, and overseen by our astute President of the Board of Trustees, Mrs. Nazik Rafik Hariri, RHU affirms its determination to embrace best practices and to become a leading institution of higher learning in Lebanon, the region, and beyond.

At RHU, the student is our number one priority. We are determined to shape futures, impact life journeys, create hopes and opportunities, and inspire students to achieve their dreams. We want students to lead full, rich lives while radiating positive energies. RHU offers more than a great learning experience. RHU broadens horizons and challenges assumptions.

RHU provides world-class curricula in all its Colleges. Our academic programs are second to none and are accredited by the highest bodies in their respective disciplines (ABET, ACBSP, and many more). The RHU Board of Trustees and the institution's leadership are seriously exploring the option of obtaining university-wide accreditation from the top higher education bodies in the United States.

RHU's positive work environment has enabled us to see, hear, and engage one another in a constructive dialogue about the present and future of the university. Such a collegial environment enables our students to observe, acquire, and develop the soft skills (communication, thoughtful inquiry, and responsible leadership) that are increasingly vital in modern workplaces.

Our world-class faculty will continue to raise the bar for academic excellence. New courses, areas of study, and specializations are offered, all intended to equip our students with the necessary skill sets (quantitative, analytical, technical, digital, and intellectual) to be marketplace-ready.

Our alumni continue to make RHU proud. They are entrepreneurs, scientists, researchers, and innovators, and some assume top faculty positions at world-leading institutions. I am humbled by the collective and ongoing contributions of RHU's faculty, staff, students, alumni, and administrators to position RHU as a beacon of hope and excellence.

RHU is determined to build bridges and form new partnerships and strategic alliances with all its internal and external stakeholders. The mission of RHU shall be embedded in and integral to our souls. In due time, we shall all become the change agents Prime

Minister Hariri aspired for, capable of turning around individuals who make the world a better place for us all.

I encourage you to take advantage of our open-door policy. Visit our campus, drop by the office of the president, contact us electronically, speak with an advisor, and learn more about the opportunities available to you at Rafik Hariri University.

Sincerely,

Said M. Ladki
President

BOARD OF TRUSTEES

Dr. Nazik Rafik Hariri, Chairperson

HE Charles Rizk

HE Ghazi Youssef

Dr. Daoud Sayegh

Mr. Mohamad El-Hout

Dr. Farah Tamim Mekkawi

Mr. Fadi Fawaz

Mr. Maroun Al Asmar

Mr. Adib Bassatni

Mr. Ahmad Hijazi

Mr. Ouday El-Sheikh

Dr. Said Ladki, RHU President

ACADEMIC CALENDAR 2026-2027

Fall Semester

Thu	Aug 27	Orientation and Registration / New Students Fall 2026
Mon	Aug 31	Deadline for Payment of Fall 2026 Tuition and Fees
Tue	Sep 1	Classes Begin
Tue - Wed	Sep 8-9	Drop and Add Period
Sun	Nov 1	Founder's Day (Celebrated on Monday, November 2, 2026)
Mon - Fri	Nov 2-6	Advising Week / Spring 2027 for Continuing students
Fri	Nov 6	Last Day to Withdraw from Courses
Mon - Fri	Nov 9-13	Registration Week / Spring 2027 for Continuing students
Sun	Nov 22	Independence Day / Holiday
Tue – Wed	Dec 1-2	Orientation and Registration / New Students Spring 2027
Wed	Dec 9	Last Day of Classes
Thu - Fri	Dec 10-11	Reading Period
Sat – Sat	Dec 12-19	Final Examinations Period
Fri	Dec 25	Christmas / Holiday
Fri	Jan 1	New Year / Holiday

Spring Semester

Wed	Jan 6	Armenian Christmas / Holiday
Fri	Jan 8	Deadline for Payment of Spring 2027 Tuition and Fees
Mon	Jan 11	Classes Begin
Mon-Tue	Jan 18-19	Drop and Add Period
Tue	Feb 9	Saint Maroon's Day / Holiday
Sun	Feb 14	H.E.P.M Rafik Al Hariri Commemoration Day
Tue-Wed	Mar 9-10	Eid Al-Fitr/ Holiday*
Mon – Fri	Mar 15-19	Advising Week / Summer–Fall 2027 for Continuing students
Fri	Mar 19	Last Day to Withdraw from Courses
Mon – Wed	Mar 22-31	Registration Week / Summer –Fall 2027 for Continuing students
Thu	Mar 25	Annunciation Day / Holiday
Fri – Mon	Mar 26-29	Easter Latin / Holiday
Tue	Apr 21	Last Day of Classes
Wed-Thu	Apr 21-22	Reading Period
Fri – Tue	Apr 23 – May 4	Final Examinations Period
Fri – Mon	Apr 30-May 3	Easter Greek Orthodox / Holiday
Sat	May 1	Labor's Day/ Holiday
Fri	May 14	Deadline for Payment of Summer 2027 Tuition and Fees
Sat	Jun 19	Commencement Exercise (Tentative)

Summer Semester

Sun-Tue	May 16-18	Eid El Adha/ Holiday*
Wed	May 26	Classes and Co-op Work Experience Begin
Thu	May 27	Drop and Add Period
Sun	Jun 6	Hijra New Year/ Holiday*
Wed	Jun 30	Last Day to Withdraw from Courses / Co-op
Tues	Jun 15	Ashoura Day / Holiday*
Thu	Jul 15	Classes and Co-op Work Experience End
Mon - Tue	Jul 19-20	Final Examinations Period
Sun	Aug 15	Assumption Day / Holiday

* Tentative dates are pending moon sightings.
P.S. Any changes in public and religious holidays shall be reflected on our calendar in due time.

DIRECTORY

	Location	Extension
President	Block E	400/401
President's Office	Block E	441/442
Vice President for Academic Affairs	Block E	404
Vice President for Development and Information Technology	Block E	403
Vice President for Administration and Finance	Block E	402
Admissions Office	Block E	405/406/407
College of Business Administration	G 101 J	301
College of Engineering	C 103	501
College of Arts and Sciences	I 201	701
Communication and Alumni Relations	B 201	754/755/603
Co-op and Career Services	Block E	414
Digital Media and Design Division	Block I	711/741
Dorm Supervisor	A 112	112
Finance Department	Block E	424/417-419
Graduate Studies and Research	Block B	611
Gymnasium	Block G	330
Health Clinic	I 119	751
Help Desk	Block G	333/334
Human Resources	Block E	787/786
Library	Block E	435
Media Center	Block B	754/755/603
Operator	Block E	0
Purchasing and Procurement Department	Block E	744
Quality Assurance and Institutional Advancement	Block E	443
Registrar's Office	B 102	615/618
Security		290
Store	Block I	752
Student Affairs Office	Block E	770/777
Support Services Department	I 110	740/741

THE UNIVERSITY OVERVIEW

UNIVERSITY ADMINISTRATION

Said Ladki, President

Mahmoud Halablab, Vice President for Academic Affairs

Ibrahim Akkawi, Vice President for Administration and Finance

Board of Deans

Mahmoud Halablab, Vice President for Academic Affairs

Mohammad Taha, Dean – College of Engineering

Anis El Khatib, Dean – College of Business Administration

Houssam Salami, Dean –College of Arts and Sciences

Admission

Zeina Tannir, Director

Communication and Alumni Relations

Rafal Tabbaa Khayat, Director

Finance Department

Marwa Khanji, Assistant Director of Finance

Human Resources Department

Doriah Naboulsi, Associate Director

Information Technology

Abdul Ghani Baba, Director of IT

Wassim Mallah, Director of Systems

Lina Basho, Assistant Director of IT HelpDesk

Library

Basima Katerji, Circulation Librarian

Quality Assurance and Institutional Advancement

Mirna Talhouk, Associate Director

Registrar

Nidal Khalaf, Registrar

Student Affairs

Sahar Hallak, Associate Director of Student Affairs

Campus Facilities

Ahmad Sabeh Ayoun, Director

THE UNIVERSITY

History

Establishing an educational institution that was financially and geographically accessible to all was a pressing dream for His Excellency, Prime Minister Rafik Hariri. In the early 1980s, PM Hariri chose Kfar Falous, an area between the South, the Bekaa, and the mountains, to establish a hospital and a university in collaboration with the University of Saint Joseph. However, the 1982 Israeli invasion destroyed the university and stole its equipment.

In 1984, the dream took a different turn with the establishment of the Rafik Hariri Foundation. It is a non-profit organization whose primary goal is to provide Lebanese youth with opportunities for higher education at local and international universities. Around 33,000 Lebanese students benefited from the University Loan Program of the Rafik Hariri Foundation and pursued their education in Lebanon, Europe, North Africa, and North America.

Following the end of the civil war in Lebanon, the Rafik Hariri Foundation sought to revive the dream of establishing an affordable and accessible university. Such a massive undertaking required the expertise of a country that had made significant strides in education. Although several educationally advanced countries were considered, Canada was eventually chosen for this collaboration. The programs of study at Rafik Hariri University were developed in association with several Canadian institutions, including the Canadian Bureau of International Education, the Canadian International Development Agency, Capilano University, and Memorial University.

Rafik Hariri University opened its doors for the first time on September 15, 1999, with the College of Business Administration, in accordance with the 1947 Presidential Decree. It began with an initial enrollment of 75 students. After the Colleges of Engineering and Sciences and Information Systems were established, RHU was granted university status on June 19, 2006, by Decree Number 17192.

Vision

RHU envisions becoming a beacon of knowledge, a brand for academic distinction, and a force of meaningful transformation. It strives to foster quality, institute authentic learning conditions, and support purposeful research to advance knowledge and nurture leaders capable of driving development and contributing to the enrichment of the community - immediate and beyond.

Mission

RHU shall institute quality education by committing to due process, academic excellence, and distinction. It shall create conditions conducive to farming dreams, stimulate imagination, and cultivate a passion for lifelong learning. It shall instill the

values of responsible behavior, tolerance, and freedom of self-expression and thought and align learning outcomes with emergent community needs to transcend students' potential, possibilities, and contributions beyond time and distance.

Values

When it comes to values, RHU does not merely pay lip service. It is indeed deeply committed to several core values that it upholds and fulfills:

Shared Governance. Enable all RHU constituents to participate in shaping the provision of quality higher education.

Excellence. Offer a meaningful contribution to the development of knowledge and to the promotion of teaching excellence and administrative support.

Integrity. Sustain a community of trust and operate with fairness, honesty, openness, and the highest ethical standards.

Diversity and Tolerance. Create a community that prioritizes diversity and tolerance in academic and civic life.

Community Service. Show commitment to serving our community by responding to the needs and concerns of all stakeholders.

Purposeful Learning. Engage students in a learning experience congruent with individual aspirations and community needs.

Location

RHU enjoys a superb location atop a seaside mountain overlooking the Mediterranean Sea in Mechref, Damour, a short drive from Beirut, Saida, and Mount Lebanon, in the heart of a prestigious gated community.

The RHU campus is 20 kilometers south of Beirut. It covers an area of 54,000 square meters and comprises nine buildings. Seven buildings are used for educational purposes, one for administration, and one for faculty and student residences. The campus also features basketball and volleyball courts, parking areas, a long row of Washingtonian Palm trees, vegetable gardens, a greenhouse, and beautiful open green areas.

RHU began its vegetable gardens in 2021 and selected locations behind Block A, the Dormitory Facility building, and Block G, the College of Business Administration. The landscaping team at RHU leveled the ground reasonably to prevent soil erosion and built a greenhouse for growing warm-season vegetables.

RHU's decision to grow crops is part of its sustainable campus initiative and its efforts to care for the environment. In addition to their aesthetic benefits, plants offer numerous advantages, including carbon sequestration, oxygen production, flood mitigation, and improved air quality. This also helps reduce the impact of global warming.

RHU's electricity supply comes via three routes: Mains Electrical from the EDL Grid, self-run diesel-oil generators, and the daytime-use RHU Solar-PV plant. A 100-kW-rated solar PV plant is located on the top of three of RHU's buildings, Blocks F, G, and H.

Climate

The Mechref Village in Damour is a residential project spanning elevations ranging from 30 to 560 meters above sea level. The climate is moderately cold from December to March and hot from June to September. Although the classrooms have heaters and air conditioners, students are advised to wear warm clothing during the cold season.

The lush greenery of the RHU campus creates a refreshing atmosphere and further motivates students to pursue their academic goals. The green lawns provide ample space for students to study, relax, or socialize. The mosaic of blue, green, and white—a blue sky, green lawns, and white buildings—of the RHU campus is itself magnificent.

Directions and Map

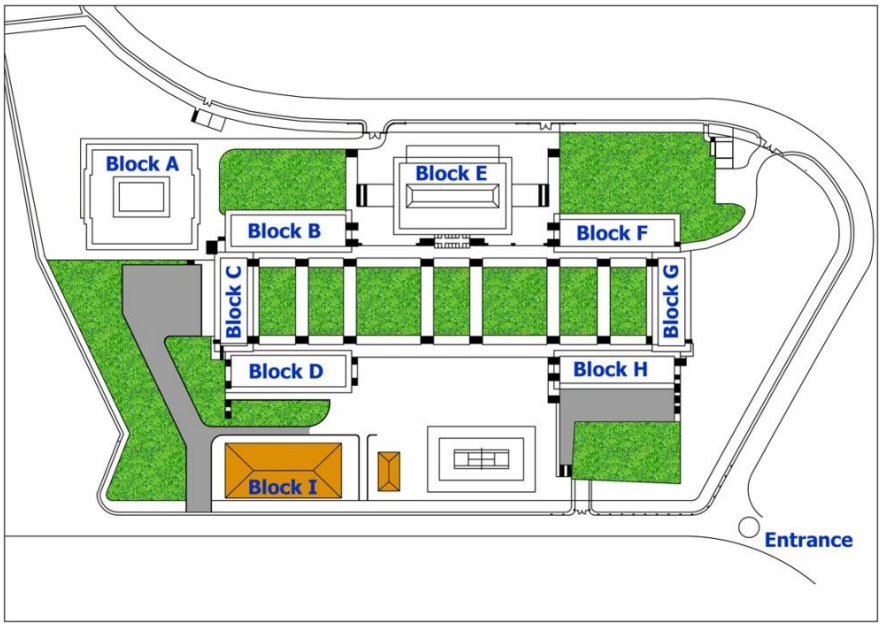
From Beirut

Take the Beirut Saida Highway south (towards Saida)
Take the Damour main exit (towards Deir Al-Qamar)
Go left at the roundabout (follow the Mechref Village road sign)
Drive for 300 meters
Take a right at the Mechref Village main gate
Follow the RHU signs leading to the Campus

From Saida

Take the Beirut Saida Highway north (towards Beirut)
Take the Damour main exit (towards Deir Al-Qamar)
Go left at the roundabout (follow the Mechref Village road sign)
Drive for 300 meters
Take a right at the Mechref Village main gate
Follow the RHU signs leading to the Campus

Map of the RHU Campus



ACADEMIC PROGRAMS AND DEGREES

RHU offers 17 accredited programs in three Colleges: Business Administration, Engineering, and Arts and Sciences. The language of instruction is English. Below is a summary of the programs offered, degrees awarded, and the number of credits required to complete a degree. All programs include 30 credits for general education courses.

College/ Program	Degree	Credits
College of Arts and Sciences		
English Language (currently frozen)	BA	93
Computer Science	BS	93
Healthcare Information Systems (Currently Frozen)	BS	100
Graphic Design	BS	111
Freshman Arts/Science		30
College of Business Administration		
Accounting	BBA	99
Business IT Management	BBA	99
Management	BBA	99
Human Resources Management	BBA	99
Marketing and Advertising	BBA	99
Finance and Banking	BBA	99
College of Engineering		
Civil Engineering	BS, BE, MS	118, 152
Biomedical Engineering	BS, BE, MS	118, 152
Computer and Communications Engineering	BS, BE, MS	118, 152
Electrical Engineering	BS, BE, MS	118, 152
Mechanical Engineering	BS, BE, MS	118, 152
Mechatronics Engineering	BS, BE, MS	118, 152

ADMISSION

Zeina Tannir, Director of the Admissions Office
Sawsan Sheikh Younes, Associate Director of the Admissions Office
Block E
Phone: 00961 25 603090; Ext 405/406/407
E-mail: admissions@rhu.edu.lb
www.rhu.edu.lb/admission

The RHU Admissions Office manages all admission-related functions. The spectrum of responsibilities includes managing admission applications; developing, maintaining, and implementing admissions policies and procedures; working closely with the academic units and University committees on admission issues; continuously engaging with prospective students by helping them complete their application dossier, responding to their inquiries and supporting them in making informative academic choices, and informing them of the admission decision and related stipulations; managing enrollment campaigns and meetings with school principals, parents and community leaders; preparing and updating admission materials; participating in the organization of students' open-day and orientation programs; and generating admission statistics and reports.

Applications for admission may be submitted via mail, email, online, or in person. Admission to RHU is competitive; students with strong high school records may receive early admission. For details on admission requirements, application deadlines, procedures, and fees for various University programs, please refer to the Admission section in this catalog or visit www.rhu.edu.lb/admission

Undergraduate Admission

General Information

Rafik Hariri University admits academically qualified and morally sound students irrespective of origin, color, gender, religion, or disability. RHU notifies students of its decision to accept them within four weeks of receiving their complete application and supporting documents. Outstanding students may receive early admission to RHU based on their academic achievement in high school.

Fulfilling the following admission requirements does not guarantee admission to a program. These requirements constitute a threshold for inclusion in the competition pool. Admission is competitive and subject to the availability of spaces in the desired program. The following stipulations pertain to admission to undergraduate programs. Admission requirements for graduate studies are in the Graduate Catalog.

Admission Requirements

Applicants to RHU must satisfy the following basic requirements: (1) attained the prerequisite education for the program of interest; (2) have not been expelled from another institution for academic or misconduct reasons; (3) completed and submitted the

application form and all application materials to the Admissions Office by the stated deadlines and paid the application fee (4) submitted all required certified documents; and (5) provided photocopies of passport and residence permit (for non-Lebanese applicants).

Students may register for their courses after completing all RHU admission requirements.

Academic Preparation

To be admitted, the student must have completed at least 12 years of schooling before joining the University and provide proof of English-language proficiency as detailed below.

Lebanese Baccalaureate. Students with a Lebanese High School Certificate may be admitted to their desired undergraduate program, subject to the requirements specified in the following table.

College	Baccalaureate Study Track
Arts and Sciences	General and Life Sciences, Sociology & Economics*, Humanities & Literature**
Business Administration	General and Life Sciences, Sociology & Economics, Humanities & Literature***,
Engineering	General and Life Sciences, Sociology & Economics*
* Sociology & Economics track students must take two remedial Math courses as a prerequisite to joining Computer Science offered by the College of Arts and Sciences, or four remedial courses, two Math and two Physics, as a prerequisite to joining any program provided by the College of Engineering, provided that grade 11 is in the scientific track. ** Humanities and Literature track students can enroll in all majors offered by the College of Arts and Sciences except for Computer Science. *** Humanities and Literature track students must take one remedial Math course as a prerequisite for admission to the Business Administration program.	

Lebanese Technical Baccalaureate. Students with a Technical Baccalaureate Certificate may pursue a degree in a major corresponding to the technical degree area of specialization, as stipulated by the Lebanese Ministry of Higher Education Decree #779/M/2013. The student must complete the remedial courses specified in the table below.

College	Majors	Course Category (# credits)
Arts and Sciences	• Computer Science	• Mathematics (6 credits) • Physics (3 credits) • Chemistry (3 credits)
Business Administration	All majors	• Mathematics (3 credits) • Statistics (3 credits) • Economics (3 credits)
Engineering	All majors	• Mathematics (6 credits)

		• Physics (6 credits) • Chemistry (3 credits)

Foreign Secondary Level Certificate. Holders of a foreign certificate equivalent to the Lebanese Official High School diploma, such as the Official Secondary Certificate from Arab Countries, the French Baccalaureate, the International Baccalaureate, or the German Abitur, may enroll in an appropriate program, provided that the Ministry of Education validates the certificate.

High School Diploma. A student who holds a High School Diploma (American system) may be admitted to the freshman program if the following conditions are met: (1) The student has completed 12 years of study; (2) the certificate is recognized by the public authorities in the country of origin; and (3) certified copy of the permission to join the Freshman Program from the Equivalence Committee of the Lebanese Ministry of Higher Education.

General Certificate of Education (GCE). An applicant with a British Certificate may be admitted to the freshman program if s/he has completed 5 Ordinary Level courses, 1 Advanced Level, and 12 years of study. Applicants with A-levels who have completed a minimum of five subjects at the Ordinary Level and three subjects at the Advanced Level, or six Advanced Supplementary (AS) subjects, excluding languages, can be admitted to the sophomore program.

Placement Exams

Students are required to provide evidence of English language proficiency and college-level skills in key areas relevant to the major they wish to pursue. This evidence may be submitted in the form of standardized test scores such as TOEFL or SAT, or by taking the RHU placement exams as indicated in the table below. Depending on their performance, students may be required to take remedial courses to improve their competencies and prepare them for the rigor of related college-level courses.

College	Placement Exam
Arts and Sciences	• English
Business Administration*	• English
Engineering*	• English

English Language Proficiency Requirements

English is the language of instruction in all programs offered at RHU. Applicants must, therefore, demonstrate English language proficiency by passing the RHU English Entrance Exam with a minimum score of 380 or by achieving a minimum score on Standardized English competency exams as follows:

Standardized English Exam	Minimum Scores
Institutional TOEFL (ITP)/ RHU In-House TOEFL	550
Internet-Based TOEFL (IBT)	80
IELTS	6.5
SAT: Reading & Writing Evidence-Based	530+

Students who fail to attain the requisite score must enroll in the Intensive English Program (IELP) to achieve the required level of English proficiency. The IELP consists of three levels, and the student is placed at the appropriate level based on the English proficiency examination score.

RHU recognizes an Institutional TOEFL score as valid for one year from the test date. However, the IBT, SAT, and IELTS scores are valid for two years.

Application Procedure

Prospective students may submit applications via one of the following methods.

Apply in person by visiting the Admissions Office:

Rafik Hariri University
Block E
Mechref Village
Damour- Lebanon

Apply online: https://www.rhu.edu.lb/apply

Required Documents

Applications for undergraduate admission must include the following officially certified documents:

1. Original grades of the last three high school years.
2. Certified copy of high school certificate or diploma: Lebanese Baccalaureate, Technical Baccalaureate, French Baccalaureate, IB Diploma, etc.
3. Certified copy of the Lebanese Baccalaureate equivalency for non-Lebanese degrees or diplomas. (French Baccalaureate; International Baccalaureate; etc.)
4. Scores of one of the English proficiency exams (see above).
5. Original or certified copy of personal civil status record (for Lebanese applicants) or photocopies of passport and residence permit (for non-Lebanese Applicants).
6. Three identical recent passport-size, colored photos.
7. Copy of the Lebanese identity card.
8. A non-refundable application fee of USD 30 is paid in cash at the RHU Finance Office or any branch of BankMed.

Admission to the Freshman Program

In addition to the documents required for undergraduate admission to RHU, applicants to the freshman programs must also:

a. Provide a certified copy of Permission from the Equivalence Committee of the Lebanese Ministry of Education to join the Foreign Program/Freshman class
b. Submit the following scores in one of the aptitude tests below before you can enroll at RHU:

Aptitude Test	Freshman Arts	Freshman Sciences
SAT-Scholastic Aptitude Test	870	950
ACT - American College Test	15	17

Requirements of the Freshman Program Equivalency

1- **If your school is accredited by one of the following accrediting boards**, then the above requirements are enough to grant you admission to the sophomore year after you complete the Freshman Program and get the equivalency from the Lebanese Ministry of Higher Education

Accrediting Boards:
- Western Association of Schools and Colleges (WASC)
- Southern Association of Colleges and Schools (SACS)
- AdvancED (Cognia)
- New England Association of Schools and Colleges (NEASC)
- North Central Association of Schools and Colleges (NCASC)
- Northwest Association of Colleges and Schools (NACS)
- Council of International Schools

2- **If your school is not accredited by one of the above accrediting boards**, then you need, in addition to the above two requirements (a & b), to submit ACT Subject Tests during your first year at RHU and before you can start your sophomore year.

Required subjects in the ACT Subject Tests exam are:
- Mathematics I and any other two subjects for Freshman Arts
- Mathematics II and two of the following subjects: Biology, Chemistry, and Physics for Freshman Science

Required scores in the ACT Subject Test:
- A minimum total score of 51 for Freshman Arts with a minimum score of 15 in each subject
- A minimum total score of 60 for Freshman Sciences, with a minimum score of 17 in each subject

Please note the following:

Lebanese students cannot declare a major and join the sophomore class until they satisfy the requirements of the Lebanese Ministry of Education and Higher Education and the RHU freshman program.

Transfer Admission

A student enrolled in an accredited institution of higher education that requires full-time attendance may be admitted as a transfer student to RHU if space in the aspired program is available and the student meets the following conditions:

- Has studied at least one semester in the old institution and has good academic standing
- Was not dismissed from the old institution for disciplinary reasons; and
- Satisfies the admission requirements of the desired program.

A transfer student must submit official transcripts attested by the Ministry of Higher Education in the country of origin, along with a certified copy of detailed course descriptions from the previous institution. Students transferring from outside Lebanon must also submit a letter of accreditation from the Lebanese Ministry of Higher Education.

If it is later discovered that the transfer student was dismissed from their former university for disciplinary reasons, the acceptance will be revoked retroactively.

The Admission Committee of the concerned College reviews the transfer applications. It forwards its recommendations to the Dean, who informs the Director of Admission of the final decision to take action.

Regarding probation and dismissal rules, a transfer student shall be treated as a new student. A regular semester equals a minimum of 12 earned credits for transfer purposes. To earn a bachelor's degree, a transfer student must study at least three regular semesters at RHU for a three-year program or four regular semesters for a four-year program and acquire Co-op work experience.

Transfer Credits

Courses earned by a transfer student before joining RHU may count as RHU credits, subject to the following conditions:

a. The student has attained a minimum Cumulative GPA of 70% or its equivalent, and a grade of 75 or above in the course.
b. The applicable course is deemed equivalent to the course offered at RHU; at least 80% of its contents match, it involves the same components (Lecture, Lab), and it has the same number of credits or more.
c. The credits were earned within three years of the effective transfer date.
d. A transfer student accepted at RHU is required to complete a minimum of half of the credit requirements toward a bachelor’s degree

A “P” grade shall appear on the student's record for each transferred course but will not be included in the calculation of the cumulative GPA.

A student may take a course at RHU for which equivalent course credits were transferred. In this case, either the transferred course or the RHU course, along with the associated grade, shall appear on the student's transcripts.

Non-Degree Admission

Visiting Student

Visiting student status allows students actively pursuing a degree at another college or university to take courses at Rafik Hariri University and transfer credits back to their home institution.

Prospective undergraduate visiting students must meet the following minimum admission requirements:

- Be in good academic standing (With a minimum CGPA of 70% or its equivalent) from a university recognized and approved by RHU.
- Demonstrate English Language Proficiency:
  TOEFL-IBT minimum 65 or above
  IELTS minimum 5.5-6 or above
  RHU in-house TOEFL minimum score 514

The above requirement may be waived for applicants from institutions where English is the language of instruction.

Required Documents:

1. Fill out and submit a relevant application form (a non-refundable application fee applies)
2. Provide an official college/university transcript.
3. Submit written permission from the home institution for the courses to be taken at RHU.
4. Three passport-size photos and a copy of the ID.
5. A non-refundable reservation deposit, which fully applies towards tuition, is to be paid upon acceptance.

Non-degree Student

Students with a recognized secondary school diploma or a university degree who are interested in taking specific courses to enhance their professional growth and skills can join RHU for academic credit.

Prospective undergraduate non-degree-seeking students must meet the following minimum admission requirements:

- Students must have a high school leaving certificate or higher level of education recognized by RHU.

- Demonstrate English Language Proficiency:
  TOEFL-IBT minimum 65 or above
  IELTS minimum 5.5-6 or above
  RHU in-house TOEFL minimum score 514

The above requirement may be waived for applicants from higher education institutions where the language of instruction is English.

Required Documents:

1. Fill out and submit a relevant application form (A non-refundable application fee applies)
2. Provide an official high school leaving certificate or college/university transcript and a copy of the degree/diploma.
3. Three passport-size photos and a copy of the ID.
4. A non-refundable Reservation Deposit, which fully applies towards tuition, to be paid upon acceptance.

Admission to RHU as a visiting or non-degree student doesn't guarantee admission to any of RHU's regular degree programs. Credits taken at RHU may be applied toward an RHU degree if the student reapplies for admission as a regular student and is accepted under the admission criteria.

Auditors

Individuals wishing to attend classes without receiving academic credit may enroll at RHU as auditing students.

Auditors should submit the relevant application form, a non-refundable application fee, a passport-size photo, a copy of their ID, and a non-refundable reservation deposit that applies towards tuition.

Part-Time Students

Students may wish to pursue a degree at RHU part-time, taking less than the full load of 12 credits per semester. This option is possible, provided the maximum duration for receiving a degree is not exceeded. Accordingly, students may be required to enroll full-time for a few semesters to complete their degrees within the sanctioned period.

Admission to Graduate Studies

Admission to any graduate program at RHU is based on merit. It is offered to applicants holding undergraduate degrees from RHU or other accredited institutions. Students can apply to graduate studies for the fall or spring semesters. Applications are available at the Admissions Office or may be downloaded from RHU's website: www.rhu.edu.lb. Please refer to the Graduate Catalog for details on graduate study programs and related regulations.

Fees and Expenses

Tuition Fees

The Board of Trustees determines the University's tuition and fees annually. Tuition and fees may increase annually without prior notice.

Tuition	LBP	USD
College of Business Administration		
Undergraduate Program		
Per credit	4,000,000	176
Graduate Program		
Per credit	5,300,000	226
College of Engineering		
Undergraduate Program		
Per credit	5,800,000	268
Graduate Program		
Per credit	8,100,000	295
College of Arts and Sciences		
Undergraduate Program		
Per credit		
Graphic Design	4,500,000	205
Computer Science; Health Care Information Systems	4,000,000	189
Other Programs		
Freshman Program (Sciences /Arts)		
Per credit	4,000,000	173
Intensive English Language Program (IELP)		
Per level		
IELP 085	5,800,000	340
IELP 095	5,800,000	340
IELP 100	4,000,000	200
Fees		
Enrollment [1]		150
Deposit [2]		150
NSSF [3]	8,400,000	
Service Fees		
Per semester		250
Per summer session		125
Deferred Payment [4]		
Per semester		20
CO-OP		220
Late Registration		20
Late Payment [5]		
On each installment	3,000,000	30

Transportation / Parking (Optional)		
Regular Shuttle [6]		
Per semester (Deferred over two installments)		480
Per summer session		240
Parking Fee		
Per semester Car/Motorbike		80/20
Per summer session Car/Motorbike		40/10
Dormitory/Housing (Optional)		
Double room in a shared apartment, fall or spring semesters (Block- A)		850
Double room in a shared apartment- summer semester (Block A)		425
Single room -fall or spring semesters (Block A)		1100
Single room - summer semester (Block A)		550
Single room -fall or spring semester (Block F)		1200
Single room - summer semester (Block F)		600
Dorm Deposit [7]		100
Dorm Late Payment Fees (minimum)[8]		25

[1] Enrollment: fee paid by new students upon the decision to join RHU and credited to the statement of fees of the first semester.
[2] Deposit: Paid by new students upon the decision to join RHU and refunded only upon graduation.
[3] NSSF subscription: An annual fee payable by Lebanese students who are not registered with the NSSF.
[4] Deferred Payment: Payments of tuition fees will be scheduled over four installments for the fall and spring semesters and two installments for the summer semester
[5] Late Payment Fee: An additional charge for any installment not paid by the established deadline.
[6] Regular Shuttle Fee: Beirut <-> RHU ; Saida <-> RHU ; (Mokhtara <-> RHU ; Iqlim <-> RHU when applicable).Non-refundable from week two.
[7] Dorm Deposit Fee: Refunded to the student upon the decision to leave the facility and only when the room furniture and assets are handed in free of damage.
[8] Dorm Late Payment: An additional Charge that may reach $100 upon excessive delay in payment. If the due fees are not paid, the residency will not be renewed.

Other Notes:

1- Students with financial holds will not have access to their grades and cannot register during the regular registration period.
2- Students with outstanding financial holds are not eligible to get any official document from the university unless the holds are cleared.

Payment Methods

Students must pay all tuition balances and other university fees by the announced deadlines to avoid late payment fees and the risk of courses being dropped. Once the student registers for a course, he/she shall remain accountable for all applicable tuition and fees until he/she drops the course formally by the drop/add deadline. Refer to the academic calendar at www.rhu.edu.lb for relevant deadlines.

Payments must be made in cash by the announced deadlines through one of the following methods:

1. At any branch of BankMed
2. RHU Finance Department

Students must download their statement of fees from RHU's Student Information System (SIS), print it, and present it to the Bank/Finance Department at the time of payment.

All student payments will be automatically deferred over four payments for the fall and spring semesters and over two for the summer semester.

Tuition Refund

In case any student decides to drop an already paid course (s), the following will be applied:

1. During the drop-and-add period, paid fees for the dropped course(s) will be credited to the student's statement of fees in full.
2. No refund will be applied after the drop and add period unless the student withdraws due to emergent circumstances.
   Refer to the withdrawal section of the catalog for more information.

If a student decides to withdraw from the University officially, then the following will be applied:

1. If a student has paid the total tuition and fees:
   - Before the official beginning of classes, 75% of the tuition and fees are refunded.
   - On the first day of classes and during the Drop and Add period, 50% of the tuition and fees are refunded.
   - After the Drop and Add Period, no refunds.
2. If a student benefits from the deferred payment program, no refund of tuition and fees is applicable.

Clearance and General Deposit Refund

Upon completion of the graduation requirements, the student must fill out a clearance form, which can be obtained from the Registrar's Office. The general deposit can be refunded upon completing the form and securing the required signatures.

The dorm deposit is refundable upon graduation and leaving the dormitory facility.
P.S. All deposits will be refunded after deducting any outstanding balances.

LIBRARY

Block E
Phone: 00961 25 603090; Ext: 435
E-mail: library@rhu.edu.lb
www.rhu.edu.lb/library

The Library at Rafik Hariri University supports students, faculty, and staff with the resources and services they need to succeed in their educational and research undertakings. The library was founded in 1999. The building occupies more than 650 square meters of space, offering users ample study and computing commons, halls for meetings, seminars, conferences, and classrooms. The library adheres to international standards and implements modern technologies across all functions and services. The RHU library provides academic programs with books, journals, and multimedia resources. It also has a modern IT infrastructure, which includes:

- An integrated library system, WMS (WorldShare Management Services)/ OCLC, that embeds all library functions, services, and auxiliary tools that can be accessed on and off campus.
- Workstations located in the reference area and connected to the Internet to access online catalogs, e-databases, Turnitin, and other resources
- A multimedia room equipped with state-of-the-art modern tools, computers, video projectors, data shows, CD writers, scanners, etc.
- Wireless connectivity covers all library spaces.

RHU continues to develop branding and library resources and services in response to changes and emergent needs, participates proactively in the enrichment of the educational experience, improves information technology and multimedia assets, repurposes available space to accommodate changing library roles, improves community outreach and branding, and improves administrative structure and commits to due process.

Information Resources

The RHU Library collections include a wide range of information resources to support all academic disciplines and meet the needs of university faculty and students. In addition to maintaining traditional printed resources, such as books and periodicals, the library holds a collection of electronic resources and multimedia materials (including open-access databases). The library home page provides links to full-text online databases, e-journals, e-books, research databases, and online catalogs. The homepage also includes information about library policies, service hours, auxiliary tools, and services. See https://rhu.edu.lb/Library.

Organization of Materials

Library resources are organized in accordance with international standards and best practices. All print resources are arranged and cataloged according to Dewey Decimal Classification 21 and MARC21 formats and standards for building bibliographic databases.

Library Services

The library staff assists users in using library resources and responds to inquiries received in person, by phone, by email, or through the library home page electronic forms. The staff also organizes specific training sessions to help users improve their skills in searching for and retrieving relevant information. Faculty, students, and staff with valid ID cards may borrow library materials in accordance with the circulation policy. Faculty members may request that books and other materials be placed on reserve for students to use within the library premises for a limited time. Computer commons provide users with on-site access to and retrieval of resources. Media resources, copiers, and printers are available inside the RHU Library. Prepaid cards may be purchased to use the printing and photocopying machines.

Also, to expand resources beyond those available on campus, improve services, and support research, the RHU Library joined the Interlibrary Loan Consortium and signed an agreement with the Baakline National Library. These agreements will facilitate resource sharing among libraries by supplying books, e-book chapters, articles, and other materials.

Interlibrary Loan

RHU students can borrow, free of charge, books, documents, and articles that are available in the libraries of the American University of Beirut, Beirut Arab University, Haigazian University, Lebanese American University, Middle East University, Notre Dame University-Louaize, United Nations - Economic and Social Commission for Western Asia, Université Saint-Esprit de Kaslik, Université Saint-Joseph de Beyrouth, and University of Balamand.

Baakline National Library

RHU students can visit the Baakline National Library and borrow books, documents, and articles free of charge.

Library Clearance

To obtain library clearance, all graduates must submit a soft copy of their senior projects in RHU Library format (Turabian format). Graduates should also ensure that all books and other borrowed materials are returned to the library before clearance.

INFORMATION TECHNOLOGY DEPARTMENT

IT Helpdesk
Phone: 00961 25 603090 Ext: 333
E-mail: helpdesk@rhu.edu.lb
www.rhu.edu.lb

The Information Technology Department exists to provide optimal IT solutions to support the University students, faculty, and staff in their quest to achieve excellence in teaching, learning, research, and service. It actively engages the teaching and learning processes by introducing multimedia solutions and mobile technologies, creating a smart classroom environment, and supporting effective utilization of the learning management system.
The IT Department also supports the administrative divisions in adopting technologies that improve users' productivity in executing University functions and managing all University information systems. Additionally, the IT Department maintains and upgrades all hardware, software, and applications that support the University's academic and administrative functions. The IT Department also manages infrastructure planning and maintenance and supervises related projects.

Operation and Network Services

The IT Department runs a state-of-the-art Data Center that houses computers and related telecommunications and storage systems, including the latest-generation servers, Virtual Desktop Infrastructure, Storage Area Networking, and Intrusion Prevention System. The center operates redundant power sources, data communications connections, and security devices.

The center includes numerous servers that provide all students, faculty, and staff with storage space and privileged access to network-based data, software resources, and Internet and e-mail services.

The network serves all students, faculty, and staff. It covers over nine buildings of classrooms, offices, and student and faculty dorms. The Gigabit Ethernet fiber-optic backbone network is connected to the Internet via a 150 Mbps line to provide bandwidth for quality services. The network has over 840 voice/data points and 40 or more wireless access points covering all university buildings. At present, nearly 400 computers make up the University network.

Management Information System (MIS)

The Management Information Systems (MIS) Section includes a highly competent team that manages the University Information Systems. The MIS is responsible for developing, administering, supporting, and providing service management for the University Enterprise Resource Planning (ERP), including the Student Information System (CampusVue), student accounting, finance, payroll, human resources systems, library and archiving systems, staff work log system, and an ID card system. Furthermore, the

MIS manages information flow and generates reports for various university divisions to support planning and decision-making.

University Website

The RHU website is a vital communication platform for the University and a primary source of information for prospective and current students, faculty, staff, and other audiences. Managed via a responsive content management system (CMS) under the system administrator's supervision, the website ensures that all content is accurate, up to date, and easily accessible.

In addition to serving as an information hub, the RHU website provides users with a wide range of online services, including the Student Information System, Moodle, an Online Payment System, Surveys and Voting tools, an Alumni Section, the latest Press Releases, Events and News, Career opportunities, Downloadable Applications, and more.

Instructional and Classroom Technologies

A team of dedicated specialists provides technology-aided teaching that meets the learning needs of faculty and students. It maintains multimedia assets in more than 32 classrooms and intelligent classrooms. The team manages Moodle – a Learning Management System (LMS) - which provides the platform through which faculty and students actively engage with course material and improve students' ability to attain expected learning outcomes. New tools simplify the creation of online courses and enable the conversion of documents into dynamic objects.

IT Helpdesk Support Services

The IT Helpdesk team provides a wide range of support services for RHU students, faculty, and staff to ensure that technology-aided teaching, learning, and administrative processes run smoothly. These services include, but are not limited to, support for all IT peripherals (hardware and software) in offices, classrooms, and computer labs; user account management; e-mail services; internet and Wi-Fi services; file-sharing management; surveillance camera management; access control management; ID card printing; etc.

The Helpdesk team uses the Ticket Management System (TMS) as its primary communication channel with RHU faculty and staff. TMS provides a transparent and interactive platform for staff and faculty members to send and save requests (tickets) to report incidents and follow up on them. Also, TMS provides a space for end-users to provide satisfaction feedback on the service they receive from the ITC personnel in each ticket.

QUALITY ASSURANCE AND INSTITUTIONAL ADVANCEMENT

Ms. Mirna Talhouk, Associate Director
Block E
Phone: 00961 25 603090, Ext. 443
E-mail: qaia@rhu.edu.lb
www.rhu.edu.lb/qaia

Established in 2014, the Quality Assurance and Institutional Advancement (QAIA) Office plays a pivotal role in fostering a culture of excellence, accountability, and continuous improvement across the university. The office is dedicated to enhancing institutional effectiveness through evidence-based decision-making, strategic planning, and rigorous quality assurance practices.

QAIA serves as a central hub for collecting, analyzing, and utilizing institutional data to support academic and administrative units in achieving their goals. By aligning quality assurance efforts with institutional priorities, QAIA ensures consistency, transparency, and innovation in all aspects of university operations.

Core Functions

- **Institutional Research & Data Analytics**
  Design and implement surveys, focus groups, and assessment tools to gather and analyze data that informs strategic decisions.
- **Reporting & Insights**
  Develop comprehensive reports for university leadership to guide planning, policy development, and performance evaluation.
- **Institutional Effectiveness Coordination**
  Harmonize assessment activities across academic and administrative units to ensure alignment with institutional outcomes and standards.
- **Data Quality & Governance**
  Identify gaps in data collection and utilization and establish protocols to ensure data integrity and reliability.
- **Knowledge Management**
  Promote the effective use of internal and external assessment data to support continuous improvement and informed decision-making.
- **Evaluation & Assessment Frameworks**
  Design and update evaluation methodologies tailored to different program categories and validate development plans for quality assurance.
- **Accreditation Support**
  Provide guidance and resources to academic units in preparing for national and international accreditation processes.

STUDENT AFFAIRS

Mrs. Sahar Hallak, Associate Director
Block E
Phone: 00961 25 603090 Ext. 770/777
Email: studentaffairs@rhu.edu.lb
www.rhu.edu.lb/studentaffairs

The Student Affairs Office is responsible for many functions that support the non-academic aspects of university life, improve students' potential for academic success, and help them attain a meaningful, holistic experience. The Office manages student clubs, promotes athletic and creative activities, organizes cultural events, oversees housing facilities, provides counseling services, holds student elections, and organizes student orientation programs, in addition to handling Work-Study and Financial Aid Programs. A brief overview of the various functions and services is presented below.

Cultural, Social, and Artistic Activities

RHU provides a supportive environment that encourages and enables students to participate actively in extracurricular activities. The Student Affairs Office works closely with student clubs, societies, and councils to enrich students' lives outside the classroom. By organizing a broad spectrum of athletic, cultural, social, artistic, and theatrical activities and events that are physically, psychologically, and socially constructive, students can advance a cause and practice a hobby they are passionate about. In doing so, students can connect with like-minded peers, build friendships, and develop leadership skills.

Counseling and Coaching

The Student Affairs Office aims to create conditions that promote students' social and psychological comfort, which is essential to achieving academic success. The Student Affairs Office supports students in overcoming obstacles and challenges during their university experience. The office organizes seminars and workshops to raise students' awareness and help them develop skills to cope with social, psychological, and emotional challenges. The office also provides one-to-one counseling to help students address personal issues and anxieties and make appropriate decisions. For more information, please contact the Student Affairs Office at studentaffairs@rhu.edu.lb or 00961 25 603090, Ext: 777.

Student Representations

We at RHU strongly believe in students' participation in the decision-making process on issues that matter to them. The Student Representative Committee (SRC) was

established to ensure that students' voices, opinions, and concerns are heard and influence the outcomes of related decisions. Elections are held annually, and students choose their representatives on the SRC transparently and democratically.

Student Housing

There are two dormitories at Rafik Hariri University, one for the female students and the other for the male students.

The employees working in the students' dormitory yearn to provide comfortable, safe housing for international students and dorm residents who study away from home. Each dorm features a social sit-in area, a study room with computers connected to the internet and library resources, and broadband Wi-Fi coverage.

The University is keen to make the on-campus living experience an opportunity to strengthen ties and build lifelong friendships among all students. The Student Affairs Office organizes sports, cultural, and social events exclusively for students living on campus. For more information on housing services, please contact us at studentaffairs@rhu.edu.lb or 00961 25 603090, Ext: 777.

Athletics and Recreation

RHU's indoor sports center and outdoor tennis, football, and basketball courts provide ample opportunities for students, faculty, and staff to participate in their favorite sports and hobbies and to meet their training needs. A team of specialists organizes athletic activities and tournaments to engage students and help them develop skills in the sports they are passionate about. Varsity teams are formed and supported to participate in local and international competitions. For more information, please contact the Athletic Department at 00961 25 603090, Ext: 330.

Ushers

This program trains student volunteers and involves them in organizing university-sponsored events and conferences on and off campus. Ushers play a vital role in the success of open-house activities, graduation ceremonies, and orientation for new students. Ushers are trained to guide students, parents, and guests and make them feel at ease during their visit to the campus. Contact us at studentaffairs@rhu.edu.lb or 00961 25 603090; Ext: 777.

Transportation

Free bus shuttles transport students from the dormitory to the highway in Damour. Shuttle bus services are available to transport students to and from the RHU campus from several areas in Lebanon.

The buses run the routes several times per day. For more information about the bus schedule, please contact the Student Affairs Office at studentaffairs@rhu.edu.lb or 00961 25 603090, Ext: 777.

Student Centers

There are student lounges where students can relax and enjoy activities such as chess, ping-pong, and other forms of entertainment. Each dorm also has two common rooms, one for social gatherings and the other for studies.

Food Services

The cafeteria at Rafik Hariri University serves a variety of traditional and international cuisines at reasonable prices throughout the day and into the evening. RHU also has ample, comfortable seating areas to enjoy your meal in the picturesque outdoor setting.

Health Care

The University Medical Clinic has a resident nurse who provides preliminary healthcare services to RHU students, faculty, and staff around the clock. The clinic is in Block "I" next to the bookstore. Ambulance services are also available around the clock if a situation needs attention beyond the clinic's available resources.

Student Rights and Responsibilities

Rafik Hariri University (RHU) pledges to students ***the right to***:

- be challenged to enrich the capacity for critical judgment;
- receive support in pursuit of answers and meanings;
- study in an inspiring and supportive learning environment;
- receive proper academic advising and mentoring;
- be treated with dignity and respect by all;
- receive a clear and informative course outline at the start of the semester and be informed of any modifications after that,
- receive constructive feedback on coursework within a reasonable timeframe;
- be assessed on the merits of academic performance without prejudice or other discrimination;
- ensure the confidentiality of academic and personal information;
- access all information and documents;
- express an opinion and convey grievances without fear of retribution; and
- protest the violation of rights, appeal a judgment deemed unfair, and receive and understand related decisions;

And expects students to accept ***responsibility to***:

- understand, comply with, and safeguard the University bylaws and student code of conduct;
- set a purpose and drive to achieve it;
- study hard to learn and commit to forever learning;
- know the program study plan and graduation requirements;
- follow the course outline, attend all classes, and accomplish the course requirements in a timely and honorable manner;
- respect and adhere to established University deadlines;
- be courteous, respectful of diversity, and tolerant of others' beliefs and concerns;
- express concerns and grievances within the confines of civility;
- follow due process and react with reason in the face of conflict;
- protect University property and preserve campus beauty;
- embrace sustainable practices and use natural resources wisely; and
- Represent the University with honor and professionalism.

Expected Conduct

Students must behave morally, act responsibly, and work diligently to advance their lives and communities. In doing so, they will:

1. Work hard and intently engage in the educational experience.
2. Speak out respectfully and feel free to propose initiatives to improve the university experience.
3. Refrain from smoking except in designated spaces.
4. Help maintain the cleanliness and beauty of the premises.
5. Place trash and recyclable items in designated receptacles.
6. Protect University facilities that are built for their benefit.
7. Never cheat or claim credit for work that is not your own.
8. Respect diversity and be tolerant of all points of view that are different from their own.
9. Understand the rights and responsibilities and appreciate the premise on which they are founded.
10. Adhere to the university bylaws and policies and assist in implementing them.

Work Study Program

The University allows students to partially support themselves by working up to 8 hours per week. In addition to earning a modest income, the Work Study (WS) program helps students gain work experience, improve communication skills, and increase future job prospects. For more information, please get in touch with the Office of Student Affairs at studentaffairs@rhu.edu.lb or 00961 25 603090 , Ext: 777.

Financial Aid Program

RHU believes every student should have the full opportunity to continue their education, regardless of their living conditions. For that purpose, RHU offers financial assistance to qualified students who cannot afford their tuition. For inquiries about this program, please contact the Student Affairs Office at studentaffairs@rhu.edu.lb or call 00961 25 603090, Ext: 777.

REGISTRAR'S OFFICE

Nidal Khalaf, Registrar
Amani Baasiri, Associate Registrar
Block B, Room 102.
Phone: 00961 25 60390, Ext: 615/618.
Email: registrar@rhu.edu.lb
www.rhu.edu.lb/registrar

The Registrar's Office administers academic processes, enforces academic regulations, and tracks students' academic progress. The office coordinates closely with academic advisors and academic units to guarantee the accomplishment of the following responsibilities: implementing updated study plans; maintaining student's academic records; enforcing University academic regulations, policies, and bylaws; tracking students' performance and generating students' statistical reports; preparing courses and exams schedules; assigning classrooms; registering students for the correct classes; issuing transcripts; verifying graduation lists; ensuring up-to-date CampusVue records and information; and updating and publishing University calendar and catalog.

The Registrar's Office uses the CampusVue Student Information System to implement academic processes and maintain students' records in coordination with the MIS team at the IT Center. Students and faculty have CampusVue personal secure accounts to access relevant documents, courses, schedules, grades, etc.

For registration procedures, deadlines, and related issues, please refer to the Registration section in this catalog or visit www.rhu.edu.lb/registrar

Registration

After consulting with their academic advisors, students must register for the courses they intend to take during the registration period noted in the academic calendar, presented in this catalog, and posted on the University website: www.rhu.edu.lb/. During the registration period, academic advisors assist students in preparing course schedules by completing the Course Registration Forms and activating their personal online CampusVue registration. Copies of the Course Registration Forms must be sent to the Registrar's Office after the drop/add period to verify students' schedules and for archiving. Any course registered online by the student without prior approval of the academic advisor shall be automatically dropped.

The course schedule must comply with the following rules: the prerequisite(s) of the courses the student intends to take were completed, the courses are required by the program and are taken in the order noted in the program's study plan, failed courses are taken when first offered, and the maximum and minimum load rules are met. After completing registration, the student must print the statement of fees and pay all applicable tuition and fees by the announced deadlines. Failure to meet financial commitments may result in the class being dropped from the schedule.

National Social Security Fund (NSSF)

The Lebanese Law mandates that all Lebanese students be registered with the NSSF. This requirement does not apply to non-Lebanese students. A Lebanese student admitted to RHU must complete the Social Security Application form dispatched with the admission letter. On the form, the student indicates:

- NSSF status (beneficiary or not)
- NSSF number if already registered (employed students only)
- NSSF number of parents (if beneficiary)
- An official attestation of the type and source of all benefits, including NSSF.

Refer to the Student Services section for more information on NSSF matters.

Passport and Visa

Expat students admitted to RHU must have their passports valid for 13 months from the date of joining RHU and obtain a visa from the Lebanese Embassy in their country of origin.

ACADEMIC REGULATIONS

The following section presents the academic regulations applied to bachelor's degree programs. These regulations are meant to assist students, academic advisors, administrators, and staff in making appropriate decisions. The President, Vice Presidents, Deans, and the Registrar's Office shall be responsible for their implementation. The University Academic Board is authorized to rule on cases not covered in these regulations and mitigate problems arising from their application.

Pleading ignorance of these regulations or of related publications and announcements posted on catalog boards in various campus buildings should not exonerate students from the consequences of violating them.

Student petitions to circumvent academic rules must be verified by the academic advisor to ensure the information's authenticity before being forwarded to the responsible entity for further action.

Duration of Study

Minimum Period

To earn a bachelor's degree, a student must study for a minimum period while enrolled at the RHU, as indicated in the table below.

Minimum duration to earn a bachelor's degree		
College of Arts and Sciences	Bachelor of Arts Bachelor of Science	Three years
College of Business Administration	Bachelor of Business Administration	Three years
College of Engineering	Bachelor of Science	Three years (equivalent to 6 regular semesters plus three summer semesters)
	Bachelor of Engineering	Four years (equivalent to 8 regular semesters plus three summer semesters)

A regular semester equals a minimum of 12 earned credits for transfer purposes. To earn a bachelor's degree, a transfer student must study at least three regular semesters at RHU for a three-year program or four regular semesters for a four-year program and acquire Co-op work experience.

Maximum Period

The maximum study period to earn a bachelor's degree, excluding semesters of approved deferment of study, is ten regular semesters for the Bachelor of Science,

Bachelor of Arts, and Bachelor of Business Administration degrees and 12 regular semesters for the Bachelor of Engineering degree.

If, for compelling reasons, a student cannot complete the bachelor's degree within the allotted period, a request for extension may be submitted through the college council for approval by the University Academic Board.

Student Class Level

The following criteria define the student's class level:

First year: 00-29 credit hours
Second year: 30-65 credit hours
Third year: 66-99 credit hours
Fourth year: 100 or more credit hours

Course Load

Minimum Load

The minimum semester load is 12 credits, unless the student requires fewer credits to graduate. In exceptional cases, the College Dean may approve a reduced load of nine credits upon the recommendation of the Academic Advisor.

Maximum Load

The maximum load in a regular semester shall be 18 credits, which may be increased to 19 upon the Dean's approval. A student may be allowed to take a maximum of 21 credit hours if s/he: (1) has a CGPA of no less than 85, (2) needs 21 credits to graduate, and has a CGPA of 73 or above.

The maximum load in a summer semester shall not exceed nine credits. A 10-credit load may be allowed if the student: (1) has a CGPA no less than 85, (2) needs ten credits to graduate and has a CGPA of 73 or above, or (3) the schedule includes a one-credit course or lab course.

The maximum load for a first-year student is 15 credits in the fall or spring semesters and nine credits in the summer semester.

Intensive English Program (IEP) Students

In addition to the English courses, students in the IEP may take courses required by the intended program of study as listed below.

IELP 085	IELP 095	IELP 100
A maximum of 10 Credits	A maximum of 10 Credits	A maximum of 13 Credits
College of Business Administration		
BITM 200; or MATH 189 or MATH 207; BADM 225; and ARAB 212	BITM 200; or MATH 189 or MATH 207; BADM 225; and ARAB 212	BITM 200; or MATH 189 or MATH 207; BADM 225; BACC 205; BADM 250; ARAB 212
College of Engineering		
MATH 190 or MATH 215; PHYS 190/ PHYS 191 or PHYS 210/ PHYS 211; and CIVE 211 or CCEE 221	MATH 190 or MATH 215; PHYS 190/ PHYS 191 or PHYS 210/ PHYS 211; and CIVE 211 or CCEE 221	MATH 190 or MATH 215; PHYS 190/ PHYS 191 or PHYS 210/ PHYS 211; CIVE 211 and/or CCEE 221; and CCEE 214
College of Arts and Sciences		
COSC: MATH 190 or MATH 210; COSC 214; and Sciences / Technology elective **GRDS:** GRDS 220; FADR 220; and FADR 200 **ENGL:** ARAB 212 and any two college-required courses	**COSC:** MATH 190 or MATH 210; COSC 214; and Sciences / Technology elective **GRDS:** GRDS 220; FADR 220; and FADR 200 **ENGL:** ARAB 212 and any two college-required courses	**COSC:** MATH 190 or MATH 210; COSC 214; Sciences/ Technology Elective; and BMKA 200 **GRDS:** GRDS 220; FADR 220; FADR 200; and FADR 215 **ENGL:** EDUC 222; PSYC 301; and any two college-required courses
Freshman Program		
Arts: MATH 189; ARAB 180; or BECN 190, Natural Science Elective **Science**: MATH 190, PHYS 190, ARAB 180, or Natural Science	**Arts**: MATH 189; ARAB 180; or BECN 190, Natural Science Elective **Science**: MATH 190, PHYS 190, ARAB 180, or Natural Science	**Arts**: MATH 189; ARAB 180; or BECN 190, Natural Science Elective; and Social Science Elective **Science**: MATH 190, PHYS 190, ARAB 180, or Natural Science; Social Science Elective

Schedule Modification

The student may modify the course schedule after registration by adding or dropping courses during the official drop-and-add period noted in the academic calendar. The student completes the Schedule Modifications Form, obtains approval from the academic advisor and the department chairperson, and settles any resulting financial obligations immediately after completing the transaction in RHUSIS. If the course modification results in a load of less than 12 credits or more than 18 credits, approval of the Dean is also required.

Dropping Courses

Students who do not plan to continue in a course must drop it during the drop/add period; otherwise, they remain financially accountable for the course. A course dropped during the official drop/add period will be deleted from the student's schedule. The student will be relieved from the financial liability associated with the deleted course. If a refund is due, the provisions of the refund policy will apply.

Adding Courses

Students may add one or more courses during the drop/add period. The student must settle the financial obligations for the course(s) by the prescribed deadlines.

Late Registration

Students who could not register during the regular registration period may register during the drop/add period and pay a late registration fee. Due to urgent circumstances, the fee may be waived for new students or those who were unable to register earlier.

Repeating Courses

The student must retake all failed courses when they are first offered. The student may also repeat a course in which s/he passed with a grade of “70” or lower only once to improve the CGPA. While all repeats shall remain on the student’s record, the credits of a repeated course shall count only once, and the highest grade shall be used to compute GPA. A course may not be repeated more than two times, including withdrawals.

Substituting Courses

Upon the approval of the college Dean, a student may be allowed to take a substitute for a required major course if:

- The required course is not offered or conflicts with another required course in the semester a student is expected to graduate.
- The student failed the course three times, provided that the student is not on probation. The required and the substituted course grades shall appear on the student’s transcripts and be used to calculate the student’s CGPA.

The substitute and required courses shall meet the following criteria: the number of credits and level of the substitute course must be the same or higher; the course is in the same or a closely related field; and the course content and expected competencies are equivalent.

Undergraduate Students Taking Graduate Courses

An undergraduate student may be allowed to register for a graduate course in substitution for a required undergraduate course not offered in the semester in which the student graduates, provided that the student's CGPA is not less than 75.

In all cases, the student must obtain the college dean's approval.

Course Syllabus

The student is entitled to receive a detailed course syllabus at the beginning of the first class meeting. The syllabus informs students of the course's salient features and rules, helping them plan accordingly. The syllabus should include the following components: Information about the instructor – Name, contact information, office location and hours, and the manner and medium of communicating with the students; information about the course - title, brief description, prerequisites, topics covered, learning outcomes; course resources - textbook, web links, references, technology tools, and instructional materials; course activities and related deadlines - homework, quizzes, projects, research papers, presentations, group work; etc.; assessment tools and grade distribution; mapping of assessment tools with learning outcomes ensuring assessment triangulation; statement to inspire students' engagement, communication, motivation, and self-expression; and class policy on attendance, exam make-up, etc. Course instructors are expected to use Moodle resources to manage the learning process.

Attendance

The instructor must state the attendance policy in the course syllabus. The student must attend all class meetings and course-related activities and is ultimately responsible for truancy consequences. Instructors must take attendance regularly and advise truants of the ramifications of missing classes. Suppose the number of absences reaches the 15 percent mark. In that case, the student must voluntarily withdraw from the course by following the proper withdrawal procedure and receive a W grade (see the Withdrawal section). If the student does not withdraw willingly after accruing 15% of the semester's absences, the instructor completes a ***forced withdrawal form*** and submits it to the dean for approval by the end of the tenth week of classes. The Dean's Office forwards the form to the Registrar's Office for action. Students representing the country or the University in official activities shall be allowed up to 20% absences. Suppose the absence is due to ill health or other difficult circumstances. In that case, the student must provide the course instructor with evidence and a plan to meet the course requirements by completing the ***attendance exemption form***. All absences, excused or otherwise, shall be counted, and the student will ultimately be accountable for the missed work.

If the course instructor does not require attendance at all class sessions, the applicable attendance policy for that course must be approved by the college dean and clearly stated in the course syllabus.

Assessment and Examinations

The College Council shall establish, regularly review, and update the course evaluation and examination policies and procedures governing the college's courses. A summary of the current policies and procedures is outlined below.

Performance Assessment

Students are allowed to assess their work and receive continuous, constructive feedback in line with university guidelines. The assessment of student performance is based on the extent to which the course outcomes stated in the syllabus are achieved. Assessment instruments may include, but are not limited to, class participation, blogs, wikis, homework, examinations, quizzes, research papers, projects, and practical work. Instructors are required to complete a midterm performance assessment and report the results before the withdrawal deadline. This process provides underperforming students with the opportunity to withdraw if necessary and enables faculty advisors to offer more effective guidance during the registration period.

Examinations

The course syllabus must state the number and dates of exams to be given during the semester. At least one major exam should be given before the withdrawal deadline to allow the student to decide whether to continue or withdraw from the course.

The instructor is responsible for preparing written, properly weighted exam questions aligned with the course content, language of instruction, learning outcomes, and the allotted exam time stipulated in the course syllabus. An appropriate answer booklet should be given to the students with clear instructions on exam rules. The instructor shall correct the exam booklets, return them to the students within a week after the exam, and report the grades in letter and percentage form to the Department Chair a week later.

The final examinations are given during the period approved by the University and noted in the academic calendar. Instructors may need to make individual arrangements for students with more than two final examinations scheduled on the same day. A student shall not be allowed to sit in for the final examinations without settling all his/her financial obligations to the University.

Absence from the Final Exam

If a student fails to attend a final examination without a valid excuse, the student shall receive a zero on the exam. If the student's absence from the exam is due to circumstances beyond the student’s control, the student may submit a petition to the Dean of the relevant college. If the Dean concludes that the reasons are justifiable, s/he informs the Registrar’s Office to record an incomplete “I” grade for the course. In coordination with the relevant department and course instructor, the student prepares to take a make-up exam to replace the “I” before the drop/add period of the following

semester begins. If these circumstances persist, the Dean may recommend changing the “I” grade to “WE.” For more details, refer to this catalog's ‘Incomplete Work’ section.

Make-up Examination

The course syllabus shall state the policy on missed examinations and quizzes so that students are fully aware of it and its consequences. Usually, a student shall receive a grade of zero for the exam or quiz s/he misses. Suppose the absence is due to a legitimate excuse for which verifiable evidence is presented. In that case, the course instructor may give the student a make-up exam or shift the weight of the missed exam (except the final exam) to other course assessment items.

Course Grades

Course Grade Distribution

The syllabus must clearly state the course components, topics, associated assessment criteria, and grade distribution. The course grade is distributed such that appropriate weights are given to exams, quizzes, homework, projects, research papers, and student participation. In all cases, the final exam grade should not exceed 35%.

Submission of Final Grades

The instructor shall enter the grades into RHUSIS, prepare the final course grade report in accordance with the established format, and submit the report to the Department Chair within two working days of the scheduled final exam time. Within 24 hours afterward, the department chair verifies course grade reports, secures the dean’s approval, and forwards them to the Registrar’s Office. The Registrar’s Office audits the grade reports and records them.

Appeal of Course Grade

If a student feels that the grade s/he has attained in a course was unfair, s/he should discuss the matter with the instructor within five days of the grade being posted by the Registrar’s Office. Suppose the student and the instructor are unable to resolve the issue. In that case, the student may submit a “**Course Grade Review Petition Form**” to the Chairperson of the concerned department within two weeks after the grade is posted. If the student’s concern is legitimate, the Dean shall form a committee consisting of the department chair, the course instructor, and one other faculty member from the same department to investigate the request and adjust the grade in accordance with standard procedures.

Incomplete Coursework

Incomplete coursework is subject to the following rules:

- A student who has completed most of the coursework satisfactorily but, for a verifiable compelling reason, is unable to complete all course requirements within the standard period may request, before the beginning of the final exam period, an extension to complete the unfinished work by submitting a "**Request for Time Extension Form**." If approved, an "I" grade will be temporarily assigned to the course. The instructor computes the course grade, with a zero assigned to the incomplete work, which shall be transferred to the course if the unfinished coursework is not completed by the deadline.
- The incomplete grade must be removed before the beginning of the drop/add period of the following semester in which the student is registered. After the unfinished work is completed and evaluated, the course instructor completes a change-of-grade form, submits it to the dean for approval, and then sends it to the Registrar's Office for final action. The final semester GPA will be calculated as soon as the final grades for the incomplete courses are submitted.
- Credit for an incomplete course will be assigned to the semester in which the course was taken.
- Graduation requirements shall be considered complete, and the student is cleared for graduation when all Incompletes are resolved.
- If the circumstances for receiving an "I" grade persist, then a grade of "WE" will be assigned to the course upon the recommendation of the Dean of the concerned College.

Change of Grade

A course grade cannot be changed after it has been submitted. If extreme circumstances warrant a grade change, the course instructor explains the reasons on a "**Change of Grade Form**" and submits it through the department chairperson to the dean for approval, then to the Registrar's Office for final action within a maximum of two weeks of grade posting. Grades cannot be changed after the bachelor's degree is awarded.

Grade Point Average (GPA)

Semester GPA

The semester GPA is computed as follows:

1. Determine the course quality points by multiplying the number of course credit hours by the numeral (see the "Grading System" section below)
2. Add the total quality points and the total number of corresponding credit hours.
3. Divide the total quality points by the total credit hours.

The higher grade of a repeated course is used in the computations. A "W" grade is not included in the GPA computations.

Cumulative GPA (CGPA)

The cumulative GPA is computed as follows:

1. Determine the quality points of all courses taken
2. Add the total quality points and the total number of corresponding credit hours.
3. Divide the total quality points by the total credit hours.

Grading System

Numeric grades are used to represent student performance in a course. The letter grades are calculated according to the scheme outlined below. The minimum passing letter grade is D.

Numeral Grade	Letter Grade	Grade Value	Description
≥ 95	A+	4	Distinguished
≥ 90 and < 95	A	3.85	Excellent
≥ 85 and < 90	B+	3.75	Very Good
≥ 80 and < 85	B	3.40	Good
≥ 77 and < 80	C+	3.00	Fair
≥ 73 and < 77	C	2.70	Average
≥ 70 and < 73	C-	2.35	Satisfactory
≥ 65 and < 70	D+	1.85	Minimal Pass
≥ 60 and < 65	D	1.25	Weak
< 60	F	Zero	Fail
Other grade designations			
I	Incomplete		
P	Passing grade		
PH	Pass with Honor		
W	Withdrawal		
WE	Excused withdrawal		
CIP	Course in progress		
CR	Credit Granted		

Honors and Distinction

Semester Honor's Lists

At the end of each semester, the University recognizes students with distinguished academic achievements by enlisting their names on the Dean's Honor List and President's Honor List. To be eligible for a mention on the honor list, a student must satisfy the following ***general honor conditions***:

1. Is a full-time student

2. Has no “F” grade in any course
3. Does not have any incomplete grades
4. Has not been subject to any disciplinary action

Students whose semester GPA (SGPA) is 85 or above shall be placed on the Dean’s Honor List and receive a commendation from the College Dean. Students whose semester GPA is 90 or above are placed on the President’s Honor List and shall receive a commendation from the University President.

Graduating with Distinction

Students graduating with a cumulative GPA (CGPA) of 85 or above shall receive the designation “Distinction,” and those graduating with a CGPA of 90 or above shall receive the designation “High Distinction,” provided they satisfy the general honor conditions stated above. The designation shall be conferred on the students’ graduation certificates and official University transcripts.

Nazik Rafik Hariri Graduate Study Award

The two students among the graduating class with the highest GPA in the College of Business Administration and the College of Engineering shall receive the Nazik Rafik Hariri Graduate Study Award. The award is a full-tuition scholarship to pursue a master’s degree at RHU.

Withdrawal

Withdrawal from Courses

With the academic advisor's approval, a student may drop one or more courses during the drop/add period listed in the academic calendar. No record of the dropped course(s) shall appear on the student’s record.

A student, upon approval of the academic advisor, may withdraw from one or more courses between the second and the tenth week of classes of a regular semester or between the second and the sixth week of a summer semester, provided that the student’s load (excluding summer) does not fall below 12 credits. A grade of “W” shall be recorded on the student’s transcript for each withdrawn course, and the course shall be counted as a course attempt for purposes of the course repeat policy. Approval from the college dean is required if the withdrawal results in a load of less than 12 credits.

Students who withdraw from classes after the drop period are responsible for all related tuition and applicable fees.

Withdrawal from a Semester

Formal Withdrawal. In verifiably difficult circumstances, a student may petition to withdraw from all courses for the semester at any time after the drop/add period and before the final exams begin. After consulting with the concerned department

chairperson, the student's academic advisor, and the instructors of the courses, the College Dean may approve the petition if the reasons for withdrawal are legitimate. A grade of "WE" shall appear on the transcripts for each withdrawn course and shall not be counted as a course attempt in applying the course repeat policy. A student returning the following semester may register by following the routine registration procedures.

Informal Withdrawal. Suppose a student withdraws from the semester without following proper withdrawal procedures as described above. In that case, a grade of "F" shall be recorded on the student's transcripts for each registered course, and the course shall be counted as a course attempt in applying the course repeat policy.

If a student withdraws informally and later decides to return, approval from the relevant college is required. The college may approve the petition if convincing evidence is presented that the reasons for the withdrawal were beyond the student's control. If the college approves the request, the "F" grades recorded on the transcripts shall be replaced with a "WE." A "WE" grade is not considered when calculating the student's GPA, and the affected courses shall not be counted as course attempts for purposes of the course repeat policy.

A withdrawn student may only transfer credits for courses s/he has studied at another institution during that period if prior approval from the concerned college is obtained.

Withdrawal from the University

Students who wish to withdraw from the University are required to complete established clearance procedures.

Leave of Absence

During the first week of a semester, a student may petition for a leave of absence (LOA) for one or two consecutive semesters by submitting the Continuous Enrollment Registration Form to the Dean for approval and paying the USD 20 LOA fee for each semester on leave.

Students may accrue up to 3 semesters of LOA while pursuing a degree. The LOA semesters shall count toward decisions regarding the maximum number of semesters of allowed study deferment. A student returning immediately after the approved LOA period may register by following the routine registration procedures.

Suppose exigent circumstances prevent a student from returning immediately after an approved LOA period ends, and the student decides to reenroll in the University later. In that case, the student must seek approval for reinstatement from the college concerned. Re-enrolled students must follow the study plan and the catalog in effect at the time of re-enrollment.

A student on an approved LOA may transfer credits only for courses s/he studies at other institutions if prior approval to study those courses is obtained from the college concerned.

Special Enrollment

The Special Enrollment (SE) status applies to a student who has completed course requirements but needs to remain an active RHU student for a legitimate reason. A student who does not register as a special enrollment student for one or more regular semesters should apply for readmission to the program of interest. Refer to the Informal Withdrawal policy in this catalog.

To register under the SE status, the student should:

- Complete the Continuous Enrollment Registration Form stating the pertinent reasons and submit it to the Registrar's Office before the end of the Drop and Add Period. The academic advisor, the department chairperson, and the college dean must approve the form.
- Pay a Continuous Enrollment Fee of 20 USD at the RHU Finance Office for each semester under this status. A penalty of 10 USD shall be added if the fees are not paid during a regular payment period.

Non-Degree Students

A student who is not on track to earn a specific degree at RHU but wishes to register for courses they are eligible to take may do so as a non-degree student.

Further Studies

RHU offers students the opportunity to extend their knowledge by studying more than one major or earning more than one degree. Available options are:

1. Study a double major concurrently;
2. Earn two degrees simultaneously;
3. Return to the University after graduation for a second degree

Double Major

Students may concurrently study two majors within the same college that share similar degree structures. A student may apply for permission to study a second major if s/he is in good academic standing after completing at least 24 credits of coursework in the first major. Permission is granted if the student meets the admission requirements for the second major. Students must complete a minimum of 24 credits of core courses in the second major, in addition to the requirements of the first major, including all mandatory core courses of the second major. Upon graduation, a single diploma specifying both majors is awarded, provided that all requirements for each major have been completed. Credits for common courses are given to both majors.

Dual Degrees

Students may simultaneously pursue two different degrees from programs offered by other colleges, each with a different degree structure. Students may apply for permission to pursue a second degree if they are in good academic standing and have completed at least 24 credits of coursework in their first-degree major. The application is approved if the student meets the admission requirements for a second-degree major. Upon graduation, a diploma is granted for each degree major, provided that all the requirements of the respective degrees are completed, including the co-op experience. Credits for common courses are granted to both degrees, and the higher tuition rates for the two degrees shall be applied.

Second Degree

Students who have earned a degree and opt to return to the University at a later date to pursue a second degree may do so under the following conditions: (1) the admission requirements of the new degree are met; (2) the applicant may be required to repeat courses if they were taken more than three years ago or if the acquired grade is low as recommended by the department of the second degree.

Minors

Students can pursue a minor in an area different from their major under the following conditions:

- Obtain a Minor Cumulative Grade Point Average of no less than 70%
- Overlapping courses between student major requirements and minor requirements are counted to fulfill minor requirements
- A student may pursue up to two minors
- A maximum of nine credits can be counted towards the requirements of two minors

For further details, please refer to the specific college section.

Allowable Study Deferment Period

The permissible period of study deferment for a bachelor student, intentional or otherwise, shall be at most six semesters (summer excluded). If the six-semester limit is surpassed, the student shall be dismissed from the University. If the student decides to return later, s/he needs to apply for reinstatement, subject to approval by the relevant college. Reinstated students must follow the study plan and the catalog in effect at the time of reinstatement.

Change of Major

A student who has completed at least one semester of coursework in a major may request a major change at any time before the thirteenth week of a regular semester (fall or spring). The dean of the college (if in case of college change), in consultation with the department chairperson of the aspired program, decides on the request before the start of the following semester based on the following conditions: the request meets college admission requirements; a seat in the desired major is available; the student has not changed major before; the number of earned credits in the old major is less than 64 credits unless the student was forced out of the major for academic reasons. All required earned credits for the new major will be transferred and used to calculate the CGPA.

Usually, all earned credits required in the new major will be transferred and used to calculate the CGPA. However, if the student wishes, six credits may be deducted from the CGPA if the transfer is within the same college, or a maximum of twelve credits if the transfer is to a new college.

Intensive English Program students in IELP 095/100 Levels may request a major reassignment before completing 15 credit hours. Reassignment of major shall not be considered a transfer or a change of major.

Academic Complications

RHU is deeply committed to providing students with every opportunity to thrive, as their future success is our top priority. To ensure you can make the most of these opportunities, we encourage you to maintain open lines of communication with your instructors, whether in the classroom, via email, phone, social media, or our Learning Management System (Moodle). We also recommend regular visits to our Learning Support Centers, where you can seek help from your peers to overcome any difficulties you may face in your academic journey.

While RHU is dedicated to helping students achieve academic success, students must manage their time effectively and utilize available resources. Students can avoid the potential complications and consequences outlined below by taking responsibility for their academic responsibilities.

Academic Probation

Students receive academic probation if their CGPA falls below 70 after attempting at least 24 credits by the second or any subsequent matriculated semester, excluding summer sessions. Students with two consecutive probations may choose to:

a) Continue in the same major with the approval of the College Council, but risk academic critical standing if, by the end of the following semester, the student accumulates three consecutive probations.
b) Change major: If accepted into the new major, the transferred courses and CGPA will be determined according to the latest major's requirements. If the CGPA for the

transferred courses is 70 or higher, the probation status is removed; otherwise, it remains in effect.

The Registrar's Office shall generate a report of students on probation at the end of every semester and send it to the colleges, which will inform the students and the relevant department of their probation status.

A student may remove probation in a summer semester, but the results of that semester shall not result in probation.

Course Load. Usually, the maximum study load for a student on probation is 12 credits (6 in a summer semester). The load may be increased to 13 credits (7 in a summer semester) if one of the courses is a 4-credit course or a 1-credit required lab course. The load may also be increased to 15 credit hours (9 in a summer semester) if the student is expected to graduate at the end of the semester, subject to the approval of the concerned dean. A student on probation shall be denied access to online registration and must register through the academic advisor. A student on probation must repeat at least two previously taken courses with a grade of 70 or lower and must repeat failed courses when first offered. A student continuing probation must repeat at least three previously taken courses with a grade of 70 or lower and must repeat failed courses when first offered.

Critical Academic Standing

A student shall be placed in *Critical Academic Standing* if s/he:

- Accrues three consecutive probations or four probations throughout the entire period of study, or
- Attains a CGPA less than 60 more than once as a matriculated student.

A student placed on *Critical Academic Standing* by the end of a regular semester must register for up to 15 credits of repeated courses. These courses must be previously taken courses with a grade of 70 or lower, or failed courses.

Based on the student's performance at the end of the semester, the student shall:

1. Resume studying in Good Standing if s/he attains a CGPA of 70 or higher;
2. Continue at RHU on CAS if s/he attains a semester GPA of 75 or above;
3. Dismissed from RHU. The college shall inform the student and the concerned parties of the dismissal decision.

A student of critical academic standing shall be denied access to online registration and must register through the academic advisor.

Forced Change of Major

A student whose CGPA falls below 60 at the end of any semester after the first semester (at least 12 credits) as a matriculated student must transfer to another program and shall not be allowed to return to the old program.

Academic Conduct

RHU embraces the values of academic honesty and integrity and expects all to uphold strict ethical and professional standards. The University forbids any unauthorized use of others' work. Acts of plagiarism, cheating on exams, or other types of work submitted for assessment as part of a course grade shall be subject to possible disciplinary action. Please refer to the Conduct Policy section in this catalog.

RHU Student Taking Courses Elsewhere

Students formally enrolled at RHU may take up to 15 credit hours of courses at a recognized university within Lebanon accredited by the MoHE or at a university outside Lebanon accredited by the country's higher education authority, subject to the following conditions:

- The student is registered at RHU
- The student has completed at least 30 credit hours at RHU
- The courses to be taken elsewhere are equivalent to courses required by the student's major
- The student receives prior approval of the courses to be taken from the student's department chairperson and college dean
- A course is transferred to RHU credit if the grade attained is equivalent to 75 or better. The transfer credits do not count when calculating the CGPA.
- The student must submit certified transcripts to receive credits for transferred courses.

Usually, students are not allowed to take courses elsewhere during the last two semesters before graduation. However, a student may be allowed to take up to 7 credits in the previous semester of studies if the courses are not offered at RHU. Students cannot transfer credits for courses taken elsewhere while on a leave of absence or during formal or informal withdrawal periods.

Student Records and Transcripts

The University is committed to protecting the rights of its students and the privacy and confidentiality of their personal and academic records kept at the Registrar's Office. Only authorized personnel are entitled to access secured Students' records. Professors must post exam results using student ID numbers, not student names. Faculty advisors and academic administrators may access students' records to advise or make academic decisions. Official transcripts may be issued to a third party only if a signed authorization from the student is presented or in compliance with a judicial order. The Registrar signs official transcripts. Students may request an unofficial copy of the transcript or a record of their grades after two weeks of grade posting and at any time they need it. Students can access their records and see their grades anytime through CampusVue.

Student Petitions

If students need to submit a petition, it must be initiated with the help of their advisor. The appropriate form must be typed and completed using proper language. The advisor must verify the accuracy of the information before signing the petition and forwarding it to the department chairperson for approval. A student's transcript should be attached to the petition if the decision is hinged on the student's record.

Commencement Exercises

Commencement exercises are held after the conclusion of the spring semester. Students who complete graduation requirements in the summer or the previous fall may attend the commencement exercises.

After the commencement date, graduates who opt not to participate in the graduation ceremony may receive their diplomas at the Registrar's Office.

Graduate Names on Diplomas and Degrees

Names on diplomas and degrees are spelled exactly as they appear on passports, as required by the Ministry of Education and Higher Education (MoEHE). If, after admission, the name on the passport changes, the student must update the records by submitting certified evidence of the change to the Registrar's Office before graduation; otherwise, the old name will appear on the diploma and cannot be changed thereafter.

Co-op and Career Services

Phone: 00961 25 603090 Ext: 414
E-mail: coop@rhu.edu.lb
www.rhu.edu.lb/coop

Co-operative Training Program

The Co-op training program bridges the intense learning experience of university and the practical, results-oriented, highly competitive professional world. The program is designed to allow the student to test and apply knowledge, skills, and competencies acquired in classrooms and labs, and to develop a clear perspective on real-world expectations and a sharper focus on career choices. It allows employers and educators to collaborate in mentoring the prospective workforce to become productive and responsible citizens. The hope is that the student, after completing the Co-op experience, will develop a clear purpose, become more engaged during the final year of studies, and adopt a pragmatic perspective on their role in a changing world.

Significance

While the Co-op work experience mutually benefits the student, the university, and the employer, it will help the student in:

- Clarifying career goals;
- Understanding of the workplace demands and constraints;
- Pursuing continuous learning and self-improvement undertakings;
- Cultivating self-reliance and self-confidence;
- Refining time management, interpersonal, and communication skills;
- Building contacts with potential employers;
- Acquiring marketable skills that improve employability prospects;
- Experiencing practical aspects of the field of study.

General Policy

1. The logistical aspects of the Co-op program shall be managed by the Co-op Coordinator associated with the Office of Academic Affairs.
2. Training opportunities shall be approved and thereafter managed by the Co-op Coordinator in collaboration with the academic advisor.
3. Students must meet the eligibility criteria outlined in this policy to participate in the Co-op program.
4. Students may arrange for a training opportunity on their own, or the Co-op and Career Services Assistant will assist the student in finding a suitable training venue.
5. A student may opt to complete the Co-op training at a reputable institution abroad, subject to pertinent approvals.

6. Each trainee shall be assigned a Co-op academic advisor to follow up on the trainee's work progress throughout the Co-op period and assess the overall training experience afterward.
7. A student must commit to completing the training at the mutually agreed-upon company.
8. Students may register for one course during the Co-op period if it is the only course needed to graduate and the employer approves the timing in writing.
9. The Co-op and Career Services Assistant completes the Co-op training offers in consultation with the Co-op work HR/supervisor, considering the employer's needs, the student's interest and eligibility, and the date of the student's application.
10. The trainee shall work for the employer for a period equivalent to eight weeks, 8 hours per day, on mutually agreeable tasks related to the trainee's field of study.
11. Trainees must conform to the employer's work rules and standards and execute their tasks professionally and ethically.
12. The student shall directly report to and coordinate with the Co-op work supervisor on all tasks and activities.
13. The Co-op work supervisor cannot be a direct relative of the trainee;
14. The Trainee shall keep a daily record of tasks performed and lessons learned;
15. The trainee and the Co-op work supervisor shall submit the training assessment forms to the Co-op academic advisor one week after completing the training.
16. The trainee shall submit to the Co-op academic advisor the daily tasks log, a report, a poster, and a PowerPoint presentation immediately after completing the Co-op work.
17. The trainee must deliver a presentation about the training experience when requested as part of the Co-op Inspiration Seminar.
18. Students can only withdraw from the Co-op course upon approval from the Co-op academic advisor and the Co-op and Career Services Assistant.
19. Failing to comply with the Co-op rules and the employer's work standards will result in a failing grade.
20. The Co-op and Career Services Assistant and the Co-op academic advisor are always available to support the student in attaining a rewarding Co-op experience.
21. The Chairperson of the academic department is responsible for implementing the provisions in this policy related to the academic advisor and student responsibilities.

Requirements

Duration and Timing

The Co-op program runs for at least eight weeks, 8 hours per day, and is normally completed during the summer term of the third year of studies. Students engage in practical training under the mentorship of professionals in areas related to their field of interest.

Eligibility Criteria

A Bachelor's student is eligible to sign up for the Co-op program if the following eligibility criteria are met:

1. Completing ENGL 217 and the requisite number of credits with a GPA of at least 70.
2. Enrolling at the university during the training period.
3. Completing all college pre-training course requirements as stipulated below.

College	Credits	Prerequisite
Engineering	1	• Passes ENGL 217 • Senior Standing • Has a CGPA of 70 or higher
Business Administration	1	• Passes ENGL 217 • Senior Standing • Has a CGPA of 70 or higher
Arts and Sciences	1-3	• Passes ENGL 217 • Senior Standing • Has a CGPA of 70 or higher

Course Preparations

In preparation for the Co-op training, students must attend the Co-op Policies and Procedures, CV Writing, and Interview Skills seminars. The Co-op Policies and Procedures seminar introduces the Co-op Training Program for all Co-op Students and outlines the required policies and procedures. The CV Writing seminar helps students hone their CV writing skills.

Students must also attend an Interview Skills seminar covering how to prepare for an interview, what to expect, and how to make a good impression.

Co-op Assignment

The Co-op and Career Services Assistant maintains a database of prospective employers and attempts to match the employers' needs with the students' fields of study and personal preferences. However, a student may secure a training venue through personal communications and agree with the employer on the conditions of employment. In this case, students in the early spring semester must provide the Co-op and Career Services Assistant with the contact information for the prospective employer and work supervisor, as well as the conditions of employment, for approval. Once approved, the Co-op and Career Services Assistant follows up with the prospective employer in accordance with the established Co-op procedures. Applicable rules: the organization is reputable and not owned by the student's parent; a student may not be supervised by a family member or assigned to a department where a family member works. Required Information for Co-op requests outside the offerings of the Co-op and Career Services Assistant: Organization name, company profile (website), Job description/training plan, supervisor name, contact person, and contact details.

Procedures

1. The Co-op and Career Services Assistant creates a co-op Moodle course for eligible students in each college.
2. The student should read, understand, and complete the Co-op Code of Conduct before the beginning of the Co-op period.
3. The Co-op and Career Services Assistant prepares a database of prospective employers, posts job opportunities and application deadlines, and communicates them to students via email and Moodle.
4. The Co-op and Career Services Assistant prepares a list of eligible students with the help of the Registrar's Office (RO).
5. Students may suggest an employer not on the posted list as long as they get tentative approval from the employer, the College academic advisor, and the Co-op and Career Services Assistant.
6. A trainee seeking a training opportunity abroad must complete the ***Training Abroad Application Form*** (Co-op-12) and submit it to the Co-op and Career Services Assistant for further consideration.
7. The Co-op and Career Services Assistant communicates with prospective employers and provides them with necessary information about the potential trainees' profiles.
8. The Co-op and Career Services Assistant invites eligible students to attend the Co-op orientation seminar in early spring to explain co-op policies and procedures.
9. Each eligible student completes the ***Co-op Application Form*** (Co-op-3), indicating three preferred Co-op venues ranked in order of priority, and submits a CV (soft copy) within the assigned deadlines;
10. The Co-op and Career Services Assistant and academic units shall attempt to match students' aspirations and fields of study with the employer's needs within two weeks of the submitted applications. If multiple trainees compete for a specific training opportunity, students who apply first and meet the eligibility criteria shall be prioritized.
11. The Co-op assignments and all related information are posted by the Co-op and Career Services Assistant on the Co-op Moodle course page and via email.
12. Each academic unit informs the students of the Co-op assignment via email. The student must know that the employer may contact them for an interview before committing to the assignment.
13. The academic advisor bears the responsibility to make at least one field visit and complete the visit assessment form; continuously follow up with the trainee and the work supervisor during the training period; collect the trainee's performance assessment forms from the trainee and the work supervisor; guides the student in preparing the training report, poster and presentation and assess the submitted material; and submits a final report to the Co-op and Career Services Assistant.

Trainee's Responsibilities

Students must fulfill the following responsibilities by the indicated dates to guarantee Co-op placement and avoid any delay toward graduation.

1. Maintain contact with the Co-op and Career Services Assistant to inquire about the

application status before training begins and update the office immediately on the work conditions during the training period.
2. Attend the orientation workshop and get familiar with the Co-op rules and regulations;
3. Set training goals in collaboration with the work supervisor and academic advisor, and drive to achieve them;
4. Perform all tasks to the highest ethical and professional expectations;
5. Complete all necessary documents by the established deadlines. Students must submit in the last week of the Co-op period the following:
    a) Trainee's Assessment of Training Form
    b) Training Tasks log
    c) A formal report
    d) A poster
    e) A PowerPoint presentation
6. Complete all necessary documents by the established deadlines;
7. Comply with the employer's work rules, policies, and standards;
8. Closely coordinate with the work supervisor to execute tasks as planned;
9. Discuss concerns and work-related inquiries with the academic advisor;
10. Complete and submit the Trainee's Assessment of Training Form;
11. Submit a formal report, a daily log of tasks, a poster, and a presentation directly after completing the training period using appropriate forms;
12. Participate in the Co-op poster exhibition;
13. Complete the entire training period agreed to.

Assessment

Based on the employer's evaluation, academic advisors' assessment, and the submitted formal report, daily log of tasks, poster, and presentation, the Co-op academic advisor will assign the student a Pass with Honors "PH," Pass "P," or Fail "F" grade. "PH" is reserved for a trainee who receives an outstanding performance evaluation from the work supervisor and the co-op academic advisor. A grade of "F" is given to a trainee who receives an unsatisfactory evaluation by the work supervisor and the Co-op academic advisor. In this case, the trainee must repeat the whole Co-op experience.

Career Services

Rationale

The Co-op and Career Services Assistant relays information about available job vacancies to RHU alumni to assist them with the employment process, career changes and transitions, and networking opportunities.

Policy

1. Communicate and build a database of prospective employers

2. Compile and announce job vacancies to students and alumni.
3. Conduct seminars to help students prepare CVs, interview skills, etc.
4. Maintain a CV data bank and send CVs to concerned employers.
5. Organize an annual job fair.
6. Create a database of job market needs to help in curricular planning and selection of Summative Learning projects.

Procedure

1. The Co-op and Career Services Assistant receives information on available job vacancies from companies in and outside Lebanon.
2. The Co-op and Career Services Assistant posts the job vacancies with a brief description of the job requirements and the application deadlines on the **Official Rafik Hariri University Alumni LinkedIn page.**
3. The company name is kept confidential in most cases based on the request of employers;
4. Alumni who wish to apply to an announced job vacancy are required to send an updated CV to the Co-op and Career Services Assistant at Coop@rhu.edu.lb;
5. The Co-op and Career Services Assistant communicates with prospective employers, provides them with the CVs of the potential candidates, and follows up with them until the vacancy is generally filled.
6. The Co-op and Career Services Assistant maintains a database of all the job vacancies received, the names of the companies, major areas, and position levels (beginner, supervisory, middle management, and top management);
7. The Co-op and Career Services Assistant supplements, at the end of every academic year, a report of all the vacancies received with insights on the job market needs to help in curricular planning and selection of Summative Learning projects.

FRESHMAN PROGRAM

RHU offers a freshman program for students who have not earned a Lebanese Baccalaureate or its equivalent, serving as a bridge to eligibility for a bachelor's degree. The freshman program covers two tracks: the Freshman Arts track and the Freshman Science track. Each track encompasses 30 credits that may be completed in two semesters. The required subjects and corresponding credit hours are summarized in the table below.

Program	Humanities/ Social Sciences	Math	Natural Sciences	Free Elective	English	Arabic
Freshman Arts	12	3	3	6	3	3
Freshman Sciences	6	6	9	3	3	3

All students whose native language is Arabic must take an Arabic language course. Students exempted from the Arabic language requirement must take a substitute course to fulfill the 30-credit requirement.

Admission Requirements

To be eligible to join the Freshman Program, an applicant must hold one of the following certificates:

1- A high school certificate based on 12 years of schooling, starting with Grade One, awarded by a government or private school and recognized by the Lebanese Ministry of Education and Higher Education (MoEHE). High school certificates based on less than 12 years of schooling, starting with grade one, are not recognized.
2- A General Certificate of Education (GCE). This applies to applicants with an A-level certificate who have completed a minimum of five subjects at the Ordinary Level and one subject at the Advanced Level or two Advanced Supplementary Levels (AS), excluding languages.

The Lebanese Ministry of Higher Education announced the following requirements to join the Freshman Program at any institute of higher education:

a. Certified copy of the permission from the Lebanese Ministry of Education Equivalence Committee to join the Foreign Program/Freshman class.
b. The following scores are required in one of the aptitude tests below before they can enroll at RHU:

Aptitude Test	Freshman Arts	Freshman Sciences
SAT-Scholastic Aptitude Test	870	950
ACT - American College Test	15	17

Requirements of the Freshman Program Equivalency

1. **If your school is accredited by one of the following accrediting boards,** then the above requirements are enough to grant you admission to the sophomore year after your Freshman Program, and to get the equivalency from the Lebanese Ministry of Higher Education

   Accrediting Boards:

   - Western Association of Schools and Colleges (WASC)
   - Southern Association of Colleges and Schools (SACS)
   - AdvancED (Cognia)
   - New England Association of Schools and Colleges (NEASC)
   - North Central Association of Schools and Colleges (NCASC)
   - Northwestern Association of Colleges and Schools (NACS)
   - Council of International Schools

2. **If your school is not accredited by one of the above accrediting boards,** then you need, in addition to the above two requirements (a & b), to submit:

   ACT Subject Tests during your first year at RHU, before your sophomore year begins.

   Required subjects in the ACT Subject Tests exam are:

   - Mathematics I and any other two subjects for Freshman Arts
   - Mathematics II and two of the following subjects: Biology, Chemistry, and Physics for Freshman Science

   Required scores in the ACT Subject Test:

   - A minimum total score of 51 for Freshman Arts with a minimum score of 15 in each subject
   - A minimum total score of 60 for Freshman Sciences, with a minimum score of 17 in each subject

Please note the following:

Lebanese students cannot declare a major and join the sophomore class until they satisfy the requirements of the Lebanese Ministry of Education and Higher Education and the RHU freshman program.

Study Plan

The following study plan summarizes the courses and credit distribution for the freshman program. Freshman courses are identified by having a “1” as the first digit of the course number.

Freshman Arts

Course No.	Course Name	Credits	Prerequisite
Fall Semester (15 Credits)			
BECN 190	Introduction to Economics*	3	
ENGL 101	Freshman English	3	
MATH 189	Fundamentals of Algebra	3	
	Free Elective	3	
	Humanities/ Social Science Elective	3	
Spring Semester (15 Credits)			
ARAB 180	Arabic Reading and Writing	3	
	Humanities/ Social Science Elective	3	
	Humanities/ Social Science Elective	3	
	Natural Science Elective	3	
	Free Elective	3	

* Required for application to the College of Business.

Freshman Science

Course No.	Course Title	Credits	Prerequisite
Fall Semester (15 Credits)			
ENGL 101	Freshman English	3	
MATH 190	Calculus I	3	
PHYS 190	General Physics I	3	
	Humanities/Social Science Elective I	3	
	Free Elective	3	
Spring Semester (15 Credits)			
ARAB 180	Arabic Reading and Writing	3	
MATH 191	Calculus II	3	MATH 190
PHYS 191	General Physics II	3	PHYS 190
	Natural Science Elective	3	
	Humanities/Social Science Elective II	3	

Courses Description

I. Mandatory Courses

Descriptions of the program's mandatory courses are given below.

ARAB 180	Arabic Reading and Writing	3(3,0)
This freshman-level course is designed for native speakers of Arabic to develop their reading and writing skills. This course is conducted in Arabic. Students will read and discuss essays, articles, and other readings, and write essays. It fulfills the Freshman Arabic language requirement. The course is mandatory for all native Arabic speakers.		

BECN 190	Introduction to Economics	3(3,0)
This course is designed to help students develop a basic general understanding of economic principles, issues, history, and institutions. It develops their economic literacy and teaches them how economics relates to the everyday lives of individuals, businesses, and society. Topics include markets, firms, economic systems, the role of government in the economy, capitalism, macroeconomic indicators, and the basics of supply and demand.		

ENGL 101	Freshman English	3(3,0)
This course is designed to help students develop their reading and writing skills. It focuses on the interrelatedness between reading and writing, whereby students critically read and analyze passages in different rhetorical modes (classification, cause/effect analysis, and comparison/contrast analysis) and learn to write similar essays in these modes. Some grammatical and structural elements are reviewed to help students improve the accuracy of their writing. In this course, students are also introduced to research skills such as evaluating sources, paraphrasing, summarizing, and quoting while following the APA style.		

MATH 189	Fundamentals of Algebra	3(3,0)
Real number systems, radicals and rational exponents, polynomials, factoring, fractional expressions, lines in the plane, functions and their graphs, inverse functions, solving equations and inequalities, real zeros and the fundamental theorem of Algebra, exponential functions and their graphs, logarithmic functions and their graphs.		

MATH 190	Calculus I	3(3,0)
Functions and their graphs, limits and continuity, differentiation, applications of derivatives, Extreme values and mean value theorem, definite and indefinite integrals, and the substitution method.		

MATH 191	Calculus II	3(3,0)
Covers limits and indeterminate forms with L'Hôpital's Rule, optimization, antiderivatives, and the Fundamental Theorem of Calculus. Topics include substitution methods, areas, volumes (cross-sections, shells), arc length, and surfaces of revolution. It introduces parametric and vector functions with applications to motion, arc length in space, curvature, and normal vectors. **Prerequisite**: Math 190.		

PHYS 190	General Physics – Mechanics	3(3,0)
This course introduces students to the basic concepts and principles of Mechanics: Vectors, displacement, velocity, acceleration, motion in 1D, projectile motion, forces, Newton's laws of motion, work, energy, momentum, collision, torque, and rotational motion.		

PHYS 191	General Physics - Electricity and Magnetism	3(3,0)
This course is an introduction to electricity and magnetism. In this course, students will explore electric charges, electric forces and fields, electrical energy and potential, capacitance, direct-current circuits, magnetic force, magnetic field, induced voltage and inductance, alternating-current circuits, and electromagnetic waves. **Prerequisite**: PHYS 190.		

II. Electives

Descriptions of the possible elective courses are given below.

BACC 150	Introduction to Accounting	3(3,0)
Students in this course will learn the basic concepts and techniques of accounting, including accounting information systems, costing, estimation, and some aspects of management accounting.		

BADM 155	Introduction to Law	3(3,0)
This course covers the basic tenets of the legal system and how they may apply to various business sectors and industries.		

BIOL 102	Introduction to Biology	3(3,0)
This is an introductory course to the fundamental principles of biology that covers the chemical basis of life, the structure and function of cells and systems, genetics, and human body systems.		

BIOL 170	Basic Concepts in Biology	3(3,0)
A freshman-level course that introduces students to the basic concepts of genetics, evolution, and ecology and explores the forms and functions of plants and animals.		

BMGT 150	Basics of Management	3(3,0)
This course provides a general overview of management basics, including principles of management science, buying and selling, and management.		
BMGT 155	**Introduction to Hospitality Management**	**3(3,0)**
Students will learn the basics of different aspects of hospitality, such as food, lodging, beverages, and pastries. They will also be exposed to hotel, restaurant, and resort hospitality operations.		

BMKT 150	Introduction to Business and Commerce	3(3,0)
This course introduces students to the nature of business, the types of commercial activities, market analysis, and the interrelationships among commercial activities.		

CCEE 111	Computer Programming	3(3,0)
This course introduces students to fundamental programming concepts such as variables, loops, conditional statements, and event handling. It also helps students use math and computer code to think creatively.		

CCEE 112	Internet Development and Support	3(3,0)
This course covers issues related to the development and implementation of internet-related tools and services. Topics include Internet organization, site registration, e-mail servers, Web servers, Web page development, legal issues, firewalls, multimedia, TCP/IP, service providers, FTP, list servers, and gateways.		

CCEE 121	Computer Technology	3(3,0)
This course is designed to help students understand the basic functions of a computer as a human-machine interacting **system**. It covers the basics of computer technology in today's world. Students will focus on the following components of computers: keyboards, operating systems, word processing, spreadsheets, database management, presentation graphics, and the internet.		

CCEE 122	Computer Hardware	3(3,0)
This course introduces the technical aspects of computer hardware units. Topics covered are computer organization, communication inside the computer, and communication with the outside world.		

CCEE 123	Introduction to Computer Troubleshooting	3(3,0)
In this course, students will learn how to set up a computer, diagnose simple faults, and troubleshoot them.		

CCEE 131	Introduction to Audio-Visual Technology	3(3,0)
In this course, the students will be introduced to the technologies involved in the Audio and Video Communication systems. The course includes video production, multimedia, sound, animation, and photography.		

CCEE 141	Switching Systems Technology	3(3,0)
This course includes the public switched telephone network; the telephone and the local loop architecture; inter-exchange networks and signaling; the evolution of switching technology and architectures, and a comparison of various systems; and traffic statistics and the theory of space-division and time-division switching networks.		

CCEE 142	Basics of Analog and Digital Communications	3(3,0)
This course covers analog and digital communication systems: basic principles of telecommunication systems, operation and application of various transducers, and basic principles of modulation and demodulation.		

CHEM 160	Chemistry I	3(3,0)
Students study the principles and methods of chemistry, including atomic structure, periodic trends, writing formulas, and determining the names of both inorganic and organic formulas.		

CHEM 170	Chemistry II	3(3,0)
Students learn about acids, bases, and chemical equilibria; the groups in the periodic table, organic chemistry, and nuclear chemistry will also be surveyed.		

CIVE 105	History of Architectural Technology	3(3,0)
The study of architectural technology from the Greek civilization to the present stresses the development of structural systems and the exploration of materials. This course will explore the interaction of building design and historic socio-economic determinants.		

CIVE 120	Introductory Strength of Materials	3(3,0)
The study of stress and strain, deformation, simplified analysis of torsion, shear, and stresses in beams, beam design, beam deflection, and columns. Reference to applications for civil and mechanical engineering technology, introduction to materials testing.		

CIVE 121	Concrete Construction Methods and Concrete Structures	3(3,0)
Emphasis is placed on applying structural design principles for new and existing concrete structures. Study of concrete structures using basic physical laws. The course covers the basics, from foundation to roof, and the properties and applications of concrete, emphasizing concrete, masonry, and shallow foundation systems. Conditions encountered during renovations and their solutions will be included.		

CIVE 135	Plumbing and Air Conditioning	3(3,0)
Review of water distribution plumbing and related fundamentals. This course covers the basics of piping, valves, backflow prevention, water quality, and hydraulic fundamentals. Review of air-conditioning fundamentals, piping, capacity requirements, and recent developments in materials, fixtures, and appliances.		

CIVE 140	Introduction to Surveying	3(3,0)
A study of topographic surveying and mapping. Determination of land areas, construction surveys and layout, control surveys, boundary surveys, route locations, and street layout. Provides experience with the use of equipment, instruments, and the fundamental techniques of surveying		

CIVE 141	Fundamentals of Topometry	3(3,0)
Measurement and graphic description of topography, topometric descriptions of objects using various techniques in mapping		

CIVE 142	Photogrammetry	3(3,0)
Principles and techniques of vertical aerial photograph photogrammetry, distance, direction, area, and photograph scale calculations and interpretation.		

CIVE 143	Geodesy	3(3,0)
A horizontal and vertical datum, gravitation and centrifugal forces; measurements of gravity and reduction of gravity, geoid, and ellipsoid, deflection of the vertical; orthometric and dynamic heights		

CIVE 144	Mapping	3(3,0)
Topographic mapping, spatial analysis, map production and cartography, spatial reference systems, scale, projections, generalization, and symbolization.		

CIVE 145	Geomorphology	3(3,0)
Surface processes and landforms; their implications for land use and for present, past, and future landscape development; weathering; glacial and fluvial processes; mass wasting; and tectonic controls on landforms.		

CIVE 146	Construction and Surveying Regulations and Laws	3(3,0)
Construction and surveying regulations and laws in Lebanon and their applications to land usage, urban development, and residential and commercial buildings, with emphasis on the construction field, the law of practice, and professional registration		

CIVE 148	Introduction to Public Works	3(3,0)
This course introduces students to construction, alteration, repair, or improvement on public land that is executed at the cost of the state or any other local public agency. Skills in Public administration and supervision of public works are emphasized.		

CVLN 160	Monotheistic Religions: A Comparative Study	3(3,0)
This course covers the three monotheistic religions. It compares the idea of God and the law in Judaism, Christianity, and Islam.		

CVLN 161	The Arabs	3(3,0)
This course reviews the history of the Arabs from the pagan days through the twentieth century.		

CVLN 170	God and Creation: East and West	3(3,0)
A literary attempt to understand the universe's origins as found in texts from various world cultures.		

CVLN 180	Youth and the Growing Up Process	3(3,0)
An introduction to the themes and challenges of autonomy and independence as experienced by youth, studied through major literary works of the past centuries.		

CVLN 181	Arab Renaissance	3(3,0)
The Arab Renaissance is studied through historical and socio-political texts.		

CVLN 190	Freshman Art History	3(3,0)
This course provides an overview of art and its development in the Western world. It covers the main art periods from “Prehistoric Art” through “Contemporary Art,” as well as the effects of the socio-political cultures of the times.		

ELEC 101	Electrical Drawing	3(3,0)
This course covers reading and interpreting electrical installation drawings and performing electrical installation, machines, and electronic circuit drawings using standard symbols. This course also covers the use of standard computer packages for electrical drawings. This course will also enable students to understand the installation of various electrical and electronic systems and wiring in buildings.		

ELEC 111	Industrial Electricity	3(3,0)
The course will introduce and cover circuit variables and elements; methods and techniques for analyzing DC and AC circuits; 3-phase systems; and power calculations. Fundamentals of industrial devices and equipment will be introduced, along with topics related to their safety.		

ELEC 121	Digital Electronics	3(3,0)
This course introduces different logic gates, develops truth tables for combinational logic circuits, covers simple Boolean expressions for logic gates, and describes the fundamental concepts of ICs.		

ELEC 122	Analog Electronics	3(3,0)
This course aims to provide a basic understanding of the principles of analog electronics and the operation of diodes, transistors, and Op Amps. This course also covers the basic design of analog amplifiers and gain stages.		

ELEC 123	Industrial Electronics	3(3,0)
This course provides a basic knowledge of circuitry for the control and conversion of electrical power with high-efficiency converters. Converters convert and regulate voltage, current, or power; examples include DC-DC converters, AC-DC rectifiers, and DC-AC inverters.		

ELEC 131	Electrical Power Generation and Distribution	3(3,0)
This course covers the principles of electrical power generation and distribution, as well as the methods used to protect electrical power systems.		

ELEC 132	Electric Machines	3(3,0)
This course covers the construction, principles of operation, characteristics, and applications of DC machines; the basic principles of transformers; and the construction, principles of operation, and applications of AC machines.		

FADR 100	Basic Drawing	3(2,2)
This course teaches students the basics of drawing with lines, shapes, tones, shadows, and colors. Still life set-ups are used to develop the student's observational skills and drawing techniques. Students experiment with various drawing media (charcoal, pencil, graphite, and ink) to develop their technical abilities and creativity.		

FADR 115	History of Art Overview	3(3,0)
This course is a comprehensive overview of artistic movements throughout the centuries, with an emphasis on the 20th century. It explores the forms, purposes, meanings, and principles of the artistic styles that influenced cultures across the globe. The course also introduces students to basic analytical tools for critically analyzing art within its cultural, political, and historical contexts.		

FADR 120	Introduction to Color Theory	3(3,0)
This course focuses on the principles, theories, and applications of additive and subtractive colors in the visual communication design process. Students investigate color schemes, color mixing, color properties, and color relationships, and examine the psychological, cultural, and symbolic aspects of color and their relation to visual communication.		

FADR 130	Basic Digital Photography	3(2,2)
This course introduces students to the basic concepts and practices of digital photography. Students learn to use the camera, lenses, and other basic photographic equipment to create professional images. It introduces students to the aesthetic principles of composition, space, exposure, light, and color. Students also acquire skills in digital manipulation and learn the technological requirements of digital formats and resolution.		

GRDS 150	Introduction to Arabic Calligraphy	3(2,2)
This course introduces students to Arabic Calligraphy, one of the highest art forms in the Arab world and a primary medium of Islamic visual expression and creativity. Students will learn about the history of Islamic Calligraphy, its origin, tools, and styles (Riqa, Naskh, Deewani, Thuluth, and Kufi). In addition, students will gain hands-on experience with basic lettering and Kufic compositions.		

GRDS 160	Principles of Advertising	3(3,0)
This course introduces the basic principles of advertising and their role in media and society. It provides an overview of the advertising industry, functions, and practices within the broader communications context. This course examines advertising agencies, advertising professionals' roles, agency-client relationships, target audiences, media, ethics, and creative and strategic thinking. A brief historical analysis of advertising is included to shed light on the important role social forces have played in its evolution.		

HIST 100	History of the Ancient World	3(3,0)
This course examines various topics, including the origins of civilization in Egypt and Mesopotamia, the civilizations of Ancient Greece and Rome, the origins and development of Judaism, and the early Christian Church. **Co-requisite**: ENGL 101.		

HIST 102	Europe from the Reformation to the French Revolution	3(3,0)
A survey of European history begins with the 16th-century Protestant Reformation and includes absolutism, the scientific revolution, the Enlightenment, the Old Regime, and the French Revolution. **Co-requisite**: ENGL 101.		

HIST 103	Europe in the 19th and 20th Centuries	3(3,0)
Topics include the Industrial Revolution; the rise of political ideologies, social classes, nationalism, and imperialism; the Russian Revolution; the two world wars; the interwar period; and the Cold War era. **Co-requisite**: ENGL 101.		

HIST 108	Issues and Themes in U.S. History: 1607-1877	3(3,0)
American history from 1607 to 1877 examines the development of the United States from early settlement through the Civil War era. Emphasis is placed on the evolution of American politics, race relations, economic and social trends, and Western expansionism. **Co-requisite**: ENGL 101.		

GEOL 160	Earth Geology	3(3,0)
This introductory course is designed to provide a general overview of some aspects of the Earth's natural environment and system. It introduces the processes and physical systems of the Earth that are important in controlling its environment. In addition, the course discusses the relationships among Earth systems and processes and their environmental effects.		

PHIL 102	Freshman Philosophy	3(3,0)
This course is an introduction to the part of philosophy concerned with questions about the ultimate nature of our world and our knowledge of it. The issues discussed include the existence and nature of God, the place of human beings in the universe, the nature of reality, human nature, human knowledge and skepticism, freedom and determinism, and the relationship between mind and body. The emphasis of the course varies from instructor to instructor. **Co-requisite**: ENGL 101.		

PHIL110	Critical Thinking	3(3,0)
This course introduces the basic rules of clear and rational thought. Students are offered an extensive analysis of examples on how to detect false reasoning, illegitimate appeals to emotions, inconsistencies, and contradictions. This course aims to develop logical awareness to the point that the rhetoric of everyday life can no longer victimize a person, and can develop arguments with clarity and confidence. **Prerequisite**: Native English speakers or English 098. **Co-requisite**: ENGL 101.		

POSC 190	Freshman Politics	3(3,0)
This survey course is recommended for students interested in discovering how politics impacts their lives and how they, in turn, can affect politics. Students are introduced to the basic concepts of Political Theory, Comparative Politics, International Relations, and Public Policy. **Co-requisite**: ENGL 101.		

PSYC 190	Freshman Psychology	3(3,0)
This course aims to develop an understanding of human psychology by focusing on the factors that influence human behavior and mental processes. It covers cognition, emotions, learning, human development, biological bases of behavior, personality, psychological disorders, psychotherapy and behavior change, and social behavior. **Co-requisite:** ENGL 101.		

PSYC 195	Freshman Stress Management	3(3,0)
This course addresses the basic principles, theories, and techniques to manage personal stress effectively. Students will better understand the mind-body relationship, learn to employ a holistic approach to stress, and adopt effective cognitive, coping, and relaxation techniques. **Co-requisite**: ENGL 101.		

SOCI 190	Freshman Sociology	3(3,0)
This course is designed to give freshman students an overview of the field and familiarize them with essential sociological concepts. It focuses on the individual in society, the study of society, social institutions, and human agency. In particular, the course will focus on how people's lived experiences are shaped by social forces and reshaped through human action. **Co-requisite**: ENGL 101.		

Academic Guidelines

Courseload:

1. Students are expected to finish the Freshman Program in two regular semesters.
2. Students can register for up to 15 credits per semester.
3. Students on probation can register for up to a maximum of 12 credits.

Transfer from Freshman Science to Arts:

Students must take BECN 190 and achieve a minimum grade of 70.

Repeating courses:

A course may not be repeated more than two times, including withdrawals.

English courses:

Students' English level upon acceptance determines the number of credits students can register for per semester, along with the English course as follows:

English level	Number of Credits
IELP 085	10
IELP 095	10
IELP 100	13

If the students' English level upon acceptance is ENGL 210, ENGL 210 replaces ENGL 101, and ENGL 210 should be replaced with a humanities elective in their degree program.

Humanities/Social Sciences Courses:

To take humanities or social sciences courses, students should consider the following:

Humanities/Social Sciences course code starting with	Co-requisite	Pre-requisite
100	ENGL 101	
200		ENGL 101
300	ENGL 210	

Joining Degree Programs

Students may join a degree-granting program at RHU after completing the freshman program if they obtain the freshman-year-to-Lebanese Baccalaureate equivalency certificate from the Equivalence Committee of the Lebanese MoEHE. Students on the Freshman Arts track may join any program offered at the College of Business Administration, or the Graphic Design Program or English Program offered at the College of Arts and Sciences. Students on the Freshman Science track are eligible to join any degree-granting program offered at the University. Additional requirements to join specific programs are summarized below.

College of Arts and Sciences:

Graphic Design Program. A GPA of at least 70 and a passing score on the RHU Graphic Design Aptitude Test.

Computer Science Program: A GPA of at least 70 and a minimum grade of 70 in MATH 190 and MATH 191.

Healthcare Information Systems Program: A GPA of at least 70 and a minimum grade of 70 in MATH 190 and MATH 191.

College of Business Administration:

Business Administration programs. A GPA of at least 70 and a grade of at least 70 in MATH 189 or MATH 190 and BECN 190.

College of Engineering:

Engineering Programs. A GPA of at least 70 and a minimum grade of 70 in MATH 190, MATH 191, PHYS 190, and PHYS 191.

GENERAL EDUCATION

Mission

In line with the University's mission, the General Education program aims to improve students' awareness and intellectual capacity to address the complex and diverse issues and challenges confronting the world around them, and to empower them to thoughtfully consider those issues in their personal and professional decisions and choices.

Learning Outcomes

Upon completion of the General Education Program, students will demonstrate:

- Critical thinking in evaluating scholarly sources and arguments;
- Information literacy and skills in applying modern computing tools and techniques;
- Ability to make informed decisions based on ethical principles and reasoning;
- Aptitude for aspiring to continuous improvement and lifelong learning.

Domains of Knowledge

Every RHU student must take 30–33 credit hours of general education courses, distributed across six domains described below. Twenty-one mandatory credits are selected from domains 1, 4, 5, and 6, and 9-12 elective credit hours are selected from domains 2 and 3:

Domain		Credits	Courses
1	Communication Competency*	8	ENGL 210: English Composition and Rhetoric ENGL 217: Professional English Communication ARAB 212**:** Arabic Language and Communication
2	Humanities/Fine Arts	3-6	Selected from an approved list
3	Social Sciences/Culture	6	Selected from an approved list
4	Natural Sciences and Technology	6	Per major requirement*
5	Quantitative Reasoning	3	Per major requirement*
6	Community and Sustainability	4	Per major requirement*

* Refer to the introductory section of the college of interest for a description of courses specific to domains 4, 5, and 6.

Domain Specific Courses

Communication Courses

ARAB 212	Arabic Language & Communication	2(2,0)
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need for their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite**: ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

Elective Courses

Description of the Courses that satisfy Domain 2 and 3 requirements is given below.

Humanities/Fine Arts

Descriptions of elective courses that may be chosen to satisfy the Humanities/Fine Arts domain are given below.

BADM 355	Business Ethics and Social Responsibility	3(3,0)
This course introduces students to contemporary business principles of ethics and social responsibility. Students learn to make ethical judgments about important issues they face daily by relating them to a framework of ethical principles, including utilitarianism, justice, moral rights, the ethics of care, and virtues and vices. Ethical dilemmas related to conflicts of interest, sustainability in business strategy, and corporate governance are among the issues discussed in this course. Prerequisite: Junior Standing.		

CMNS 360	Ethics in Communication	3(3,0)
This course explores the theoretical and practical ethical questions about communication in four contexts: interpersonal communication, workplace		

communication, communication within one's community and society, and mass and digital communication. Discussion of ethical theories in communication studies will provide a basis to explore case studies and contemporary communication dilemmas, heightening personal sensitivity to the underlying ethical implications of human communication. **Co-requisite:** ENGL 210.

CMNS 470	**Videography in Media**	**3(3,0)**
This course will explore the world of cinema, video shooting, and editing. Students will dive into the preproduction, production, and post-production phases. Throughout the course, students will be expected to complete assignments in shooting, editing, and creating different types of narratives. They will understand and learn how to shoot and handle different types of videos. Students will also apply these skills to create trailers, shoot interviews, and create an advertisement reel.		

CMNS 320	**Creative Nonfiction**	**3(3,0)**
This course teaches students rhetorical modes in short essays and journalism, including memoirs, reportage, and reflective essays. Students will practice a writing style focused on conciseness and clarity, deliver effective oral presentations, and read articles, essays, and other increasingly complex materials. **Co-requisite**: ENGL 210.		

CVLN 301	**Civilization Studies I**	**3(3,0)**
This course surveys key figures, movements, and texts in cultural studies from the birth of civilization through the Middle Ages. The concentration will be on the historical, political, religious, cultural, and institutional aspects during this period. Students will read, analyze, and interpret certain primary works to gain insight into the contexts in which they originated. **Co-requisite**: ENGL 210.		

CVLN 302	**Civilization Studies II**	**3(3,0)**
This course surveys some key figures, movements, and texts in cultural studies from the 14th through the 18th centuries. The concentration will be on the historical, political, religious, cultural, and institutional aspects during this period. Students will read, analyze, and interpret certain primary works to gain insight into the contexts in which they originated. **Co-requisite**: ENGL 210.		

CVLN 303	**Civilization Studies III**	**3(3,0)**
This course surveys some key figures, movements, and texts in cultural studies from the 19th century to the present. The concentration will be on the historical, political, religious, cultural, and institutional aspects during this period. Students will read, analyze, and interpret certain primary works to gain insight into the contexts in which they originated. **Co-requisite**: ENGL 210.		

CVLN 320	**Feminism Beyond Borders**	**3(3,0)**
This course introduces students to feminism by exploring its historical development, major waves, and key issues. At its core, this course aims to challenge students' understanding of feminism and its relevance today. The course focuses on the		

intersection of feminism with other social justice movements, including race, class, and culture, to challenge universal or “one-size-fits-all” feminist narratives.

EDUC 201	**Fundamentals of Education**	**3(3,0)**
This course studies modern education principles and their application in elementary and secondary schools. It includes examinations and discussion of teaching procedures and techniques, as well as considerations of the historical, cultural, and societal views of schooling.		

EDUC 211	**Fundamentals of Special Education**	**3(3,0)**
This course introduces Special Education, with a particular focus on current practices. Students are introduced to common learning disabilities and intervention programs, as well as a historical overview of the development of Special Education.		

EDUC 312	**Educational Philosophy**	**3(3,0)**
This course provides a survey of the development of thought on education by examining major philosophical writings in historical context. **Prerequisite**: EDUC 210.		

EDUC 360	**Children’s Literature**	**3(3,0)**
This course draws on the theoretical and practical aspects of the study of literature for children. Students develop a wide familiarity with children's books and understand how children's literature fits into the elementary school curriculum.		

EDUC 365	**Art Education**	**3(3,0)**
This course explores concepts in art, music, drama, and movement for young children. The methods cover various activities that develop children's physical-motor, social-emotional, and cognitive skills, providing conditions for creativity and aesthetic awareness.		

ENGL 214	**Introduction to English Poetry**	**3(3,0)**
This course introduces first-year students to poets with intrinsic literary merits. In addition to broadening students’ understanding and appreciation of literature, the course's chief aim is to develop students’ critical thinking and analysis and to encourage original responses to literary expression through well-developed oral and written criticism. Co-requisite: ENGL 210.		

ENGL 220	**Introduction to Literature**	**3(3,0)**
This course introduces literature through fiction, poetry, and drama. It introduces students to the pleasures of reading literature and to interpretative approaches to literature. It aims to provide students with competence in critical thinking and practice in close reading and analysis, knowledge of the formal characteristics of each genre, and an appreciation of literary excellence. Readings include a variety of short stories, poems, and plays selected from a wide range of British, American, and World literature. Co-requisite: ENGL 210.		

ENGL 222	Introduction to Language	3(3,0)
This survey course examines current theoretical and applied areas of linguistics, including the different levels of structure, the nature of language acquisition, language variation and evolution, and language teaching. **Co-requisite**: ENGL 210.		

ENGL 223	Introduction to Drama	3(3,0)
In this course, you will explore what drama is, its basic elements, and the evolution of its dramatic structure over centuries. You will study selected and representative plays from different periods that will help you broaden your understanding and appreciation of literature. The course also aims to develop your critical thinking and analysis, and encourages you to respond critically to dramatic works through well-developed oral and written criticism.		

ENGL 230	Shakespeare's Plays	3(3,0)
Students study Shakespeare's representative comedies, histories, and tragedies in this course. The plays are read intensively and understood in the context of the period's theatrical conventions, the culture of play in early modern England, and the period's social, cultural, religious, and intellectual history. Co-requisite: ENGL 210.		

ENGL 235	Creative Writing	3(3,0)
This course introduces students to the process and practice of imaginative writing. Students will read works of fiction, creative non-fiction, and poetry and learn writing techniques in those genres. **Co-requisite**: ENGL 210.		

ENGL 252	Introduction to American Literature	3(3,0)
This course explores the works of major literary figures and authors representing different literary schools from the sixteenth century to modern times. Students will closely examine selected texts as literary achievements and situate them within the historical context and framework of American culture. Co-requisite: ENGL 210.		

ENGL 300	Introduction to World Literature	3(3,0)
The course surveys selected works in English or in translation from non-Anglo-American cultural traditions. Texts can be drawn from African, Asian, European, Latin American, and Middle Eastern literature, with a focus on their historical and cultural contexts. Co-requisite: ENGL 210.		

ENGL 330	Language Acquisition	3(3,0)
This course will survey theories of language acquisition. It will address human language processing, learners' motivational factors, and contextual factors that influence language learning. It will uncover the principles of first- and second-language acquisition. **Co-requisite**: ENGL 210.		

FADR 215	History of Art	3(3,0)
This course explores the major forms of artistic expression from the ancient world to the present, from various cultural perspectives. Students learn to examine and analyze		

works of art in their historical context and to articulate what they see or experience in a meaningful way. This course covers the pre-Renaissance, Renaissance, and post-Renaissance periods, as well as the rich layers of 19th- and 20th-century modern art up to the present day.

GRDS 230	Photography I	3(2,2)
This course teaches students the basics of black-and-white digital photography. It is a lecture-based course with extensive hands-on practical training. Students learn how to develop an eye for photography and for framing objects. They also enhance their knowledge of the different parts of a camera and the mechanisms of film printing and development while continuously scanning other photographers' work in contemporary and historical scenes.		

HIST 210	History of England	3(3,0)
This broad survey course provides a narrative of British history from 1066, including political, social, and cultural developments. Co-requisite: ENGL 210.		

HIST 350	History of the Arabs in the Classical Period	3(3,0)
This is an elective course whose primary objectives are to familiarize students with the history of the Arab peoples. The course's emphasis will be on explaining the early Arab conquests of Bilad al-Sham, Egypt, Persia, and North Africa; the internal civil strife within the early Muslim community; the emergence and development of the Umayyad caliphate; and the emergence and development of the Abbasid Empire.		

HIST 360	Contemporary History	3(3,0)
This is an elective course whose major objectives are to familiarize students with contemporary issues and to enhance their understanding from historical and cultural perspectives. Co-requisite: ENGL 210.		

HIST 365	History of Europe in the Middle East	3(3,0)
This course examines the ties between Europe and the Middle East in the 19th and 20th centuries, focusing on the reaction of Middle Eastern societies to European intervention and influence. **Co-requisite**: ENGL 210.		

HIST 370	The Economic and Social History of the Middle East	3(3,0)
This elective course aims to familiarize students with key events that shaped the evolution of social and economic developments in the Middle East, enhancing their understanding of the historical and economic dimensions of these developments. **Co-requisite:** ENGL 210.		

HIST 480	Special Topics in History	3(3,0)
This course provides an in-depth look at selected issues related to historical events across ages and stages of development. Current issues related to historical events		

across ages and development are thoroughly explored through readings, research, discussion, and guest lectures. Prerequisite: Consent of the instructor.

MUSC 201	Introduction to Music	3(3,0)
This introductory music appreciation course is intended for students who wish to broaden their knowledge of music. It addresses the basics of music theory and history. Students learn to listen to and analyze music and to apply these skills to musical masterpieces across various styles, including classical, romantic, flamenco, blues, and jazz. **Co-requisite**: ENGL 101.		

MUSC 301	Introduction to the World of Western and Arabic Music: Theory and Practice	3(3,0)
Music is considered one of the essential humanities courses at most Universities worldwide. This course will cover a brief historical review of Western and Arabic music; the Art of listening to music; important musical terminology; fundamentals of tuning and pitch frequencies in Western and Arabic music; the seven most important elements of music: Rhythm, Melody, Harmony, Timbre, Dynamics, Texture, and Form; the technical interference behind the development of Western and Arabic musical scale-structures; introduction to reading Western and Arabic sheet music; Chords structure, Chords progress, and Arpeggios; the use of the guitar in the Arabic music.		

PHIL 301	Major Trends in Philosophy	3(3,0)
This is a study of some major topics in philosophy. The course emphasizes theory of knowledge, theory of mind, determinism and free will, and morality and ethical values. Along with some assigned readings, students will watch movies whose analysis will help them better comprehend the course content. **Co-requisite**: ENGL 210.		

PHIL 303	Introduction to Ethics	3(3,0)
People often wonder about what makes for good ways to live and the right ways to act. They also speculate about the best way of life, what action is right, and what authority moral claims have over us. The course introduces students to the major moral theories and thinkers addressing these questions. **Co-requisite**: ENGL 210.		

PHIL 307	Business Ethics	3(3,0)
This course introduces contemporary philosophical thinking on ethical issues in business. Students will be exposed to important ethical issues they might face, asked to give sound ethical judgment to problems they might face in their line of work, helped to become armed with a set of codes that will prepare them to confront and resolve ethical dilemmas they might encounter at work, and enabled to apply the techniques for analyzing and resolving ethical problems when they arise. **Co-requisite**: ENGL 210.		

PHIL 310	Philosophers of Peace	3(3,0)
This course introduces students to peacebuilding and conflict prevention by examining the main theories on peace and security. It examines the ideas of Thucydides, Aristotle,		

Erasmus Grotius, Kant, Gandhi, and other thinkers who believed that peace as a whole was highly valued. It develops students' critical skills and understanding necessary to translate their academic learning into specific practical situations, such as those posed by peacebuilding with the UN, governments, or NGOs. It helps to understand the complex and interconnected challenges to peace and provides knowledge to meet them. **Co-requisite**: ENGL 210.

PHIL 480	**Special Topics in Philosophy**	**3(3,0)**
This course provides an in-depth look at selected issues in philosophy and philosophical thinking. Through readings, research, discussion, and guest lectures, current issues related to philosophy and philosophical thinking are thoroughly explored. **Prerequisite**: Consent of the instructor.		

THEA 301	**Introduction to Theatre**	**3(3,0)**
This course will allow students to experience the process of making theater. Its outcome is a theatrical production involving students in different roles: scriptwriters, actors, and production crew, such as house managers, designers, and operators. The work is conducted under the supervision of the course instructor. **Co-requisite**: ENGL 210.		

Social Sciences/Culture

Descriptions of elective courses that may be chosen to satisfy the Social Sciences/Culture domain are given below.

BECN 301	**Microeconomics**	**3(3,0)**
Students will study the general principles of microeconomics. Included are the theoretical constructs of consumer behavior, cost structure, and the operations of business firms in the market economy under conditions of perfect competition, oligopoly, monopoly, and monopolistic competition. **Prerequisite**: Junior Standing.		

BECN 302	**Macroeconomics**	**3(3,0)**
Students will study the general principles of macroeconomics. This course presents the formal Keynesian theory of income determination and its contemporary critiques, including the possible causes and solutions to unemployment and inflation, as well as the importance of the international economy. Government fiscal and monetary policies are examined in detail. **Prerequisite:** Junior Standing.		

BMGT 200	**Introduction to Management**	**3(3,0)**
Students study the basic functions of management and are exposed to modern management practices, current events, problem-solving, and ethical dilemmas. Topics include decision-making, strategic and operational planning, organizational structure, Human Resources management, leadership, and control techniques. The instructor facilitates discussion and integrates these topics using contemporary business issues and case studies. **Co-requisite**: ENGL 210.		

CMNS 310	Persuasion in a Mediated World	3(3,0)
Through reading, writing, and speech, students will become familiar with the persuasive strategy described in Aristotle's Rhetoric and consider how to employ it in professional and personal situations, including writing, public speaking, and negotiation. **Co-requisite**: ENGL 210.		

CMNS 350	Mass and Digital Communication Ethics	3(3,0)
This course addresses the ethical problems in journalism, public relations, and advertising, particularly in today's digital media environment. **Prerequisite**: CMNS 301.		

CMNS 380	Social Media Campaigns and Strategies	3(3,0)
This course examines the strategic uses, impacts, and implications of emerging social media and how it affects the daily lives of our society and its members. It expands the student's knowledge of social media applications in business, advertising, and public relations, as well as their use in political movements and the development of communities. **Co-requisite:** ENGL 210.		

CMNS 401	Peace Communication: Media and Conflict Resolution	3(3,0)
This course integrates methods, perspectives, and concepts of Peace Studies, Cultural Communication, and Peace Journalism to explain contemporary debates in the global public sphere. It considers how cultural stereotypes about ideology, religion, and culture frame Arabs and Americans in the mass media. RHU students will engage in online, asynchronous conversations with students at collaborating universities in the United States and participate in shared projects exploring the role of the media in shaping Arab-American relations. **Co-requisite:** ENGL 210.		

CMNS 410	Public Speaking	3(3,0)
This course is designed to provide a practical introduction to the fundamental principles of public speaking and a forum for practicing public speaking skills. Students will learn how effective speeches are conceived, prepared, and delivered through a range of instructional strategies (discussions, class workshops, readings, lectures, and presentations). **Co-requisite:** ENGL 210.		

EDUC 210	Educational Psychology	3(3,0)
This course draws on psychological principles, cognitive development, and learning theories applicable to teaching/learning situations. Focus areas include human growth and development, specifically adolescent development and developmentally appropriate practices, learning theory, motivation theory, instructional and college practices, individual differences, student interpersonal and group behavior, classroom management, and organization. **Co-requisite:** ENGL 210.		

EDUC 351	Behavior Management and Motivation for Special Learners	3(3,0)
This course provides students with practice in applying behavioral psychology techniques to modify the behavior of and motivate students with special needs. It also addresses the limits of behavior modification. **Co-requisite:** ENGL 210.		

EDUC 415	Factors in Student Motivation	3(3,0)
This course will examine motivational factors that influence children's learning. Topics include attribution theory, teacher influences (expectancy effects), learning structures (competitive vs. cooperative learning), family influences (the socialization of achievement), cultural influences (how cultural background impacts college performance), and individual differences (intrinsic motivation and sex differences in learning). **Co-requisite:** ENGL 210.		

EDUC 480	Special Topics in Education	3(3,0)
This course provides students with advanced study on selected topics or emerging issues in Arab or international education. It is designed for seminar format, multimedia presentation, experiential learning activities, interactive assignments, and cooperative group work. **Prerequisite**: EDUC 201 or consent of advisor.		

JRSM 210	Arabic News Writing & Reporting	3(3,0)
This course emphasizes defining news and its importance in a democratic society; the news-gathering process; news elements; an introduction to basic news reporting and writing for print and broadcast; the use of the Internet as a reporting and research tool; and accuracy and fairness as journalistic imperatives. Outside community research and reporting time is required. **Co-requisite:** ENGL 210.		

JRSM 240	Media and Public Relations Writing	3(3,0)
This introductory-level course is designed to help students learn the formats and conventions of media and public relations writing, including newspapers, magazines, web and interactive media, advertising copy, public relations writing, and social media. It also addresses ethical issues related to writing in these fields. **Co-requisite:** ENGL 210.		

JRSM 300	Mass Media Essentials	3(3,0)
This course familiarizes students with concepts and terminology used in various communication fields and helps them understand the development of mass communication in political, economic, social, and cultural contexts. **Co-requisite:** ENGL 210.		

JRSM 310	Advanced Arabic News Writing and Reporting	3(3,0)
This course aims to provide students with the theory and practice of news story development, with special emphasis on interviewing, observation, document research, source development, and other standard reporting techniques. Students cover		

community beats, report on them, and write news stories from those beats, primarily in public affairs reporting. **Co-requisite:** ENGL 210.

JRSM 320	**Media and Society**	**3(3,0)**
This course examines the relationship between the media and society, considering how each impacts the other. Students will be introduced to various theories that consider this relationship and its implications. **Co-requisite:** ENGL 210.		
JRSM 380	**Arab and International Media**	**3(3,0)**
This course offers students a comparative survey of international media, examining the basic principles and concepts of news and media developed in the West, as applied primarily in Arab countries. **Co-requisite**: ENGL 210.		

POSC 301	**Introduction to Political Science**	**3(3,0)**
This course aims to familiarize students with the basic concepts and disciplines of political science. It focuses on the nature of politics as a science, political ideologies, state and state-society relations, political parties, electoral systems, democracy, and international organizations. A significant emphasis is placed on students' ability to apply these concepts to the analysis of current political issues, disputes, and trends. The course combines lectures and class discussions, debates, and presentations. **Co-requisite**: ENGL 210.		

PSYC 301	**Introduction to Psychology**	**3(3,0)**
This course introduces the theories, concepts, and viewpoints of the discipline of psychology. The course is designed to help students understand human behavior by addressing topics including the history of psychology, learning, personality, motivation, perception, social psychology, mental health, and related areas. **Co-requisite**: ENGL 210.		

PSYC 302	**Social Psychology**	**3(3,0)**
This course studies the dynamics and effects of social influences on individual human behavior. In this course, students will explore topics such as attitude change, social beliefs, roles, conformity within group processes, and prejudices, with special attention to the Middle Eastern perspective. **Co-requisite**: ENGL 210.		

PSYC 303	**Psychology of Personality**	**3(3,0)**
This course presents theories of personality, including psychoanalytic, behaviorist, humanist, and others, while considering different factors that shape personality and behavior. The study includes methods of assessing personality. **Co-requisite**: ENGL 210.		

PSYC 304	**Child Psychology**	**3(3,0)**
This course will introduce students to the psychological development of children from conception to adolescence (with a focus on infancy through middle childhood). Topics, including perceptual, cognitive, personality, and social development, focus on age-		

related issues and events that may affect individuals at different ages. **Co-requisite**: ENGL 210.

PSYC 306	**Psychology of Women**	**3(3,0)**
This course examines the biological, sociological, and cultural influences on the psychology of women. It includes such topics as gender stereotypes, the development of gender roles, gender comparisons, women and work, love relations, women's physical and mental health, violence against women, and women in later adulthood. **Co-requisite**: ENGL 210.		

PSYC 310	**Positive Psychology**	**3(3,0)**
This course examines scientific research on happiness and its practical application in three main areas: positive emotions, positive individual traits, and positive institutions. This course also presents practical ways to promote and maintain well-being and happiness across all aspects of one's life. **Co-requisite**: ENGL 210.		

PSYC 312	**Children and War**	**3(3,0)**
This course is intended to help students understand the consequences of war and conflict on children's mental health and well-being. The course will examine a variety of effects, emphasizing posttraumatic stress disorder (PTSD) and other psychological conditions that result from war. The course will empower students with techniques for working with children affected by war. **Co-requisite**: ENGL 210.		

PSYC 313	**Adolescent Psychology**	**3(3,0)**
This course is designed to examine adolescents' physical, cognitive, social, and moral development within the family, among peers, at school, at work, and in the media. It discusses major theories, methods of studying adolescents, adolescent development, and contemporary adolescent issues and concerns. **Co-requisite**: ENGL 210.		

PSYC 400	**Conflict Transformation**	**3(3,0)**
This course explores practical theories for understanding congregational conflict and various perspectives on conflict transformation. Students will reflect on their styles of dealing with conflict and learn new ways of responding. They will explore practices of dialogue and mediation to address conflict in interpersonal, small-group, and congregational settings, developing an approach to navigating inevitable differences and change. **Co-requisite**: ENGL 210.		

PSYC 401	**Emotional Intelligence**	**3(3,0)**
Intelligence quotient (IQ) gets you hired, but emotional quotient (EQ) gets you promoted". This course explores basic concepts of emotional intelligence and its application to education, the workplace, and personal psychological adjustment. It will offer useful lessons through practical exercises, games, audiovisual tools, case studies, and classroom interaction, providing a roadmap for fostering emotional intelligence in the college and office to achieve health, happiness, and optimal performance at work. **Co-requisite:** ENGL 210.		

PSYC 410	Abnormal Psychology	3(3,0)
An introduction to the theories of abnormal behaviors, including addiction and substance abuse, bipolar disorder, eating disorders, depression, obsessive-compulsive behaviors, phobias, schizophrenia, and sexual disorders. Students will consider the theoretical implications and treatments for various abnormal behaviors. **Co-requisite**: ENGL 210.		

PSYC 480	Special Topics in Psychology	3(3,0)
This course provides an in-depth look at selected issues in psychology and individual psychology. Current psychology-related issues are thoroughly explored through readings, research, discussion, and guest lectures. **Prerequisite**: Consent of the instructor.		

SOCI 210	Research Methods in the Social Sciences	3(3,0)
This course is designed for psychology, sociology, communications, and other social sciences students. It introduces them to advanced research design and methodology, including statistical analysis. **Prerequisite**: MATH 220.		

SOCI 301	Introduction to Sociology	3(3,0)
This course explores a variety of sociological theories and approaches to understanding society. The student will learn about human social life, groups, and societies, and be helped to detect the social influences shaping our lives. The approach used in this course is a self-exploratory process that develops individual and social awareness. A secondary objective is to discuss methods common to sociology, conflict theory, symbolic interactionism, and feminism. **Co-requisite**: ENGL 210.		

SOCI 302	Political Sociology	3(3,0)
This course examines the social issues surrounding the historical and contemporary evolution of various political ideologies. It aims to develop students' understanding of the social bases of politics and political ideologies, as well as recent changes in the socio-political world. Different economic and political philosophies are considered, and their effects on the fate of nations and their citizens are detected. **Co-requisite**: ENGL 210.		

SOCI 304	Introduction to Educational Sociology	3(3,0)
This course introduces the study of sociological concepts as they apply to the institution of education. Students will examine issues of race, class, status, poverty, and bureaucracy in the educational setting. They will also examine current perspectives and research about the social aspects of the learning process. **Co-requisite**: ENGL 210.		

SOCI 310	Intercultural Communication	3(3,0)
This course deals with cultural issues like cultural awareness, enculturation, acculturation, and culture shock. It involves hands-on activities to promote cultural awareness and equip students with skills for dealing with people from diverse cultural backgrounds. It includes team-oriented research and presentation that reflects all the		

cultural aspects and some minute details of the culture they choose to work on. **Co-requisite**: ENGL 210.

SOCI 313	**Interpersonal Communication**	**3(3,0)**
Students in this course examine communication theory and case studies to gain insight into the factors that affect human communications. It aims to improve students' communication skills and equip them with techniques to enhance the quality of their relationships. The course also offers strategies for managing interpersonal conflicts. **Co-requisite**: ENGL 210.		

SOCI 340	**Arab Culture and Society**	**3(3,0)**
This course examines contemporary Arab society, considering its social structures and groups, cultural patterns, and other aspects. Students consider current issues related to Arab culture and society. **Co-requisite** ENGL 210.		

SOCI 410	**Culture and Identity**	**3(3,0)**
This course analyzes the role of culture in an individual's sense of identity, emphasizing how it manifests in the Middle East. **Co-requisite** ENGL 210.		

SOCI 420	**Social Inequalities**	**3(3,0)**
The course addresses class, status, and gender inequalities, examining how they affect individuals and society. **Co-requisite** ENGL 210.		

SOCI 480	**Special Topics in Sociology**	**3(3,0)**
This senior course provides an in-depth look at selected issues in sociology and society. Through readings, research, discussion, and guest lectures, current issues in sociology are thoroughly explored. **Prerequisite**: Consent of the instructor.		

Natural Science and Technology Electives

Descriptions of elective courses that may be chosen to satisfy the Natural Sciences domain are given below.

BIOL 210	**Human Anatomy and Physiology and Lab**	**3(2, 2)**
A general overview covers the structure, function, physiology, and anatomy of the human body systems. These include cardiovascular, central nervous, respiratory, urinary, digestive, immune, and musculoskeletal systems. The course also offers a set of experiments that deal with basic biological processes and advanced biological assessments. **Prerequisite**: None.		

BIOL 211	**General Biology**	**3(3,0)**
Topics covered are small/large molecules, cell organization, membranes, energy, enzymes and catalysts, basic cell functions, and biological control systems.		

BIOL 222	Introduction to Food Safety and Lab	3(2, 2)
This course aims to teach the general principles of food microbiology and food safety. It covers the biology and epidemiology of foodborne microorganisms of public health significance, including bacteria, fungi, protozoa, and viruses, and food spoilage microorganisms; chemical pollution of foodstuffs; safety and preservation of food commodities; Principal methods for examination of foods; food quality control and implementation of the Hazard Analysis Critical Control Point (HACCP) scheme. The course also offers a set of experiments to cover the laboratory methods used in the microbiological analysis of foods and to identify the characteristics of the major groups of microorganisms associated with food spoilage, foodborne disease, and food fermentations. **Prerequisite**: None.		

BITM 300	Business IT Management	3(3,0)
This course provides a comprehensive introduction to information systems and their application. It explains how to use and manage information technology to revitalize business processes, conduct e-commerce, improve business decision-making, and gain a competitive advantage. **Prerequisite**: BITM 200		

CHEM 210	Principles of Chemistry	3(3,0)
This course emphasizes the fundamental principles of chemistry, including atomic structure, bonding, stoichiometry, gases, solutions, acids and bases, and solution equilibria. **Prerequisite**: None.		

CHEM 211	Environmental Chemistry and Lab	3(2,2)
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.		

COSC 214	Introduction to Programming	3(3,0)
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

EDUC 317	Neurological Development and Cognitive Change	3(3,0)
This course provides a unified view of neurological development, beginning with the growth of the first neural cells and extending through the emergence of complex thought and behavior. It considers brain mechanisms underlying cognitive, perceptual, social, and emotional development during infancy and childhood.		

EDUC 222	Computer Applications in Education	3(3,0)
This course focuses on general knowledge of the use of microcomputers in education, including common computer applications such as word processing, spreadsheet, database, and presentation software in teaching/learning activities, as well as evaluating the effectiveness of educational software in teaching/learning across various subject matter areas.		

GEOL 300	Human Geography: People and Culture	3(3,0)
Different cultures in different landscapes are compared to those in the Middle East. This course examines the many issues surrounding human population, perceptions of the environment, the diffusion of ideas, and the cultural aspects of resources and urban growth. **Co-requisite**: ENGL 210.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

Quantitative Reasoning Electives

BADM 250	Business Statistics	3(3,0)
Business Statistics introduces students to the fundamentals of applied statistics. Accordingly, students are exposed to statistics as they are directly applied in solving business problems. The course will cover random variables, sampling, probability distributions, expectation, hypothesis testing and confidence intervals, analysis of variance, correlation, and simple linear regression.		

BADM 420	Quantitative Methods for Business	3(3,0)
This course introduces students to managerial decision analysis using quantitative tools. The course will introduce students to the use and development of mathematical models to help managers make informed decisions. The focus is on the applied aspects of statistics and math. As such, the course will cover the basics of probabilistic and statistical techniques, decision analysis, linear programming, optimization, forecasting, and waiting-line theory. **Prerequisite:** BADM 250, MATH 351, or Senior Standing.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

GRDS 220	Introduction to Digital Media	3(3,0)
In this course, students will develop a solid foundation in the tools and techniques of Adobe Illustrator and Adobe Photoshop. Students will learn the difference between vector-based and pixel-based programs and how to integrate both work fields to produce creative graphics. Moreover, students will be briefly introduced to Adobe InDesign, including its workspace, functions, and tools.		

GRDS 335	Production	3(3,0)
In this course, students will develop an understanding of the “process flow” of contemporary print production methods. The course covers the entire production timeline: Prepress, Press, and Postpress. In addition to printing technologies, students will learn about various types of ink and paper, binding, varnishes, finishing techniques, and the effect of the printing process on the product. Students will also learn how to handle projects and deal with clients as freelance designers or as designers working for companies. Prerequisite: GRDS 222.		

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, structural induction, sets and set operations, functions, growth of functions, relations and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 220	Statistics for Social & Behavioral Sciences	3(3,0)
This course introduces statistical techniques for analyzing social science research data. Topics include frequency distribution, central tendency, variability, probability theory, Bayes’ rule, mathematical expectation, variance and covariance, binomial, Poisson, hypergeometric and normal distributions, sampling distributions, estimation problems, and hypothesis testing. **Prerequisite**: None.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215.		

PHIL 302	Introduction to Formal Logic	3(3,0)
This course is a standard introduction to the formal techniques of argument analysis. Formal logic was invented to mirror and evaluate mathematical reasoning; however, this course will concentrate on the relationship between formal logic and everyday reasoning and language usage in general. **Co-requisite**: ENGL 210.		

Community and Sustainability Electives

BADM 215	Personal Development and Management	1(1,0)
This course engages the student in self-reflection, self-discovery activities, and lectures. It is intended to encourage students to develop themselves and plan for the present and the future by becoming more self-aware. Topics and issues include personality types, communication styles, personal performance factors, career choices, and personal development planning.		

BADM 290	Community Engagement Experience	0(0,1)
The Community Engagement experience consists of a specific number of volunteer hours by students at community organizations under the supervision of a faculty member. Community organizations may include governmental agencies, non-governmental organizations, and civil society organizations.		

CMNS 200	Etiquette	1(1,0)
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, regardless of the industry, organization, or sector in which they are employed. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentations, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	First Aid	1(1,0)
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

EDUC 315	Children's Health and Safety in the School Environment	3(3,0)
This course introduces students to childcare facilities and to environmental health concepts related to children's health and well-being. It examines current policies and practices of environmental health. It covers indoor and outdoor safety concerns, safe food handling, Universal Precautions, and other aspects of disease prevention, as well as the recognition and reporting of child abuse. It includes specific strategies to		

strengthen families to reduce the risk of child abuse and neglect. **Co-requisite** ENGL 210.

EDUC 480	**Special Topics in Education**	**3(3,0)**
This course provides students with advanced study on selected topics or emerging issues in Arab or international education. It is designed for seminar format, multimedia presentation, experiential learning activities, interactive assignments, and cooperative group work. **Prerequisite**: EDUC 201 or consent of advisor.		

ENVS 201	**Introduction to Environmental Science**	**3(3,0)**
The main goal of the Introduction to Environmental Science (ENVS201) course is to provide students (except engineering) with the scientific principles and concepts required to understand the interrelationships of the natural world, to identify and analyze environmental problems, both natural and man-made; to evaluate the relative risks associated with these problems; and to examine alternative solutions for resolving and preventing environmental problems. **Prerequisite**: None.		

GRDS 440	**Design and Social Impact**	**3(3,0)**
This course prepares students to become leading agents of social change through community engagement. It aims to promote collaboration between computer science and graphic design students to develop effective solutions to existing community issues. The course is structured around a balance of classroom theory, extensive field research, and the implementation of a real-world solution that draws on both. **Prerequisite:** Junior Standing.		

INDS 335	**Design Issues and Sustainability**	**3(3,0)**
This course investigates theoretical and philosophical constructs, as well as design issues, related to the study of interior architecture and design. It examines philosophical and practical principles of sustainable design by exploring environmental issues, sustainable materials and methods, and professional practice. Students will develop an awareness of the environmental implications of design decisions. They will gain a foundation for evaluating materials, processes, and practices in accordance with sustainable and environmentally responsible design principles.		

PSYC 314	**Psychology for Sustainability**	**3(3,0)**
This course will take a holistic approach to the topic, drawing on most major areas of psychology to explore environmental sustainability from a psychological perspective. We will use psychological principles, theories, and methods to examine the reciprocal relationship between human beings and the natural world, discuss the complex nature of environmental problems, and review important psychological and social factors that underlie sustainable and non-sustainable behaviors. **Co-requisite**: ENGL 210.		

College of Arts and Sciences

COLLEGE OF ARTS AND SCIENCES (CAS)

College Officers

Said Ladki	President
Mahmoud Halablab	Vice President for Academic Affairs
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History and Overview

RHU's College of Arts and Sciences (CAS) is uniquely positioned to meet the growing need in Lebanon for education, language, media, computer science, math, and design fields. We focus on graduating 21st-century professionals who excel in their fields and contribute significantly to their communities. CAS graduates have the opportunity to play pivotal roles in advancing civil society and raising civic awareness. We are committed to developing students' leadership skills and empowering them to make a real difference in the community.

At the College of Arts and Sciences, our faculty members are not just researchers; they are practitioners in their varied fields. They bring their hands-on professional experience into the classroom, enriching our students' learning. Our faculty members' publications, including books, academic journal articles, conference papers, magazine and newspaper articles, broadcast media scripts, and more, are a testament to their expertise and commitment to their disciplines.

Vision

The College of Arts and Sciences (CAS) aims to provide excellent educational standards, purposeful research, and interdisciplinary investigation into present and future problems facing humanity in Lebanon, the MENA region, and the World.

Mission

As educators committed to Rafik Hariri University, our mission in the College of Arts and Sciences is to provide students with a holistic, integrated education across the arts, English, humanities, social sciences, design, math, sciences, and computing. We are

dedicated to excellence in teaching, scholarly work, academic quality, professional programs, technological innovations, and friendly faculty-student relationships. We prepare our graduates to be highly productive in their prospective careers.

Core Values

The College of Arts and Sciences is committed to acting by the following guiding values:

Professional Excellence
We, faculty and students alike, strive to consistently do our utmost to reach high standards of excellence in our performance.

Collaboration
We value and benefit from each other's unique perspectives by engaging in mutual respect. We share expertise, resources, and time to pursue common goals.

Integrity
We are committed to transparent decision-making in all College of Arts and Sciences processes. We insist on treating every faculty member, staff member, and student fairly and compassionately.

Celebration
We recognize, appreciate, and celebrate the achievements of all members of the College of Arts and Sciences (faculty, students, and staff).

Compassion
We are concerned about the whole person and their well-being, whether they are within our college or the greater community. That concern translates into compassionate actions towards individuals and the larger community.

Innovative Teaching
We engage faculty and students in innovative teaching methodologies, including the effective use of technology.

Learning Objectives

Students in the College of Arts and Sciences will:

Develop professional background and skills in their chosen fields.
Students will become proficient in theory and application relevant to their majors and learn to pursue research in their chosen fields.

Integrate academic learning with community experiences.
Students will apply what they have learned from their workplace experiences to enter their professions with confidence upon graduation.

Value and practice reflective thinking
Students will engage in reflective, critical, and independent thinking.

Develop communication skills
Students will engage in classes that emphasize oral and written communication skills.

Develop personal values
Programs and activities at the college will provide opportunities for students to develop their values by engaging with community social issues.

Develop digital media literacy.
Students will develop the ability to "read" and "write" in today's media, a key skill for all 21st-century citizens.

Academic Programs

The College of Arts and Sciences offers an undergraduate program leading to a Bachelor of Arts (BA) degree and three undergraduate programs leading to a Bachelor of Science (BS) degree. The programs are as follows:

1. Bachelor of Arts in English Language
2. Bachelor of Science in Computer Science (ABET Accredited)
3. Bachelor of Science in Healthcare Information Systems
4. Bachelor of Science in Graphic Design
5. Minors in Computer Science, Data Analytics, Graphic Design, Mathematics, English Language and Literature, and Psychology

Program Codes

The following table lists the code used for each program. This code is the prefix of each course designation in the program.

Program Code	Program
EDUC	Education
ENGL	English Language
COSC	Computer Science
HCIS	Healthcare Information Systems
MATH	Mathematics
GRDS	Graphic Design

Admission Requirements

Students wishing to enter the College of Arts and Sciences must fulfill the University admission requirements. Students aspiring to major in the English language must attain a minimum score of 550 on their English placement test. Students aspiring to study in the Design Department must take an aptitude test to get accepted into the major. Students are strongly advised to review the University Catalog carefully for admission and degree requirements, as well as related academic policies.

Graduation Requirements

Each program in the College of Arts and Sciences grants students a bachelor's degree. The BA program encompasses 93 credits for the English Language. The BS program comprises 93 credits in Computer Science, 100 in Healthcare Information Systems, and 111 in Graphic Design. The credit hour allocations for each program are shown in the following tables:

BA in English Language (93 credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	9	-	6	15	30	32
College Requirements	-	-	9		9	10
Program Requirements	27	18	3	6	54	58
Credits	54		39		93	100

BS in Computer Science (93 credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	6	–	15	9	30	32
College Requirements	-	–	–	–	-	0
Program Requirements	36	9	18	–	63	68
Credits	42	9	33	9	93	100

BS in Healthcare Information Systems (100 credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	9	-	13	9	31	31
College Requirements	-	-	-	-	-	-
Program Requirements	42	9	18	-	69	70
Credits	51	9	31	9	100	100

BS in Graphic Design (111 credits)

Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	12	-	9	9	30	27
College Requirements	-	-	-	-	-	-
Program Requirements	51	6	21	3	81	73
Credits	63	6	30	12	111	100

A student in any of the CAS programs is eligible for graduation if s/he has:

1) Completed all the requirements of the degree
2) Attained a cumulative GPA of 70 or higher
3) Attained a major GPA of 70 or higher
4) Attained at least a 70 grade on the summative learning project/ senior project
5) Completed the mandatory Co-op training program.

I. University Requirements (General Education)

Every RHU student must take 30–33 credit hours of general education courses, distributed across six domains described below. Twenty-one mandatory credits are selected from domains 1, 4, 5, and 6, and 9-12 elective credit hours are selected from domains 2 and 3.

Domain		Credits	Courses
1	Communication Competency*	8	ENGL 210: English Composition and Rhetoric ENGL 217: Professional English Communication ARAB 212**:** Arabic Language and Communication
2	Humanities/Fine Arts	3-6	Selected from an approved list
3	Social Sciences/Culture	6	Selected from an approved list
4	Natural Sciences and Technology	6	Selected from an approved list
5	Quantitative Reasoning	3	Selected from an approved list
6	Community and Sustainability	4	Selected from an approved list

* Refer to the introductory section of the college of interest for a description of courses specific to domains 1, 4, 5, and 6.

Descriptions of the mandatory general education courses are given below:

BITM 200	**Information Technology Essentials**	**3(3,0)**

This course introduces MS Office products (Word, Excel, PPT, and ACCESS), Web page design and server upload, Internet use, and how PCs work.

CMNS 355	**Digital Media Literacy**	**3(3,0)**
The class aims to produce media-literate students who successfully use media to voice their ideas, become better citizens, and engage in social, political, and cultural change. It focuses on how media communicate meanings that influence our perceptions of reality, including our perceptions of ourselves, society, politics, and culture. It teaches students to effectively use critical thinking to deconstruct media messages, analyze and interpret intent, and generate constructive responses. It provides students with essential new media production skills and knowledge to create digital media messages, including digital design, photo manipulation, video/audio production, blogging, podcasting, and screencasting.		

EDUC 222	**Computer Applications in Education**	**3(3,0)**
This course focuses on general knowledge of the use of microcomputers in education, including common computer applications such as word processing, spreadsheet, database, and presentation software in teaching/learning activities, as well as evaluating the effectiveness of educational software in teaching/learning across various subject matter areas.		

EDUC 480	**Special Topics in Education**	**3(3,0)**
This course provides students with advanced study on selected topics or emerging issues in Arab or international education. It is designed for seminar format, multimedia presentation, experiential learning activities, interactive assignments, and cooperative group work. **Prerequisite**: EDUC 201 or consent of advisor.		

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGL 420	**Literature and Culture**	**3(3,0)**

This course considers major works of literature, specifically in the context of twentieth-century cultural theory, including Marxism, post-colonialism, national literature, ethnic writings, and feminist theory. The primary intention of this course is to explore how various texts interact with their societies or how those societies influence the construction of literary works. **Prerequisite**: Consent of the instructor.

GRDS 220	**Introduction to Digital Media**	**3(3,0)**
In this course, students will develop a solid foundation in the tools and techniques of Adobe Illustrator and Adobe Photoshop. Students will learn the difference between vector-based and pixel-based programs and how to integrate both work fields to produce creative graphics. Moreover, students will receive a brief introduction to Adobe InDesign, including its workspace, functions, and tools.		

GRDS 335	**Production**	**3(3,0)**
In this course, students will develop an understanding of the "process flow" of contemporary print production methods. The course covers the entire production timeline: Prepress, Press, and Postpress. In addition to printing technologies, students will learn about various types of ink and paper, binding, varnishes, finishing techniques, and the effect of the printing process on the product. Students will also learn how to handle projects and deal with clients as freelance designers or as designers working for companies. **Prerequisite**: GRDS 222.		

GRDS 360	**Visual Perception & Infographics**	**3(3,0)**
Visual perception is the ability to interpret information from our surroundings through visible light that reaches the eye. This course will explore the various theories of visual perception with an emphasis on Gestalt theory and its relation to Graphic Design. It will create an understanding of our human visual perception and its limitations. Students will also be given projects in which they will apply Gestalt principles and the theories they have learned. In addition to visual perception theories, students will learn two key design disciplines: Infographics and Wayfinding. **Prerequisite**: GRDS 311.		

GRDS 440	**Design and Social Impact**	**3(3,0)**
This course prepares students to become leading agents of social change through community engagement. It aims to promote collaboration between computer science and graphic design students to develop effective solutions to community issues. The course is structured around a balance of classroom theory, extensive field research, and the implementation of a real-world solution that draws on both. **Prerequisites:** Junior Standing.		

HCIS 201	**Environmental Health**	**3(3, 0)**
This course is designed to provide students with an introduction to the key areas of environmental health. The course will cover factors associated with the development of environmental health problems from population and community perspectives. **Prerequisite***:* None*.*		
INDS 220	**Computer Design I**	**3(3,0)**

This course introduces the principles and applications of computer-aided drafting and design using the latest version of AutoCAD. Students will acquire the technical skills to draft 2D architectural plans, sections, elevations, and details. In addition, students will learn computer-rendering techniques using Adobe Photoshop.

INDS 325	**Lighting & Mechanics**	**3(3,0)**
This course comprehensively studies lighting systems, mechanics, principles, and theories. Students will learn about lighting design applications, lighting specifications, in-depth lighting mathematical calculations, available equipment and fixtures, and aesthetic considerations needed to complete the installation. Students will also become familiar with different heating, ventilation, and AC systems, including their installation, design considerations, and needs, as well as plumbing and sanitary systems. **Prerequisite**: INDS 315.		

INDS 335	**Design Issues and Sustainability**	**3(3,0)**
This course investigates theoretical and philosophical constructs, as well as design issues, related to the study of interior architecture and design. It examines philosophical and practical principles of sustainable design by exploring environmental issues, sustainable materials and methods, and professional practice. Students will develop an awareness of the environmental implications of design decisions. They will gain a foundation for evaluating materials, processes, and practices in accordance with sustainable and environmentally responsible design principles.		

MATH 207	**College Algebra**	**3(3,0)**
Solving linear and non-linear equations, modeling with equations, functions, and their graphs, increasing and decreasing functions-transformation, quadratic functions and minima, modeling with functions, combining functions, polynomial functions and their graphs, dividing polynomials, real zeros of polynomials, complex numbers, complex zeros of polynomials, exponential and logarithmic functions, sequences and summation notation-arithmetic sequences, counting and probability.		

MATH 220	**Statistics for Social & Behavioral Sciences**	**3(3,0)**
This course introduces statistical techniques for analyzing social science research data. Topics include frequency distribution, central tendency, variability, probability theory, Bayes' rule, mathematical expectation, variance and covariance, binomial, Poisson, hypergeometric and normal distributions, sampling distributions, estimation problems, and hypothesis testing. **Prerequisite**: None.		

MATH 351	**Probability and Statistics**	**3(3,0)**
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215.		
PHYS 211	**Physics: Electricity and Magnetism and Lab**	**3(3,1)**

This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.

PHYS 312	**Modern Physics**	**3(3,0)**
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

II. College Requirements

A. Remedial Courses

Proficiency in the English Language is a requirement for admission to any program in the College of Arts and Sciences. Students who do not meet the University English proficiency admission requirements may gain admission through the Intensive English Program, in which they can develop their English proficiency. In addition, incoming students in the Computer Science, Healthcare Information Systems, and Mathematics programs must pass the mathematics placement exam. Those who fail the mathematics placement exam and obtain a low score must take one or two remedial Mathematics courses before taking program-level calculus courses. The possible remedial courses are:

- MATH 190 - Calculus I
- MATH 191 – Calculus II

These remedial courses do not count toward fulfilling the degree requirements. A description of the remedial courses follows.

MATH 190	**Calculus I**	**3(3,0)**
Functions and their graphs, limits and continuity, differentiation, applications of derivatives, Extreme values and mean value theorem, definite and indefinite integrals, and the substitution method.		

MATH 191	**Calculus II**	**3(3,0)**
Covers limits and indeterminate forms with L'Hôpital's Rule, optimization, antiderivatives, and the Fundamental Theorem of Calculus. Topics include substitution methods, areas, volumes (cross-sections, shells), arc length, and surfaces of revolution. It introduces parametric and vector functions with applications to motion, arc length in space, curvature, and normal vectors. **Prerequisite**: Math 190.		

B. Mandatory Courses

All students are required to take nine credit hours of mandatory foundational courses. A list of these courses and their descriptions follows.

Course	Title	Credits	Prerequisite
ARAB 212	Arabic Language and Communication	2	
	Community & Sustainability	1	
ENGL 210	English Composition and Rhetoric	3	
ENGL 217	Professional English Communication	3	

Descriptions of these courses are given below.

ARAB 212	**Arabic Language and Communication**	**2(2,0)**
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need in their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

CMNS 200	**Etiquette**	**1(1,0)**
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, no matter what industry, organization, or sector they are employed in. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentations, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	**First Aid**	**1(1,0)**
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite**: ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails,		

memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.

III. Program Requirements

Requirements for the bachelor's degree are program-specific. They encompass two categories: Major and non-major. Each category consists of mandatory and elective courses. The program requirements for the bachelor's degree in the various CAS majors are provided below. Details and titles of relevant courses are included in the Student's Study Plan (SSP), which every CAS student will have.

Course Coding

Each course offered by the College of Arts and Sciences is designated by a four-letter code representing the program or elective category, followed by a three-digit number denoting the course number; for example, "**EDUC 301**."

The four-letter codes below refer to the respective course categories as follows:

ADPR: Advertising and Public Relations
ARAB: Arabic
CMNS: Communications
COSC: Computer Science
EDUC: Education
ENGL: English Language
GRDS: Graphic Design
HCIS: Healthcare Information Systems
HIST: History
IELP: Intensive English Language Program
INDS: Interior Design
JRSM: Journalism – Media and Digital Communication
MATH: Mathematics
POSC: Political Science
PSYC: Psychology
SOCI: Sociology

The first digit in the three-digit course number indicates the course level: **1** for freshman, **2** for sophomore, **3** for junior, **4** for senior, and **5** for graduate.

The following example demonstrates the coding system: EDUC 201 is a sophomore-level education course.

The designation used to represent the credit-hour breakdown (t, p) of a course is as follows: "c" is the total credit hours, "t" is the theoretical component of the course, and "p" is the practical or laboratory component. For example, 3(3, 0) denotes a 3-credit-hour course with 3 contact lecture hours and 0 laboratory hours.

Learning Support Center

The Learning Support Center (LSC) at the College of Arts and Sciences is another manifestation of RHU's resolute commitment to students' academic success. The LSC offers the students a peer-to-peer learning opportunity that complements classroom learning. The Center aims to support students in advancing their knowledge and skills, reinforcing what they learn in the classroom or overcoming learning and performance difficulties. In addition to serving as a focal point for inquiry and two-way "asker-replier" learning, the LSC provides space for academic interactions that can significantly enhance learning. The Center is in Block I and is open to all students throughout the week.

The LSC is staffed by Teaching Assistants, upper-level undergraduate students, who are always available to assist their fellow students in overcoming academic challenges. Additionally, "on-call" faculty members are assigned to ensure that the LSC is acting in the best interests of the students. The center's resources include books, study guides, course materials, boards, computers, large-screen LCD panels, and internet-connected PCs.

LANGUAGES AND LIBERAL ARTS UNIT (LLA)

Faculty Members

Professor:	Najwa Saba ‘Ayon Fares Mahmoud
Associate Professor:	Grasiella Harb,
Assistant Professor:	Maysaa Banat, Naziha Shamseddine
Academic Coordinator:	Najwa Saba ‘Ayon Fares Mahmoud
Senior Lecturer:	Dina Baba
Adjunct Faculty:	Mira Alameddine, Kamal Nahas, Rola Danab, Likaa Abou Hadir, Mirna Talhouk, Houda Arkadan, Nisreen Sinjab, Wafaa Sheaib, Fawzieh Orabi, Fayza Hazimeh, Nashrawan Mourad, Maya Hamzeh, Abir Oweini, Claudia Hassan, Farid Khoury, Khaled Ghattas.

Programs Offered

The Languages and Liberal Arts Unit offers an undergraduate program leading to a Bachelor of Arts (BA) degree in English Language and two minors: one in English Language and Literature and one in Psychology. It also offers all RHU students the Intensive Language Program and core education courses. The details of each program will follow.

ENGLISH LANGUAGE PROGRAM

Learning Objectives

The main goal of the Bachelor of Arts in English Language Program is to prepare interested students in the English major to become 21st-century competent English Language graduates by providing them with the essential linguistic, literary, fiction/non-fiction writing, and intercultural backgrounds needed to pursue careers and graduate work in their field. Following the latest teaching methods, the BA English Language Program helps students deepen their knowledge in the field, develop critical thinking, and learn to work collaboratively. It also equips students with strong research, literacy, and critical-thinking skills to meet the needs of 21st-century English-language graduates.

Learning Outcomes

After completing all the required courses in the English Language program, students should be able to:

1. Identify different subfields in linguistics;
2. Use linguistic and literary terminology in the analysis of literary texts;
3. Differentiate between different theories in linguistics;
4. Analyze and critique different literary texts;
5. Communicate competently in English with speakers from diverse cultures;
6. Synthesize information obtained from different texts;
7. Conduct research related to the English language;
8. Take up a profession in the English language.

Career Opportunities

English Language graduates find fulfilling careers in various public and private professions, such as schools, translation firms, publishing houses, tourism-industry public relations firms, voluntary and charitable organizations, and media organizations, to name a few. English Language graduates may become English teachers, editors, journalists, public relations practitioners, writers, translators, and many other professionals where skill in using language is essential. Many also pursue graduate studies in fields such as linguistics, education, literature, and communication.

Program Overview

The Bachelor of Arts in English Language Program aims to provide students with a theoretical and practical understanding of the many functions of English, including its structure, use, and learning. This program also equips students with linguistic foundations and a general background in literature and fiction/non-fiction writing.

Through effective teaching, students develop their own critical and analytical thinking. Finally, this program prepares students to either pursue graduate studies or take up a profession in language-related fields.

To obtain a Bachelor of Arts degree in English Language, the student must complete 93 credit hours. These hours cover University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BA in English Language (93 credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	9	-	6	15	30	32
College Requirement	-	-	9		9	10
Program Requirement	27	18	3	6	54	58
Credits	54		39		93	100

I. University Requirements (General Education)

The list of the University's required courses and their descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented in the introductory pages of the College of Arts and Sciences section of this catalog.

III. Program Requirements

A. Mandatory Requirements

The English Language Program’s mandatory courses are listed in the table below.

Course #	Title	Credits	Prerequisites
ENGL 222	Introduction to Language	3	
ENGL 250	Introduction to English Literature I	3	
ENGL 251	Introduction to English Literature II	3	ENGL 250
ENGL 342	Introduction to Shakespeare	3	ENGL 210
ENGL 350	History of the English Language	3	ENGL 222
ENGL 360	Introduction to Phonetics	3	ENGL 222
ENGL 370	Modern English Grammar	3	ENGL 210
ENGL 400	Advanced English Grammar	3	ENGL 370
ENGL 450	Advanced Academic Writing	3	ENGL 210

ENGL 462	Introduction to Psycholinguistics	3	
ENGL 463	Introduction to Sociolinguistics	3	
ENGL 465	Senior Project	3	ENGL 217; senior standing

B. Major Electives

As part of the Bachelor of Arts in English Language program, the student must study 18 credit hours of major electives. Major electives include upper-level English language, literature, or fiction/non-fiction writing courses. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility, allowing students to tailor the program to their interests.

The student should select elective courses that best meet their needs and aspirations in cooperation with the academic advisor. It is highly recommended that the students register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Co-requisites
ENGL 221	English Composition and Rhetoric II	3	ENGL 210
ENGL 223	Introduction to Drama	3	ENGL 210
ENGL 235	Creative Writing	3	ENGL 210
ENGL 252	Introduction to American Literature	3	ENGL 210
ENGL 300	Introduction to World Literature	3	ENGL 210
ENGL 320	Creative Nonfiction	3	ENGL 210
ENGL 330	Language Acquisition	3	
ENGL 365	Applied Linguistics	3	
ENGL 420	Literature and Culture	3	
ENGL 460	Discourse Analysis	3	
ENGL 497I	Advanced Topics in English	3	Consent of instructor

Study Plan

The Bachelor of Arts in English Language encompasses 93 credit hours over six semesters and two summer sessions. The following study plan serves as a roadmap for smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
EDUC 222	Computer Applications in Education	3	
ENGL 210	English Composition and Rhetoric	3	ITP TOEFL 550/ or SAT 380+ or IELTS 6.5 or ENGL 101
ENGL 222	Introduction to Language	3	
ENGL 250	Introduction to English Literature I	3	
	Community & Sustainability	1	

Year 1, Spring Semester (18 Credits)			
ENGL 217	Professional English Communication	3	ENGL 210
ENGL 251	Introduction to English Literature II	3	ENGL 250
ENGL 360	Introduction to Phonetics	3	ENGL 222
MATH 220	Statistics for Social and Behavioral Sciences	3	
	Community and Sustainability	3	Co-req.: ENGL 210
	Natural Science Elective	3	
Year 2, Fall Semester (15 Credits)			
ENGL 342	Introduction to Shakespeare	3	ENGL 210
ENGL 350	History of the English Language	3	ENGL 222 or consent of the instructor
	English Linguistics Elective	3	
	Technology Elective	3	
	Social Sciences	3	Co-req.: ENGL 210
Year 2, Spring Semester (15 Credits)			
ENGL 370	Modern English Grammar	3	ENGL 210
PSYC 301	Introduction to Psychology	3	Co-req.: ENGL 210
	Fiction | Nonfiction Writing Elective	3	ENGL 210
	Humanities/Culture Elective I	3	ENGL 210
	Social Sciences	3	Co-req.: ENGL 210
Year 3, Fall Semester (15 Credits)			
ENGL 400	Advanced English Grammar	3	ENGL 370
ENGL 450	Advanced Academic Writing	3	ENGL 210
ENGL 462	Introduction to Psycholinguistics	3	
	English Linguistic Elective	3	
	Humanities/Culture Elective II	3	ENGL 210
Year 3, Spring Semester (15 Credits)			
ENGL 463	Introduction to Sociolinguistics	3	
ENGL 465	Senior Project	3	ENGL 217; Senior Standing
	English Linguistic Elective	3	
	English Literature Elective	3	
	Fiction | Nonfiction Writing Elective	3	ENGL 210

Courses Description

Mandatory Courses

Major Courses

Descriptions of the major mandatory courses are given below.

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGL 222	**Introduction to Language**	**3(3,0)**
This course examines current theoretical and applied areas of linguistics, including the different levels of structure, the nature of language acquisition, language variation and evolution, and language teaching.		

ENGL 342	**Introduction to Shakespeare**	**3(3,0)**
A course in which students study Shakespeare's representative comedies, histories, and tragedies. The plays are read intensively and understood in the context of the period's theatrical conventions, the culture of play in early modern England, and the period's social, cultural, religious, and intellectual history. **Prerequisite**: ENGL 210		

ENGL 250	**Introduction to English Literature I**	**3(3,0)**
This course covers English literature from Anglo-Saxon times to the late eighteenth century. Specific texts by the principal writers of these periods will be examined against the social, historical, and philosophical background of their periods.		

ENGL 251	**Introduction to English Literature II**	**3(3,0)**
This course continues the survey of English literature, which began in Introduction to Literature I, moving from Blake to Eliot, through examining specific texts by major authors against the period's social, historical, and philosophical background. **Prerequisite** ENGL 250		

ENGL 350	History of the English Language	3(3,0)
This is an introductory survey of the history of the English language from its earliest Indo-European origins to the present day. The nature and changes of the language are presented by reviewing the shifts from Indo-European, Germanic, Old English, Middle English, and Early Modern English. **Prerequisite**: ENGL 222 or consent of the instructor.		

ENGL 360	Introduction to Phonetics	3(3,0)
This is a course in which students study the articulatory, auditory, and acoustic descriptions of English sounds. This course also includes descriptions of the sounds of other languages. The practical component of this course involves practice in the transcription and production of English sounds. **Prerequisite:** ENGL 222		

ENGL 370	Modern English Grammar	3(3,0)
This course teaches grammar through exploration and analysis. It provides a more detailed study of word and phrase formation, pragmatics, and a critical analysis of descriptive uses of grammar. **Prerequisite**: ENGL 210		

ENGL 400	Advanced English Grammar	3(3,0)
An advanced course in English grammar that equips students to analyze the differences among major English grammars. Traditional, descriptive, and transformational grammar are examined. **Prerequisite**: ENGL 370		

ENGL 450	Advanced Academic Writing	3(3,0)
This course allows students to further their academic writing skills through advanced independent research, writing, and discussion of various academic topics. Students will receive rigorous research training, critiquing and analyzing texts and resources, synthesis, and composition. This required course helps students to pursue graduate studies. **Prerequisite**: ENGL 210		

ENGL 465	Senior Project	3(3,0)
In this course, students must conduct a group research study on English language learning or teaching. **Prerequisite**: Senior standing.		

Non-Major Courses

ARAB 212	Arabic Language and Communication	2(2,0)
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need in their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

CMNS 200	Etiquette	1(1,0)
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, no matter what industry, organization, or sector they are employed in. Students will improve their professional style as they study topics including polite conversation, personal appearance, office		

politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentations, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.

EDUC 222	**Computer Applications in Education**	**3(3,0)**
This course focuses on general knowledge of the use of microcomputers in education, including common computer applications such as word processing, spreadsheet, database, and presentation software in teaching/learning activities, as well as evaluating the effectiveness of educational software in teaching/learning across various subject matter areas.		

MATH 220	**Statistics for Social & Behavioral Sciences**	**3(3,0)**
This course introduces statistical techniques for analyzing social science research data. Topics include frequency distribution, central tendency, variability, probability theory, Bayes' rule, mathematical expectation, variance and covariance, binomial, Poisson, hypergeometric and normal distributions, sampling distributions, estimation problems, and hypothesis testing. **Prerequisite**: None.		

PSYC 301	**Introduction to Psychology**	**3(3,0)**
This course introduces the theories, concepts, and viewpoints of the discipline of psychology. The course is designed to help students understand human behavior by addressing topics including the history of psychology, learning, personality, motivation, perception, social psychology, mental health, and related areas. **Co-requisite**: ENGL 210.		

Major Elective Courses

Descriptions of some major elective courses are given below.

ENGL 223	**Introduction to Drama**	**3(3,0)**
In this course, you will explore what drama is, its basic elements, and the evolution of its dramatic structure over centuries. You will study selected, representative plays from different periods to broaden your understanding and appreciation of literature. The course also aims to develop your critical thinking and analysis, and encourages you to respond critically to dramatic works through well-developed oral and written criticism.		

ENGL 235	**Creative Writing**	**3(3,0)**
This course introduces students to the process and practice of imaginative writing. Students will read works of fiction, creative nonfiction, and poetry and learn writing techniques in those genres.		

ENGL 252	**Introduction to American Literature**	**3(3,0)**
This course explores the works of major literary figures and authors representing different literary schools from the sixteenth century to modern times. Students will closely examine selected texts as literary achievements and situate them within the historical context and framework of American culture. **Co-requisite**: ENGL 210.		

ENGL 300	Introduction to World Literature	3(3,0)
The course surveys selected works in English or in translation from non-Anglo-American cultural traditions. Texts can be drawn from African, Asian, European, Latin American, and Middle Eastern literature, with a focus on their historical and cultural contexts. **Prerequisite**: ENGL 210.		

ENGL 330	Language Acquisition	3(3,0)
This course will survey theories of language acquisition. It will address human language processing, learners' motivational factors, and contextual factors that influence language learning. It will uncover the principles of first- and second-language acquisition.		

ENGL 365	Applied Linguistics	3(3,0)
This course is intended to give a broad overview of Applied Linguistics. It introduces learners to key areas that deepen their understanding of how language works across personal, social, and professional environments. The course touches on practical issues such as the relationships among languages, cultures, and societies; the acquisition of second and foreign languages; and the teaching and learning of languages.		

ENGL 420	Literature and Culture	3(3,0)
This course considers major works of literature, specifically in the context of twentieth-century cultural theory, including Marxism, post-colonialism, national literature, ethnic writings, and feminist theory. The primary intention of this course is to explore how various texts interact with their societies or how those societies influence the construction of literary works.		

ENGL 430	Advanced Language Acquisition	3(3,0)
An in-depth exploration of current issues in language acquisition in relation to recent developments in linguistic theories. Diverse topics in language acquisition, such as children's acquisition of pragmatics, school-aged acquisition, bilingualism, and foreign language acquisition, will be discussed. **Prerequisite**: ENGL 365.		

ENGL 460	Discourse Analysis	3(3,0)
This course considers several approaches to analyzing written, spoken, and signed language use in social science disciplines, including linguistics, sociology, anthropology, social work, psychology, and others. Students will learn discourse analysis methods, including how to work with various data. They will also consider various schools of thought that have evolved in discourse analysis.		

ENGL 462	Introduction to Psycholinguistics	3(3,0)
This course addresses the psychological and neurobiological aspects of language acquisition and how they influence humans' ability to learn, use, and understand language. It also considers the relationship between language and thought.		

ENGL 463	Introduction to Sociolinguistics	3(3,0)
This course addresses the ways language serves and is shaped by society. It considers topics such as dialect variation across regions, gender differences in language use, and the social functions of language.		

ENGL 497	Advanced Topics in English	3(3,0)
This course offers students an in-depth study of linguistics, literature, or fiction/nonfiction writing. Students will be involved in research and teamwork activities. Prerequisite: Consent of the instructor		

Freshman English

ENGL 101	Freshman English	3(3,0)
This course helps students reinforce their reading and writing skills. It focuses on the interrelatedness of reading and writing, in which students critically read and write essays using different rhetorical modes (classification, cause-and-effect analysis, comparison-and-contrast analysis, and persuasion). Some structural elements are reviewed to ensure accuracy in students' writing. Students are also introduced to research skills and APA documentation. **Prerequisite**: TOEFL Score of 514-549 or IELP 095.100		

Intensive English Language Program (IELP)

The mission of the Intensive English Language Program (IELP) at Rafik Hariri University is to provide incoming students with high-quality English language instruction to help them read, write, speak, and understand English at increasingly high levels of proficiency. We prepare students to communicate effectively in English in a friendly and supportive learning environment.

English Placement Chart

Students are placed in levels based on their English proficiency and progress through them as their English skills grow. The contents of any level address a variety of language skills. The following table summarizes the number of credits per level, the number of courses a student can take, and the corresponding scores on standard tests (TOEFL, SAT, and IELTS) required for placement at the IELP level.

IELP Level	# of Credits	Allowable Major Credits	TOEFL IBT	TOEFL ITP*	Old SAT Writing	New SAT Writing	IELTS
IELP 085	12	10	30 - 40	380-439**	320-339	18	3 – 3.5
IELP 095	12	10	41-52	440-479	340-359	19	4
IELP 100	6	13	53-64	480-513	360-379	20	4.5 –5
ENGL 101	3	Regular Student	65-79	514-549	380-399	21	5.5 -6
ENGL 210	3		80+	550+	400+	22+	6.5+

* These scores are compatible with the comparison table done by the English Testing Service (ETS) and the Common European Framework (CEFR).

Passing Criteria

The passing grade for any IELP level is 60.

Promotion Criteria

Current IELP Level	TOEFL Skipping Criteria	Level Skipped to
085	480-513	IELP 100
095	514-549	ENGL 101
IELP 100	550+	ENGL 210

IELP Course Descriptions.

The following are the descriptions of the IELP courses.

IELP 085	Elementary English	12(12,0)
This non-credit, elementary-level English course focuses on reading, writing, speaking, and listening. The course consists of twelve hours of classroom instruction over a 15-week semester. In this course, students develop elementary language skills in listening, speaking, reading, and writing. Students' work mostly integrates reading-writing and speaking-listening skills into direct class activities. Students prepare for the TOEFL as part of the curriculum. **Prerequisite:** TOEFL score 380-439 or its equivalent.		

IELP 095	Intermediate English	12(12,0)
This non-credit, intermediate-level English course focuses on reading, writing, speaking, and listening. The course consists of twelve hours of classroom instruction over a 15-week semester. Students develop intermediate language skills in listening, speaking, reading, and writing in this course. Students' work mostly integrates reading-writing and speaking-listening skills into direct class activities. Students prepare for the TOEFL as part of the curriculum. **Prerequisites:** TOEFL score of 440 – 479 or its equivalent, or IELTS band score of 085.		

IELP 100	Advanced English	6(6,0)
This course helps students develop their reading and writing skills. It teaches students to read critically and write paragraphs/essays in different rhetorical modes. It gives special attention to sentence structure and editing skills. It introduces students to research skills such as evaluating sources, summarizing, paraphrasing, and quoting. **Prerequisite**: TOEFL score of 480–513 or IELTS score of 95.		

Minor in English Language and Literature

Program Objectives

This minor aims to help interested students across diverse majors advance their knowledge of the English language and literature. The primary objectives of this minor program are:

a. Familiarize students with linguistic and literary concepts,
b. Equip students with skills to interpret and analyze English texts and utterances;
c. Improve students' English language competencies.

Learning Outcomes

Upon completion of this minor, students will be able to:

a. Use linguistic and literary concepts and terminology correctly,
b. Analyze different linguistic and literary texts;
c. Produce essays/papers in the discipline that follow professional rules and conventions.

Admission Requirements

1. To be eligible to apply for a minor in English Language and Literature, students should achieve a minimum of 70 in ENGL 210
2. To graduate with a minor in English Language and Literature, students must fulfill the following requirements:
    a. Completing a minimum of 18 credits offered in the English Language Program, two of which are mandatory: ENGL 222: Introduction to Language and ENGL 250: Introduction to Literature. The other four courses, which are minor electives, should be chosen from the list of elective courses below.
    b. Achieving a minimum CGPA of 70 in all 18 credits.

Career Options

A minor in English Language and Literature is a great way to complement a variety of majors. An English Language and literature minor can be paired with business majors such as HR, management, and marketing. It could also be paired with majors in graphic design, computer science, and education.

Graduates with a minor in English Language and Literature can take teaching, administrative, and editing positions. It can open several opportunities for these graduates, including pursuing a degree in computational linguistics (when coupled with computer science), a TD, and an MA in TESOL/TEFL.

Curriculum

In addition to the required courses, students can choose any 12 credit hours. Still, they are encouraged to consult with the program advisor regarding appropriate choices given their educational and career goals.

Elective Courses (12cr.):

a. ENGL 251 Introduction to English Literature II
b. ENGL 342 Introduction to Shakespeare
c. ENGL 350 History of the English Language
d. ENGL 360 Introduction to Phonetics
e. ENGL 370 Modern English Grammar
f. ENGL 400 Advanced English Grammar
g. ENGL 450 Advanced Academic Writing
h. ENGL 462 Introduction to Psycholinguistics
i. ENGL 463 Introduction to Sociolinguistics
j. ENGL 223 Introduction to Drama
k. ENGL 235 Creative Writing
l. ENGL 252 Introduction to American Literature
m. ENGL 300 Introduction to World Literature
n. ENGL 320 Creative Nonfiction
o. ENGL 330 Language Acquisition
p. ENGL 365 Applied Linguistics

Minor in Psychology

Program Objectives

This minor aims to help interested students across diverse majors deepen their knowledge of Psychology. The main objectives of the minor program are:

a. Familiarize students with theories of psychology
b. Equip students with the skills in psychological research.
c. Improve students' competencies in understanding human behavior.

Learning Outcomes

Upon completion of this minor, students will be able to:

a. Differentiate between psychology theories
b. Analyze human behavior by referring to varied theories
c. Interpret human behavior in different situations.

Admission Requirements

1. To be eligible to apply for a minor in Psychology, students should achieve a 70 in PSYC 301: Introduction to Psychology.
2. To graduate with a minor in Psychology, students must fulfill the following requirements:
   a. Completing a minimum of 18 credits offered in the Languages and Liberal Arts Unit, one of which is mandatory, PSYC 301: Introduction to Psychology
   b. Achieving a minimum GPA of 70 in all 18 credits.

Career options

A minor in Psychology is a good option to complement a variety of majors. The Psychology major can be paired with any major at any college. Students in this field learn how humans behave, think, and react, key skills in several professions, especially those that work directly with clients. College graduates with a psychology minor have many career options, ranging from nursing and social work to teaching, police work, and business.

Curriculum

In addition to the required course, students can choose any 15 credit hours, but are encouraged to consult their advisor about appropriate choices.
Elective Courses (15 cr.)

PSYC 302	Social Psychology
PSYC 303	Psychology of Personality
PSYC 305	Introduction to Special Needs

PSYC 306	Psychology of Women
PSYC 310	Positive Psychology
PSYC 312	Children and War
PSYC 314	Psychology for sustainability
SOCI 210	Research Methods in Social Science
PSYC 400	Conflict Transformation
PSYC 410	Abnormal Psychology
PSYC 315	Emotional Intelligence
	*Any other approved PSYC elective course

COMPUTER, MATH, AND SCIENCES DEPARTMENT (CMS)

Faculty Members

Chairperson:	Lara Abou Orm
Professor:	Mahmoud Halablab
Associate Professor:	Houssam Salami
Assistant Professor:	Lara Abou Orm, Ali Rammal, Roaa Soloh
Instructors:	Roba Al-Hajj
Adjunct Faculty:	Mohamad El-Abed, Elissar Nasreddine, Ahmad Tarhini, Lama Abdallah, May Abdallah, Inas Abdelmalak, Rawan Abdallah, Khaled Abo Alfa, Lama Abdallah, Hiba El Hajj Chehade, Tala Alaaedine.

Programs Offered

The CMS Department offers two programs – Computer Science and Healthcare Information Systems. Each program grants a Bachelor of Science degree. The details of each program will follow.

Vision

The vision of the Computer, Math, and Sciences (CMS) Department is to achieve academic excellence beyond accreditation standards in computing technologies and Information Systems.

COMPUTER SCIENCE PROGRAM

Mission

The mission of the Computer Science program is to provide students with a state-of-the-art curriculum; maintain enduring relationships with industry to create opportunities for students; cultivate learners with a sense of responsibility towards the profession and a passion for lifelong learning; and provide educational experiences that enable our students to be innovative leaders.

Objectives

The educational objectives of the Computer Science program intend to enable its graduates, within a few years of graduation, to:

- Solve computing problems of varying complexity that meet local, regional, or global markets
- Demonstrate professional skills and ethical behavior in practicing their profession
- Progress in computing and other professionally related fields at the level of professional employment or graduate studies

Program Outcomes

Upon successful completion of the BS program in Computer Science, an RHU graduate in Computer Science should demonstrate the following:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of Computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Accreditation

The Bachelor of Science program in Computer Science is accredited by the Computing Accreditation Commission (CAC) of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Computer Science.

Career Opportunities

A Computer Science major at Rafik Hariri University opens up diverse opportunities and prepares students for careers across various sectors. Our graduates are in industries, governmental and academic units, TV stations, the banking sector, and more. They hold roles such as games developer, system analyst, software engineer, web designer, mobile application developer, database analyst/developer/administrator, network engineer/administrator, data analyst, AI developer, and IT consultant. The work is challenging but intellectually stimulating as you get to work on interesting problems, and it is financially rewarding.

Program Overview

The Computer Science Program at Rafik Hariri University develops students' computational and critical-thinking skills and teaches them how to create, not simply use, new technologies. Students learn to think logically and solve problems by acquiring the fundamental principles of science and advanced techniques currently used in practical systems development. They also learn oral communication and presentational proficiencies that enable them to work with multicultural teams. Furthermore, they learn to work on individual and team projects to develop new systems, thereby enhancing time management and team coordination skills. RHU's Computer Science Department graduates are lifetime learners; they can adapt quickly to this ever-changing, challenging field.

To obtain a Bachelor of Science degree in CS, the student must complete 93 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BS in Computer Science (93 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	6	–	15	9	30	32
College Requirement	–	–	–	–	-	0
Program Requirement	36	9	18	–	63	68
Credits	42	9	33	9	93	100

I. University Requirements (General Education)

The list of the University's required courses and descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

III. Program Requirements

A. Mandatory Requirements

Mathematics requirements
The mathematics requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
MATH 210	Discrete Mathematics	3	
MATH 215	Calculus III	3	
MATH 311	Linear Algebra	3	
MATH 421	Numerical Analysis	3	MATH 311

Business requirements
The business requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
BMGT 300	Project Management	3	Junior Standing

Major requirements
The CS mandatory core courses are listed in the table below.

Course #	Title	Credits	Prerequisites
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 341	Software Engineering	3	COSC 214
COSC 351	Logic Design	3	
COSC 351L	Logic Design Lab	1	Co-req: COSC 351
COSC 353	Computer Organization	3	COSC 351
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req: COSC 360
COSC 399	Co-op Training Experience	1	ENGL 217; Senior Standing
COSC 421	Theory of Computation	3	COSC 215 and MATH 210

COSC 451	Operating Systems	3	COSC 214
COSC 490	Summative Learning Experience	3	ENGL 217

B. Technical Electives

As part of the Bachelor of Science in Computer Science program, students must take 9 credit hours of technical electives. These courses allow students to focus on a specific area for in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to the various Computer Science disciplines. In cooperation with the academic advisor, the student should select electives that best align with their needs and aspirations. The listed technical elective courses and other courses from engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

1. Internet and Computer Networks
2. Game Design and Development
3. Artificial Intelligence and Data Mining
4. Mobile/Web Design and Development

The CS elective courses are listed in the table below:

Course #	Title	Credits	Prerequisites
BITM 320	Enterprise Resource Planning	3	COSC 231
CCEE 426	Design of Embedded Systems	3	COSC 351
CCEE 561	Computer Vision	3	COSC 214
CCEE 566	Natural Language Processing	3	COSC 214
COSC 434	Advanced Web Programming	3	COSC 231 and COSC 333
COSC 435	Mobile Application Development	3	COSC 214
COSC 436	Game Programming	3	COSC 214
COSC 438	Introduction to Software Testing	3	COSC 341
COSC 461	Advanced Networking	3	COSC 360
COSC 463	Cybersecurity Operations	3	COSC 360
COSC 480	Machine Learning and Data Mining	3	COSC 214; MATH 351 or BADM 250
COSC 481	Artificial Intelligence	3	COSC 214
COSC 482	Data Science and Web Scraping	3	Co-req.: COSC 333 and MATH 351
COSC 496	Independent Study I	3	Senior Standing and a CGPA of 80 and above.
COSC 497	Independent Study II	3	Senior Standing and a CGPA of 80 and above.
COSC 498	Special Topics in Computer Science	3	Senior standing
HCIS 341	Health Information Systems I	3	
HCIS 342	Health Information Systems II	3	HCIS 341

Or any other approved CS elective course.

In addition, the students can take the following three labs instead of one elective.

Course #	Title	Credits	Prerequisites
Networking			
COSC 460L	Networking Lab 2	1	Co-req.: COSC 360L
COSC 461L	Networking Lab 3	1	COSC 460L
COSC 462L	Networking Lab 4	1	Co-req.: COSC 461L or CCEE 555L

Or any other approved CS elective course.

C. Summative Learning Project

Students must complete a 3-credit-hour course (COSC 490 Summative Learning Experience) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or on an innovative idea. The SLP could be an extension of the projects the students had worked on in computing courses, leading to more useful outcomes.

D. Co-op Training Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students must submit a formal report and a poster and make a formal presentation about their co-op experience. While ENGL 217 and senior standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

The courses offered in the CS program are designated code numbers in the form of (COSC ABC), where:

a	Designates year/level
b	Designates are as follows: 1. Computer Software; 2. Computer Science Theory; 3. Mobile/Web Development; 4: Systems Analysis and Design; 5: Computer Hardware; 6: Networking and Security
c	Designates the course sequence in an area

Study Plan

The Bachelor of Science in Computer Science encompasses 93 credit hours over six semesters and two summer sessions. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Semester 1 (15 Credits)			
COSC 214	Introduction to Programming	3	
COSC 351	Logic Design	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
	Natural Sciences and Technology Elective	3	
Year 1, Semester 2 (16 Credits)			
ARAB 212	Arabic Language and Communication	2	
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
COSC 351L	Logic Design Lab	1	COSC 351
	Community & Sustainability	1	Co-req,: ENGL 210
ENGL 217	Professional English Communication	3	ENGL 210
MATH 210	Discrete Mathematics	3	
Summer Session (3 Credits)			
	Humanities/Fine Arts Elective	3	
Year 2, Semester 1 (16 Credits)			
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req,: COSC 360
MATH 351	Probability and Statistics	3	MATH 215
	Social Science/Culture Elective I	3	
Year 2, Semester 2 (15 Credits)			
BMKA 200	Introduction to Marketing	3	ENGL 210
COSC 341	Software Engineering	3	COSC 214
MATH 311	Linear Algebra	3	
	CS Major Elective I	3	
	CS Major Elective II	3	
Summer Session (1 Credit)			
COSC 399	Co-op Training Experience	1	ENGL 217; Senior Standing
Year 3, Semester 1 (15 Credits)			
COSC 353	Computer Organization	3	COSC 351

COSC 421	Theory of Computation	3	COSC 215 and MATH 210
COSC 451	Operating Systems	3	COSC 214
GRDS 440	Design and Social Impact	3	Junior Standing
	Social Sciences/Culture Elective II	3	
Year 3, Semester 2 (12 Credits)			
BMGT 300	Project Management	3	Junior Standing
COSC 490	Summative Learning Experience	3	ENGL 217
MATH 421	Numerical Analysis	3	MATH 311
	CS Major Elective III	3	

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

COSC 215	**Advanced Programming and Data Structures**	**3(2,2)**
Continuing COSC 214, this course covers the basics of Object-Oriented Programming (OOP) languages and data structures. It covers pointers, classes, encapsulation, constructors/destructors, object instantiation, and templates. The course also details data structures and applications such as stacks, queues, lists, and trees. **Prerequisite***:* COSC 214. Equivalent to CCEE 216.		

COSC 231	**Database Management Systems**	**3(3,0)**
This course exposes students to the fundamental concepts for designing, using, and implementing database systems. It covers database concepts, data modeling, relations, normalization, and data manipulation languages. **Prerequisite***:* COSC 214. Equivalent to CCEE 315.		

COSC 316	**Design and Analysis of Algorithms**	**3(3,0)**
This course consolidates algorithm design and programming techniques. It provides an extended study of object-oriented programming properties, data structures, and data abstraction, as well as an introduction to complexity considerations. **Prerequisite***:* COSC 215.		

COSC 333	Web Programming	3(3,0)
This course teaches students how to develop and implement web-based programs, with an emphasis on front-end programming. It introduces students to web development, client-side languages, and the styles needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. **Prerequisite***:* COSC 214. Equivalent to CCEE 411.		

COSC 341	Software Engineering	3(3,0)
This course provides a deep and comprehensive understanding of the fundamentals of software engineering. The course introduces students to software process models, project management, software requirements, and engineering processes. The course covers SDLC stages, UML diagrams, and the use of CASE tools. **Prerequisite***:* COSC 214. Equivalent to CCEE 310.		

COSC 351	Logic Design	3(3,0)
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices, including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices**. Prerequisite***:* None. Equivalent to CCEE 221.		

COSC 351L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits, including combinational circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: COSC 351. Equivalent to CCEE 221L.		

COSC 353	Computer Organization	3(3,0)
This course exposes students to computer system organization and design; implementation of CPU data paths and control; instruction set design; memory hierarchy (caches, main memory, virtual memory) organization and management; input/output subsystems (bus structures, interrupts, DMA); performance evaluation; and pipelined processors. **Prerequisite**: COSC 351. Equivalent to CCEE 324.		

COSC 360	Networking	3(3,0)
This course enables students to gain fundamental knowledge of computer networks, appreciate various trade-offs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field of networks. **Prerequisite***:* COSC 214. Equivalent to CCEE 354.		

COSC 360L	Networking Lab	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. **Co-requisite**: COSC 360 or CCEE 354. Equivalent to CCEE 354L.		

COSC 399	Co-op Training Experience	1(1,0)
Each student must complete 8 weeks of practical training in a field related to their interests. This Co-op work experience is usually fulfilled during the summer semester of the third year in the program. Students must submit a formal report and poster and make a formal presentation about their Co-op experience. **Prerequisite***:* ENGL 217; Senior Standing.		

COSC 421	Theory of Computation	3(3,0)
This course covers theoretical principles embodied in automata and grammars. Topics include Deterministic, Non-deterministic, and Epsilon Non-deterministic Finite Automata (DFA, NFA, and e-NFA), DFA minimization, equivalence of NFAs and DFAs, Moore and Mealy machines, closure properties, regular expressions, regular languages, regular grammar, pumping lemma, context-free languages, context-free grammar, pushdown automata, Turing machines, and other selected topics as time permits. **Prerequisites***:* COSC 215 and MATH 210		

COSC 451	Operating Systems	3(3,0)
Students learn the concepts of operating systems theory and their implementations. The course covers computer and operating system structures, process and thread management, process synchronization, deadlocks, security and protection, inter-process communication, memory management, and I/O control systems. A case study of a UNIX operating system will accompany the course. **Prerequisite***:* COSC 214. Equivalent to CCEE 513.		

COSC 490	Summative Learning Experience	3(3,0)
This is a yearlong graduation project focused on individual problems and research. Each student must complete it in their emphasis area. **Prerequisite***:* ENGL 217.		

Business Required Courses

As described below, the COSC program requires two business courses, BMKA 200 and BMGT 300.

BMGT 300	Project Management	3(3,0)
This course provides students with the skills to manage their business projects using effective techniques for leading, organizing, scheduling, and controlling tasks that advance project goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling and budgeting. **Prerequisite:** Junior Standing.		

BMKA 200	Introduction to Marketing	3(3,0)
This course introduces the basic principles, theories, and marketing practices in our modern, ever-changing business environment. The course covers the marketing process activities of creating value for customers and capturing value from customers in return. It also discusses the marketing mix and how to build long-term customer relationships. Students will analyze case studies about a “real-life” product or service. Videos and in-class discussions on current marketing topics will assist in the learning experience. **Co-requisite**: ENGL 210		

II. Elective Courses

Descriptions of the major elective courses are given below.

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

COSC 434	Advanced Web Programming	3(3,0)
This course builds on the web programming course and prepares students to become full-stack developers. It allows students to learn how to develop back-end programs, connect their website or web application to a database, use regular expressions, implement asynchronous client access to server data, work with structured data formats such as JSON, and work with third-party media like web services. **Prerequisites:** COSC 231 and COSC 333. Equivalent to CCEE 514.		

COSC 435	Mobile Application Development	3(2,2)
This course examines the principles of mobile application design, development, and testing. It covers memory management, user interface design, input methods, data handling, and network techniques. Students are expected to work on a project that produces a professional-quality mobile application. Projects will be deployed in real-world applications. **Prerequisite***:* COSC 214, Equivalent to CCEE 511		

COSC 436	Game Programming	3(2,2)
This course provides an overview of software technologies for game development. It gives students a conceptual understanding of game design and practical experience in game creation. **Prerequisites***:* COSC 214; Senior standing.		

COSC 438	Introduction to Software Testing	3(3,0)
This course will introduce students to software testing, its importance, goals, approaches, and the roles of major players (developers, testers, and users). Students will learn about the software testing life cycle, standards, types of testing (manual vs. automated), testing methods (black-box vs. white-box), testing levels (functional, unit		

testing…), and documentation. This course will also cover hands-on experience with automated testing tools. **Prerequisite***:* COSC 341

COSC 460L	**Networking Lab 2**	**1(0,2)**
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will acquire knowledge of the functionalities of network equipment and protocols, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: COSC 360L or CCEE 354L. Equivalent to CCEE 554L.		

COSC 461	**Advanced Networking**	**3(3,0)**
This course prepares students to act as System and Network Administrators by implementing Active Directory Domain Services (AD DS) in distributed environments that can include complex network services and domain controllers. The covered materials help students efficiently automate the administration of users, groups, and computers. **Prerequisite***:* COSC 360 or CCEE 354*.*		

COSC 461L	**Networking Lab 3**	**1(0,2)**
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 2) Certification. The primary focus of this Lab is routers and switches in small networks. The student will learn the architecture, components, and operations of routers and switches, as well as their basic configuration and functionality. They will also learn how to configure and troubleshoot static and dynamic routing protocols, access-control lists, VLANs, DHCP for IPv4 and IPv6 networks, and NAT operations. **Pre-requisite**: COSC 460L or CCEE 554L. Equivalent to CCEE 555L.		

COSC 462L	**Networking Lab 4**	**1(0,2)**
This lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn to manage CISCO IOS software licensing and configuration files. **Co-requisite**: COSC 461L or CCEE 555L. Equivalent to CCEE 556L.		

COSC 463	**Cybersecurity operations**	**3(3,0)**
This hands-on, career-oriented course focuses on practical topics to prepare students to acquire the skills required to work in the cybersecurity domain. The course will help students seeking to pass the Cisco Cybersecurity Operations Associate certification. **Pre-requisite**: COSC 360 (or CCEE 354).		

COSC 480	Machine Learning and Data Mining	3(3,0)
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process, applying appropriate approaches to solve practical problems, and developing intelligent software applications. The course covers several topics about classification, prediction, and clustering. **Prerequisites***:* COSC 214 and MATH 351 or BADM 250. Equivalent to CCEE 564.		

COSC 481	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include Intelligent agents, informed and uninformed search strategies, and adversarial search. The Python language libraries will also be introduced. **Prerequisite**: COSC 214. Equivalent to CCEE 562.		

COSC 482	Data Science and Web Scraping	3(3,0)
Data is becoming the fuel of the 21st century, and acquiring data processing and analysis skills is becoming necessary. This course introduces data science processes, with a focus on web scraping as an application. The course will combine different domains, i.e., web programming, system programming, and machine learning. In particular, the course focuses on analyzing webpage HTML using Python, extracting available information, and generating dashboards. Co-requisite: COSC 333 (or CCEE 411) and MATH 351 (or BADM 250).		

COSC 496	Independent Study I	3(3,0)
This course allows students to explore new academic research/study that is unavailable in regular courses. It might be an interdisciplinary research project in a field of special interest. **Prerequisites**: Senior Standing and a CGPA of 80 and above.		

COSC 497	Independent Study II	3(3,0)
This course allows students to explore new academic research/study that is unavailable in regular courses. It might be an interdisciplinary research project in a field of special interest. **Prerequisites**: Senior Standing and a CGPA of 80 and above.		

COSC 498	Special Topics in Computer Science	1-3 credits
A special topic course to be defined based on current trends and departmental special interests. It may incorporate technical seminars, advanced readings, and special projects as required/approved by the department. **Prerequisite**: Senior standing.		

MINOR IN COMPUTER SCIENCE

Rationale

The CMS department offers a Computer Science minor for RHU students. It is designed for students who want to include basic proficiency in computer science to enhance or supplement their major program.

Program Objectives

The aims of a minor in Computer Science are:

- Provide RHU graduates with basic proficiency in computer science
- Support RHU graduates with essential computing skills desired/demanded by the market

Learning Outcomes

At the end of this minor, the student is expected to demonstrate:

- An ability to analyze a problem, identify and define the computing requirements appropriate to its solution
- An ability to use current techniques, skills, and tools necessary for computing practices
- An ability to develop programming skills using modern programming languages

Entrance Requirements

This minor is offered to all RHU students except those in CCE and Business IT Management.

Early in their major, interested RHU students must complete the appropriate form declaring their intent to minor in Computer Science while completing their regular major.

Computer Science Minor Policies

- Obtain a Minor Cumulative Grade Point Average of no less than 70%
- Although common courses between student major requirements and Computer Science minor requirements are counted to fulfill the minor requirements, the student must complete a minimum of 6 credit hours of coursework that is not counted toward the requirements for their major or any other minor.

Career Options

This minor allows its holders to pursue careers in various sectors, regardless of their major. Graduates of this minor can seek jobs in software engineering, web and mobile development, database/system administration, network engineering/administration, and IT consultancy.

Curriculum and Program

The Computer Science minor consists of six courses (18 credits), of which two are mandatory and four are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. Mandatory Courses

The two mandatory courses (6 credits) are:

COSC 214	Introduction to Programming	3(2,2)
COSC 215	Advanced Programming and Data Structures	3(2,2)

II. Elective Courses

The elective courses will be selected in consultation with the Computer Science minor advisor based on the student's needs and background. Students may choose four elective courses (12 credits) from the following list.

COSC 231	Database Management Systems	3(3,0)
COSC 316	Design and Analysis of Algorithms	3(3,0)
COSC 333	Web Programming	3(3,0)
COSC 341	Software Engineering	3(3,0)
COSC 351	Logic Design	3(3,0)
COSC 434	Advanced Web Programming	3(3,0)
COSC 435	Mobile Application Development	3(2,2)
COSC 436	Game Programming	3(3,0)
COSC 438	Introduction to Software Testing	3(3,0)
COSC 451	Operating Systems	3(3,0)
COSC 480	Machine Learning and Data Mining	3(3,0)
COSC 481	Artificial Intelligence	3(3,0)
COSC 482	Data Science and Web Scraping	3(3,0)
Any other approved CS elective course		

MINOR IN DATA ANALYTICS

Rationale

"Information is the oil of the 21st century, and analytics is the combustion engine" - P. Sondergaard.

Are you a student in Business, Engineering, or Science? Do you want to add a new dimension to your areas of expertise to boost your career? You can do more by enrolling in the new DATA ANALYTICS (DA) minor. The demand for data analytics expertise has grown rapidly over the past few years and is expected to grow even faster in the coming years. Students graduating with a data analytics minor are well positioned to bridge the gap in their chosen field.

Program's Purpose

Graduates of this minor will be able to use their acquired skills across various industries and in the non-profit and government sectors. Telecommunications, banking, financial services, the environmental sector, energy, biomedical, police, biology, bioinformatics, and physics are just a few areas in which you can operate. The minor shall also serve as good preparation for further and more advanced graduate and professional studies in Data Analytics and related areas.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:

- Use current techniques, architectures, and tools to collect and store data;
- Apply data analytics tools and techniques to support and improve decision-making;
- Design systems and tools to interpret and visualize data.

Entrance Requirements

This minor is offered to all RHU students except CCE students.

Early in their major, interested RHU students must complete the appropriate form declaring their intent to minor in Data Analytics while completing their regular major.

Program Requirements

This minor is structured to accommodate undergraduate RHU students in the BE or BS program from different disciplines (engineering, computer science, business). It may also

be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their studies, interested students must declare their intent to pursue a minor in DA by completing the relevant minor declaration form and informing their advisor.

To complete the DA minor, a student must

1) Declare a minor in Data Analytics by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the student's college major and the dean of the College of Arts and Sciences;
3) Earn a Minor Cumulative Grade Point Average of no less than 70%;
4) Complete 18 credits of coursework as specified below.

It should be noted that common courses between students' major requirements and the data analytics minor requirements count toward both the major and the minor. The student must complete at least 6 credit hours of coursework that are not counted toward the requirements of their major or any other minor.

Career Options

In today's world, most sectors and industries involve some form of data analysis. Therefore, minor holders would be well-suited to work in various sectors, including but not limited to telecommunication, banking, financial services, environmental, energy, biomedical, police, biology, and physics.

Moreover, minor holders may assume specialized roles such as data scientists, data analysts, and data engineers.

Curriculum and Program

The data analytics minor consists of six courses (18 credits), of which three are mandatory and three are elective, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. Mandatory Courses

The three mandatory courses (9 credits) are:

BADM 250/Math 351	Business Statistics/Probability and Statistics	3(3,0)
COSC 480	Machine Learning and Data Mining	3(3,0)
CCEE 567 / BADM 420	Optimization / Quantitative Methods for Business	3(3,0)

II. Elective Courses

Students may choose three elective courses (9 credits) from the following list or any other approved course.

COSC 316	Design and Analysis of Algorithms	3(3,0)
COSC 481	Artificial Intelligence	3(3,0)
COSC 482	Data Science and Web Scraping	3(3,0)
CCEE 516	Advanced Programming and Database Management Systems	3(2,2)
CCEE 568	Big data and analytics/Big data system	3(3,0)
CCEE 561	Computer Vision	3(3,0)
CCEE 566	Natural Language Processing	3(3,0)
CCEE 612	Advanced-Data Mining	3(3,0)
BITM 320	Enterprise Resource Planning	3(3,0)
BITM 415	Business Intelligence	3(3,0)
BMKA 420	Digital and Social Media Marketing	3(3,0)

It should be noted that CBA students (excluding BITM students) can also choose from the following list:

COSC 214	Introduction to Programming	3(3,0)
COSC 231	Database Management Systems	3(3,0)
COSC 333	Web Programming	3(3,0)

III. Courses Description

BADM 250	**Business Statistics**	**3(3,0)**
This course introduces students to the fundamentals of applied multivariate statistics. As such, the course covers factor analysis, multiple regression analysis, discriminant analysis, analysis of variance, and hypothesis testing.		

BADM 420	**Quantitative Methods for Business**	**3(3,0)**
This course introduces students to managerial decision analysis using quantitative tools. The course will introduce students to the use and development of mathematical models to help managers make informed decisions. The focus is on the applied aspects of statistics and math. As such, the course will cover the basics of probabilistic and statistical techniques, decision analysis, linear programming, optimization, forecasting, and waiting-line theory. **Prerequisite**: BADM 250, MATH 351, or Senior Standing.		

BITM 350	**Fundamentals of Data Analytics**	**3(3,0)**
This course introduces students to statistical techniques for analyzing large datasets. The course covers the theory and application of both parametric and nonparametric methods. Students will learn to visualize the data using univariate and bivariate plots and to investigate whether correlations exist in a multidimensional space using factor		

and cluster analyses. They will also build and test predictive models, including linear, logistic, and time-series models. **Prerequisite**: BADM 250 or MATH 351.

BITM 415	**Business Intelligence**	**3(3,0)**
This course introduces business intelligence as computerized support for managerial decision-making. It concentrates on the theoretical and conceptual foundations of business intelligence and commercial tools and techniques available for effective decision support. It focuses on extracting business intelligence from data sets for various applications, such as reporting and visual analytics across domains like web and business analytics, to support decision-making. Provides hands-on experience with various business intelligence software for reporting and building visualizations and dashboards**. Prerequisite**: Senior Standing & BADM 350		

BMKA 420	**Digital and Social Media Marketing**	**3(3,0)**
Digital marketing has evolved from a peripheral element of organizational marketing to the hub of customer-centric communications in an increasingly multi-channel environment. This course covers digital marketing topics like social media, email and mobile marketing, search engine optimization, paid search, and content marketing. It explains the principles of digital marketing and the major factors involved in implementing, measuring, and evaluating successful campaigns that utilize digital marketing channels. **Prerequisite**: Senior Standing		

CCEE 516	**Advanced Programming and Database Management Systems**	**3(3,0)**
This course will introduce students to database management, querying, and data warehousing. Students will also learn advanced programming tools, including Bigtable, NoSQL, R, Python, Scala, MapReduce, and Elasticsearch, and apply them to address big data challenges. **Prerequisite**: CCEE 315 or equivalent.		

CCEE 561	**Computer Vision**	**3(3,0)**
This course introduces the principles, models, and applications of computer vision. The course will cover image structure, projection, stereo vision, and the interpretation of visual motion. Case studies of industrial (robotic) computer vision applications, including visual navigation for autonomous robots, robot hand-eye coordination, and novel man-machine interfaces. **Prerequisite**: CCEE 214 or COSC 214.		

CCEE 566	**Natural Language Processing**	**3(3,0)**
This course introduces the student to natural language processing (NLP). The student is first introduced to word and sentence tokenization. The student then uses the learned skills to implement systems for text classification and sentiment analysis, spelling correction, information extraction, parsing, meaning extraction, and question answering, Machine learning algorithms, as well as algorithms like n-gram language modeling, naive Bayes and maxent classifiers, sequence models like Hidden Markov		

Models, probabilistic dependency and constituent parsing, and vector-space models of meaning will be introduced as needed for the above NLP applications. **Prerequisite**: CCEE 214 or COSC 214.

CCEE 567	**Optimization**	**3(3,0)**
This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. It includes a team project in which students select and solve a problem in practice. **Equivalent to** BADM 420 and MECH 571. **Prerequisite**: MATH 215 or equivalent.		

CCEE 568	**Big Data and Analytics/Big Data System**	**3(3,0)**
This course enables students to understand why the Big Data Era has emerged. Students will become conversant in the terminology and core concepts of big data problems, applications, and systems. Students will learn how to make Big Data useful in their business or career. Students will be introduced to one of the most common frameworks, Hadoop, which has made big data analysis easier and more accessible -- increasing the potential for data to transform our world. **Prerequisite**: CCEE 564 or equivalent.		

CCEE 612	**Advanced-Data Mining**	**3(3,0)**
In this course, advanced data mining topics will be covered, namely, classification (linear discriminant analysis, quadratic discriminant analysis, neural networks), combining multiple learners (bagging, boosting, cascading, stacking), dimensionality reduction (principal component analysis, linear discriminant analysis, subset selection), deep learning, anomaly detection, and reinforcement learning. **Prerequisite**: CCEE 564 or equivalent.		

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

COSC 316	**Design and Analysis of Algorithms**	**3(3,0)**
This course consolidates algorithm design and programming techniques. It provides an extended study of object-oriented programming properties, data structures, and data abstraction, as well as an introduction to complexity considerations. **Prerequisite***:* COSC 215.		

COSC 333	**Web Programming**	**3(3,0)**
This course teaches students how to develop and implement web-based programs, with an emphasis on front-end programming. It introduces students to web development, client-side languages, and the styles needed to build functional,		

responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. **Prerequisite***:* COSC 214. Equivalent to CCEE 411.

COSC 480	**Machine Learning and Data Mining**	**3(3,0)**
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process, applying appropriate approaches to solve practical problems, and developing intelligent software applications. The course covers several topics about classification, prediction, and clustering. **Prerequisites:** COSC 214 and MATH 351 or BADM 250. Equivalent to CCEE 564.		

COSC 481	**Artificial Intelligence**	**3(3,0)**
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include Intelligent agents, informed and uninformed search strategies, and adversarial search. The Python language libraries will also be introduced. **Prerequisite**: COSC 214. Equivalent to CCEE 562.		

COSC 482	**Data Science and Web Scraping**	**3(3,0)**
Data is becoming the fuel of the 21st century, and acquiring data processing and analysis skills is becoming necessary. This course introduces data science processes, with a focus on web scraping as an application. The course will combine different domains, i.e., web programming, system programming, and machine learning. In particular, the course focuses on analyzing webpage HTML using Python, extracting available information, and generating dashboards. **Co-requisite:** COSC 333 or CCEE 411 and MATH 351 or BADM 250.		

MATH 351	**Probability and Statistics**	**3(3,0)**
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

HEALTHCARE INFORMATION SYSTEMS PROGRAM (HCIS)

Mission

The mission of the Healthcare Information Systems program is devoted to providing students with a state-of-the-art curriculum, maintaining persistent relationships with industry to create opportunities for students, cultivating learners with a sense of responsibility towards the profession and a passion for lifelong learning, and providing the educational experiences that will enable our students to be innovative leaders.

Objectives

The educational objectives of the Healthcare Information Systems program intend to enable its graduates, within a few years of graduation, to:

- Solve healthcare computing problems that meet local, regional, or global markets.
- Demonstrate professional skills and ethical behavior in practicing their profession
- Progress in computing and other professionally related fields at the level of professional employment or graduate studies

Program Outcomes

Upon successful completion of the BS degree in Healthcare Information Systems, graduates should demonstrate:

a. An ability to apply knowledge of health information systems and use current technologies, techniques, skills, and tools necessary for effective delivery of healthcare services appropriate to healthcare organizations;
b. An ability to apply health analytic methods to influence the planning, organizing, administering, and policy creation of healthcare organizations;
c. An ability to design, implement, and evaluate healthcare systems, processes, and procedures to meet desired needs;
d. An ability to evaluate healthcare processes, practices, management of data, technologies, and conditions to ensure compliance with established quality and professional standards;
e. An ability to function effectively on teams to accomplish a common goal;
f. An understanding of professional, ethical, legal, security, and social issues and responsibilities;
g. An ability to communicate effectively with a range of audiences.
h. Recognition of the need for and an ability to engage in continuing professional development.

Career Opportunities

With a Healthcare Information Systems degree, students will gain foundational knowledge to develop and apply information technologies and systems to meet the growing needs of healthcare organizations. Given the importance of information accuracy, privacy, and security, HCIS graduates will be prepared for involvement in regulatory compliance and quality assessment activities designed to ensure that health information systems support patient care and safety. HCIS graduates can work with nurses, physicians, other healthcare providers, managers, and technical specialists in a variety of settings such as hospitals, long-term-care facilities, insurance and managed care organizations, government agencies, and vendor firms.

Program Overview

RHU's Healthcare Information Systems (HCIS) program combines concepts from healthcare, information technology, and leadership practice to prepare graduates with the knowledge and skills needed to be effective designers and developers of health information systems. The HCIS program at RHU develops students' computational and critical thinking skills to create, not simply use, new healthcare technologies and systems. Students learn to think logically and solve problems by acquiring the fundamental principles of science and advanced techniques currently used in practical systems development. They also learn oral communication and presentational skills and how to work on individual and team projects to develop new systems that enhance time management and team coordination. RHU's HCIS program graduates are lifetime learners; they can adapt quickly to this ever-changing, challenging field.

To obtain a Bachelor of Science degree in HCIS, the student must complete 100 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BS in Healthcare Information Systems (100 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	9	–	13	9	31	31
College Requirement	–	–	–	–	–	–
Program Requirement	42	9	18	–	69	69
Credits	51	9	31	9	100	100

I. University Requirements (General Education)

The list of the University's required courses and descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented in the introductory pages of the College of Arts and Sciences section of this catalog.

III. Program Requirements

A. Mandatory Requirements

Mathematics
The mathematics requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
MATH 210	Discrete Mathematics	3	
MATH 215	Calculus III	3	
MATH 311	Linear Algebra	3	
MATH 421	Numerical Analysis	3	MATH 311

Business Requirements
The business requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BMGT 300	Project Management	3	Junior Standing
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210

Major Requirements
The HCIS mandatory core courses are listed in the table below.

Course #	Title	Credits	Prerequisites
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 341	Software Engineering	3	COSC 214
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req.: COSC 360
COSC 480	Machine Learning and Data Mining	3	COSC 214; MATH 351 or BADM 250
HCIS 240	Medical Terminology	3	
HCIS 341	Health Information Systems I	3	BIOL 210
HCIS 342	Health Information Systems II	3	HCIS 341

HCIS 343	Information Security and Privacy in Healthcare	3	HCIS 341
HCIS 399	Co-op Training Experience	1	ENGL 217; Senior Standing.
HCIS 444	Healthcare Business Intelligence & Data Analysis	3	HCIS 341
HCIS 445	Research in Healthcare Information Systems	1	Senior Standing
HCIS 490	Summative Learning Experience	3	ENGL 217

B. Technical Electives

As part of the Bachelor of Science in Healthcare Information Systems (HCIS) program, students must complete 9 credit hours of technical electives. These courses allow students to focus on a specific area for in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to the various HCIS disciplines. In cooperation with the academic advisor, the student should select electives that best align with their needs and aspirations. The listed technical elective courses and other courses from engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in areas of interest.

Course #	Title	Credits	Prerequisites
BIOM 520	Health-care Facility Management	3	HCIS 341
BIOM 565	Introduction to E-Healthcare	3	HCIS 341
COSC 434	Advanced Web Programming	3	COSC 231 and COSC 333
COSC 435	Mobile Application Development	3	COSC 214; Senior Standing
COSC 436	Game Programming	3	COSC 214
COSC 438	Introduction to Software Testing	3	COSC 341
COSC 451	Operating Systems	3	COSC 214
COSC 461	Advanced Networking	3	COSC 360 or CCEE 354
COSC 463	Cybersecurity Operations	3	COSC 360
COSC 481	Artificial Intelligence	3	COSC 214
COSC 482	Data Science and Web Scraping	3	Co-req.: COSC 333 and MATH 351
COSC 498	Special Topics	3	
HCIS 498	Special Topics	3	Senior Standing
CCEE 561	Computer Vision	3	COSC 214
CCEE 566	Natural Language Processing	3	COSC 214 or COSC 214.

* The list of elective courses is subject to change.

In addition, the students can take the following three labs instead of one elective.

Course #	Title	Credits	Prerequisites
Networking			
COSC 460L	Networking Lab 2	1	Co-req.: COSC 360L
COSC 461L	Networking Lab 3	1	COSC 460L
COSC 462L	Networking Lab 4	1	Co-req.: COSC 461L or CCEE 555L.

C. Summative Learning Project

Students must complete a 3-credit-hour course in which they work, preferably in groups, on a problem of concern to industry or the community at large, or on a promising idea to innovate. The SLP could be an extension of the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Training Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member is assigned to mentor the student and follow up on their progress throughout the Co-op period by conducting field visits and ensuring that the student's performance aligns with their aspirations and the employer's needs. Students must submit a formal report and a poster and make a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are prerequisites for co-op training, the program must stipulate additional conditions.

Course Coding

Code numbers designate the courses offered in the HCIS program in the form of (HCIS abc), where:

a	Designate the year (level)
b	Designate the focus area as follows: 1. Software Engineering and Quality Assurance; 2. Network Administration and Security; 3. Healthcare
c	Designate the course sequence in a focus area.

Study Plan

The Bachelor of Science in Healthcare Information Systems encompasses 100 credit hours over six semesters and two summer sessions. The following study plan serves as a roadmap for smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	

BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
	Community & Sustainability	1	Co-req.: ENGL 210
COSC 214	Introduction to Programming	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
Year 1, Spring Semester (16 Credits)			
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
CMNS 205	First Aid	1	
ENGL 217	Professional English Communication	3	ENGL 210
HCIS 240	Medical Terminology	3	
MATH 210	Discrete Mathematics	3	
Year 1, Summer Semester (6 Credits)			
BIOL 210	Human Anatomy and Physiology and Lab	3	
	Social Science/Culture Elective I	3	
Year 2, Fall Semester (16 Credits)			
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req.: COSC 360
HCIS 341	Health Information Systems I	3	
MATH 351	Probability and Statistics	3	MATH 215
Year 2, Spring Semester (15 Credits)			
COSC 341	Software Engineering	3	COSC 214
HCIS 342	Health Information Systems II	3	HCIS 341
HCIS 343	Information Security and Privacy in Healthcare	3	HCIS 341
MATH 311	Linear Algebra	3	
	HCIS Major Elective I	3	
Year 2, Summer Semester (1 Credit)			
HCIS 399	Co-op Training Experience	1	ENGL 217; Senior Standing
Year 3, Fall Semester (15 Credits)			
MATH 421	Numerical Analysis	3	MATH 311
GRDS 440	Design and Social Impact	3	
HCIS 444	Healthcare Business Intelligence and Data Analysis	3	HCIS 341
	Humanities/Fine Arts Elective	3	
	HCIS Major Elective II	3	
Year 3, Spring Semester (16 Credits)			
BMGT 300	Project Management	3	Junior Standing

COSC 480	Machine Learning and Data Mining	3	COSC 214; MATH 351 or BADM 250
HCIS 490	Summative Learning Experience	3	ENGL 217
HCIS 445	Research in Healthcare Information Systems	1	Senior Standing
	HCIS Major Elective III	3	
	Social Sciences/Culture Elective II	3	

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

COSC 215	**Advanced Programming and Data Structures**	**3(2,2)**
Continuing COSC 214, this course covers the basics of object-oriented programming (OOP) languages and data structures. It covers pointers, classes, encapsulation, constructors/destructors, object instantiation, and templates. The course also details data structures and applications such as stacks, queues, lists, and trees. **Prerequisite***:* COSC 214. Equivalent to CCEE 216.		

COSC 231	**Database Management Systems**	**3(3,0)**
This course exposes students to the fundamental concepts for designing, using, and implementing database systems. It covers database concepts, data modeling, relations, normalization, and data manipulation languages. **Prerequisite***:* COSC 214. Equivalent to CCEE 315.		

COSC 316	**Design and Analysis of Algorithms**	**3(3,0)**
This course consolidates algorithm design and programming techniques. It provides an extended study of object-oriented programming properties, data structures, and data abstraction, as well as an introduction to complexity considerations. **Prerequisite***:* COSC 215.		

COSC 333	**Web Programming**	**3(3,0)**
This course teaches students how to develop and implement web-based programs, with an emphasis on front-end programming. It introduces students to web development, client-side languages, and the styles needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. **Prerequisite***:* COSC 214. Equivalent to CCEE 411.		

COSC 341	Software Engineering	3(3,0)
This course provides a deep and comprehensive understanding of the fundamentals of software engineering. The course introduces students to software process models, project management, software requirements, and engineering processes. The course covers SDLC stages, UML diagrams, and the use of CASE tools. **Prerequisite***:* COSC 214. Equivalent to CCEE 310.		

COSC 360	Networking	3(3,0)
This course enables students to gain fundamental knowledge of computer networks, appreciate various tradeoffs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field of networks. **Prerequisite***:* COSC 214. Equivalent to CCEE 354.		

COSC 360L	Networking Lab	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. This course is an accompanying Lab for COSC 360. **Co-requisite**: COSC 360 or CCEE 354. Equivalent to CCEE 354L.		

COSC 480	Machine Learning and Data Mining	3(3,0)
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process, applying appropriate approaches to solve practical problems, and developing intelligent software applications. The course covers several topics about classification, prediction, and clustering. **Prerequisites**: COSC 214; MATH 351 or BADM 250. Equivalent to CCEE 564.		

HCIS 240	Medical Terminology	3(3,0)
This course introduces elements of medical terminology, including the foundations of words used to describe the human body and its conditions, terminology for medical procedures, and the names of commonly prescribed medications. Spelling, pronunciation, and meanings of terms used in a professional healthcare system are covered, as well as the recognition of common abbreviations. **Prerequisite**: None.		

HCIS 341	Health Information Systems I	3(3,0)
This course introduces healthcare, medical, and business processes from a software design perspective. Topics include the history of – and current topics related to – the healthcare delivery process, healthcare functions supported by hospital IT departments, and interactions between healthcare and business data domains, as well as between medical and allied health professionals.		

HCIS 342	Health Information Systems II	3(3,0)
In this course, students explore current technologies, regulations, and standards, including picture archiving and communication systems (PACS); the Health Insurance Portability and Accountability Act (HIPAA); 21CFR Part 11; FDA General Principles of Software Validation; and Health Level Seven (HL7), and examine their effects on software development. Other topics include information technologies for storing and maintaining data quality, ensuring safety, enforcing security, and electronic health record systems. **Prerequisite:** HCIS 341.		

HCIS 343	Information Security and Privacy in Healthcare	3(3,0)
This course introduces students to emerging issues in healthcare information security, privacy, and regulatory compliance. It also provides the students with a substantive overview and analysis of relevant information security subject matter that directly and materially impacts the healthcare system. **Prerequisite**: HCIS 341.		

HCIS 399	Co-op Training Experience	1(1,0)
Each student must complete 8 weeks of practical training in a field related to their interests. This Co-op work experience is usually fulfilled during the summer semester of the third year in the program. Students must submit a formal report and poster and make a formal presentation about their Co-op experience. **Prerequisites**: ENGL 217; Senior Standing.		

HCIS 444	Healthcare Business Intelligence and Data Analysis	3(3,0)
This course enables students to learn how Healthcare Analytics and Health Information Exchange (HIE) solutions can help uncover new growth opportunities and gain an advantage in the profitable healthcare technology market. Other areas covered include providing quality patient care while complying with government regulations, controlling costs, and increasing your organization's visibility in a crowded healthcare field. **Prerequisite**: HCIS 341.		

HCIS 445	Research in Healthcare Information Systems	1(1,0)
This course focuses on current research trends in healthcare information systems. It may incorporate technical seminars, advanced readings, and special projects as required/approved by the department. **Prerequisite:** Senior Standing.		

HCIS 490	Summative Learning Experience	3(3,0)
This is a graduation project directed at individual problems and research. Each student must complete it in their emphasis area. **Prerequisite:** ENGL 217.		

Business Required Courses

As described below, the HCIS program requires two business courses, BMKA 200 and BMGT 300.

BMGT 300	Project Management	3(3,0)
This course provides students with the skills to manage their business projects using effective techniques for leading, organizing, scheduling, and controlling tasks that advance project goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling and budgeting. **Prerequisite:** Junior Standing.		

BMKA 200	Introduction to Marketing	3(3,0)
This course introduces the basic principles, theories, and marketing practices in our modern, ever-changing business environment. The course covers the marketing process activities of creating value for customers and capturing value from customers in return. It also discusses the marketing mix and how to build long-term customer relationships. Students will analyze case studies about a “real-life” product or service. Videos and in-class discussions on current marketing topics will assist in the learning experience. **Co-requisite**: ENGL 210		

II. Elective Courses

Descriptions of the major elective courses are given below.

COSC 434	Advanced Web Programming	3(3,0)
This course builds on the web programming course and prepares students to become full-stack developers. It allows students to learn how to develop back-end programs, connect their website or web application to a database, use regular expressions, implement asynchronous client access to server data, work with structured data formats such as JSON, and work with third-party media like web services. **Prerequisites:** COSC 231 and COSC 333. Equivalent to CCEE 514		

COSC 435	Mobile Application Development	3(2,2)
This course examines the principles of mobile application design, development, and testing. It covers memory management, user interface design, input methods, data handling, and network techniques. Students are expected to work on a project that produces a professional-quality mobile application. Projects will be deployed in real-world applications. **Prerequisite**: COSC 214, Equivalent to CCEE 511		

COSC 436	Game Programming	3(2,2)
This course provides an overview of software technologies for game development. It gives students a conceptual understanding of game design and practical experience in game creation. **Prerequisites**: COSC 214, Senior standing.		

COSC 438	Introduction to Software Testing	3(3,0)
This course will introduce students to software testing, its importance, goals, and approaches, as well as the roles of the major players (developers, testers, and users). Students will learn about the software testing life cycle, standards, types of testing (manual vs. automated), testing methods (black-box vs. white-box), testing levels (functional, unit testing…), and documentation. This course will also provide hands-on experience with automated testing tools. **Prerequisite***:* COSC 341.		

COSC 451	Operating Systems	3(3,0)
Students learn the concepts of operating systems theory and their implementations. Covered topics include computer and operating system structures, process and thread management, process synchronization, deadlocks, security and protection, inter-process communication, memory management, and I/O control systems. A case study of a UNIX operating system will accompany the course. **Prerequisite:** COSC 214. Equivalent to CCEE 513.		

COSC 460L	Networking Lab 2	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will learn the functionalities of network equipment and protocols, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: COSC 360L or CCEE 354L. Equivalent to CCEE 554L.		

COSC 461	Advanced Networking	3(3,0)
This course prepares students to act as a System and Network Administrator by implementing Active Directory Domain Services (ADDS in distributed environments that can include complex network services and domain controllers. The covered materials help students efficiently automate the administration of users, groups, and computers. **Prerequisite**: COSC 360 or CCEE 354.		

COSC 461L	Networking Lab 3	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn to manage CISCO IOS software licensing and configuration files. **Prerequisite**: COSC 460L or CCEE 554L. Equivalent to CCEE 555L..		

COSC 462L	Networking Lab 4	1(0,2)
This lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn to manage CISCO IOS software licensing and configuration files. **Co-requisite**: COSC 461L or CCEE 555L. Equivalent to CCEE 556L.		

COSC 463	Cybersecurity operations	3(3,0)
This hands-on, career-oriented course focuses on practical topics to prepare students to acquire the required skills to work in the cybersecurity domain. The course will help students seeking to pass the Cisco Cybersecurity Operations Associate certification. **Pre-requisite**: COSC 360 (or CCEE 354).		

COSC 481	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include Intelligent agents, informed and uninformed search strategies, and adversarial search. The Python language libraries will also be introduced. **Prerequisite**: COSC 214. Equivalent to CCEE 562.		

COSC 482	Data Science and Web Scraping	3(3,0)
Data is becoming the fuel of the 21st century, and acquiring data processing and analysis skills is becoming necessary. This course introduces data science processes, with a focus on web scraping as an application. The course will combine different domains, i.e., web programming, system programming, and machine learning. In particular, the course focuses on analyzing webpage HTML using Python, extracting available information, and generating dashboards. Co-requisite: COSC 333 (or CCEE 411) and MATH 351 (or BADM 250).		

HCIS 498	Special Topics in Healthcare	1-3 credits
This special topic course will be defined based on current trends and departmental special interests. It may incorporate technical seminars, advanced readings, and special projects as required/approved by the department. **Prerequisite**: Senior standing.		

MINOR IN MATHEMATICS

Rationale

The CMS Department offers a Mathematics minor for RHU students. It is designed to give students a solid foundation in mathematics and some advanced-level experience in the discipline.

Program Objectives

The aims of a minor in Mathematics are:

- Provide RHU graduates with a basic proficiency in Mathematics to compete at the university level.
- Support RHU graduates with essential mathematical skills to enhance their knowledge and understanding of their majors.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate:

- An ability to analyze mathematically experimental or physical results
- An ability to use mathematical techniques, skills, and facts in their respective research

Entrance Requirements

Interested RHU students must complete the appropriate form declaring their intent to minor in Mathematics while completing their regular major.

Mathematics Minor Policies

- Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
- Up to four courses between student major requirements and mathematics minor requirements are counted to fulfill the mathematics minor requirements

Career Options

This minor allows its holders to pursue careers across various sectors, regardless of their major. Graduates of this minor can seek jobs in teaching, banking and finance, computing, and statistics.

Program Requirements

To complete the Minor in Mathematics, a student must

1) Declare a Minor in Mathematics by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the college major and the dean of CAS
3) Obtain a Minor Cumulative Grade Point Average of no less than 70%.
4) Complete 18 credits of Mathematics coursework as specified below.

It should be noted that the student must complete at least six credit hours of coursework that are not counted toward the requirements for their major or any other minor.

Curriculum and Program

The mathematics minor consists of six courses (18 credits), of which three are mandatory and three are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. Mandatory Courses

The three mandatory courses (9 Cr.) are:

Course #	Title	Credits	Prerequisites
MATH 215	Calculus III	3	
MATH 216	Calculus IV	3	MATH 215
MATH 311	Linear Algebra with Applications	3	

II. Elective Courses

The Mathematics Minor Advisor will choose the elective courses based on students' needs and backgrounds. Students may choose three elective courses (9 credits) from the following list.

Course #	Title	Credits	Prerequisites
MATH 210	Discrete Mathematics	3	
MATH 217	Advanced Calculus	3	MATH 215
MATH 314	Ordinary Differential Equations	3	MATH 216
MATH 316	Introduction to Analysis	3	MATH 217
MATH 317	Partial Differential Equations	3	MATH 314
MATH 318	Vector Calculus	3	MATH 217 & MATH 311
MATH 351	Probability and Statistics	3	MATH 215
MATH 421	Numerical Analysis	3	MATH 311
MATH 425	Introduction to Complex Variables	3	MATH 217
MATH 442	Introduction to Graph Theory	3	MATH 316, MATH 210

Courses Description

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, structural induction, sets and set operations, functions, growth of functions, relations and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	Partial Differential Equations	3(3,0)
Lagrange theorem, boundary conditions of first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables, introduction to Fourier series and integrals, boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the topics mentioned above. **Prerequisite**: MATH 311.		

MATH 316	Introduction to Analysis	3(3,0)
Ordered, finite, countable and uncountable sets, sequences, subsequences, Cauchy sequences, upper and lower limits, series, limits of sequences of functions, continuity and compactness, connectedness, infinite limits, and limits at infinity, differentiation of vector-valued functions, series of functions, uniform convergence and continuity, functions of several variables, the inverse function and the implicit function theorems, the rank theorem. **Prerequisite**: MATH 217.		

MATH 318	Vector Calculus	3(3,0)
Theory of vector-valued functions, divergence, gradient, curl, vector fields, path integrals, surface integrals, constrained extrema, and Lagrange multipliers. Implicit function theorem. Green's and Stokes' theorems: an introduction to differential geometry. **Prerequisites**: MATH 217 and MATH 311.		

MATH 425	Introduction to Complex Variables	3(3,0)
Complex numbers, geometric representation, analytic functions, real line integrals, complex integration, power series, residues, poles, and conformal mappings. **Prerequisite**: MATH215.		

MATH 442	Introduction to Graph Theory	3(3,0)
Combinatorics through graph theory. Topics include connectedness, factorization, Hamiltonian graphs, network flows, Ramsey numbers, graph coloring, graph automorphisms, and Pólya's Enumeration Theorem—prerequisites: MATH 316 and Math 210.		

DESIGN DEPARTMENT (DD)

Faculty Members

Chairperson:	Serene Srouji
Associate Professor:	Serene Srouji
Assistant Professor:	Noura Sammoura, Nadine Zahreddine
Adjunct Faculty:	Lina Hassoun, Mohamad Darwish, Myriam Chamoun, Sobhi Farroukh, Rihab Zebian, Saber Abdallah, Sara Makki, Lana Charara.

Vision

The Design Department at Rafik Hariri University strives to be the leader in high-standard, progressive education in Lebanon and the region. It aims to develop creative, well-informed, skilled professionals aware of their responsibility to solve visual problems in a dynamic global society. Our vision is a future in which RHU faculty, students, and alumni become trendsetters and innovators in the design world, both theoretically and practically.

Programs Offered

The Department of Design offers one program leading to a Bachelor of Science degree in Graphic Design. It also offers a minor in Graphic Design. Details of each program will follow.

GRAPHIC DESIGN PROGRAM

Mission

Our mission is to educate students to become successful thinkers and creative designers who can enhance human experience and improve communication. This is achieved through:

- Providing high-standard teaching methods and progressive curricula that deliver professional excellence.
- Educating designers to be socially and ethically responsible to serve a better community.
- Promote growth and innovation personally and academically to become successful leaders in the workplace.
- Mentoring students to tackle design challenges creatively and innovatively while adhering to design principles.
- Providing students coverage through diverse media forums, exhibitions, and events.
- Providing students with design perspectives from designers and thinkers outside RHU through juries, workshops, lectures, and seminars.
- Establishing and maintaining close relationships with the industry and other academic institutions to improve the curriculum, expand collaboration, and increase job opportunities for our students.

Objectives

- Provide students with both print and multimedia design courses.
- Provide students with the necessary thinking, planning, and execution skills to succeed in the workplace.
- Provide students with technologically advanced skills in new fields such as VR, AR, 3D Printing…
- Provide students with the key elements to develop a portfolio and CV that stand out in the market.
- Allow students to explore various platforms through personal creative skills.
- Emphasize theoretical and practical design aspects, completely understanding the design field.
- Explore the diverse roles of designers as effective visual creators, ethical and social beings, and problem solvers.

Program Outcomes

Upon successful completion of the BS program in Graphic Design, graduates will:

- Effectively communicate messages by juxtaposing images and type to a specific target audience.
- Acquire, articulate, and apply specialized terminology and knowledge relevant to graphic design, including relationships with other disciplines and contemporary global issues.
- Implement creative solutions from concept through completion using a formal process.
- Display competency in technical skills applicable to graphic design.
- Demonstrate the ability to use design-thinking strategies in an iterative design process.
- Apply design principles to develop strategic marketing and communication solutions for companies, products, and services.
- Be able to analyze, synthesize, and develop successful solutions.
- Become conceptual thinkers aware of social, cultural, and ethical issues…
- Access information through traditional and new technologies and synthesize this information for problem-solving activities.
- Understand the relationship of graphic design to other disciplines and society.
- Develop a professional resume, business card, and portfolio, and learn how to conduct a job search. They will also gain familiarity with the types of jobs available in the graphic design industry.

Career Opportunities

Graphic designers gain a broad range of knowledge and skills that equip them well to enter the professional field. Graphic Design offers vast opportunities in various fields, including TV stations, advertising agencies, design and packaging firms, editorial firms and magazines, branding agencies, interactive design firms (web and applications), 3D animation studios and gaming, photography studios, printing presses, illustration firms, and publishing houses. Furthermore, many NGOs, banks, and other business companies require in-house graphic designers. In addition to working full-time, local and international freelance projects are available for extra income and independent work, driven by high market demand.

Graphic Design is also an excellent foundation for careers in product management, marketing, and social media.

Program Overview

The RHU Graphic Design (GRDS) Program provides excellent teaching standards based on academic and creative research and professional practice. The graphic design department at RHU offers students a great opportunity to grow as individuals, nurture their unique style, and build self-confidence. Various design-related forums are explored, and students are taught skills for client communication and workplace success.

The graphic design program is based on a student-centered approach in which students gain extensive knowledge of design theory and a thinking approach essential to all

problem-solving. The program's curriculum, facilities, and committed faculty members ensure students' academic and personal growth in an environment that inspires learning and drives creativity.

BS in Graphic Design (111 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	12	-	9	9	30	27
College Requirement	-	-	-	-	-	0
Program Requirement	51	6	21	3	81	73
Credits	63	6	30	12	111	100

I. University Requirements (General Education)

The list of the University's required courses and their descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

II. College Requirements

The diversity of the programs precludes the identification of college requirements common to all CAS programs.

III. Program Requirements

A. Fine Arts Requirements (18 credits)

As part of the Bachelor of Science in Graphic Design program, students must take 18 credit hours of Fine Arts requirement courses listed in the table below. These courses build a solid artistic and technical foundation that helps students excel in design.

Course #	Title	Credits	Prerequisites
FADR 200	Drawing I	3	
FADR 202	Drawing II	3	FADR 200
FADR 210	Rendering and Perspective Techniques	3	
FADR 215	History of Art	3	
FADR 220	Fundamentals of 2D	3	
FADR 222	Fundamentals of 3D	3	FADR 220

B. Business Requirement

The business requirement is a three-credit course listed in the table below.

Course #	Title	Credits	Prerequisites
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210

C. Mandatory Requirements

The set of mandatory graphic design courses encompasses the 51 credits listed in the table below.

Course #	Title	Credits	Prerequisites
GRDS 203	Introduction to Graphic Design	3	GRDS 205
GRDS 205	Type I	3	GRDS 220
GRDS 222	Advanced Digital Media	3	GRDS 220
GRDS 230	Photography I	3	
GRDS 300	Illustration	3	GRDS 220, FADR 202
GRDS 305	Type II	3	GRDS 205
GRDS 306	Type III	3	GRDS 222, GRDS 305
GRDS 310	Design I	3	GRDS 203, GRDS 220
GRDS 311	Design II	3	GRDS 310
GRDS 345	History of Graphic Design	3	FADR 215
GRDS 399	Co-op Training Experience	1	ENGL 217; Senior Standing.
GRDS 409	Design Writing Research	3	GRDS 310
GRDS 411	Design III	3	GRDS 311, GRDS 306
GRDS 412	Design IV	3	GRDS 411, GRDS 409
GRDS 423	Motion Graphics	3	GRDS 220
GRDS 424	Web and Interactive Design	3	GRDS 423
GRDS 450	Branding and Advertising	3	GRDS 411
GRDS 452	Digital Illustration	2	GRDS 222, GRDS 300

D. Major Elective Requirements

Students must take 6 credit hours of major elective courses to deepen their knowledge of innovative disciplines and current issues in the design field.

Course #	Title	Credits	Prerequisites
GRDS 320	UI/UX Design	3	GRDS 220
GRDS 330	Photography II	3	GRDS 230
GRDS 331	Digital Processing	3	GRDS 222
GRDS 350	3D Printing & Prototyping	3	GRDS 222
GRDS 355	Calligraphy	3	GRDS 305
GRDS 375	3D Animation	3	GRDS 423
GRDS 385	VR Illustration and Sculpting	3	GRDS 300
GRDS 381	Book Binding	3	Junior Standing
GRDS 402	Silkscreen	3	GRDS 300
GRDS 403	Painting	3	FADR 202
GRDS 436	Game Design	3	GRDS 220
GRDS 498	Special Topics in Graphic Design	3	

E. Social Science/Culture Requirement

Students must take three extra-credit hours in social science/culture to fulfill the five

humanities and social sciences courses required for accreditation.

F. Co-op Training Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits to ensure that the student's performance is aligned with their aspirations and the employer's needs. Students must submit a formal report, a poster, and a presentation on their Co-op experience. While ENGL 217 and Senior Standing are the prerequisites for pursuing co-op training, the program is mandated to stipulate additional conditions.

Study Plan

The Bachelor of Science in Graphic Design encompasses 111 credit hours spread over six semesters and three summer sessions, with the last dedicated to the CO-OP experience. The following study plan is a roadmap for the student's smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Semester 1 (15 Credits)			
ENGL 210	English Composition & Rhetoric	3	Placement
GRDS 220	Introduction to Digital Media	3	
FADR 200	Drawing I	3	
FADR 215	History of Art	3	
FADR 220	Fundamentals of 2D	3	
Year 1, Semester 2 (15 Credits)			
GRDS 205	Type I	3	Co-req.: GRDS 220
GRDS 222	Advanced Digital Media	3	GRDS 220
FADR 202	Drawing II	3	FADR 200
FADR 210	Rendering and Perspective Techniques	3	
FADR 222	Fundamentals of 3D	3	FADR 220
Summer Session (9 credits)			
ARAB 212	Arabic Language and Communication	2	
GRDS 203	Introduction to Graphic Design	3	GRDS 205
GRDS 230	Photography I	3	
	Community & Sustainability	1	Co-req.: ENGL 210
Year 2, Semester 1 (18 Credits)			
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
ENGL 217	Professional Communication Skills	3	ENGL 210
GRDS 300	Illustration	3	GRDS 220, FADR 202
GRDS 305	Type II	3	GRDS 205

GRDS 310	Design I	3	GRDS 203, GRDS 220
	Social Sciences/Culture Elective I	3	
Year 2, Semester 2 (15 Credits)			
GRDS 306	Type III	3	GRDS 222, GRDS 305
GRDS 311	Design II	3	GRDS 310
GRDS 423	Motion Graphics	3	GRDS 220
GRDS 335	Production	3	GRDS 222
GRDS 345	History of Graphic Design	3	FADR 215
Summer Session (9 Credits)			
GRDS 360	Visual Perception & Infographics	3	GRDS 311
	Social Sciences / Culture Elective II	3	
	Humanities/Fine Arts Elective	3	
Year 3, Semester 1 (15 Credits)			
GRDS 409	Design Writing Research	3	GRDS 310
GRDS 411	Design III	3	GRDS 311, GRDS 306
GRDS 424	Web and Interactive Design	3	GRDS 423
GRDS 440	Design & Social Impact	3	Junior standing
	Major Elective I	3	
Year 3, Semester 2 (14 Credits)			
GRDS 412	Design IV	3	GRDS 411, GRDS 409
GRDS 452	Digital Illustration	2	GRDS 222, GRDS 300
GRDS 450	Branding & Advertising	3	GRDS 411
	Social Sciences/Culture Elective III	3	
	Major Elective II	3	
Summer Session (1 Credit)			
GRDS 399	Co-op Training Experience	1	ENGL 217; Senior Standing

Courses Description

I. Mandatory Requirements

Fine Arts Courses

Descriptions of the Fine Arts Requirement courses are given below.

FADR 200	Drawing I	3(2,2)
This course is a comprehensive introduction to the art of drawing. Students learn how to represent objects, planes, and volumes accurately and proportionally by developing hand-to-eye coordination with lines, shades, tones, and wet and dry media. The course introduces drawing techniques, composition principles, perspective basics, and various media to prepare students with the essential ingredients needed to grow in the design field.		

FADR 202	Drawing II	3(2,2)
This drawing course teaches students the basics of drawing the human figure through an understanding of its anatomy and proportions. It emphasizes the use of the human figure in space as a compositional element. Students will explore the potential and limits of media and materials. Students will begin to form a personal approach to drawing and even develop a personal drawing/illustrating style. **Prerequisite**: FADR 200.		

FADR 210	Rendering and Perspective Techniques	3(2,2)
This course introduces students to perspective techniques in rendering three-dimensional objects and scenes on two-dimensional surfaces. Students learn to apply perspective drawing, composition, and conceptualization to develop visual communication skills. Design, composition, light rendering, and perspective are explored to enhance the students' drawing and rendering techniques.		

FADR 215	History of Art	3(3,0)
This course explores the major forms of artistic expression from the ancient world to the present, from various cultural perspectives. Students learn to examine and analyze works of art in their historical context and to articulate what they see or experience in a meaningful way. This course covers the pre-Renaissance, Renaissance, and post-Renaissance periods, as well as the rich layers of 19th- and 20th-century modern art up to the manifestations of our present day.		

FADR 220	Fundamentals of 2D	3(2,2)
This course focuses on the student's visual awareness by introducing the fundamentals of 2-D design elements, including line, texture, pattern, tone, form, color, light, partial illusion, balance, and proportion. Students also explore and experiment with design principles such as repetition, variety, emphasis, and movement. This course broadens the understanding of compositional devices, dynamics, and methods. It also aims to		

develop a clear understanding of visual organization, shape interaction, and color theory.		
FADR 222	**Fundamentals of 3D**	**3(2,2)**
This course extends two-dimensional design concepts into volumetric relationships, emphasizing design concepts through structural and sculptural forms. Students will work in various media, including paper, cardboard, wood, wire, metal, and plaster, to explore concepts of modularity, sequence and series, relief, contour, structure, symmetry, and asymmetry as they relate to the study of forms in nature. Students will also examine the function of space, volume, mass, plane, and line. The main emphasis of this course is the development of critical thinking, technical, and visual aesthetic skills. **Prerequisite**: FADR 220.		

Business Course Requirement

The Graphic Design program requires one business course, BMKA 200, as described below.

BMKA 200	**Introduction to Marketing**	**3(3,0)**
This course introduces the basic principles, theories, and marketing practices in our modern, ever-changing business environment. The course covers the marketing process activities of creating value for customers and capturing value from customers in return. It also discusses the marketing mix and how to build long-term customer relationships. Students will analyze case studies about a “real-life” product or service. Videos and in-class discussions on current marketing topics will assist in the learning experience. **Co-requisite**: ENGL 210		

Major Graphic Design Courses

A description of the Graphic Design courses follows.

GRDS 203	**Introduction to Graphic Design**	**3(3,0)**
Graphic design is a creative process that combines art and technology to communicate ideas visually. This course introduces students to graphic design by exploring the elements, principles, and design process. It focuses on the design process, from the initial stage of choosing a topic through the intermediate working stages to the final presentation. It is a theoretical course that includes practical exercises in visual communication, organization, and perception. Projects explore the various design dimensions to fully comprehend their expressive power as a visually communicative tool and a problem-solving method. **Prerequisite**: GRDS 205.		

GRDS 205	**Type I**	**3(3,0)**
Type, characters, and letterforms are the essential building blocks of visual communication. In this course, students learn to express ideas not only through word meaning but also by manipulating the shape of a typographic character, thereby improving the quality of information and communication. This introductory course teaches students the anatomy of type, essential type terminology, the history of type categorization and standardization, and the difference between calligraphy and typography. **Prerequisite**: GRDS 220.		

GRDS 220	Introduction to Digital Media	3(3,0)
In this course, students will develop a solid foundation in the tools and techniques of Adobe Illustrator and Adobe Photoshop. Students will learn the difference between vector-based and pixel-based programs and how to integrate both work fields to produce creative graphics. Moreover, students will be briefly introduced to Adobe InDesign, including its workspace, functions, and tools.		

GRDS 222	Advanced Digital Media	3(3,0)
The first section of the course teaches students advanced tools and techniques in Adobe Illustrator (mesh 3D) and Adobe Photoshop (masks). The second part emphasizes learning technical skills in Adobe InDesign, a computer-based page layout software. Students will learn to use master pages, style sheets, typographic controls, and flow and format text; place and manipulate images; combine images and text; create tables, gradients, and PDFs; and correctly prepare digital files for offset printing. **Prerequisite**: GRDS 220.		

GRDS 230	Photography I	3(3,0)
This course teaches students the basics of black-and-white digital photography. It is a lecture-based course with extensive hands-on practical training. Students learn how to develop an eye for photography and for framing objects. They also enhance their knowledge of the different parts of a camera and the mechanisms of film printing and development while continuously scanning other photographers' work in contemporary and historical scenes.		

GRDS 300	Illustration	3(2,2)
This class is designed to introduce students to the art of illustration and visual storytelling. It teaches students a range of technical skills to enhance their ideas and creativity. This class allows students to discover their style through line quality, colors, and shapes. Students will be aided in developing their own ‘style’ through visual analysis and personal visions. Hands-on projects will help enhance ideas and techniques such as ink and brush, scraperboard, monoprints, watercolor, relief printing, and collage. **Prerequisites**: GRDS 220; FADR 202.		

GRDS 305	Type II	3(3,0)
This intermediate-level course studies the marriage of Latin and Arabic typography. This course develops students' ability to create typographic designs by visually and aesthetically merging Arabic and Latin Fonts. This multi-script combination targets the needs of our direct Arab and Lebanese markets. Students will learn to dissect fonts, create modules, experiment with type, generate Arabic adaptations, and construct typographical patterns that evolve into 3D objects. The student further learns the rules of the typographic grid system and experiments with breaking the Grid when designing layouts, spreads, and posters. **Prerequisite**: GRDS 205.		

GRDS 306	Type III	3(3,0)
This course focuses on Arabic typography, its history, and the modern approaches to enhance our Arabic fonts. Students will learn the anatomy of the Arabic type to manipulate its parts creatively. Projects will include creating an experimental display of Arabic font, arabesques, Arabic graffiti, and Arabic adaptations. The skills acquired in previous type classes will equip students with the techniques needed to explore further how typography, particularly Arabic, can serve as a communicative tool. **Prerequisites**: GRDS 222; GRDS 305.		

GRDS 310	Design I	3(2,2)
This course investigates the creative process the designer goes through while designing the main elements of visual communication: symbols, pictograms, icons, and logos. Students will be asked to create a visual vocabulary (corporate identity) that is credible enough to convince the target audience to purchase a specific product or service. It is an in-depth insight into planning and developing a corporate image, including logos and other business communication applications. In addition, students learn to design a complete restaurant branding image from the initial concept to the final execution. **Prerequisites**: GRDS 203; GRDS 220.		

GRDS 311	Design II	3(2,2)
This course provides a general overview of design principles for packaging design's structure and visual aspects, packaging history, and consumer decision-making psychology. The use of graphic imagery and innovative die-cuts across various packages and packaging materials will be examined. Key course concepts will include researching and designing 3D packages, understanding eco-friendly and sustainable packaging, experimenting with playful and conceptual packaging design, and defining the target audience to design the best packaging solution. **Prerequisite**: GRDS 310.		

GRDS 335	Production	3(3,0)
In this course, students will develop an understanding of the “process flow” of contemporary print production methods. The course covers the entire production timeline: Prepress, Press, and Postpress. In addition to printing technologies, students will learn about various types of ink and paper, binding, varnishes, finishing techniques, and the effect of the printing process on the product. Students will also learn how to handle projects and deal with clients as freelance designers or as designers working for companies. **Prerequisite**: GRDS 222.		

GRDS 345	History of Graphic Design	3(3,0)
This course covers the evolution of graphic communication from prehistory through postmodern design and the digital revolution. This course provides graphic design students with the knowledge and understanding of the places, people, and events, as well as historical and cultural factors and technological innovations that have influenced the development of graphic design into the practice known today. Historical awareness provides a meaningful context for young designers to evolve and contribute positively to the cultures in which they live and work. **Prerequisite**: FADR 215.		

GRDS 360	Visual Perception & Infographics	3(3,0)
Visual perception is the ability to interpret information from our surroundings through visible light that reaches the eye. This course will explore the various theories of visual perception with an emphasis on Gestalt theory and its relation to Graphic Design. It will create an understanding of our human visual perception and its limitations. Students will also be given projects in which they will apply Gestalt principles and the theories they have learned. In addition to visual perception theories, students will learn two key design disciplines: Infographics and Wayfinding. **Prerequisite**: GRDS 311.		

GRDS 399	Co-op Training Experience	1(1,0)
Each student must complete 8 weeks of practical training in a field related to their interests. This Co-op work experience is usually fulfilled during the summer semester of the third year in the program. Students must submit a formal report and poster and make a formal presentation about their Co-op experience. **Prerequisites***:* ENGL 217; Senior Standing.		

GRDS 409	Design Writing Research	3(3,0)
In this course, students will explore various research methodologies to develop the skills needed to write a well-rounded research paper for their final-year design project. They will be exposed to visual analysis tools, enabling them to articulate their understanding and appreciation of a given visual. Students will engage in writing exercises focused on various topics in the design field. By the end of this course, students should develop their senior project research proposal. **Prerequisite**: GRDS 310.		

GRDS 411	Design III	3(2,2)
This course prepares students for the design challenges of creating professional-looking layouts for editorial content (newsletters, reports, books, & magazines) and on-screen (websites and web applications). Students learn to create multi-page publications and on-screen interfaces with effective typography, images, illustrations, and layouts. The core focus of this course is working with user interactivity, composition, layout, format, negative space, grids, type, and graphical elements. **Prerequisite**: GRDS 311, GRDS 306.		

GRDS 412	Design IV	3(2,2)
This course focuses on creating a final-year project that bridges theory and practice in visual communication through an extensive exploration of the design process. Students will research, develop, and design their senior projects to demonstrate their visual communication ability. This class provides students with a forum to showcase their talent and skills. Emphasis is placed on creativity and quality of work. **Prerequisite**: GRDS 411, GRDS 409.		

GRDS 423	Motion Graphics	3(3,0)
A rapidly expanding and thriving field, motion graphics gives designers new opportunities to work in television, film, and web design. This course focuses on motion		

graphics and its diverse methodologies to lay down the necessary specialized essentials. Adobe After Effects and Flash teach the foundations of compositing, video editing, and special effects. Assignments center on upholding a certain amount of experimentation and are process-oriented. Students will learn to animate graphics and type to communicate ideas better visually. **Prerequisite**: GRDS 220.

GRDS 424	**Web and Interactive Design**	**3(3,0)**
As the World Wide Web became a fundamental fixture of life, it became essential for design to ensure that any interface is accessible, exciting, and effective. This course examines the ever-evolving field of User Interface and User Experience Design. It combines web and application design as a communication medium with a distinctive user experience. Instruction focuses on achieving a crucial balance among form and function, visual design and effectiveness, and accessible navigation and communication. **Prerequisite**: GRDS 423.		

GRDS 440	**Design and Social Impact**	**3(3,0)**
This course prepares students to become leading agents of social change through community engagement. It aims to promote collaboration between computer science and graphic design students to develop effective solutions to existing community issues. The course is structured around a balance of classroom theory, extensive field research, and the implementation of a real-world solution that draws on both. **Prerequisite:** Junior standing.		

GRDS 450	**Branding & Advertising**	**3(3,0)**
This course explains both branding & advertising and their relationship to each other. It explores the history of advertising and discusses the most influential and transformative periods it has undergone (from billboards to radio, TV, and social media). It also emphasizes the importance of advertising in defining and differentiating a product within our contemporary mass market. This course includes hands-on experience applying advertising and branding concepts and techniques in a complete campaign: creating a product, identifying competitors, defining the target audience, conducting marketing research, designing the package, and developing the ad campaign, from press ads to billboards. **Prerequisite**: GRDS 411.		

GRDS 452	**Digital Illustration**	**2(1,2)**
Digital illustration uses digital tools, software, and applications to create complex images and illustrations. This course teaches students a range of techniques and approaches to producing fresh, creative, and aesthetically pleasing illustrations. Students will receive hands-on training in basic drawing and painting, as well as in manipulating and creating vector-based graphics, illustrations, and digital paintings. Projects include various areas, from comic strips to 3-D rendering. **Prerequisites**: GRDS 222; GRDS 300.		

II. Elective Courses

Descriptions of the major elective courses are given below.

GRDS 320	UI / UX Design	3(3,0)
This course combines digital technology and creative design to create an interactive App prototype. It emphasizes the efficient use of Adobe XD as a testing and prototyping tool. Students will be asked to implement digital design research methods in the study of User Experience, including forming proto-personas, journey maps, site maps, red routes, sketches, and wireframes, which will then be transformed into a clickable application. In addition, students will learn to design a complete app design (Logo and In-App Screen designs) from A to Z. After the application design, students will implement the evaluation criteria to user-test the app for improvement. **Prerequisite**: GRDS 220.		

GRDS 330	Photography II – Photography for Social Media	3(3,0)
This course emphasizes developing a critical eye and using photography as a form of self-expression and an artistic medium for social media posts. Students are expected to have a working knowledge of the photographic process. Students will produce photographic posts for real-life brand products. **Prerequisite**: GRDS 230.		

GRDS 331	Digital Processing	3(2,2)
Digital photography is now a standard set, ranging from compacts to medium-format sensor-size cameras. Through Adobe Photoshop and Lightroom, this course will give students a wider understanding of digital processing and important attributes. An image taken from the camera certainly needs modifications to get the best out of it. With that, pictures will look the way we intend them to, following precise adjustments and an understanding of why and what the result will be. **Prerequisite**: GRDS 222.		

GRDS 350	3D Printing & Prototyping	3(2,2)
This course focuses on the fundamental 3D design and model-making process. It will follow the process that industrial designers use to create 3D-printed products. The course will cover the design thinking process for bringing initial sketch ideas to life while considering limitations. Multiple software applications will be used to design the 3D models, including Adobe Photoshop, Autodesk Maya, and Pixologic ZBrush. **Prerequisite**: GRDS 222.		

GRDS 355	Calligraphy	3(2,2)
In this course, students will learn the basics of Arabic calligraphy as defined by Ibn Muqla. Variations in the main Arabic scripts are explained through a set of visual examples. The course touches upon Islamic calligraphy as a communicative and embellishing technique in the visual arts. The practical aspect of the course explores the techniques and rules of drawing letters of the main standardized Arabic Script. Calligraphy will also be used to create modern visual expressions. **Prerequisite**: GRDS 305.		

GRDS 375	3D Animation	3(3,0)
In the first section of this course, students will learn Autodesk Maya. They will explore 3D form and space, emphasizing digital modeling of real-life objects. They will develop		

and construct 3D objects intended graphically for any design or animation use. The course covers basic to intermediate 3D modeling skills, with an emphasis on texturing and lighting. Maxwell Realflow, an industry standard for fluid simulation, will be covered in the second part of this course. Water simulation and dynamics will be taught and integrated into Maya for rendering. **Prerequisite**: GRDS 423.

GRDS 385	**VR Illustration & Sculpting**	**3(3,0)**

Virtual Reality is a new technology tool used for game and art production. Its concept has been around for a long time, but it was only recently viewed as a novelty. In this course, students can acquire this new skill for the fast-growing design market. They will learn illustration in virtual space using the Oculus app and TiltBrush, and sculpting in Oculus Medium, which replaces the usual computer software, ZBrush or Photoshop. Future developments in Oculus will allow students to print their models on 3D printers. **Prerequisite***:* GRDS300

GRDS 381	**Bookbinding**	**3(2,2)**

This class teaches students various bookbinding techniques: Japanese binding, accordion folding, and signature binding. This course will cover the bookbinding process from the initial concept to the finished, bound book. Multiple types of bookmaking will be covered, along with detailed explanations of the advantages and disadvantages of each binding. A creative approach to a design problem will be addressed, using book design as a form of art.

GRDS 402	**Silkscreen**	**3(2,2)**

Silkscreen is one of the most flexible and widely used types of printmaking. The class focuses on various silkscreen techniques taught through demonstrations and specific projects. Images and graphics will be created using hand-drawn separations, photographic film, digital separations, and photocopied images. Water-based silkscreen inks are used, allowing for soap-and-water cleanup. Students will be encouraged to experiment with multiple techniques and combinations of traditional and contemporary methods. **Prerequisite**: GRDS 300.

GRDS 403	**Painting**	**3(2,2)**

This course introduces various painting materials, skills, and techniques. It helps students develop artistic potential to create a range of moods. Students will learn to use the brush to create different strokes and manipulate colors to suit their concepts. Color theory will be discussed in class. The course includes detailed studies of still life, landscape, and the human figure. **Prerequisite**: FADR 202.

GRDS 431	**Critical Theory**	**3(3,0)**

This course is an introduction to graphic design theory. It explores the aesthetic and social purposes of design practice through written selections spanning a century of design evolution. Students will read about design to stimulate growth and change in their work. Students will also learn how to analyze and critically write design essays.

This course also puts theory into practice, and students can manually experiment with past and present theories. **Prerequisites**: GRDS 305; GRDS 310.

GRDS 436	**Game Design**	**3(2,2)**
Turning ideas into reality, the game design course is a hands-on experience in translating concepts, character sketches, storyboards, and animations into finalized elements for use in the game industry. Students will explore the process of designing meaningful experiences for the players. Working in groups with the computer science students, students will design and develop a 2D/3D game that the public can download and test. **Prerequisites**: GRDS 220		

GRDS 498	**Special Topics in Graphic Design**	**3(3,0)**
A special topic course to be defined based on current and evolving design trends and technological advancements. Students will be able to explore various topics under the vast design umbrella.		

MINOR IN GRAPHIC DESIGN

Rationale

The CAS-GD department offers a Graphic Design minor for RHU students. It is designed for students who want to gain basic proficiency in graphic design to enhance and supplement their major program.

Program Objectives

The aims of a minor in Graphic Design are to:

- Provide RHU graduates with basic proficiency in graphic design
- Support RHU graduates with essential design skills desired/demanded by the market

Learning Outcomes

At the end of this minor, the student is expected to:

- Identify the key elements and principles of design
- Apply design thinking and the design process to create effective visual solutions
- Use the required technical skills learned to develop print and web design solutions
- Experiment with layouts, grids, images, graphical elements, typography, and color to communicate an intended message to a target audience

Entrance Requirements

This minor is offered to all RHU students.

Early in their major, interested RHU students must complete the appropriate form declaring their intent to minor in Graphic Design while completing their regular major.

Graphic Design Minor Policies

- Obtain a Minor Cumulative Grade Point Average of no less than 70 %
- Overlapped courses between student major requirements and GD minor requirements are counted to fulfill the GD minor requirements.

Career Options

This GD minor allows holders to pursue career opportunities in various sectors (banks, TV channels, advertising companies…) regardless of their major. Graduates of this minor can seek jobs in branding, advertising, and motion graphics.

Curriculum and Program

The Graphic Design minor consists of six mandatory courses (18 credits) selected to satisfy the proposed program objectives and learning outcomes requirements.

III. Mandatory Courses

The six mandatory courses (18 credits) are:

GRDS 203	Introduction to Graphic Design	GRDS 205	3(3,0)
GRDS 205	Type I	GRDS 220	3(3,0)
GRDS 220	Introduction to Digital Media		3(3,0)
GRDS 230	Photography I		3(3,0)
GRDS 310	Design I	GRDS 203	3(2,2)
GRDS 423	Motion Graphics	GRDS 220	3(3,0)

Faculty List

Abdallah, Lama; Instructor, Master's in Medical Physics and Imaging Technology, Lebanese University, 2018.

Abdallah, May; Lecturer, Ph.D. in Applied Mathematics, Lebanese University, 2023.

Abdallah, Rawan; Lecturer, Ph.D. in Mathematics, Politehnica University of Bucharest, 2023

Abdallah, Saber; Instructor, BA in Business Administration, RHU, 2015.

Abdelmalak, Inas; Lecturer, Ph.D. in Chemistry, Lebanese University, 2018.

Abo Alfa, Khaled; Lecturer, Ph.D. in Mathematics, Université de Nantes, Nantes, France. Thesis Title: Magnetic tunneling effects in various geometries—advisor: Frédéric Hérau, 2023.

Abo Hadeer, Likaa; Instructor, Master's in Teaching English as a Foreign Language, LIU, 2022

Abou Orm, Lara; Assistant Professor, Ph.D. in Mathematics, École Nationale Supérieure des Mines de Saint-Étienne, Saint-Étienne, 2013.

Abdallah, Lama; Instructor; MS in Medical Physics and Imaging Technologies, LU, 2018.

Al-Hajj, Roba; Instructor, Master 2 in Microbial Systems, University of Montpellier, 2011.

Alaaedine, Tala: Ph.D. student, Beirut Arab University, MS in Mathematics, Beirut Arab University, 2022.

Alameddine, Mira; Lecturer, Ph.D. in Education, Lebanese University, 2019

Arkadan, Houda; Instructor, BS in English Language, Rafik Hariri University, 2020

Baba, Dina; Senior Lecturer, MA, Business Administration, Lebanese American University, 2003.

Banat, Maysaa; Assistant Professor, MA, TESOL, Lebanese University, 1995.

Chamoun, Myriam, Instructor, MSc in Cognitive Systems and Interactive Media, Universitat Pompeu Fabra, 2023; MSc in Interaction Design, IED – Istituto Europeo di Design, 2020.

Charara, Lana, Instructor, MA Illustration, Arts University Bournemouth, 2017.

Danab, Rola; Instructor, Master's Degree in Educational Studies, The College of Teachers, London, 2016

Darwish, Mohammed. Instructor, BS in Graphic Design, Rafik Hariri University, 2022.

Shoueib, Wafaa; Instructor, Master's of Business Administration, LAU, 2003

El-Abed, Mohamad; Lecturer, Ph.D. in Computer Science, University of Caen Lower Normandy, France, 2011.

Fares, Najwa; Professor, Ed.D., Educational Research - TESOL, University of Sussex, 2013.

Farroukh, Sobhi; Instructor, MA in Graphic Design, Florence Institute of Design International, 2025.

Ghattas, Khaled; Lecturer, Ph.D. in Cell Biology, American university of Beirut, 2013.

Hajj Chehade, Hiba; Instructor, Master of Science and Technology of Information Systems, Lebanese University, 2022.

Halablab, Mahmoud; Professor, Ph.D. in Microbiology, King's College London, University of London, 1991.

Harb, Graziella; Associate Professor, Ph.D. Applied Linguistics and Literature, USEK, 2018.

Hassan, Claudia; Lecturer, Ph.D. in English Language & Literature, LU,2021.

Hassoun, Lina; Lecturer, Ph.D. in Art and Arts Sciences, LU, 2020.

Hazime Fayza; Instructor**,** Master's in Education Counseling, Haigazian University, 2010.

Khoury, Farid; Instructor, Ma in Ancient Religions and Archaeology, La Sierra University,1993.

Makki, Sara; Instructor**,** Master in Visual Design, Scuola Politecnica di Design, 2011.

Mourad, Nashrawan; Instructor, Master's in Clinical Psychology, LU, 2025**.**

Nahhas, Kamal; Instructor, MA, Counseling, Walsh University, 1983.

Nasreddine, Elissar; Lecturer, Ph.D. in Mathematics, Toulouse University, Paul Sabatier, 2013.

Orabi, Fawzieh; Instructor, Master's in English Language and Literature, LU, 2019

Oweini, Abir; Instructor, Master's in Middle Eastern Politics, School of Oriental and African Studies (SOAS), UK, 2003.

Rammal, Ali; Assistant Professor, Ph.D. in Computer Science, University of Paul Sabatier, 2010.

Salami, Houssam; Associate Professor, Ph.D. in Physics, Lyon 1 University, Claude Bernard, 2007

Sammoura, Noura; Assistant Professor, Master's in Communication Design, Politecnico di Milano, 2020.

Shamseddine, **Naziha,** Assistant Professor, Ph.D. in English Literature, BAU, 2019.

Sinjab, Nisreen; Instructor, MA in Philosophy, AUB,2013.

Soloh, Rouaa; Assistant Professor, Ph.D. Computer Science, Normandie University 2022.

Srouji, Serene; Associate Professor and Chairperson, MFA in Design, University of Texas at Austin, 2007

Talhouk, Mirna; Instructor, M.Sc., Sociology, Lebanese University, 2001.

Tarhini, Ahmad; Lecturer, Ph.D. in Theoretical Particle Physics, University Claude Bernard Lyon, 2013

Zahreddine, Nadine; Assistant Professor, Ph.D. in Fine Arts, Lebanese University, 2021
Zebian, Rihab; Instructor, B.S. in Graphic Design, Lebanese American University, 2006.

COLLEGE OF BUSINESS ADMINISTRATION

COLLEGE OF BUSINESS ADMINISTRATION (CBA)

College Officers

Said Ladki	President
Mahmoud Halablab	Vice President for Academic Affairs
Anis El Khatib	Dean

Contact Information
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Block G, Room 101-G
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History and Overview

Being the first and oldest college of the University, the College of Business Administration (CBA) at Rafik Hariri University was established in 1999. Since then, the College has grown significantly and now offers six undergraduate business programs: Accounting, Business IT Management, Finance and Banking, Human Resources Management, Management, Marketing, and Advertising. In addition, the College offers a graduate MBA program in general business administration. Moreover, the CBA takes pride in being the first in Lebanon and the Arab World to offer a graduate MBA program in Oil and Gas Management.

The faculty is active in applied research across several business domains and related areas, focusing on developing solutions to real-world community problems. To that end, and in partnership with governmental entities, civil society, trade associations, and the private sector, the College actively engages in outreach initiatives to support community development and prosperity.

In 2019, the CBA became the first Lebanese university to earn full ACBSP accreditation for all its degrees and programs. This great achievement represents an internationally recognized certificate of rigorous, high-quality education in accordance with international standards and best practices.

Finally, the CBA distinguishes itself by its systematic Co-op Work Experience program, which allows its students to gain real-world work experience, and its Community Engagement Experience program, which enables students to develop leadership skills through hands-on community service.

Vision

The College of Business Administration aims to become the region's premier institution for innovative business education.

Mission

The College of Business Administration aspires to transform students' lives through graduate and undergraduate business education, advance knowledge through applied research, and positively impact society's pursuit of development and prosperity.

We aim to foster an educational culture and environment of innovation and collaboration that enable the development of leaders and professionals who are capable and willing to become agents of principled, ethical, and socially responsible human progress.

The College undertakes its mission within a higher education philosophy that emphasizes proficient technical know-how in a specialized area, reinforced by core knowledge of key business functional areas, and grounded in the fundamentals of liberal arts education.

Core Values

The College of Business Administration is committed to maintaining a system of norms, behaviors, and conduct well-grounded in the following main guiding values:

Academic Freedom of Inquiry
Faculty and students are free to pursue knowledge and learning so long as such pursuits do not breach University and College missions, policies, and regulations.

Excellence through Innovation
The College has an unwavering commitment to pursuing excellence in everything we do. Key to this pursuit are our innovative initiatives, ideas, and efforts.

Ethical and Socially Responsible Conduct
The College makes every effort to ensure awareness of and adherence to ethical and socially responsible norms.

Tolerance and Diversity
The College is committed to highlighting and raising awareness of tolerant mentalities that accept and respect differences with others. Moreover, the College recognizes and promotes the enrichment resulting from diverse individuals, communities, ideas, and perspectives.

Personal Initiative and Individual Responsibility
Leaders and professionals dare to initiate and take responsibility for their choices. The College promotes this principle and encourages its stakeholders to learn and act in accordance with it.

Teamwork, collaboration, and cooperation
The College is committed to conducting its affairs in the spirit of teamwork and collaboration. Furthermore, the College opens opportunities for students to learn and appreciate the value of collective work.

Continuous Improvement and Innovation
The College regards continuous improvement and innovation in its programs and scholarship as necessary to meet the challenges of rapidly changing business environments.

College Learning Goals and Objectives

In line with the College's mission statement, we aim to transform the lives of our students through graduate and undergraduate business education; the College's Learning Objectives specify the expected outcome of this transformation: graduates capable of becoming agents of human progress.

Accordingly, the College's general learning objective is to equip its students with the values, knowledge, competencies, and skills needed to make a positive impact in whatever they do in life. In particular, these values, knowledge, competencies, and skills are as follows:

General and Specialized Knowledge

Knowledge of the main business functional areas (CLG1)	Students will acquire basic knowledge of the main functional areas of business, which is necessary to understand the general business environment and its interrelationships.
Specialized knowledge (CLG2)	Students will become proficient in up-to-date theories, applications, best practices, and other dimensions of their chosen area of specialization, including hands-on applications.

Communication

Effective communication (CLG3)	Students will learn to communicate effectively, orally and in writing, in various professional environments and settings.

Critical Thinking

Analysis and critical thinking (CLG 4)	Programs and activities in the College will involve learning settings that require students to apply analytical and critical thinking tools and methods.

Ethics and Social Responsibility

Ethical and socially responsible conduct (CLG 5)	The college will ensure that its students are exposed to learning opportunities that will allow them to

	improve their recognition and awareness of ethical dilemmas and socially responsible behaviors.

Academic Programs

The College of Business Administration offers six undergraduate programs leading to a Bachelor of Business Administration (BBA) degree, two minor programs, and two graduate programs leading to a Master of Business Administration (MBA) degree. The programs are as follows:

1. Bachelor of Business Administration in Accounting
2. Bachelor of Business Administration in Business IT Management
3. Bachelor of Business Administration in Finance and Banking
4. Bachelor of Business Administration in Human Resources Management
5. Bachelor of Business Administration in Management
6. Bachelor of Business Administration in Marketing and Advertising
7. Minor in Business Administration
8. Minor in Human Resources Management
9. Master of Business Administration in General Business Management
10. Master of Business Administration in Oil and Gas Management

Program Codes

The following table lists the code used for each program. This code is the prefix of each course designation in the program.

Program Code	Program
BACC	Accounting
BFIN	Finance and Banking
BADM	General Business Administration
BECN	Economics
BITM	Business IT Management
BMGT	Management
BHRM	Human Resources Management
BMKA	Marketing and Advertising

Admission Requirements

In addition to fulfilling the University admission requirements, students aspiring to study one of the business majors may be required to take a mathematics placement examination, depending on their Lebanese baccalaureate math score. Students who fail to attain a passing score must take and pass one or two related courses, depending on their score, to ensure their mastery of basic skills and improve their ability to handle the rigor of college-level subjects. Students are strongly advised to review the University

Catalog carefully for admission and degree requirements, as well as related academic policies.

Graduation Requirements

Each Business program awards a Bachelor of Business Administration (BBA) degree, which comprises 99 credits distributed across three categories: University Requirements (UR), College Requirements (CR), and Program Requirements (PR). The University and College requirements are common to all College of Business Administration programs. Each department has its own required and elective courses. The credit hour allocations for each program are shown in the following tables:

BBA in Accounting (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	6	-	35	-	41	41
Program Requirement	25	3	-	-	28	28
Credits	31	3	53	12	99	100

BBA in Business IT Management (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	3	-	15	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Credits	31	3	53	12	99	100

BBA in Finance and Banking (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	28	3	56	12	99	100

BBA in Human Resources Management (99 Credits)

Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	-	-	41	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	25	3	59	12	99	100

BBA in Management (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	28	3	56	12	99	100

BBA in Marketing and Advertising (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	28	3	56	12	99	100

A business student in any of the programs is eligible for graduation if s/he has:

1) Completed all the requirements of the degree
2) Attained a cumulative GPA of 70% or higher
3) Attained a major GPA of 70% or higher
4) Attained at least a 70% grade in the BADM 490 course
5) Completed the mandatory COOP Work Experience and the Community Engagement Experience.

I. University Requirements (General Education)

Every student is required to take 30 credit hours of general education courses distributed over six domains. Eighteen mandatory credit hours are selected from four domains, and twelve elective credit hours are selected from three domains, as indicated on the following pages.

Domain		Credits	Courses
1	Communication Competency*	8	ENGL 210: English Composition and Rhetoric ENGL 217: Professional English Communication ARAB 212: Arabic Language and Communication
2	Humanities/Fine Arts	3-6	Selected from an approved list
3	Social Sciences/Culture	6	Selected from an approved list
4	Natural Sciences and Technology	6	Natural Sciences: One 3-credit course selected from an approved list. Technology: BITM 200: Information Technology Essentials
5	Quantitative Reasoning	3	BADM 225: Business Math
6	Community and Sustainability	4	BADM 355: Business Ethics and Social Responsibility One 1-credit course selected from an approved list

[1] Non-native Speakers of Arabic are allowed to take either SOCI 310 or CMNS 410 as a substitute for ARAB 212.

Description of the specific courses in this group is given below:

ARAB 212	Arabic Language and Communication	2(2,0)
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need for their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

BADM 225	Business Math	3(3,0)
Linear equations, supply and demand analysis, non-linear equations, quadratic functions, exponential and logarithmic functions, compound interest, geometric series, investment appraisal, Derivatives and marginal revenue-cost-profit, elasticity, functions of several variables, partial elasticity and marginal functions, simple optimizations, indefinite and definite integration, matrix operations and Cramer's rule.		

BADM 355	Business Ethics and Social Responsibility	3(3,0)
This course introduces students to the contemporary principles of ethics and social responsibility in business. Students learn to make ethical judgments about important issues they face daily by relating them to a framework of ethical principles, including utilitarianism, justice, moral rights, the ethics of care, and virtues and vices. Ethical dilemmas related to conflicts of interest, sustainability in business strategy, and corporate governance are among the issues discussed in this course. Prerequisite: Junior Standing.		

BITM 200	Information Technology Essentials	3(3,0)

This course is an introduction to the general and business use of computers. It covers the basics of hardware, the operating system, the internet, and software, with an emphasis on the MS Office Suite.

CMNS 200	**Etiquette**	**1(1,0)**
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, regardless of the industry, organization, or sector in which they are employed. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentations, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	**First Aid**	**1(1,0)**
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 100+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

II. College Requirements

A. Remedial Courses

Proficiency in the English Language is a requirement for admission to any program in the College of Business Administration. The student may be required to take a remedial mathematics course, depending on the placement exam or the Baccalaureate mathematics score. This remedial course does not count toward fulfilling the degree requirements. A description of the remedial course follows.

MATH 189	Fundamentals of Algebra	3(3,0)
Real number systems, radicals and rational exponents, polynomials, factoring, fractional expressions, lines in the plane, functions and their graphs, inverse functions, solving equations and inequalities, real zeros and the Fundamental Theorem of Algebra, exponential functions and their graphs, logarithmic functions and their graphs.		

B. Mandatory Courses

All College of Business Administration students must take 41 credit hours of mandatory foundational and business functional courses. Marketing and Advertising Program students take Introduction to Digital Media instead of Quantitative Methods for Business. A list of these courses and their descriptions follows.

Course	Title	Credits	Prerequisite
BACC 205	Financial Accounting	3	
BACC 255	Managerial Accounting	3	BACC 205
BADM 215	Personal Development and Management	1	
BADM 235	Business Research Methodology	1	
BADM 230	Business Law	3	
BADM 250	Business Statistics	3	
BADM 290	Community Engagement Experience	0	
BADM 420[1]	Quantitative Methods for Business	3	BADM 250 or MATH 351 or Senior Standing
BECN 301	Microeconomics	3	Junior Standing
BECN 302	Macroeconomics	3	Junior Standing
BFIN 300	Financial Management I	3	BACC 205
BITM 300	Business IT Management	3	BITM 200
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
BMGT 200	Introduction to Management	3	Co-req. ENGL 210.
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
MATH 207[2]	College Algebra	3	Placement

[1] Students in the Marketing and Advertising program take GRDS 220 – Introduction to Digital Media instead of BADM 420- Quantitative Methods for Business

[2] Students who do not pass the Mathematics Placement Test must take MATH 207. Students who pass the test take a free elective instead of MATH 207.

Descriptions of these courses are given below.

BACC 205	Financial Accounting	3(3,0)

This course introduces the basic concepts and principles of financial accounting. It covers major areas of financial accounting: generally accepted accounting principles, the accounting cycle, financial reporting, and the accrual basis of accounting. It provides the necessary understanding of basic accounting principles and procedures for recording the financial assets, inventories, noncurrent assets, liabilities, and owner's equity.

BACC 255	Managerial Accounting	3(3,0)
This course introduces students to the basic concepts, analyses, uses, and procedures of management accounting. It would enable students to understand and view cost as part of activities planned and implemented by a company. It aims to develop managerial decision-making skills by covering the following topics: cost categories, cost-volume-profit analysis, master and flexible budgets, direct costs, and manufacturing overhead variances and relevant costs. **Prerequisite**: BACC 205.		

BADM 215	Personal Development and Management	1(1,0)
This course engages the student in self-reflection, self-discovery activities, and lectures. It is intended to encourage students to develop themselves and plan for the present and the future by becoming more self-aware. Topics and issues include personality types, communication styles, personal performance factors, career choices, and personal development planning.		

BADM 230	Business Law	3(3,0)
This course introduces the legal framework of business, emphasizing Lebanese law governing business conduct. Topics include legal concepts, sources of law, types and classes of contracts, legal forms of business organizations, the commercial code, and labor issues.		

BADM 235	Business Research Methodology	1(1,0)
This course exposes students to business research methodology and the steps for systematically solving a business problem or addressing a market opportunity. The course's main topics include problem definition, research design and methodology, data collection, and sampling techniques. The course concludes with students writing a research proposal.		

BADM 250	Business Statistics	3(3,0)
Business Statistics introduces students to the fundamentals of applied statistics. Accordingly, students are exposed to statistics as they are directly applied in solving business problems. The course will cover random variables, sampling, probability distributions, expectation, hypothesis testing and confidence intervals, analysis of variance, correlation, and simple linear regression.		

BADM 290	Community Engagement Experience	0(0,1)
The Community Engagement experience consists of a specific number of volunteer hours by students at community organizations under the supervision of a faculty		

member. Community organizations may include governmental agencies, non-governmental organizations, and civil society organizations.

BADM 420	Quantitative Methods for Business	3(3,0)
This course introduces students to managerial decision analysis using quantitative tools. The course will introduce students to the use and development of mathematical models to help managers make informed decisions. The focus is on the applied aspects of statistics and math. As such, the course will cover the basics of probabilistic and statistical techniques, decision analysis, linear programming, optimization, forecasting, and waiting-line theory. **Prerequisite:** BADM 250, MATH 351, or Senior Standing.		

BECN 301	Microeconomics	3(3,0)
Students will study the general principles of microeconomics. Included are the theoretical constructs of consumer behavior, cost structure, and the operations of business firms in the market economy under conditions of perfect competition, oligopoly, monopoly, and monopolistic competition. **Prerequisite**: Junior Standing.		

BECN 302	Macroeconomics	3(3,0)
Students will study the general principles of macroeconomics. This course presents the formal Keynesian theory of income determination and its contemporary critiques, including the possible causes and solutions to unemployment and inflation, as well as the importance of the international economy. Government fiscal and monetary policies are examined in detail. **Prerequisite:** Junior Standing.		

BFIN 300	Financial Management I	3(3,0)
This course will expose students to the foundational principles, theories, and applications of corporate and business finance and financial management. It covers simple and compound interest, risk and rates of return, the time value of money, stock and bond valuations, discounted cash flow analysis, and financial ratios. **Prerequisite:** BACC 205.		

BITM 300	Business IT Management	3(3,0)
This course provides a comprehensive introduction to information systems and their application. It explains how to use and manage information technology to revitalize business processes, conduct e-commerce, improve business decision-making, and gain a competitive advantage. **Prerequisite:** BITM 200		

BITM 350	Fundamentals of Data Analytics	3(3,0)
This course introduces students to statistical techniques for analyzing large datasets. The course covers the theory and application of both parametric and nonparametric methods. Students will learn to visualize the data using univariate and bivariate plots, factor analysis, and cluster analysis to investigate whether correlations exist in a multidimensional space and to build and test predictive models such as linear, logistic, and time-series models. **Prerequisite:** BADM 250 or MATH 351.		

BMGT 200	Introduction to Management	3(3,0)
Students study the basic functions of management and are exposed to modern management practices, current events, problem-solving, and ethical dilemmas. Topics include decision-making, strategic and operational planning, organizational structure, Human Resources management, leadership, and control techniques. The instructor facilitates discussion and integrates these topics through contemporary business issues and case studies. **Co-requisite**: ENGL 210.		

BMKA 200	Introduction to Marketing	3(3,0)
This course introduces the basic principles, theories, and marketing practices in our modern, ever-changing business environment. The course covers the marketing process activities of creating value for customers and capturing value from customers in return. It also discusses the marketing mix and how to build long-term customer relationships. Students will analyze case studies about a “real-life” product or service. Videos and in-class discussions on current marketing topics will assist in the learning experience. **Co-requisite**.: ENGL 210		

GRDS 220	Introduction to Digital Media	3(3,0)
In this course, students will develop a solid foundation in the tools and techniques of Adobe Illustrator and Adobe Photoshop. Students will learn the difference between vector-based and pixel-based programs and how to integrate both work fields to produce creative graphics. Moreover, students will receive a brief introduction to Adobe InDesign, including its workspace, functions, and tools.		

MATH 207	College Algebra	3(3,0)
Solving linear and non-linear equations, modeling with equations, functions, and their graphs, increasing and decreasing functions-transformation, quadratic functions and minima, modeling with functions, combining functions, polynomial functions and their graphs, dividing polynomials, real zeros of polynomials, complex numbers, complex zeros of polynomials, exponential and logarithmic functions, sequences and summation notation-arithmetic sequences, counting and probability.		

Program Requirements

Requirements for the Bachelor of Business Administration degree are program-specific. They encompass two categories: Major and non-major. Each category consists of mandatory and elective courses. The program requirements for the Bachelor of Business Administration degree in the different business majors are given below. Details and titles of relevant courses are included in the Student’s Study Plan (SSP), which every business student will have.

Course Coding

Each course offered by the College of Business Administration is designated by a four-letter code representing the college and program, followed by a three-digit number denoting the course number, "**BABC xyz**." The letter **"B"** refers to the College of Business Administration, and "**ABC**" refers to the program as follows:

ACC: Accounting
ADM: General Business Administration
ECN Economics
FIN: Finance
HRM: Human Resources Management
ITM: Information Technology Management
MGT: Management
MKA: Marketing and Advertising

The three-digit number "**xyz**," between 200 and 599, denotes the course number. The first digit indicates the course level: **1** for freshman, **2** for sophomore, **3** for junior, **4** for senior, and **5** for graduate.

The following example demonstrates the coding system: BMKA 200 is a sophomore-level marketing course in the College of Business Administration.

The designation used to represent the credit-hour breakdown **c (t, p)** of a course is as follows: **"c"** is the total credit hours**, "t"** stands for the theoretical component of the course, and **"p"** is the practical or laboratory component. For example, 3(3, 0) denotes a 3-credit course with 3 contact lecture hours and 0 laboratory hours.

Learning Support Center

The Learning Support Center (LSC) at the College of Business Administration is another manifestation of RHU's unwavering commitment to students' academic success. The LSC offers the students a peer-to-peer learning environment that complements classroom learning. The Center aims to support students in advancing their knowledge and skills, reinforcing what they learn in the classroom, or overcoming learning and performance difficulties. In addition to being a focal point for inquiry and two-way "questioner-replier" learning, the LSC provides space for academic interactions that can significantly enhance learning.

The Center is in room H 101 and is open to all students throughout the week. The LSC is staffed by carefully selected Teaching Assistants, who are graduate students and upper-level undergraduates. Teaching Assistants are always available and ready to assist their fellow students in overcoming their academic challenges. Additionally, an "on-call" faculty member is assigned to ensure that the LSC acts in the best interests of students and to intervene when necessary to help them.

The Center's resources include books, study guides, course materials, boards, computers, videos, DVDs, documentaries, and other learning tools.

Financial Studies Department (FSD)

Faculty Members

Chairperson:	Ghalia Nassereddine
Associate Professor:	Anis El Khatib, Mohamad Tarabay, Ghalia Nassereddine
Assistant Professor:	Zeinab Srour
Senior Lecturer:	Rima Hakim
Lecturer:	Jamil Hammoud
Adjunct Faculty:	Mohamad Al Hajj, Maha Hajj Omar, Samar Abou Lteif, Alexan Hagopian, Yara Eid

Programs Offered

The Financial Studies Department (FSD) offers three programs – Accounting, Finance and Banking, and Business IT Management. Each program leads to a Bachelor of Business Administration degree. The details of each program follow.

Accounting Program

Program Overview

If you want to be financially literate and play a key role in business decision-making, Accounting is the right career choice for you.

Accounting is about the measurement, analysis, and communication of financial information pertinent to the economic health of business entities. Accountants devise and use financial information systems to enable investors, creditors, managers, and regulators to make sound decisions.

There are two main tracks in accounting: Financial Accounting and Management Accounting. If the purpose is to provide information to investors and creditors for resource allocation and decision-making, we are talking about financial accounting. But if the purpose is to provide information to managers to plan, evaluate performance, and make decisions, we are talking about management accounting.

To obtain a Bachelor of Business Administration degree in Accounting, the student must complete 99 credit hours. These span University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BBA in Accounting (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	6	-	35	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	31	3	53	12	99	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

II. College Requirements

The list of the college-required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

III. Program Requirements

A. Mandatory requirements

The mandatory courses for the Accounting program are listed in the table below.

Course #	Title	Credits	Prerequisites
BACC 305	Intermediate Accounting I	3	BACC 205
BACC 310	Cost Accounting	3	BACC 255
BFIN 350	Credit and Financial Analysis	3	BFIN 300
BACC 405	Intermediate Accounting II	3	BACC 305
BACC 450	External Auditing	3	BACC 405
BACC 460	Advanced Financial Accounting	3	BACC 405
BACC 465	Taxation	3	BACC 405
	Major Elective	3	
BADM 485	Co-op Work Experience	1	ENGL 217 Senior Standing
BADM490	Entrepreneurship	3	Senior Standing

B. Major Electives

As part of the Bachelor of Business Administration in Accounting program, the student must study 3 credit hours of major electives. Major electives could be upper-level courses in accounting or closely related business areas. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility into the program, allowing the integration of emerging topics.

In consultation with the academic advisor, the student should select the elective course that best aligns with their needs and aspirations. It is highly recommended that the student register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Prerequisites
BACC 350	Accounting Information Systems	3	BACC 205, BITM 300
BACC 455	Internal Auditing	3	BACC 405
BACC 470	Forensic Accounting and Fraud Detection	3	BFIN 300, BACC 205
BADM 480	Independent Studies	3	Advisor's approval

Moreover, under certain conditions specified by the department and subject to the advisor's approval, a student may take BADM 480 (independent studies) as a substitute for a program's mandatory course. The course's description is below.

C. Entrepreneurship

The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. Prerequisite: Senior Standing.

D. Co-op Work Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students must then submit formal reports and posters and make formal presentations about their Co-op experience. While ENGL 217 and Senior Standing are prerequisites for co-op training, the program must stipulate additional conditions.

Career Opportunities

Accounting offers a wide range of careers in various types of organizations. In addition to the traditional roles of financial reporters and analysts, internal and external auditors, cash managers, and cost controllers, accountants can serve as advisors and strategic contributors across different areas of the organization. Moreover, accountants typically work in government agencies, businesses, industries, and organizations.

Program Goals and Student Learning Outcomes

The purpose of the Accounting Program at the College of Business Administration of Rafik Hariri University is an integrative composite of four main dimensions: General and Specialized Business Knowledge, Communication, Critical Thinking, and Growth Potential.

General & Specialized Business Knowledge

Goal 1	Prepare graduates with effective professional competence in critical accounting tasks and activities.
Outcome 1	Apply essential business knowledge and skills in problem-solving and decision-making.
Outcome 2	Prepare Financial Statements in accordance with International Financial Reporting Standards, Generally Accepted Accounting Principles, and best practices.

Communication

Goal 2	Equip graduates with the knowledge and skills to secure placements in financial and managerial accounting positions.
Outcome 3	Demonstrate written and oral English communication skills adequate for challenging entry- and middle-level professional positions and self-employment.

Critical thinking

Goal 3	Promote analytical and critical thinking in approaching business issues and problems.
Outcome 4	Identify business problems, analyze them, and present plausible solutions.

Ethics and Social Responsibility

Goal 4	Raise awareness about ethics and social responsibility to promote better business for a better community.
Outcome 5	Recognize ethical dilemmas in business and respond to them in accordance with standard codes of conduct, ethical principles, and best practices.

Study Plan

The Bachelor of Business Administration in Accounting encompasses 99 credit hours over six semesters and two summer sessions. The second summer session should be dedicated to the Community Engagement Experience. Meanwhile, the third session should be for the Co-op work experience. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (16 Credits)			
BACC 205	Financial Accounting	3	
BADM 215	Personal Development and Management	1	
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210.
BITM 200	Information Technology Essentials	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 207	College Algebra*	3	Placement
Year 1, Spring Semester (16 Credits)			
BACC 255	Managerial Accounting	3	BACC 205
BADM 225	Business Math	3	Placement
BADM 235	Business Research Methodology	1	
BADM 250	Business Statistics	3	
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
ENGL 217	Professional English Communication	3	ENGL 210
Year 2, Fall Semester (18 Credits)			
BACC 305	Intermediate Accounting I	3	BACC 205
BACC 310	Cost Accounting	3	BACC 255
BADM 230	Business Law	3	
BECN 301	Microeconomics	3	Junior Standing
BFIN 300	Financial Management I	3	BACC 205

Year 2, Spring Semester (18 Credits)			
BACC 405	Intermediate Accounting II	3	BACC 305
BECN 302	Macroeconomics	3	Junior Standing
BFIN 350	Credit and Financial Analysis	3	BFIN 300
BITM 300	Business IT Management	3	BITM 200
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
	Humanities / Fine Arts Elective	3	
Year 2, Summer Semester (0 Credits)			
BADM 290	Community Engagement Experience	0	
Year 3, Fall Semester (15 Credits)			
BACC 450	External Auditing	3	BACC 405
BACC 465	Taxation	3	BACC 405
BADM 355	Business Ethics and Social Responsibility	3	Junior Standing
BADM 420	Quantitative Methods for Business	3	BADM 250 or MATH 351 or Senior Standing
	Social Science Elective	3	
Year 3, Spring Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
BACC 460	Advanced Financial Accounting	3	BACC 405
BADM 490	Entrepreneurship	2	Senior Standing
	Community & Sustainability	1	Co-req.: ENGL 210
	Major Elective	3	
	Social Science Elective	3	
Year 3, Summer Semester (1 Credit)			
BADM 485	Co-op Work Experience	1	ENGL 217 Senior Standing

* Students exempted from this course based on their score on the mathematics placement exam will select a free elective course as a substitute for it.

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

BACC 305	**Intermediate Accounting I**	**3(3,0)**
This course is an in-depth study of accounting issues related to the measurement and reporting of assets, liabilities, and income in accordance with IFRS. It enables students to evaluate and understand financial accounting concepts and practices. The course covers the accounting framework, the use of the time value of money in accounting, and the preparation of financial statements. The course also includes a detailed accounting study of current and noncurrent assets. **Prerequisite:** BACC 205.		

BACC 310	Cost Accounting	3(3,0)
This course is a continuation of BACC 255. It offers students comprehensive knowledge of cost allocation, process costing, and joint and product costing. Moreover, the course deals with the costs of spoilage and waste. It allows students to better understand the relationships among quality, time, and cost. The course also concentrates on capital budgeting and cost control system analysis issues. **Prerequisite:** BACC 255.		

BACC 405	Intermediate Accounting II	3(3,0)
This course is a continuation of BACC 305. It concentrates on measuring and reporting liabilities and the various components of stockholders' equity. It introduces students to dilutive securities and their effect on earnings per share. The course emphasizes fair value, proper accounting for financial instruments, and new developments related to revenue recognition and reporting accounting changes and errors. Moreover, the course covers the preparation and presentation of the statement of cash flows in accordance with IFRS. **Prerequisite:** BACC 305.		

BACC 450	External Auditing	3(3,0)
This course introduces the students to international financial accounting standards and general auditing standards. It will enable the students to develop and apply auditing programs and procedures to various financial statement items. It concentrates on auditing the main business cycles such as the inventory, revenue, expenditure, and investment cycles. **Prerequisite:** BACC 405		

BACC 460	Advanced Financial Accounting	3(3,0)
This course is designed to enable students to address specialized financial accounting topics, including consolidated financial statements; accounting practices for forming and liquidating partnerships; accounting for multinational corporations; accounting for branches and segments; accounting for not-for-profit organizations; and governmental accounting. **Prerequisite:** BACC 405.		

BACC 465	Taxation	3(3,0)
This course covers various aspects of the Lebanese taxation system, including methods of tax imposition, taxes based on real profit, computation of taxable profit, tax rates and tax due, taxes on non-residents, fixed assets, holding companies, offshore companies, Insurance companies, financial institutions, tax on employees, tax on movable capital (stocks, interests, dividends, etc.) **Prerequisite:** BACC 405		

BADM 485	Co-op Work Experience	1(0,1)
The Co-op work experience is designed to provide students with full-time work terms in business, industry, or government. The main thrust of this course is the opportunity to put into practice the major aspects of the student's business education. Special emphasis will be placed on assessing the student's attitude and work ethics. Students will be encouraged to network in the industry and to participate in professional organizations. **Prerequisite:** ENGL 217; Senior Standing.		

BADM 490	Entrepreneurship
The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing	

BFIN 350	Credit and Financial Analysis	3(3,0)
Financial fitness and performance evaluation are core activities for credit officers, loan officers, and financial managers. This course will equip students with the knowledge and tools to conduct sound financial analysis using both public and non-public statements and reports. The main areas of analysis include liquidity, profitability, solvency, leverage, and market performance. **Prerequisite:** BFIN 300.		

II. Elective Courses

Descriptions of major elective courses are given below.

BACC 350	Accounting Information Systems	3(3,0)
This course is designed to give students an in-depth understanding of accounting information systems. It introduces the student to file and database organization, business processes and internal controls, the systems development process, and the management of information systems. Moreover, end-user application software includes "off-the-shelf" accounting software packages and spreadsheets. **Prerequisite:** BACC 205, BITM 300.		

BACC 455	Internal Auditing	3(3,0)
The course develops an understanding and appreciation of the role of internal auditing in an organization. Topics include internal auditing standards, risk assessment, governance, ethics, and audit techniques. **Prerequisite:** BACC 405		

BACC 470	Forensic Accounting and Fraud Detection	3(3,0)
This course will cover the basic concepts of forensic accounting. Topics include detecting, preventing, investigating, and resolving various types of fraud. The course covers many types of financial statement fraud, including asset misappropriation, fraudulent financial statements, tax fraud, and electronic fraud. **Prerequisite:** BFIN 300 and BACC 205.		

BADM 480	Independent Studies	3(3,0)
This course focuses on advancing the student's knowledge in their specialization through independent research and application on current or emerging topics, as approved by the assigned faculty member. **Prerequisite:** Advisor's approval.		

IV. Non-Business Programs Courses

BACC 210	SME's Financial Planning and Accounting	3(3,0)
This subject provides an overview of today's accounting and financial planning techniques for facility management for professionals who intend to operate small businesses, work as independent contractors, or work as freelancers. Using computer software applications, technical professionals should be able to perform budgeting, materials procurement and management, work order systems, work planning and scheduling, cost control techniques, preparation, and understanding of simplified financial statements. At least one-third of the coverage in this course is devoted to hands-on practical applications.		

Business IT Management

Program Overview

Success in the workplace in the twenty-first century can hardly be achieved without knowledge, skills, and experience in using modern technology and its components, such as information and communication technology (ICT), computer-based systems, and business applications.

The Business IT Management program combines business and information technology to enable students to gain a thorough understanding of how IT and computer applications improve effectiveness, increase efficiency, and facilitate business operations.

Students are thoroughly exposed to the business uses of database management, programming, networking, telecommunications, e-commerce, the internet, and online development.

To obtain a Bachelor of Business Administration degree in Business IT Management, the student must complete 99 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BBA in Business IT Management (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	3	-	15	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	31	3	53	12	99	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented on the introductory pages of the College of Business Administration section of this catalog.

III. Program Requirements

A. Mandatory Requirements

The mandatory Business IT Management program courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing
BADM 490	Entrepreneurship	3	Senior Standing
BITM 305	Introduction to Programming	3	BITM 200
BITM 310	Database Management Systems	3	BITM 305
BITM 340	The Development Tools of Information Systems	3	BITM 305
BITM 355	Networking	3	Junior Standing
BITM 415	Business Intelligence	3	Senior Standing
BITM 450	Big Data Analytics	3	BITM 350
BMGT 300	Project Management	3	Junior Standing
	Major Elective Course	3	

B. Major Electives

As part of the Bachelor of Business Administration in Business IT Management program, the student must study 3 credit hours of major electives. Major electives could be upper-level courses in Accounting or closely related business or computing areas. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility into the program, allowing the integration of emerging topics.

In consultation with the academic advisor, the student should select the elective course that best aligns with their needs and aspirations. It is highly recommended that the student register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Prerequisites
BACC 350	Accounting Information Systems	3	BACC 205, BITM 300
BADM 480	Independent Studies	3	Advisor's approval
BITM 320	Enterprise Resource Planning	3	BITM 300
BITM 402	Advanced Web Programming	3	BITM 400
BITM 410	Advanced Programming and Data Structures	3	BITM 305
BITM 455	Advanced Networking	3	BITM 355

Moreover, under certain conditions specified by the department and subject to the advisor's approval, a student may take BADM 480 (independent studies) as a substitute for a program's mandatory course. The course's description is below.

C. Entrepreneurship

The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.

D. Co-op Work Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students must then submit formal reports and posters and make formal presentations about their Co-op experience. While ENGL 217 and Senior Standing are prerequisites for co-op training, the program must stipulate additional conditions.

Career Opportunities

An increasing number of companies and entrepreneurial ventures in the Middle East and North Africa (MENA) region have sought to capitalize on the computing and information technology revolution to enable and facilitate business operations in areas such as telemarketing, e-commerce, e-government, and social media. This creates numerous job opportunities for people with skills in online sales, electronic market research, e-commerce, online transaction security, mobile application development, and information systems management.

Program Goals and Student Learning Outcomes

The purpose of the Business IT Management Program at the College of Business Administration of Rafik Hariri University is an integrative composite of four main dimensions: General and Specialized Business Knowledge, Communication, Critical Thinking, and Growth Potential.

General & Specialized Business Knowledge

Goal 1	Prepare graduates with effective professional competence in conducting critical business activities, particularly those related to Business Information Technology Management.
Outcome 1	Apply essential business knowledge and skills in problem-solving and decision-making.
Outcome 2	Use ICT tools and programs to enhance performance and improve operational business efficiency.

Communication

Goal 2	Equip graduates with the knowledge and skills to secure placements in business, IT management, and systems positions.
Outcome 3	Demonstrate written and oral English communication skills adequate for challenging entry- and middle-level professional positions and self-employment.

Critical thinking

Goal 3	Promote analytical and critical thinking in approaching business issues and problems.
Outcome 4	Identify business problems, analyze them, and present plausible solutions.

Ethics and Social Responsibility

Goal 4	Raise awareness about ethics and social responsibility to promote better business for a better community.
Outcome 5	Recognize ethical dilemmas in business and respond to them in accordance with standard codes of conduct, ethical principles, and best practices.

Study Plan

The Bachelor of Business Administration in Business IT Management encompasses 99 credit hours over six semesters and two summer sessions. The second summer session should be dedicated to the Community Engagement Experience. Meanwhile, the third session should be for the Co-op work experience. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (16 Credits)			
BACC 205	Financial Accounting	3	
BADM 215	Personal Development and Management	1	
BITM 200	Information Technology Essentials	3	
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210.
ENGL 210	English Composition and Rhetoric	3	ENGL 101
MATH 207	College Algebra*	3	Placement
Year 1, Spring Semester (16 Credits)			
BACC 255	Financial Accounting	3	BACC 205
BADM 225	Business Math	3	Placement
BADM 235	Business Research Methodology	1	
BADM 250	Business Statistics	3	
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
BITM 305	Introduction to Programming	3	BITM 200

Year 2, Fall Semester (18 Credits)			
BADM 230	Business Law	3	
BECN 301	Microeconomics	3	Junior Standing
BFIN 300	Financial Management I	3	BACC 205
BITM 300	Business IT Management	3	BITM 200
BITM 310	Database Management Systems	3	BITM 305
ENGL 217	Professional English Communication	3	ENGL 210
Year 2, Spring Semester (18 Credits)			
BECN 302	Macroeconomics	3	Junior Standing
BITM 340	Development Tools of Information Systems	3	BITM 305
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
BMGT 300	Project Management	3	Junior Standing
	Science Elective	3	
	Humanities / Fine Arts Elective	3	
Year 2, Summer Semester (0 Credits)			
BADM 290	Community Engagement Experience	0	
Year 3, Fall Semester (15 Credits)			
BADM 355	Business Ethics and Social Responsibility	3	Junior Standing
BITM 415	Business Intelligence	3	Senior Standing
BITM 355	Networking	3	Junior Standing
BITM 450	Big Data Analytics	3	BITM 350
	Social Science Elective	3	
Year 3, Spring Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
BADM 420	Quantitative Methods for Business	3	BADM 250 or MATH 351 or Senior Standing
BADM 490	Entrepreneurship	3	Senior Standing
	Community & Sustainability	1	Co-req.: ENGL 210
	Major Elective	3	
	Social Science Elective	3	
Year 3, Summer Semester (1 Credit)			
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing

* Students exempted from this course based on their score on the mathematics placement exam will select a free elective course as a substitute for it.

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

BADM 485	Co-op Work Experience	1(0,1)
The Co-op work experience is designed to provide students with full-time work terms in business, industry, or government. The main thrust of this course is the opportunity to put into practice the major aspects of the student's business education. Special emphasis will be placed on assessing the student's attitude and work ethics. Students will be encouraged to network in the industry and to participate in professional organizations. **Prerequisite:** ENGL 217; Senior Standing.		

BADM 490	Entrepreneurship
The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.	

BITM 305	Introduction to Programming	3(2,2)
This is a foundation course for all computer programming courses. It enhances programming skills and presents the fundamentals of structured programming concepts in C. The course covers primitive data types, expressions, control statements, functions, and arrays. It also provides a hands-on experience with MATLAB. **Prerequisite:** BITM 200 (equivalent to COSC 214).		

BITM 310	Database Management Systems	3(3,1)
Students will explore advanced database concepts, including automation techniques, using popular Windows-based DBMS software. The course includes the following topics: the planning, creation, and maintenance of databases; the development of information retrieval techniques; the design and customization of forms; and the creation and use of macros and modules in an automated system. **Prerequisite:** BITM 305; equivalent to COSC 231 and CCEE 315.		

BITM 340	The Development Tools of Information Systems	3(3,0)
This course offers a traditional look at the system's life cycle process. Modeling and charting will be included. The tools and methodology applied by the systems analyst will be studied. The course covers an overview of the system life cycle, UML Diagrams, feasibility analysis, project management, HCI, and the golden rules of user interface design. **Prerequisite:** BITM 305; equivalent to COSC 341 and CCEE 510.		

BITM 355	Networking	3(2,2)
An introduction to data communications and networking, covering networking hardware/operating system concepts, modems, WAN, and LAN standards and protocols. **Prerequisite:** Junior Standing; Equivalent to COSC 360, CCEE 354.		

BITM 401	Web Programming	3(3,1)
This course teaches students how to develop and implement web-based programs, with an emphasis on interface programming. It introduces students to web development, client-side languages, and the styles needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. Equivalent to COSC 333, CCEE 411.		

BITM 415	Business Intelligence	3(2,1)
This course introduces business intelligence as computerized support for managerial decision-making. It focuses on the theoretical and conceptual foundations of business intelligence, as well as on commercial tools and techniques for effective decision support. It focuses on extracting business intelligence from data sets for various applications, such as reporting and visual analytics across domains like web and business analytics, to aid decision-making. Provides hands-on experience with a variety of business intelligence software for reporting and building visualizations and dashboards. **Prerequisite:** Senior Standing.		

BITM 450	Big Data Analytics	3(2,1)
This course provides students with a comprehensive understanding of big data analytics from a Business Information Technology (BIT) perspective, with a strong emphasis on Apache Hadoop and its ecosystem. The course focuses on how organizations leverage Hadoop-based platforms to store, process, analyze, and derive value from large-scale business data. Students will gain hands-on experience with HDFS, MapReduce, Hive, HBase, Pig, and ZooKeeper, and learn how these technologies support business intelligence, operational analytics, and managerial decision-making. The course prioritizes practical Hadoop applications, business use cases, and the interpretation of analytical outputs over advanced algorithmic theory. Through lectures, laboratory exercises, assignments, and a group project, students will design and implement end-to-end Hadoop-based analytics solutions to address real-world business problems. **Prerequisite**: BITM 350		

BMGT 300	Project Management	3(3,0)
This course provides students with the skills to manage their business projects by effectively leading, organizing, scheduling, and controlling tasks to achieve project goals. Topics include project selection and statement of work, project manager skills, task breakdown structure, PERT/CPM scheduling, and budgeting. **Prerequisite:** Junior Standing.		

II. Elective Courses

Descriptions of some elective courses are given below.

BACC 350	Accounting Information Systems	3(3,0)
This course is designed to give students an in-depth understanding of accounting information systems. It introduces the student to file and database organization,		

business processes, internal controls, the systems development process, and the management of information systems. Moreover, end-user application software includes "off-the-shelf" accounting software packages and spreadsheets. **Prerequisite:** BACC 205; BITM 300.

BADM 480	**Independent Studies**	**3(3,0)**
This course focuses on advancing the student's knowledge in their specialization through independent research and application on current or emerging topics, as approved by the assigned faculty member. **Prerequisite:** Advisor's approval.		

BITM 320	Enterprise Resource Planning	
Enterprise Resource Planning (ERP) systems are a major investment for businesses. These systems are essential for maintaining competitiveness, meeting customer demands, and improving efficiency and flexibility in a global marketplace. By adopting ERP, companies can streamline their operations, align with industry best practices, and fully utilize integrated data resources. This course will provide students with the skills and knowledge needed to plan, design, and implement ERP systems effectively. Prerequisite: BITM 300		

BITM 402	**Advanced Web Programming**	**3(3,1)**
This course focuses on server-side programming. It allows students to learn how to connect their website or web application to a database and to save and retrieve data from it. The course exposes students to web controls, validation controls, data source controls, data binding controls, state management, and working with third-party media like XML and web services. **Prerequisite:** BITM 401; equivalent to COSC 434 and CCEE 514.		

BITM 410	**Advanced Programming and Data Structures**	**3(3,1)**
This is a continuation course using advanced C++. The student studies object-oriented programming concepts such as classes, objects, polymorphism, data hiding, and encapsulation. This course gives the student a new perspective on how to think about objects. **Prerequisite:** BITM 305; equivalent to COSC 215 and CCEE 216.		

BITM 455	**Advanced Networking**	**3(2,2)**
This course prepares students to act as System and Network Administrators by implementing Active Directory Domain Services ADDS in distributed environments that can include complex network services and domain controllers. The covered materials help students efficiently automate the administration of users, groups, and computers. **Prerequisite:** BITM 355, Equivalent to COSC 461.		

Finance and Banking Program

Program Overview

The Bachelor of Business Administration with specialization in Finance and Banking program is carefully designed to prepare graduates for successful careers in financial management, the financial services industry, and banking.

The Program offers a balanced blend of economic and financial theories, with general and industry-specific applications, carefully designed to ensure effective financial knowledge and competence.

To obtain a Bachelor of Business Administration degree in Finance and Banking, the student must complete 99 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BBA in Finance and Banking (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	28	3	56	12	99	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented on the introductory pages of the College of Business Administration section of this catalog.

III. Program Requirements

A. Mandatory Requirements

The mandatory courses in the Finance and Banking program are listed in the table below.

Course #	Title	Credits	Prerequisites
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing

BADM 490	Entrepreneurship	3	Senior Standing
BFIN 305	Introduction to Banking	3	Junior Standing
BFIN 310	Financial Markets and Institutions	3	BFIN 300
BFIN 350	Credit and Financial Analysis	3	BFIN 300
BFIN 400	Financial Management II	3	BFIN 300
BFIN 405	Bank Financial and Risk Management	3	BFIN 305
BFIN 450	Investment Analysis	3	BFIN 400
BFIN 455	Financial Derivatives	3	Senior Standing
	Major Elective	3	

B. Major Electives

As part of the Bachelor of Business Administration in Finance and Banking program, the student must study 3 credit hours of major electives. Major electives could be upper-level courses in Finance, Banking, or closely related business areas. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility into the program, allowing the integration of emerging topics.

In consultation with the academic advisor, the student should select the elective course that best aligns with their needs and aspirations. It is highly recommended that the student register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Prerequisites
BACC 310	Cost Accounting	3	BACC 255
BACC 470	Forensic Accounting and Fraud Detection	3	BFIN 300 and BACC 205
BADM 480	Independent Studies	3	Advisor's approval
BECN 305	Managerial Economics	3	BECN 301
BFIN 355	International Finance	3	BFIN 300, BECN 302

C. Entrepreneurship

The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.

D. Co-op Work Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students must then submit formal reports and posters and make formal presentations

about their Co-op experience. While ENGL 217 and Senior Standing are prerequisites for co-op training, the program must stipulate additional conditions.

Career Opportunities

The Finance and Banking program equips students with the skills and competencies to pursue and succeed in career opportunities across organizations and vibrant industries, including banking, insurance, investments, and financial services. Possible career opportunities include financial manager, credit analyst, loan officer, branch manager, trust officer, mortgage banker, financial analyst, investment advisor, and financial planner.

Program Goals and Student Learning Outcomes

The purpose of the Finance and Banking Program at the College of Business Administration of Rafik Hariri University is an integrative composite of four main dimensions: Specialized and General Business Knowledge, Communication, Critical Thinking, and Growth Potential.

General & Specialized Business Knowledge

Goal 1	Prepare graduates with professional competence in critical finance and banking tasks and activities.
Outcome 1	Apply essential business knowledge and skills in problem-solving and decision-making.
Outcome 2	Apply financial theory to evaluate investments and alternatives in terms of performance and risk.

Communication

Goal 2	Equip graduates with the knowledge and skills to secure placements in financial and banking roles.
Outcome 3	Demonstrate written and oral English communication skills adequate for challenging entry- and middle-level professional positions and self-employment.

Critical thinking

Goal 3	Promote analytical and critical thinking in approaching business issues and problems.
Outcome 4	Identify business problems, analyze them, and present plausible solutions.

Ethics and Social Responsibility

Goal 4	Raise awareness about ethics and social responsibility to promote better business for a better community.

Outcome 5	Recognize ethical dilemmas in business and respond to them in accordance with standard codes of conduct, ethical principles, and best practices.

Study Plan

The Bachelor of Business Administration in Finance and Banking encompasses 99 credit hours over six semesters and two summer sessions. The second summer session should be dedicated to the Community Engagement Experience. Meanwhile, the third summer session should be for the Co-op work experience. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (16 Credits)			
BACC 205	Financial Accounting	3	
BADM 215	Personal Development and Management	1	
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210.
BITM 200	Information Technology Essentials	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 207	College Algebra*	3	Placement
Year 1, Spring Semester (16 Credits)			
BACC 255	Managerial Accounting	3	BACC 205
BADM 225	Business Math	3	Placement
BADM 235	Business Research Methodology	1	
BADM 250	Business Statistics	3	
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
ENGL 217	Professional English Communication	3	ENGL 210
Year 2, Fall Semester (18 Credits)			
BADM 230	Business Law	3	
BECN 301	Microeconomics	3	Junior Standing
BFIN 300	Financial Management I	3	BACC 205
BFIN 305	Intro to Banking	3	Junior Standing
BITM 300	Business IT Management	3	BITM 200
	Science Elective	3	
Year 2, Spring Semester (18 Credits)			
BECN 302	Macroeconomics	3	Junior Standing
BFIN 310	Financial Markets and Institutions	3	BFIN 300
BFIN 350	Credit and Financial Analysis	3	BFIN 300
BFIN 400	Financial Management II	3	BFIN 300
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
	Humanities / Fine Arts Elective	3	

Year 2, Summer Semester (0 Credits)			
BADM 290	Community Engagement Experience	0	
Year 3, Fall Semester (15 Credits)			
BADM 355	Business Ethics and Social Responsibility	3	Junior Standing
BADM 420	Quantitative Methods for Business	3	BADM 250 or MATH 351 or Senior Standing
BFIN 405	Bank Financial and Risk Management	3	BFIN 305
BFIN 450	Investment Analysis	3	BFIN 400
	Social Science Elective	3	
Year 3, Spring Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
BADM 490	Entrepreneurship	3	Senior Standing
BFIN 455	Financial Derivatives	3	Senior Standing
	Community & Sustainability	1	Co-req.: ENGL 210
	Major Elective	3	
	Social Science Elective	3	
Year 3, Summer Semester (1 Credit)			
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing

* Students exempted from this course based on their score on the mathematics placement exam will select a free elective course as a substitute for it.

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

BADM 485	**Co-op Work Experience**	**1(0,1)**
The first Co-op work experience is designed to provide students with full-time work terms in business, industry, or government. The main thrust of this course is the opportunity to put into practice the major aspects of the student's business education. Special emphasis will be placed on assessing the student's attitude and work ethic. Students will be encouraged to network in the industry and to participate in professional organizations. **Prerequisite:** ENGL 217; Senior Standing.		

BADM 490	**Entrepreneurship**	**3(3,0)**
The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.		

BFIN 305	Introduction to Banking	3(3,0)
This course provides students with a comprehensive understanding of bank management, emphasizing key hard skills essential for success in the banking sector. Topics include asset-liability management, risk management practices, effective financial strategy formulation, and performance evaluation of banking institutions. The course also covers banking regulations, the development of banking systems, and an analysis of products and services offered by banks. Students will explore the internal organization and structure of banks, equipping them with the necessary tools to navigate the complexities of the banking environment. **Prerequisite**: Junior Standing.		

BFIN 310	Financial Markets and Institutions	3(3,0)
This course provides a comprehensive exploration of financial markets and institutions, focusing on equity market organization, security market indices, and market efficiency, and examining their implications for investors and policymakers. It also covers fixed-income securities and markets, including issuance, trading mechanisms, funding strategies, and the term structure of interest rates. In addition to traditional financial systems, the course introduces emerging topics in finance, such as machine learning and artificial intelligence in financial analysis, cryptocurrencies, and digital assets, highlighting their impact on market structure and investment strategies. By integrating fundamental concepts with evolving financial technologies, this course equips students with the knowledge and analytical tools needed to navigate today's dynamic global financial environment. **Prerequisite**: BFIN 300.		

BFIN 350	Credit and Financial Analysis	3(3,0)
This course provides a comprehensive introduction to financial reporting and analysis. It covers the structure and interpretation of financial statements, including the balance sheet, income statement, and cash flow statement, along with key accounting principles, revenue recognition, inventory valuation, and long-term asset accounting. Participants will learn financial ratio analysis, assess financial reporting quality, and explore the impact of different accounting policies under IFRS and U.S. GAAP. The course equips learners with the skills to analyze financial statements, detect red flags in reporting, and make informed investment and credit decisions. **Prerequisite**: BFIN 300.		

BFIN 400	Financial Management II	3(3,0)
This course provides a comprehensive introduction to corporate finance. It covers fundamental concepts such as capital budgeting, cost of capital, capital structure, working capital management, and measures of corporate performance. Participants will explore financial decision-making frameworks, risk and return analysis, dividend policy, and the impact of leverage on a firm's financial health. Additionally, the course will delve into environmental, social, and governance (ESG) factors and their significance in corporate governance, highlighting how these elements influence financial strategies and stakeholder expectations. By the end, learners will develop the skills to assess corporate financial strategies, optimize capital allocation, and make informed investment and financing decisions while considering ESG principles and effective governance practices. **Prerequisite:** BFIN 300.		

BFIN 405	Bank Financial and Risk Management	3(3,0)
This course provides a comprehensive introduction to financial and risk management in banking, focusing on the key principles of asset and liability management. Students will explore fixed-income risk and return, with an emphasis on duration, convexity, and the impact of interest rate movements on bond pricing and portfolio management. The course covers the fundamentals of credit analysis, including credit risk assessment, credit spreads, and default probabilities. Additional topics include liquidity and capital adequacy management, market risk, operational risk, and regulatory compliance. Students will also examine key risk measurement techniques, such as Value-at-Risk (VaR) and stress testing, as well as the role of money market operations in financial stability. By integrating theoretical concepts with real-world applications, this course prepares students to analyze and manage financial risks in banking institutions effectively. **Prerequisite**: BFIN 305		

BFIN 450	Investment Analysis	3(3,0)
This course introduces students to the fundamental principles of portfolio management, focusing on investment selection, risk assessment, and portfolio construction. Students will explore key topics such as risk-return trade-offs, diversification, and asset allocation strategies. The course covers portfolio risk management techniques, including systematic and unsystematic risk, beta, and standard deviation, and examines portfolio performance evaluation methods. Additionally, students will learn the basics of portfolio planning and construction, incorporating principles of asset allocation, rebalancing, and investment objectives. By integrating theoretical frameworks with real-world applications, this course prepares students to develop effective investment strategies in various market conditions. **Prerequisite**: BFIN 400		

BFIN 455	Financial Derivatives	3(3,0)
This course introduces students to financial derivatives and their applications in risk management, corporate strategy, and investment decision-making. Students will learn how derivatives can be used for hedging and speculation, with a focus on instruments such as forwards, futures, options, and swaps. Particular emphasis is placed on their role in foreign exchange trading and financial risk mitigation. Additionally, the course provides an introduction to alternative investments, including private equity, hedge funds, real estate, commodities, and structured products. Students will explore the risk-return characteristics of these assets and their role in portfolio diversification. By integrating derivatives and alternative investments, this course equips students with a comprehensive understanding of modern financial instruments and their strategic applications. **Prerequisite:** Senior Standing.		

II. Elective Courses

Descriptions of some major elective courses are given below.

BACC 310	Cost Accounting	3(3,0)
This course introduces students to the basic concepts, analyses, uses, and procedures of management accounting. This course would enable students to understand and view cost as part of activities planned and implemented by a company. It covers cost categories, cost behavior, cost-volume-profit analysis, master and flexible budgets, direct costs and manufacturing overhead variances, and inventory cost systems. **Prerequisite:** BACC 255.		

BACC 470	Forensic Accounting and Fraud Detection	3(3,0)
This course will cover the basic concepts of forensic accounting. Topics include detecting, preventing, investigating, and resolving various types of fraud. The course covers many types of financial statement fraud, including asset misappropriation, fraudulent financial statements, tax fraud, and electronic fraud. **Prerequisite**: BFIN 300 and BACC 205.		

BADM 480	Independent Studies	3(3,0)
This course focuses on advancing the student's knowledge in their specialization through independent research and application on current or emerging topics, as approved by the assigned faculty member. **Prerequisite:** Advisor's approval.		

BECN 305	Managerial Economics	3(3,0)
Given the constraints firms face, this course applies microeconomic theory to various management and planning decisions, such as output maximization and cost minimization. Demand analysis, cost analysis, and different market structures are studied. **Prerequisite:** BECN 301.		

BFIN 355	International Finance	3(3,0)
This course exposes students to international financial management and trade from the perspective of managers working in international corporations. Topics include the management of foreign exchange exposure, foreign investments, multinational capital budgeting, the balance of payments, determination of exchange rates, and international banking. **Prerequisite:** BFIN 300, BECN 302.		

Management and Marketing Studies Department (DMMS)

Faculty Members

Chairperson:	Mohamad Al Majzoub
Assistant Professor:	Mohamad Al Majzoub, Maya Daher, Baheej Chaaban.
Adjunct Faculty:	Adel Saheb, Buthayna Hajjar, Dorriah Itani, Kassem Masri, Nassif Al Hakim, Hana El Toqi, Maha Hajj Omar, Bassel Diab.

Programs Offered

The Management and Marketing Studies Department (MMD) offers three programs – Management, Human Resources Management, and Marketing and Advertising. Each program leads to a Bachelor of Business Administration degree. The details of each program follow.

Human Resources Management Program

Program Overview

The Human Resources Management program offers students a unique experience in the operational and strategic activities of the HR department, a thought-provoking curriculum flavored with experiential learning opportunities and taught by internationally certified faculty, and a state-of-the-art platform from which HRM graduates can launch their careers.

To obtain a Bachelor of Business Administration degree in Human Resources Management, the student must complete 99 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BBA in Human Resources Management (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	-	-	41	-	41	41
Program Requirement	25	3	-	-	28	28
Total Credits	25	3	59	12	99	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented in the introductory pages of the College of Business Administration section of this catalog.

III. Program Requirements

A. Mandatory Requirements

The HRM mandatory core courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing
BADM 490	Entrepreneurship	3	Senior Standing

BHRM 300	Human Resources Management	3	BMGT 200
BHRM 305	Labor Law & Relations	3	Junior Standing
BHRM 350	Workforce Planning, Recruitment & Selection	3	BHRM 300
BHRM 400	HR Development & Training	3	BHRM 300
BHRM 450	Performance Management, Compensation & Benefits	3	BHRM 300
BMGT 205	Organizational Behavior	3	BMGT 200
BMGT 485	Strategic Management	3	Senior Standing
	Major Elective	3	

B. Major Electives

As part of the Bachelor of Business Administration in Human Resources Management program, students must study 3 credit hours of major electives. Major electives could be upper-level courses in the major or closely related business areas, such as Project Management or E-Business. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility into the program, allowing the integration of emerging topics.

In cooperation with the academic advisor, students should select the elective course that best meets their needs, abilities, and aspirations. It is highly recommended that the student register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Prerequisites
BITM 415	Business Intelligence	3	Senior Standing
BADM 480	Independent Studies	3	Advisor's approval
BFIN 400	Financial Management II	3	BFIN 300
BMGT 300	Project Management	3	Junior Standing
BMKA 440	Social Media Marketing	3	Senior Standing

Moreover, under certain conditions specified by the department and subject to the advisor's approval, a student may take BADM 480 (independent studies) as a substitute for a program's mandatory course. The course's description is below.

C. Entrepreneurship

The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. Prerequisite: Senior Standing.

D. Co-op Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The co-op experience is usually fulfilled during the summer semester of the program's third year. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits to ensure that the student's performance is aligned with their aspirations and the employer's needs. Students must then submit formal reports and posters and make formal presentations about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may impose additional conditions.

Career Opportunities

For those who pursue a professional career, business graduates majoring in HRM typically start as an "HR specialist" in one HR function at a large organization or as an "HR generalist" in the HR department of a small firm. From there, they can advance to upper-level managerial positions. HRM graduates pursuing an academic career can advance by choosing a graduate or postgraduate path in general management or HRM, which can lead to careers in academia and research.

Program Goals and Student Learning Outcomes

The purpose of the Human Resources Management Program at the College of Business Administration of Rafik Hariri University is an integrative composite of four main dimensions: General and Specialized Business Knowledge, Communication, Critical Thinking, and Growth Potential.

General & Specialized Business Knowledge

Goal 1	Prepare graduates with effective professional competence in human resources tasks and activities.
Outcome 1	Apply essential business knowledge and skills in problem-solving and decision-making.
Outcome 2	Demonstrate knowledge of standard HR functions, such as recruitment, selection, compensation & benefits, in accordance with legal requirements and professional best practices.

Communication

Goal 2	Equip graduates with the knowledge and skills to secure placements in financial and managerial accounting positions.
Outcome 3	Demonstrate written and oral English communication skills adequate for challenging entry- and middle-level professional positions and self-employment.

Critical thinking

Goal 3	Promote analytical and critical thinking in approaching business issues and problems.
Outcome 4	Identify business problems, analyze them, and present plausible solutions.

Ethics and Social Responsibility

Goal 4	Raise awareness about ethics and social responsibility to promote better business for a better community.
Outcome 5	Recognize ethical dilemmas in business and respond to them in accordance with standard codes of conduct, ethical principles, and best practices.

Study Plan

The Bachelor of Business Administration in Human Resources Management encompasses 99 credit hours over six semesters and two summer sessions. The first summer session should be dedicated to the Community Engagement Experience. Meanwhile, the second session should be for the Co-op work experience. The following study plan serves as a roadmap for smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (16 Credits)			
BACC 205	Financial Accounting	3	
BADM 215	Personal Development and Management	1	
BITM 200	Information Technology Essentials	3	
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210.
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 207	College Algebra*	3	Placement
Year 1, Spring Semester (16 Credits)			
BACC 255	Managerial Accounting	3	BACC 205
BADM 225	Business Math	3	Placement
BADM 235	Business Research Methodology	1	
BADM 250	Business Statistics	3	
BMGT 205	Organizational Behavior	3	BMGT 200
ENGL 217	Professional English Communication	3	ENGL 210
Year 2, Fall Semester (18 Credits)			
BADM 230	Business Law	3	
BECN 301	Microeconomics	3	Junior Standing
BFIN 300	Financial Management I	3	BACC 205
BHRM 300	Human Resources Management	3	BMGT 200
BHRM 305	Labor Law and Labor Relations	3	Junior Standing
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210

Year 2, Spring Semester (18 Credits)			
BADM 355	Business Ethics and Social Responsibility	3	Junior Standing
BECN 302	Macroeconomics	3	Junior Standing
BHRM 350	Workforce Planning, Recruitment, and Selection	3	BHRM 300
BITM 300	Business IT Management	3	BITM 200
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
	Science Elective	3	
Year 2, Summer Semester (0 Credit)			
BADM 290	Community Engagement Experience	0	
Year 3, Fall Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
BADM 420	Quantitative Methods for Business	3	BADM 250 or MATH 351 or Senior Standing
BHRM 400	HR Training and Development	3	BHRM 300
	Community & Sustainability	1	Co-req.: ENGL 210
	Major Elective	3	
	Social Science Elective	3	
Year 3, Spring Semester (15 Credits)			
BADM 490	Entrepreneurship	3	Senior Standing
BHRM 450	Performance Management, Compensation & Benefits	3	BHRM 300
BMGT 485	Strategic Management	3	Senior Standing
	Humanities / Fine Arts Elective	3	
	Social Science Elective	3	
Year 3, Summer Semester (1 Credit)			
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing

* Students exempted from this course based on their score on the mathematics placement exam will select a free elective course as a substitute for it.

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

BADM 485	**Co-op Work Experience**	**1(0,1)**
The Co-op work experience is designed to provide students with full-time work terms in business, industry, or government. The main thrust of this course is the opportunity to put into practice the major aspects of the student's business education. Special emphasis will be placed on assessing the student's attitude and work ethic. Students		

will be encouraged to network in the industry and to participate in professional organizations. **Prerequisite:** ENGL 217; Senior Standing.

BADM 490	**Entrepreneurship**
The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.	

BHRM 300	**Human Resources Management**	**3(3,0)**
This course provides students with a practical understanding of current Human Resources Management concepts and applications in both small and large businesses. Topics include recruitment & selection, training & development, performance management, compensation & benefits, employee relations, and legal compliance. **Prerequisite**: BMGT 200		

BHRM 305	**Labor Law & Relations**	**3(3,0)**
This course introduces students to the fundamentals of labor law, collective bargaining, contract administration, and Social Security regulations governing employee management. Labor law history, development, and applications are aspects of Lebanese labor law covered to the extent they apply to the management of human resources. **Prerequisite:** Junior Standing		

BHRM 350	**Workforce Planning, Recruitment & Selection**	**3(3,0)**
This course aims to provide students in the HRM major with a solid foundation in workforce planning, recruitment, and selection activities within the HRM function. In particular, this course enables students to conduct effective job analyses, write job descriptions, recruit qualified candidates, and use modern selection techniques to staff the organization with the required human capital. **Prerequisite**: BHRM 300		

BHRM 400	**HR Training & Development**	**3(3,0)**
This course will introduce students in the HRM major to the HR Training & Development function of Human Resources Management. It will enable students to design and implement effective, efficient training & development plans to meet the organization's current and future needs. **Prerequisite**: BHRM 300		

BHRM 450	**Performance Management, Compensation & Benefits**	**3(3,0)**
This course will provide students in the HRM major with a solid understanding of the Compensation & Benefits function of Human Resources Management. It will teach students how to establish effective performance management processes and design strategic pay structures and performance-based incentives. This course will also enable students to develop financial and non-financial benefits as part of the compensation package. Strategic compensation plans for executives, expatriates,		

and the contingent workforce are also covered in this course. **Prerequisite**: BHRM 300.

BMGT 205	**Organizational Behavior**	**3(3,0)**
This course aims to help the student develop the necessary skills and conceptual tools to understand and effectively address human behavior in organizations. Special emphasis will be placed on employee motivation, teamwork, leadership, communication, conflict, negotiation, and an overview of attitudes, values, personality, and perception. **Prerequisite:** BMGT 200.		

BMGT 485	**Strategic Management**	**3(3,0)**
This is an advanced course for upper-level management students. This course covers basic concepts of strategic management, corporate governance and social responsibility, environmental scanning and industry analysis, internal scanning and organizational analysis, and strategy formulation within a global context. **Prerequisite:** Senior Standing.		

II. Elective Courses

Descriptions of major elective courses are given below.

BADM 480	**Independent Studies**	**3(3,0)**
This course focuses on advancing the student's knowledge in their specialization through independent research and application on current or emerging topics, as approved by the assigned faculty member. **Prerequisite:** Advisor's approval.		

BITM 415	**Business Intelligence**	**3(3,0)**
This course introduces business intelligence as computerized support for managerial decision-making. It concentrates on the theoretical and conceptual foundations of business intelligence and commercial tools and techniques available for effective decision support. It focuses on extracting business intelligence from data sets for various applications, such as reporting and visual analytics across domains like web and business analytics, to aid decision-making. Provides hands-on experience with various business intelligence software for reporting and building visualizations and dashboards**. Prerequisite**: Senior Standing & BADM 350		

BFIN 400	**Financial Management II**	**3(3,0)**
As a continuation of Financial Management I, this course exposes students to the firm's financial management for value maximization. That includes capital budgeting, capital structure and leverage, dividend policy, mergers & acquisitions, long-term debt, and financial planning and policy. **Prerequisite:** BFIN 300.		

BMGT 300	**Project Management**	**3(3,0)**
This course provides students with the skills to manage their business projects by effectively leading, organizing, scheduling, and controlling tasks to achieve project		

goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling and budgeting. **Prerequisite:** Junior Standing.

BMKA 450	**Events Marketing and Management**	**3(3,0)**
This course covers the skills and concepts necessary to create, manage, market, and finance special events. It provides a conceptual overview and systematic study of event marketing and management. Additionally, it fosters professionalism, coordination, and collaboration among team members to achieve the objectives of special events. The course concludes with the implementation and evaluation of a special event. **Prerequisite:** Senior Standing.		

MANAGEMENT PROGRAM

Program Overview

The Management program offers students an exceptional preparatory experience in management and administration. Our experienced and dedicated faculty deliver a state-of-the-art curriculum that balances theory and practice, using highly effective pedagogical tools that give students unlimited opportunities to excel.

To obtain a Bachelor of Business Administration degree in Management, the student must complete 99 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BBA in Management (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Credits	28	3	56	12	99	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented in the introductory pages of the College of Business Administration section of this catalog.

III. Program Requirements

A. Mandatory Requirements

The Management mandatory core courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BADM 410	International Business	3	Senior Standing
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing
BADM 490	Entrepreneurship	3	Senior Standing

BECN 305	Managerial Economics	3	BECN 301
BHRM 300	Human Resources Management	3	BMGT 200
BMGT 205	Organizational Behavior	3	BMGT 200
BMGT 300	Project Management	3	Junior Standing
BMGT 400	Operations Management	3	BADM 250
BMGT 485	Strategic Management	3	Senior Standing

B. Major Electives

As part of the Bachelor of Business Administration in Management program, students must study 3 credit hours of major electives. Major electives could be upper-level courses in Management or closely related business areas. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility into the program, allowing the integration of emerging topics.

In consultation with the academic advisor, the student should select the elective course that best aligns with their needs and aspirations. It is highly recommended that the student register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Prerequisites
BADM 480	Independent Studies	3	Advisor's approval
BFIN 400	Financial Management II	3	BFIN 300
BMGT 410	Quality Management	3	BADM 250
BMKA 365	Sales Management	3	BMKA 200; Junior Standing
BMKA 450	Event Marketing and Management	3	Senior Standing

Moreover, under certain conditions specified by the department and subject to the advisor's approval, a student may take BADM 480 (independent studies) as a substitute for a program's mandatory course. The course's description is below.

C. Entrepreneurship

The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.

D. Co-op Work Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is typically completed during the Summer Semester of the program's second year. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits to ensure that the student's performance is aligned with their aspirations and the employer's needs. Students must then submit formal reports and posters and make formal presentations

about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may impose additional conditions.

Career Opportunities

In pursuing a professional career, business management graduates would be highly eligible for entry-level administrative positions in banking, insurance, tourism, retail, and most support functions of the economy's health, education, consultancy, and industrial sectors. With more experience, they can advance to upper-level managerial positions. Management graduates pursuing an academic career can advance by choosing a graduate or postgraduate path in a specialization within the management field, such as HRM, Supply Chain Management, Operations, or Retail Management, among others, which can lead to careers in academia and research.

Program Goals and Student Learning Outcomes

The objectives of the Management Program at the College of Business Administration of Rafik Hariri University are integrated into four main dimensions: General and Specialized Business Knowledge, Communication, Critical Thinking, and Growth Potential.

The objectives of the Management Program at the College of Business Administration of Rafik Hariri University are integrated into four main dimensions: General and Specialized Business Knowledge, Communication, Critical Thinking, and Growth Potential.

General & Specialized Business Knowledge

Goal 1	Prepare graduates with effective professional competence in conducting critical business tasks, particularly managerial and entrepreneurial activities.
Outcome 1	Apply essential business knowledge and skills in problem-solving and decision-making.
Outcome 2	Apply managerial and entrepreneurial competencies to launch and manage a business.

Communication

Goal 2	Equip graduates with the knowledge and skills to secure placements in financial and managerial accounting positions.
Outcome 3	Demonstrate written and oral English communication skills adequate for challenging entry- and middle-level professional positions and self-employment.

Critical thinking

Goal 3	Promote analytical and critical thinking in approaching business issues and problems.
Outcome 4	Identify business problems, analyze them, and present plausible solutions.

Ethics and Social Responsibility

Goal 4	Raise awareness about ethics and social responsibility to promote better business for a better community.
Outcome 5	Recognize ethical dilemmas in business and respond to them in accordance with standard codes of conduct, ethical principles, and best practices.

Study Plan

The Bachelor of Business Administration in Management encompasses 99 credit hours over six semesters and two summer sessions. The second summer session should be dedicated to the Community Engagement Experience. Meanwhile, the second session should be for the Co-op work experience. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (16 Credits)			
BACC 205	Financial Accounting	3	
BADM 215	Personal Development and Management	1	
BITM 200	Information Technology Essentials	3	
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210.
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 207	College Algebra*	3	Placement
Year 1, Spring Semester (16 Credits)			
BACC 255	Managerial Accounting	3	BACC 205
BADM 225	Business Math	3	Placement
BADM 235	Business Research Methodology	1	
BADM 250	Business Statistics	3	
BMGT 205	Organizational Behavior	3	BMGT 200
ENGL 217	Professional English Communication	3	ENGL 210
Year 2, Fall Semester (18 Credits)			
BADM 230	Business Law	3	
BECN 301	Microeconomics	3	Junior Standing
BFIN 300	Financial Management I	3	BACC 205
BHRM 300	Human Resources Management	3	BMGT 200
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210

	Science Elective	3	
Year 2, Spring Semester (18 Credits)			
BADM 355	Business Ethics and Social Responsibility	3	Junior Standing
BECN 302	Macroeconomics	3	Junior Standing
BITM 300	Business IT Management	3	BITM 200
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
BMGT 300	Project Management	3	Junior Standing
	Humanities / Fine Arts Elective	3	
Year 2, Summer Semester (0 Credits)			
BADM 290	Community Engagement Experience	0	
Year 3, Fall Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
BADM 410	International Business	3	Senior Standing
BADM 420	Quantitative Methods for Business	3	BADM 250 or MATH 351 or Senior Standing
BMGT 400	Operations Management	3	BADM 250
	Community & Sustainability	1	Co-req.: ENGL 210
	Social Science Elective	3	
Year 3, Spring Semester (15 Credits)			
BADM 490	Entrepreneurship	3	Senior Standing
BECN 305	Managerial Economics	3	BECN 301
BMGT 485	Strategic Management	3	Senior Standing
	Major Elective	3	
	Social Science Elective	3	
Year 3, Summer Semester (1 Credit)			
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing

* Students exempted from this course based on their score on the mathematics placement exam will select a free elective course as a substitute for it.

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

BADM 410	**International Business**	**3(3,0)**
This course is a blend of lectures, case studies, and discussions of current global and international business environments. Students will learn about international business strategies and procedures, as well as comparative environmental frameworks. It familiarizes students with theories and practices of international trade, investment, and the financial environment. Students will apply strategies of international business,		

country evaluation and selection, export and import, foreign direct investment, and global marketing. **Prerequisite**: Senior Standing.

BADM 485	**Co-op Work Experience**	**1(1,0)**
The first Co-op work experience is designed to provide students with full-time work terms in business, industry, or government. The main thrust of this course is the opportunity to put into practice the major aspects of the student's business education. Special emphasis will be placed on assessing the attitude and work ethic of the "co-op" student. Students will be encouraged to network in the industry and to participate in professional organizations. **Prerequisite:** ENGL 217; Senior Standing.		

BADM 490	**Entrepreneurship**
The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.	

BECN 305	**Managerial Economics**	**3(3,0)**
Given the constraints firms face, this course applies microeconomic theory to various management and planning decisions, such as output maximization and cost minimization. Demand analysis, cost analysis, and different market structures are studied. **Prerequisite**: BECN 301.		

BHRM 300	**Human Resources Management**	**3(3,0)**
This course provides students with a functional understanding of current Human Resources Management concepts and applications in both small and large businesses. Topics include recruitment & selection, training & development, performance management, compensation & benefits, employee relations, and legal compliance. **Prerequisite:** BMGT 200.		

BMGT 205	**Organizational Behavior**	**3(3,0)**
This course aims to help the student develop the necessary skills and conceptual tools to understand and effectively address human behavior in organizations. Special emphasis will be placed on employee motivation, teamwork, leadership, communication, conflict, negotiation, and an overview of attitudes, values, personality, and perception. **Prerequisite:** BMGT 200.		

BMGT 300	**Project Management**	**3(3,0)**
This course provides students with the skills to manage their business projects by effectively leading, organizing, scheduling, and controlling tasks to achieve project goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling and budgeting. **Prerequisite:** Junior Standing.		

BMGT 400	Operations Management	3(3,0)
In addition to the principles of supply chain management, students learn advanced skills in forecasting, quality management, facility layout, inventory control systems, capacity and aggregate planning, JIT, and statistical process control (SPC). **Prerequisite:** BADM 250.		

BMGT 485	Strategic Management	3(3,0)
This is an advanced course for upper-level management students. It covers basic concepts of strategic management, including corporate and environmental scanning and industry analysis; internal scanning and organizational analysis; and strategy formulation, including situation analysis, business strategy, corporate strategy, and functional strategy. **Prerequisite**: Senior Standing.		

II. Elective Courses

Descriptions of some major elective courses are given below.

BADM 480	Independent Studies	3(3,0)
This course focuses on advancing the student's knowledge in their specialization through independent research and application on current or emerging topics, as approved by the assigned faculty member. **Prerequisite:** Advisor's approval.		

BFIN 400	Financial Management II	3(3,0)
As a continuation of Financial Management I, this course exposes students to the firm's financial management for value maximization. That includes capital budgeting, capital structure and leverage, dividend policy, mergers & acquisitions, long-term debt, and financial planning and policy. **Prerequisite:** BFIN 300.		

BITM 415	Business Intelligence	3(3,0)
This course introduces business intelligence as computerized support for managerial decision-making. It concentrates on the theoretical and conceptual foundations of business intelligence and commercial tools and techniques available for effective decision support. It focuses on extracting business intelligence from data sets for various applications, such as reporting and visual analytics across domains like web and business analytics, to aid decision-making processes. Provides hands-on experience with various business intelligence software for reporting and building visualizations and dashboards**. Prerequisite**: Senior Standing & BADM 350		

BMGT 410	Quality Management	3(3,0)
This course highlights that TQM has become crucial for business excellence in worldwide markets. It allows students to learn techniques that establish sustainable quality improvement in both product and service industries. Topics include seven quality control tools; SPC; quality function deployment and the house of quality; product reliability; Six Sigma; Benchmarking; quality certification and awards; and quality costing. **Prerequisite:** BADM 250		

BMKA 365	Sales Management	3(3,0)
Rapidly changing market conditions and customers' expectations are redefining the roles of salespeople and the sales process within a modern framework. This course helps students understand and apply practical interpersonal techniques and persuasive communication strategies in the selling process. It focuses on value-added techniques that are based on four broad strategic areas. The concepts of customer value and building and retaining long-term relationships are integrated throughout the course. **Prerequisite**: BMKA 200; Junior Standing.		

BMKA 450	Events Marketing and Management	3(3,0)
This course covers the skills and concepts necessary to create, manage, market, and finance special events. It provides a conceptual overview and systematic study of event marketing and management. Additionally, it fosters professionalism, coordination, and collaboration among team members to achieve the objectives of special events. The course concludes with the implementation and evaluation of a special event. **Prerequisite:** Senior Standing.		

III. Non-Business Programs Courses

BMGT 210	SME Management	3(3,0)
A course on starting and operating a small business. It introduces the fundamentals of business management, including planning, raising capital, using business information, managing employees, and marketing products and services. Participants learn how to start a small business or operate as independent contractors or freelancers. The course covers a small business, essential management skills, business plan preparation, marketing strategies, and legal issues. At least one-third of the coverage in this course is devoted to hands-on practical applications.		

MARKETING AND ADVERTISING PROGRAM

Program Overview

The Bachelor of Business Administration with specialization in Marketing and Advertising is a program carefully designed to prepare graduates for successful careers in marketing communications in the media services and advertising industry. These careers require expertise in both marketing and advertising.

The program is founded on a general business and marketing core knowledge base, with additional technical and specialized knowledge drawn from and integrated across the main dimensions of marketing, advertising, public relations, and marketing communications.

To obtain a Bachelor of Business Administration degree in Marketing and Advertising, the student must complete 99 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BBA in Marketing and Advertising (99 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	-	-	18	12	30	30
College Requirement	3	-	38	-	41	41
Program Requirement	25	3	-	-	28	28
Credits	28	3	56	12	99	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Business Administration section in this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented in the introductory pages of the College of Business Administration section of this catalog. Students in Marketing and Advertising take GRDS 220, Introduction to Digital Media, instead of BADM 420, Quantitative Methods for Business.

III. Program Requirements

A. Mandatory Requirements

The mandatory courses in the Marketing and Advertising program are listed in the table below.

Course #	Title	Credits	Prerequisites
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing
BADM 490	Entrepreneurship	3	Senior Standing
BMKA 310	Consumer Behavior	3	BMKA 200; Junior Standing
BMKA 365	Sales Management	3	BMKA 200; Junior Standing
BMKA 370	Marketing Research	3	BMKA 200, BADM 250
BMKA 380	Advertising Media and Strategies	3	BMKA 200
BMKA 430	Advertising Design and Creativity	3	GRDS 220, BMKA 350
BMKA 440	Social Media Marketing	3	Senior Standing
BMKA 485	Strategic Marketing	3	Senior Standing
	Major Elective	3	

B. Major Electives

As part of the Bachelor of Business Administration in Marketing and Advertising program, the student must study 3 credit hours of major electives. Major electives could be upper-level courses in Marketing and Advertising or closely related business areas. These courses allow students to focus on a specific area for in-depth knowledge and understanding. Meanwhile, they introduce flexibility into the program, allowing the integration of emerging topics.

In consultation with the academic advisor, the student should select the elective course that best aligns with their needs and aspirations. It is highly recommended that the student register for these courses after completing the Departmental requirements. The table below lists some recommendations for major elective courses:

Course #	Title	Credits	Prerequisites
BADM 480	Independent Studies	3	Advisor's approval
BMGT 300	Project Management	3	Junior standing
BMKA 320	Introduction to Public Relations	3	BMKA 200
BMKA 405	Media Planning Analysis and Management	3	Senior Standing
BMKA 450	Events Marketing and Management	3	Senior Standing

Moreover, under certain conditions specified by the department and subject to the advisor's approval, a student may take BADM 480 (independent studies) as a substitute for a program's mandatory course. The course's description is below.

C. Entrepreneurship

The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. Prerequisite: Senior Standing.

D. Co-op Work Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the Summer Semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits to ensure that the student's performance is aligned with their aspirations and the employer's needs. Students are then required to submit formal reports and posters and make formal presentations about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Career Opportunities

The program opens the doors to a range of career possibilities in media, advertising, and marketing communications. Career opportunities include such positions as account executive, communications and media planner, advertising and promotion manager, sales manager, brand manager, and marketing manager.

Program Goals and Student Learning Outcomes

The purpose of the Marketing and Advertising Program at the College of Business Administration of Rafik Hariri University is an integrative composite of four main dimensions: General and Specialized Business Knowledge, Communication, Critical Thinking, and Growth Potential.

General & Specialized Business Knowledge

Goal 1	Prepare graduates to demonstrate an effective level of professional competence in conducting critical business activities, particularly those related to marketing and advertising.
Outcome 1	Apply essential business knowledge and skills in problem-solving and decision-making.
Outcome 2	Develop integrated marketing communication strategies through advertising designs, promotional approaches, and media plans.

Communication

Goal 2	Equip graduates with the knowledge and skills to secure placements in financial and managerial accounting positions.

Outcome 3	Demonstrate written and oral English communication skills adequate for challenging entry- and middle-level professional positions and/or self-employment.

Critical Thinking

Goal 3	Promote analytical and critical thinking in approaching business issues and problems.
Outcome 4	Identify business problems, analyze them, and present plausible solutions.

Ethics and Social Responsibility

Goal 4	Raise awareness about ethics and social responsibility to promote better business for a better community.
Outcome 5	Recognize ethical dilemmas in business and respond to them in accordance with standard codes of conduct, ethical principles, and best practices.

Study Plan

The Bachelor of Business Administration in Marketing and Advertising comprises 99 credit hours spread over 6 semesters and 2 summer sessions. The first summer session should be dedicated to the Community Engagement Experience. Meanwhile, the second session should be for the Co-op work experience. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (16 Credits)			
BACC 205	Financial Accounting	3	
BADM 215	Personal Development and Management	1	
BITM 200	Information Technology Essentials	3	
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 207	College Algebra*	3	Placement
Year 1, Spring Semester (16 Credits)			
BACC 255	Managerial Accounting	3	BACC 205
BADM 225	Business Math	3	Placement
BADM 235	Business Research Methodology	1	
BADM 250	Business Statistics	3	
BMKA 200	Introduction to Marketing	3	Co-req. ENGL 210
ENGL 217	Professional English Communication	3	ENGL 210
Year 2, Fall Semester (18 Credits)			
BADM 230	Business Law	3	
BECN 301	Microeconomics	3	Junior Standing

BFIN 300	Financial Management I	3	BACC 205
BITM 300	Business IT Management	3	BITM 200
BMKA 310	Consumer Behavior	3	BMKA 200; Junior Standing
	Science Elective	3	
Year 2, Spring Semester (18 Credits)			
BECN 302	Macroeconomics	3	Junior Standing
BITM 350	Fundamentals of Data Analytics	3	BADM 250 or MATH 351.
BMKA 370	Marketing Research	3	BMKA 200, BADM 250
BMKA 380	Advertising Media and Strategies	3	BMKA 200
GRDS 220	Introduction to Digital Media	3	
	Social Science Elective	3	
Year 2, Summer Semester (0 Credit)			
BADM 290	Community Engagement Experience	0	
Year 3, Fall Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	
BMKA 430	Advertising Design and Creativity	3	GRDS 220; BMKA 380
BMKA 440	Social Media Marketing	3	Senior Standing
	Community & Sustainability	1	Co-req.: ENGL 210
	Major Elective	3	
	Humanities / Fine Arts Elective	3	
Year 3, Spring Semester (15 Credits)			
BADM 355	Business Ethics and Social Responsibility	3	Junior Standing
BADM 490	Entrepreneurship	3	Senior Standing
BMKA 365	Sales Management	3	BMKA 200 and Junior Standing
BMKA 485	Strategic Marketing	3	Senior Standing
	Social Science Elective	3	
Year 3, Summer Semester (1 Credit)			
BADM 485	Co-op Work Experience	1	ENGL 217; Senior Standing

* Students exempted from this course based on their score on the mathematics placement exam will select a free elective course as a substitute for it.

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

BADM 485	Co-op Work Experience	1(1,0)
The Co-op work experience is designed to provide students with a full-time work term in business, industry, or government. The main thrust of this course is the opportunity to put into practice the major aspects of the student's business education. Special emphasis will be placed on assessing the attitude and work ethics of the "co-op" student. Students will be encouraged to network in the industry and to participate in professional organizations. **Prerequisite:** ENGL 217; Senior Standing.		

BADM 490	Entrepreneurship	3(3,0)
The instructor leads the students through the detailed aspects of starting a business, from identifying an opportunity and conducting a feasibility study to preparing a business plan covering marketing, operations, and finance. Students also learn about funding, expansion, and franchising strategies. **Prerequisite:** Senior Standing.		

BMKA 310	Consumer Behavior	3(3,0)
This course introduces students to the world of consumer behavior. Students explore how perceptions, learning, memory, personality, and attitudes influence consumption behavior. They learn how consumption changes during one's life cycle and how powerful cultural and subcultural factors influence consumers. Theories and case study analysis are applied throughout the course. **Prerequisites**: BMKA 200; Junior Standing		

BMKA 365	Sales Management	3(3,0)
Rapidly changing market conditions and customers' expectations are redefining the roles of salespeople and the sales process within a modern framework. This course helps students understand and apply practical interpersonal techniques and persuasive communication strategies in the selling process. It focuses on value-added techniques that are based on four broad strategic areas. The concepts of customer value and building and retaining long-term relationships are integrated throughout the course. **Prerequisites**: BMKA 200 and Junior Standing.		

BMKA 370	Marketing Research	3(3,0)
This course introduces students to marketing research and its role in the marketing decision-making planning process. It explains the principles of the marketing research process and helps students apply them to real-life marketing problems and opportunities. The course will conclude with students submitting a marketing research project in which they identify a marketing problem/opportunity, develop a research methodology, and collect and analyze data using IBM SPSS© Statistics software. Students will learn how to present their research results, make appropriate interpretations, and provide recommendations based on their analysis. **Prerequisites**: BMKA 200; BADM 250.		

BMKA 380	Advertising Media and Strategies	3(3,0)
This course introduces students to the role of advertising in the business world from both theoretical and applied perspectives. It outlines the structure of the advertising		

industry and focuses on the advertising and communication process, message and creative strategy development, and campaign design. The course endorses an integrated marketing communications perspective, in which students learn strategic media planning, buying, management, and evaluation for advertising purposes. **Prerequisite:** BMKA 200

BMKA 430	**Advertising Design and Creativity**	**3(3,0)**
In this course, students explore and apply principles of graphic design to solve appropriate marketing and advertising problems. The design process is approached from both the creative and business perspectives. Students learn to develop advertising concepts and ideas and transform them into effective print ads, brochures, and online communications. Case studies from broadcast and outdoor media, print ads, and digital marketing channels showcase advertising concepts at work: persuasion, color psychology, composition, copywriting, and typography. **Prerequisites:** GRDS 220; BMKA 380		

BMKA 440	**Social Media Marketing**	**3(3,0)**
Social media marketing has become an essential component of modern marketing communications in a dynamic and customer-centric environment. This course is designed to help students master the essential skills of building social media communication strategies, managing different social media channels, and measuring and reporting results. **Prerequisite**: Senior Standing.		

BMKA 485	**Strategic Marketing**	**3(3,0)**
This course explores in depth the concepts and methods of the strategic marketing planning process. Drawing heavily on real-world marketing case studies, it covers market situational analysis, marketing strategies, and tactics for building and sustaining competitive advantage. The course concludes by developing and presenting a marketing plan**. Prerequisite**: Senior Standing.		

II. Elective Courses

Descriptions of some major elective courses are given below.

BADM 480	**Independent Studies**	**3(3,0)**
This course focuses on advancing the student's knowledge in their area of specialization through independent research and/or application work on current or emerging topics, as approved by the assigned faculty member. **Prerequisite:** Advisor's approval.		

BMGT 300	**Project Management**	**3(3,0)**
This course provides students with the skills to manage their business projects by effectively leading, organizing, scheduling, and controlling tasks to achieve project goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling; and budgeting. **Prerequisite:** Junior Standing.		

BMKA 320	Introduction to Public Relations	3(3,0)
This course introduces students to the strategies and tactics of modern public relations practices. It explains the basic concepts and theories of public relations by examining campaigns of well-known companies, institutions, and individuals. In addition, the course explains how to plan, develop, and evaluate public relations campaigns using a wide variety of communication tools. **Prerequisite:** BMKA 200.		

BMKA 405	Media Planning, Analysis, and Management	3(3,0)
This course introduces critical and strategic media planning and evaluation for advertising purposes. The course emphasizes the principles of media planning, buying, and management. Students will study audience measurement, media research, audience segmentation, and advertising strategies. **Prerequisite:** BMKA 380.		

BMKT 450	Events Marketing and Management	3(3,0)
This course covers the skills and concepts necessary to create, manage, market, and finance special events. It provides a conceptual overview and systematic study of event marketing and management. Additionally, it fosters professionalism, coordination, and collaboration among team members to achieve the objectives of special events. The course concludes with the implementation and evaluation of a special event. **Prerequisite:** Senior Standing.		

MINOR IN BUSINESS ADMINISTRATION PROGRAM

Program Overview

In today's world of globalization, interdependence, and tough competitive pressures, professional work environments have become multidimensional and multidisciplinary, regardless of one's area of expertise. Consequently, 21st-century professionals are expected to branch out beyond their areas of expertise and acquire additional competencies to meet the challenges of diverse tasks and assignments. In particular, it has become difficult today to succeed in engineering, science, medicine, the arts, and humanities without some fundamental competence in business administration.

In recognition of this development and to meet its demands and expectations, the College of Business Administration at Rafik Hariri University offers a Minor in Business Administration for non-business students. The program is designed to ensure the acquisition of a fundamental foundation of knowledge and competence in business administration, with an entrepreneurial flavor.

Program Requirements

To complete the Minor in Business Administration program, a student must

1) Declare a Minor in Business Administration by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the college major and the dean of the College of Business Administration.
3) Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
4) Complete 18 credits of business coursework as specified below

Course #	Title	Credits	Prerequisites
BACC 205	Financial Accounting	3	
BADM 490	Entrepreneurship	3	Senior Standing
BECN 301 or BECN 302	Microeconomics or Macroeconomics	3	Junior Standing
BFIN 301	Financial Management I	3	BACC 205
BMGT 200	Introduction to Management	3	Co-req.: ENGL 210
BMKA 200	Introduction to Marketing	3	Co-requisite ENGL 210
* Engineering students may take Engineering Economics instead of Microeconomics or Macroeconomics, and Engineering Project Management instead of Introduction to Management. Computer Science students may take Project Management instead of Entrepreneurship.			

Learning Goal

Provide students with a fundamental base of knowledge and competence in business administration, with a flavor of entrepreneurship.

Learning Objectives

Upon Completion of the minor program, students should be able to

1) Identify and explain basic core concepts in the main business functional areas, namely accounting, economics, finance, management, and marketing.
2) Utilize business principles to assess and consequently understand the dynamics of internal and external business environments.
3) Assess and discuss the requirements needed to launch and succeed in a startup business and entrepreneurial initiatives.

MINOR IN HUMAN RESOURCES MANAGEMENT

Program Overview

This Minor program aims to prepare students for professional certification and/or a career in human resources management. Although the program is designed to broaden the professional credentials of non-business students, these students may still find value in it.

Students in the humanities, social sciences, and languages may greatly benefit from the program by adding a practical business dimension to their career choices and employment prospects.

Program Requirements

To complete the Minor in Human Resources Management program, a student must

1) Declare a Minor in Human Resources Management by completing the Minor Declaration Form.
2) Obtain the approval of the dean of the college major and the dean of the College of Business Administration.
3) Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
4) Complete 18 credits of business coursework as specified below

Course #	Title	Credits	Prerequisites
BHRM 300	Human Resources Management	3	BMGT 200
BHRM 305	Labor Law and Labor Relations	3	Junior Standing
BHRM 350	Workforce Planning, Recruitment, and Selection	3	BHRM 300
BHRM 400	HR Training and Development	3	BHRM 300
BHRM 450	Performance Management, Compensation, and Benefits	3	BHRM 300
BMGT 200	Introduction to Management	3	Co-req. ENGL 210

It should be noted that common courses between the student's major requirements and the BHRM minor requirements count toward fulfilling the minor requirements. Nonetheless, the student must complete at least 6 credit hours of coursework that are not counted toward the requirements of their major or any other minor.

Students majoring in Management already take BMGT 200 and BHRM 300. They may also benefit from taking one of the other courses listed above as a major elective. Accordingly, they will still need to complete 3 courses (9 credits) to earn the minor.

Students majoring in business, other than BHRM or BMGT, already take BMGT 200. They may also benefit from taking one of the other courses listed above as a major elective. Accordingly, they will still need to complete 4 courses (12 credits) to earn the minor.

Learning Goal

The primary goals of this minor program are to:

a. Familiarize students with the theories, principles, and best practices of human resources management.
b. Enable students to leverage their knowledge in other domains into the best practices of employee relations.
c. Provide students with the theoretical and educational background needed for certification and/or employment in human resources

Learning Objectives

Upon completion of this minor, students will be able to:

1. Demonstrate key knowledge and skills in human resources management,
2. Apply analytical skills to HR problem-solving,
3. Contribute to individual, team, and organizational goals using interpersonal communication skills,
4. Reflect upon the basics of organizational and human resources strategic planning

Faculty List

Abou Ltaif, Samar; Lecturer, Ph.D. in Business Administration, European University Cyprus, 2024.

Al-Hakim Nassif; Instructor, Master of Business Law, LAU,2018.

Al Majzoub, Mohamad; Assistant Professor, Ph.D. in Business Management, VGTU, 2023.

Chaaban, Baheej; Assistant Professor, DBA, Grenoble École de Management, 2018.

Daher, Maya; Assistant Professor, DBA, Nottingham Trent University, 2011.

Diab, Bassel; Lecturer, Ph.D. in Economic Informatic, Bucharest University of Economic Studies, 2021.

Eid, Yara; Instructor, Master in Management Information Systems, Université la Sagesse, 2025.

El Hajj, Mohamad; Lecturer, Ph.D. in Management & Economic Sciences, Université de Poitiers, 2015.

El Khatib, Anis; Associate Professor, Ph.D. in International Business, International School of Management, 2018.

Hagopian, Alexan; Lecturer, DBA, University Lyon III Jean Moulin, 2024.

Hajj Omar, Maha; Lecturer, DBA, Paris School of Business, 2023.

Hajjar Buthayna; Instructor, MBA, Lebanese American University,2012.

Hakim, Rima; Senior Lecturer, M.S., CMA in Business Management, Beirut University College (BUC), Beirut, Lebanon, 1993.

Hammoud, Jamil; Professor, Ph.D. in Economics, École des Hautes Études en Sciences Sociales (EHESS), Paris, France, 2003.

Itani, Doriah; Instructor, PHR, MBA, Rafik Hariri University, Lebanon, 2011.

Kanaan, May; Lecturer, Ph.D. in Leadership and HRM, USJ, 2024.

Masri, Kasem; Instructor, Master's in International Business, Grenoble École de Management, 2012.

Nasreddine, Ghalia; Associate Professor, Ph.D. in Information Technology and Systems, University of Technology of Compiegne, France, 2009

Saheb, Adel; Instructor, MS in Construction Management, University of Pittsburgh, USA, 1987.

Srour, Zeinab; Assistant Professor, Ph.D. in Management (Finance) & Applied Mathematics, University of Rennes 1 & LU, 2019.

Tarabay, Mohamad; Associate Professor and Chair, DBA in Finance, Grenoble École de Management, Grenoble, France, 2010.

COLLEGE OF ENGINEERING

COLLEGE OF ENGINEERING (CE)

College Officers

Said Ladki	President
Mahmoud Halablab	Vice President for Academic Affairs
Mohamad Taha	Dean

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History

The College of Engineering at Rafik Hariri University (RHU) was established in 1999. Since then, the College has grown significantly and presently offers six undergraduate engineering programs in Biomedical, Civil, Computer and Communications, Electrical, Mechanical, and Mechatronics Engineering. Fully accredited by the Lebanese Ministry of Higher Education, the College works with each department and program towards accreditation by the Engineering Accreditation Commission of ABET – the most prestigious accrediting agency worldwide.

The College also features active research programs in several engineering areas. In this regard, master's degree programs are available in all six programs. For those students who are employed and can only be present on campus in the afternoons, a non-thesis master's degree option is also available. More than 400 students are currently enrolled in undergraduate and graduate programs.

Overview

The College of Engineering at RHU was established to address a perceived gap in high-caliber yet affordable higher education in technology and related fields. The College strives to apply learning methodologies that can transform entrant students into 21st-century leaders in their respective fields. Furthermore, the college stresses the importance of balancing knowledge, skills, and application. To that end, specialized Labs, modern classrooms, and a unique Co-op experience are implemented.

Vision

The College of Engineering aspires to be a center of excellence in engineering education and an incubator for innovation that serves Lebanon, the region, and the world.

Mission

The College of Engineering's mission is to provide quality professional education in a learning environment conducive to transformation, enabling our students to effectively innovate practical solutions to society's problems and individual needs in Lebanon, the Middle East, and the world.

Goals

We strive to achieve our mission. To this end, we are committed to continually revising our programs while keeping our senses attuned to our surroundings. More than anything else, our students - their learning and potential success - receive our greatest attention.

Values

The College of Engineering fully endorses the values of RHU of Academic Freedom of Inquiry, Excellence through Innovation, Ethical and Socially Responsible conduct, Tolerance and Diversity, Personal Initiatives and Individual Responsibility, Teamwork, Collaboration and cooperation, and Continuous Improvement.

Accreditation

The Bachelor of Engineering programs in Biomedical Engineering, Civil Engineering, Computer and Communications Engineering, Electrical Engineering, Mechanical Engineering, and Mechatronics Engineering have been accredited by the Engineering Accreditation Commission (EAC) of ABET, 415 North Charles Street, Baltimore, MD 21201; www.abet.org.

ABET is the most prestigious and internationally renowned engineering accreditation organization in the USA. ABET accreditation verifies that our educational experience meets the global standard for technical education in the engineering profession and enhances employment opportunities—multinational engineering corporations require graduation from an ABET-accredited program.

Academic Programs

The College of Engineering offers six undergraduate programs leading to a Bachelor of Science (BS) or a Bachelor of Engineering (BE) degree and six programs leading to a Master of Science (MS) degree.

Bachelor of Science Programs

1. Bachelor of Science in Civil Engineering
2. Bachelor of Science in Electrical Engineering
3. Bachelor of Science in Computer and Communications Engineering
4. Bachelor of Science in Biomedical Engineering
5. Bachelor of Science in Mechanical Engineering
6. Bachelor of Science in Mechatronics Engineering

Bachelor of Engineering Programs

1. Bachelor of Engineering in Civil Engineering
2. Bachelor of Engineering in Electrical Engineering
3. Bachelor of Engineering in Computer and Communications Engineering
4. Bachelor of Engineering in Biomedical Engineering
5. Bachelor of Engineering in Mechanical Engineering
6. Bachelor of Engineering in Mechatronics Engineering

Master of Science Programs

1. Master of Science in Civil Engineering
2. Master of Science in Electrical Engineering
3. Master of Science in Computer and Communications Engineering
4. Master of Science in Biomedical Engineering
5. Master of Science in Mechanical Engineering
6. Master of Science in Mechatronics Engineering

Program Codes

The following table lists the code used for each program. This code is used as the prefix of each course designation in the program.

Program Code	Program
CIVE	Civil Engineering
BIOM	Biomedical Engineering
CCEE	Computer and Communications Engineering
ELEC	Electrical and Electronics Engineering
MECH	Mechanical Engineering
MECA	Mechatronics Engineering

Admission Requirements

In addition to fulfilling the University admission requirements, students aspiring to study one of the engineering majors who did not submit SAT scores may be required to take placement examinations in mathematics and physics, depending on their Lebanese Baccalaureate scores in those subjects. Students who fail to attain a passing score in one of these subjects are required to take and pass a related remedial course to ensure their mastery of basic skills and improve their ability to handle the rigor of college-level subjects. Students are strongly advised to carefully review the University Catalog for admission and degree requirements, as well as all related academic policies.

Graduation Requirements

Each Engineering program offers students two undergraduate degrees: a Bachelor of Science (BS) or a Bachelor of Engineering (BE). The BS program encompasses 116 credits, and the BE 152 credits. A student must choose the degree that s/he wishes to attain before completing 60 credits. The first three years are common in both programs (except for the BS project). A student is eligible to receive a BS degree after completing the 116 credits cited in the first three years of the program's study plan. In addition to completing the credit requirements, a student is eligible to earn the BS or BE degree if s/he attains a Cumulative GPA, a Major GPA, and a Summative Learning Project Grade of 70 or above.

Students who completed and earned their BS degree from RHU and wish to pursue a BE degree in the same major may reapply for admission to the BE program. However, the student needs to take a technical elective course (3 credits) to replace the BS project.

The required credits are distributed among three categories: University requirements (UR), General Education (GE), College requirements (CR), and program requirements (PR). The University's General Education and College requirements are common to all departments in the College of Engineering. Each department has its own required, mandatory, and elective courses. The credit-hour allocations for each degree (BS or BE) within each program are shown in the following table.

Bachelor of Engineering (BE) degrees - 152 credits							
Program	General Education		College Requirements		Program Requirements		Total Credits
	Mandatory	Electives	Mandatory	Electives	Mandatory	Electives	
CIVE	20	10	27	0	80	15	152
BIOM	20	10	27	0	80	15	152
CCEE	17	13	27	0	77	18	152
ELEC	20	10	24	3	80	15	152
MECH	20	10	27	0	80	15	152
MECA	20	10	27	0	80	15	152
Credits	30		27		95		
Bachelor of Science (BS) degrees - 118 credits							

Program	General Education		College Requirements		Program Requirements		Total Credits
	Mandatory	Electives	Mandatory	Electives	Mandatory	Electives	
CIVE	20	7	24	0	67	0	118
BIOM	20	7	24	0	67	0	118
CCE	17	10	14	0	67	0	118
ELEC	20	7	21	3	67	0	118
MECH	20	7	24	0	67	0	118
MECA	20	7	24	0	67	0	118
Credits	27		23		67		118

An engineering student in all programs is eligible for graduation if s/he has:

1. Completed all the requirements of the degree
2. Attained a cumulative GPA of 70 or above.
3. Attained a major GPA of 70 or above.
4. Attained a grade of 70 or above on the Summative Learning Project.
5. Received a passing grade on the Co-op work experience.

I. General Education

Every BE (BS)- bound student is required to take 30 (27) credits of general education courses distributed across seven domains. 22 mandatory credit hours are taken from Domains 1, 4, 5, and 6 in addition to 9 (6 for B-bound students) elective credit hours from Domains 2 and 3, as indicated below.

Domain		Credits	Courses
1	Communication Competency*	9	ENGL 210: English Composition and Rhetoric ENGL 217: Professional English Communication ARAB 212: Arabic Language and Communication 1 credit- Community and Sustainability Elective
2	Humanities/Fine Arts	3	Selected from an approved list
3	Social Sciences/Culture	6	Selected from an approved list
4	Natural Sciences and Technology	6	3 Credits – All programs CCEE 214 or COSC 214 – Introduction to Programming 3 Credits Science Course: CHEM211-Environ.Chemistry-for CIVE+MECH+MECA and PHYS 312 – Modern Physics for ELEC+BIOM Science Elective for CCEE

5	Quantitative Reasoning	6	MATH 351 - Probability and Statistics ENGR 300 – Engineering Economics and Management

A description of the specific courses in this group is given below.

ARAB 212	**Arabic Language and Communication**	**2(2,0)**
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need in their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

CCEE 214	**Introduction to Programming + Lab**	**3(2,1)**
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures. Laboratory experiments in programming for concepts. Equivalent to COSC 214.		

CMNS 200	**Etiquette**	**1(1,0)**
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, regardless of the industry, organization, or sector in which they are employed. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentation, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	**First Aid**	**1(1,0)**
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to		

write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

II. College Requirements

A. Remedial Programs and Courses

Proficiency in the English Language is a requirement for admission to any program in the College of Engineering. In addition, all incoming engineering students must sit for the placement exams in mathematics and physics. Depending on placement exam scores, a student may be required to take one or two remedial courses in Physics and/or Mathematics before taking program-level calculus and physics courses. The possible remedial courses are:

- MATH 190 - Calculus I
- MATH 191 – Calculus II
- PHYS 190 - General Physics – Mechanics (for MECH/MECA/CIVE students)
- PHYS 191 - General Physics - Electricity and Magnetism (for ELEC/CCE/BIOM students)

These remedial courses do not count toward fulfilling the degree requirements. A description of the remedial courses follows.

MATH 190	Calculus I	3(3,0)
Functions and their graphs, limits and continuity, differentiation, applications of derivatives, Extreme values and mean value theorem, definite and indefinite integrals, and the substitution method.		

MATH 191	Calculus II	3(3,0)
Covers limits and indeterminate forms with L'Hôpital's Rule, optimization, antiderivatives, and the Fundamental Theorem of Calculus. Topics include		

substitution methods, areas, volumes (cross-sections, shells), arc length, and surfaces of revolution. It introduces parametric and vector functions with applications to motion, arc length in space, curvature, and normal vectors. **Prerequisite**: Math 190.

PHYS 190	**General Physics – Mechanics**	**3(3,0)**
This course introduces students to the basic concepts and principles of Mechanics: Vectors, displacement, velocity, and acceleration; motion in 1D, projectile motion, forces, and Newton's laws of motion; work, energy, momentum, collision, torque, and rotational motion.		

PHYS 191	**General Physics - Electricity and Magnetism**	**3(3,0)**
This course is an introduction to electricity and magnetism. In this course, students will explore electric charges, electric forces and electric fields, electrical energy and potential, capacitance, direct-current circuits, magnetic force, magnetic field, induced voltage and inductance, alternating-current circuits, and electromagnetic waves.		

B. Mandatory Courses

All Engineering students (BS and BE) are required to take 24 credit hours of college-required courses. The BE program includes an additional mandatory 3-credit course, Engineering Project Management. A list of these courses and their descriptions follows.

Course	Title	Credits	Notes
Mathematics (18 credits for MECH, MECA, and CCEE; 15 credits for others)			
MATH 210	Discrete Mathematics	3	CCE major only
MATH 215	Calculus III	3	All majors
MATH 216	Calculus IV	2	All majors
MATH 311	Linear Algebra and Applications	3	All majors
MATH 314	Ordinary Differential Equations	3	All majors
MATH 317	Partial Differential Equations	3	MECH & MECA majors only
MATH 421	Numerical Analysis	3	All majors
Sciences (6 credits for MECH, MECA, and CCEE; 9 credits for all others)			
BIOL 210	Human Anatomy and Physiology and Lab	3	BIOM major only
GEOL 221	Fundamentals of Geology	3	CIVE major only
PHYS 210	Fundamentals of Physics and Lab	3	MECH, MECA, and CIVE majors only
PHYS 211	Physics: Electricity and Magnetism and Lab	3	ELEC, CCE, and BIOM majors only
	Science Elective	3	ELEC major
Engineering (6 credits)			
ELEC 210	Electric Circuits	3	All majors
ENGR 510	Engineering Project Management	3	All majors (BE track only)

Descriptions of the required Engineering, mathematics, and science courses are given below.

C. Engineering Courses

Descriptions of the Engineering required courses are given below.

ELEC 210	Electric Circuits	3(3,0)
Circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; Laplace Transform; sinusoids and phasors; and an introduction to network theory. Sinusoidal (AC) steady-state, mutual inductance, transformers, and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

D. Mathematics

A description of the required mathematics courses is given below.

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, and structural induction, sets and set operations, functions, growth of functions, relations, and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and		

spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.

MATH 311	**Linear Algebra with Applications**	**3(3,0)**
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	**Ordinary Differential Equations**	**3(3,0)**
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	**Partial Differential Equations**	**3(3,0)**
Lagrange's theorem, boundary conditions for first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables, introduction to Fourier series and integrals, boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 421	**Numerical Analysis**	**3(3,0)**
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

E. Science Courses

A description of the required Sciences courses is given below.

BIOL 210	**Human Anatomy and Physiology and Lab**	**3(2,2)**
A general overview of cell structure and function, and the anatomy and physiology of human body systems. These include cardiovascular, central nervous, respiratory, urinary, digestive, immune, and musculoskeletal systems. The course also offers a set of experiments that deal with basic biological processes and advanced biological assessments. **Prerequisite**: None.		

CHEM 211	**Environmental Chemistry and Lab**	**3(2,2)**
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global		

warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.

GEOL 221	Fundamentals of Geology	3(3,0)
Physical and chemical properties of the earth, structure of the earth, plate tectonic theory and continental drift, volcanism, mountain building processes, surface processes, mineralogy, formation and classification of rocks, earth hazards, geological problems related to civil engineering such as landslides and earthquakes, geologic maps, geophysical exploration methods, earthworks and tunneling. **Prerequisite**: None. Annually.		

PHYS 210	Fundamentals of Physics and Lab	3(3,1)
Welcome to the Introductory Physics course. This course consists of: Review of Classical Mechanics; Fluid Mechanics; General Properties of Waves; Electric Forces; Electric Fields for Discrete and Continuous Charge Distributions; Gauss's Law; Electric Potential; Kirchhoff's Rules; Magnetic Fields and Forces; and Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

F. Independent Study Courses

The College of Engineering offers an Independent Study course option at the undergraduate level. Typically, a maximum of two independent studies (totaling not more than 6 credits) may be counted towards your degree as part of your technical elective credits. To begin an Independent Study, you must reach an agreement with a faculty member willing to supervise your work. The supervisor must either be a full-time College of Engineering faculty member or be affiliated with one of the college's engineering departments. The faculty member approves the specific number of credits the proposed work will qualify for, approves a reading list, and meets with you periodically to discuss the reading and your overall progress. 'Independent Study' courses are created by students who work with a faculty member to explore a topic of mutual interest. Students

and faculty in Engineering use independent study courses for several purposes, including a capstone design and/or experimental project, conducting research, or writing a substantial research paper.

Registering for an Independent Study

Follow this procedure to register for an independent study:

1. Identify a faculty advisor and work with them to decide on the content and structure for your course, including how to assess your work, and set up the work timeframe.
2. Complete the “independent study proposal form”.
3. Once your form is submitted, your faculty advisor will review the application. They will either issue approval and forward it to the corresponding department chair or ask you to revise and resubmit your application. You will be notified by email once you can register for the course, after the college issues final approval.

ENGR 597	**Independent Study I**	**3(3,0)**
This course provides students with the possibility to explore new academic research/study that may not be available in regular courses. The student needs to identify a faculty advisor within their department and work with them to decide on the content and structure of the work, including how to assess the student’s work and set the timeframe. The student needs to complete the “independent study proposal form,” which includes an abstract that identifies the nature of the research, goals, and expected outcomes, and submit it to the faculty advisor. Once the form is submitted, the faculty advisor will review the application. They will either issue approval and forward it to the corresponding department chair, ask the student to revise and resubmit the proposal form, or deny it. The student will be notified by email after the college issues final approval. The faculty member approves the specific number of credits the proposed work will qualify for, approves a reading list, and meets with the student periodically to discuss the reading and the overall progress. At the end of the semester, the student is required to submit a paper or a project report acceptable to the faculty supervisor. This course is subject to the same rules and regulations for dropping and withdrawing as any other regular course. **Prerequisite**: Senior Standing		

ENGR 598	**Independent Study II**	**3(3,0)**
This course provides students with the possibility to explore new academic research/study that may not be available in regular courses. The student needs to identify a faculty advisor within their department and work with them to decide on the content and structure of the work, including how to assess it and set the timeframe. The student needs to complete the “independent study proposal form,” which includes an abstract that identifies the nature of the research, goals, and expected outcomes, and submit it to the faculty advisor. Once the form is submitted, the faculty advisor will review the application. They will either issue approval and forward it to the corresponding department chair, ask the student to revise and resubmit the proposal form, or deny it. The student will be notified by email after the college issues final approval. The faculty member approves the specific number of credits the proposed		

work will qualify for, approves a reading list, and meets with the student periodically to discuss the reading and the overall progress. At the end of the semester, the student is required to submit a paper or a project report acceptable to the faculty supervisor. This course is subject to the same rules and regulations for dropping and withdrawing as any other regular course. **Prerequisite**: Senior Standing.

III. Program Requirements

Requirements for the Bachelor of Science and Bachelor of Engineering degrees are program-specific. They encompass two categories: Major and non-major. Each category consists of mandatory and elective courses. The program requirements for the Bachelor of Science and Bachelor of Engineering degrees across the different engineering majors are presented in the following sections. Details and titles of relevant courses are included in the Student's Study Plan (SSP) that every engineering student will have.

Course Coding

The courses offered by the College of Engineering programs are designated according to the following coding System (xxxx abc):

xxxx	Program as indicated above
abc	"a" designates the Year or level 1, 2, 3, 4; "b" program focus area; "c" course sequence - 0, 1, …, 9

The designation used to represent the credit-hour breakdown (t, p) of a course is as follows: "c" is the total credit hours, "t" is the theoretical component of the course, and "p" is the practical or laboratory component. For example, 3(3, 0) denotes a 3-credit-hour course with 3 contact lecture hours and 0 laboratory hours.

Learning Support Center

The Learning Support Center (LSC) at the College of Engineering is another manifestation of RHU's resolute commitment to students' academic success. The LSC offers the students a peer-to-peer learning opportunity that complements classroom learning. The Center aims to support students in advancing their knowledge and skills, reinforcing what they learn in the classroom or overcoming learning and performance difficulties. In addition to being a focal point for inquiry and two-way "asker-replier" learning, the LSC provides space for academic interactions that can significantly enhance learning.

The Center is in room D205 and is open to all students throughout the week. The LSC is staffed by Teaching Assistants, who are graduate and upper-level undergraduate students, and are always available and ready to assist their fellow students in overcoming their academic challenges. Additionally, "on-call" faculty members are assigned to ensure that the LSC is acting in the best interests of the students.

The center's resource portfolio includes books, study guides, course materials, boards, computers, videos, DVDs, documentaries, large-screen LCD panels, and internet-connected PCs.

My Idea to Innovate

My Idea to Innovate (ENGR 499 – 3 credits) is an open course that encourages innovation. A student or a group of students who have a passion for and a dream to nurture and pursue an innovative idea of their own may collaborate with faculty mentors to design the course to cover topics and experiences that could not be achieved within the course offering framework of the concerned program and require collaboration across disciplines and/or fields. Course contents, prerequisites, number of credits, scope, experiences, required resources, outcomes, assessment tools and rubrics, and timeline for completion will be formulated by the student (s) and the faculty mentor(s) once the University Academic Board approves the idea. Student(s) supported by faculty advisor(s) may request approval to design and offer the course after the first year of studies at RHU. The course could count as technical elective credits.

CIVIL AND ENVIRONMENTAL ENGINEERING DEPARTMENT (CEE)

Faculty Members

Chairperson: Meheddene Machaka
Associate Professor: Meheddene Machaka, Riad Al Wardany,
Assistant Professor: Rana Hajj Chehade
Instructor: May Mrad

Programs Offered

Currently, the CEE Department offers one program in Civil Engineering, leading to Bachelor of Science, Bachelor of Engineering, and Master of Science degrees.

CIVIL ENGINEERING PROGRAM

Mission

The mission of the Civil & Environmental Engineering Department at Rafik Hariri University is fourfold:

- To provide high-quality education at the undergraduate and graduate levels.
- To prepare students for successful careers in the civil engineering field and research.
- To equip students with the will and tools to engage in lifelong learning, teamwork, team leading, and continual professional development.
- To cultivate in students habitual ethical and tolerant behavior and the desire to positively contribute to the advancement of their community and their profession.

Objectives

The educational objectives of the Civil Engineering program intend to enable its graduates, within a few years of graduation, to:

- Achieve successful careers in civil engineering and related fields;
- Pursue graduate studies and research in civil engineering and other fields;
- Effectively function as team members or as team leaders;
- Continue their lifelong learning process and their professional development; and
- Participate in community service and positively contribute to their society.

Program Outcomes

The Civil & Environmental Engineering Department adopts ABET Student Learning Outcomes as the Civil Engineering (CIVE) Program outcomes. Conforming to the Program Objectives, graduates of the CIVE Program will acquire each of the following abilities and characteristics:

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Ability to communicate effectively with a range of audiences.
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Civil Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Civil and Similarly Named Engineering Programs.

Career Opportunities

Civil engineers attain a broad spectrum of skills sought after by a wide range of professions. Industrial sectors, enterprises, and services in which a civil engineer can pursue a career include construction, infrastructure, transportation, consulting, water resources, environmental, government, and other public works sectors. Most importantly, students will be equipped with the skills to pursue entrepreneurial ventures that enable them to realize their own ideas. There is so much work to be done to fashion a better future, and civil engineers can be major contributors to that objective.

Program Overview

The Civil Engineering Program at Rafik Hariri University puts students' best interests first and foremost. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Civil Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS Program in Civil Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	7	27	23
College Requirement	0	0	24	0	24	21
Program Requirement	58	0	9	0	67	56
Credits	58	0	53	7	118	100

BE Program in Civil Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	10	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	71	15	9	0	95	62
Credits	71	15	56	10	152	100

I. University Requirements (General Education)

The list of the University-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Civil Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of elective courses for the BE degree only. Both categories are summarized below.

A. Mandatory requirements

The mandatory courses for the major and non-major CIVE programs are listed in the table below.

Course #	Title	Credits	Prerequisites
CIVE 200	Introduction to Civil Engineering	2	
CIVE 202	Civil and Architectural Drawings	3	
CIVE 211	Statics	3	

CIVE 240	Surveying	3	
CIVE 312	Structural Analysis I	3	MECH 320
CIVE 314	Concrete I	3	MECH 320
CIVE 321	Construction Materials	3	
CIVE 321L	Construction Materials Lab	1	Co-req: CIVE 321
CIVE 341	Highway Engineering	3	CIVE 240
CIVE 342	Traffic Engineering	3	CIVE 341
CIVE 400*	Summative Learning (BS) Project	3	ENGL 217 Senior Standing
CIVE 413	Computer Modeling of Structures	3	CIVE 312
CIVE 415	Concrete II	3	CIVE 312 and CIVE 314
CIVE 417	Steel Structures	3	CIVE 312
CIVE 423	Soil Mechanics	3	MECH 320; GEOL 221
CIVE 423L	Soil Mechanics Laboratory	1	Co-req.: CIVE 423
CIVE 424	Foundation Engineering	3	CIVE 423
CIVE 432	Environmental Engineering	3	CHEM 211
CIVE 435	Hydraulics	3	MECH 333
CIVE 499	Co-op Work Experience	3	ENGL 217; Senior Standing
CIVE 501	Construction Planning and Management	3	Senior Standing
CIVE 507	Structural Analysis II	3	CIVE 312
CIVE 508	AI Applications in Civil Engineering	3	CCEE 214
CIVE 536	Hydrology	3	CIVE 435
CIVE 538	Water and Wastewater Treatment	3	CIVE 432, CIVE 435
CIVE 595A	BE Summative Learning Project 1	1	Senior Standing
CIVE 595B	BE Summative Learning Project 2	3	CIVE 595A
CIVE 5xx	Technical Electives I-V	15	Per course requirements
MECH 220	Dynamics	3	CIVE 211
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 333	Thermal Fluid Sciences	3	MATH 216

* Required for BS-bound students. BE students take a technical elective instead.

The civil engineering courses cover five areas, as listed in the following table.

Code	Course Title	Credits	Prerequisite
Common Courses and Construction Management			
CIVE 200	Introduction to Civil Engineering	2	
CIVE 202	Civil and Architectural Drawings	3	
CIVE 595	BE Summative Learning Project (A + B)	3	Senior standing
CIVE 501	Construction Planning and Management	3	Senior standing
CIVE 508	AI Applications in Civil Engineering	3	CCEE 214
Structural Analysis & Design			
CIVE 211	Statics	3	
CIVE 312	Structural Analysis I	3	MECH 320
CIVE 314	Concrete I	3	MECH 320

CIVE 413	Computer Modeling of Structures	3	CIVE 312
CIVE 415	Concrete II	3	CIVE 312 and CIVE 314
CIVE 417	Steel Structures	3	CIVE 312
CIVE 507	Structural Analysis II	3	CIVE 312
Construction Materials & Geotechnics			
CIVE 321	Construction Materials	3	
CIVE 321L	Construction Materials - Laboratory	1	Co-req: CIVE 321
CIVE 423	Soil Mechanics	3	MECH 320 & GEOL 221
CIVE 423L	Soil Mechanics Laboratory	1	Co-req.: CIVE 423
CIVE 424	Foundation Engineering	3	CIVE 423
Water Resources & Wastewater Treatment			
CIVE 432	Environmental Engineering	3	CHEM 211
CIVE 435	Hydraulics	3	MECH 333
CIVE 536	Hydrology	3	CIVE 435
CIVE 538	Water and Wastewater Treatment	3	CIVE 432, CIVE 435
Public Works			
CIVE 240	Surveying	3	
CIVE 341	Highway Engineering	3	CIVE 240
CIVE 342	Traffic Engineering	3	CIVE 341

B. Technical Electives

As part of the Bachelor of Engineering in Civil Engineering program, the student is required to complete 15 credit hours of technical electives. These courses allow the student to attain in-depth knowledge and understanding in a focused area of interest. The students can mix and match elective courses from different areas to gain broader knowledge across Civil Engineering disciplines. The student should, in cooperation with the academic advisor, select the set of electives that best meets their needs and aspirations. The CIVE available technical elective courses are distributed in five areas:

1. Construction Management
2. Structural Analysis and Design
3. Construction Materials and Geotechnics
4. Water Resources & Wastewater Treatment
5. Public Works

It is highly recommended that students take elective courses from the list below after completing the mandatory requirements in their respective areas. Students may select technical electives from other programs, subject to the approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
Construction Management Sequence			
CIVE 502	Engineering Risk Analysis	3	MATH 351
CIVE 503	Computer Methods in Civil Engineering	3	Senior Standing

CIVE 504	Infrastructure Asset Management	3	Senior Standing
CIVE 506	LEED Principles in Construction Management	3	Senior Standing
Structural Analysis and Design Sequence			
CIVE 510	Finite Elements Analysis	3	CIVE 312
CIVE 511	Structural Dynamics	3	CIVE 312
CIVE 514	Earthquake Engineering	3	CIVE 312
CIVE 515	Prestressed Concrete	3	CIVE 314
CIVE 519	Tall Building Structures	3	CIVE 413
Construction Material and Geotechnics Sequence			
CIVE 521	Special Concretes	3	CIVE 321
CIVE 522	Durability of Construction Materials	3	CIVE 321
CIVE 523	Repair of Concrete Structures	3	CIVE 321
CIVE 524	Nondestructive Evaluation and Instrumentation of Infrastructure	3	CIVE 321
CIVE 525	Concrete Technology	3	CIVE 321
CIVE 526	Admixtures for Concrete	3	CIVE 321
CIVE 527	Seepage and Dams	3	CIVE 424 Co-req.: CIVE 536
CIVE 528	Earth Retaining Structures	3	CIVE 314, 424
CIVE 529	Advanced Foundation Engineering	3	CIVE 424
Water and Environmental Engineering Sequence			
CIVE 571	Water Quality	3	CIVE 432
CIVE 573	Solid Waste Management	3	CIVE 432
CIVE 575	Water Supply & Wastewater Systems	3	CIVE 435
Public Works Sequence			
CIVE 540	GIS for Civil Engineering	3	Senior Standing
CIVE 541	Pavement Design	3	CIVE 423

C. Summative Learning Project

Students must complete a 3-credit-hour course for the BS (4 credits for the BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to develop a promising idea. The SLP could be an extension of the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their area of interest. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal

presentation about their Co-op experience. While ENGL 217 and senior standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in CIVE Engineering. The following study plan serves as a roadmap for smooth progression toward graduation.

Course No.	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
CCEE 214	Introduction to Programming	3	
CIVE 211	Statics	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 210	Fundamentals of Physics and Lab	3	
Year 1, Spring Semester (17 Credits)			
ARAB 212	Arabic Language & Communication	2	
MATH 216	Calculus IV	3	MATH 215
CIVE 200	Introduction to Civil Engineering	2	
	Community & Sustainability	1	Co-req.: ENGL 210
GEOL 221	Fundamentals of Geology	3	
MECH 320	Mechanics of Materials	3	CIVE 211
	Social Sciences	3	
Year 1, Summer Semester (9 Credits)			
CHEM 211	Environmental Chemistry and Lab	3	
CIVE 240	Surveying	3	
MATH 311	Linear Algebra and Applications	3	
Year 2, Fall Semester (16 Credits)			
CIVE 312	Structural Analysis I	3	MECH 320
CIVE 321	Construction Materials	3	
CIVE 321L	Construction Materials - Laboratory	1	Co-req: CIVE 321
CIVE 341	Highway Engineering	3	CIVE 240
MATH 314	Ordinary Differential Equations	3	MATH 216
MECH 220	Dynamics	3	CIVE 211
Year 2, Spring Semester (18 Credits)			
CIVE 202	Civil and Architectural Drawings	3	
CIVE 314	Concrete I	3	MECH 320
CIVE 342	Traffic Engineering	3	CIVE 341
ENGL 217	Professional English	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 215
MECH 333	Thermal Fluid Sciences	3	MATH 216

Year 2, Summer Semester (9 Credits)			
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
ENGR 300	Engineering Economics & Management	3	
	Humanities / Fine Arts Elective	3	
Year 3, Fall Semester (16 Credits)			
CIVE 413	Computer Modeling of Structures	3	CIVE 312
CIVE 415	Concrete II	3	CIVE 312 and CIVE 314
CIVE 423	Soil Mechanics	3	MECH 320 & GEOL 221
CIVE 423L	Soil Mechanics Laboratory	1	Co-req.: CIVE 423
CIVE 435	Hydraulics	3	MECH 333
CIVE 432	Environmental Engineering	3	CHEM 211
Year 3, Spring Semester (15 Credits)			
CIVE 417	Steel Structures	3	CIVE 312
CIVE 424	Foundation Engineering	3	CIVE 423
CIVE 507	Structural Analysis II	3	CIVE 312
MATH 421	Numerical Analysis	3	MATH 311
	Engineering Technical Elective I*	3	Per course requirements
Year 3, Summer Semester (3 Credits)			
CIVE 499	Co-op Work Experience	3	ENGL 217; Senior Standing
The above 118 credits complete the requirements for a Bachelor of Science degree in Civil Engineering.			
Year 4, Fall Semester (16 Credits)			
CIVE 501	Construction Planning and Management	3	Senior standing
CIVE 595A	BE Summative Learning Project 1	1	ENGL 217, senior standing
ENGR 510	Engineering Project Management	3	ENGR 300
	Engineering Technical Elective II	3	Per course requirements
	Engineering Technical Elective III	3	Per course requirements
CIVE 508	AI Applications in Civil Engineering	3	CCEE 214
Year 4, Spring Semester (18 Credits)			
CIVE 536	Hydrology	3	CIVE 435
CIVE 538	Water and Wastewater Treatment	3	CIVE 432, CIVE 435
CIVE 595B	BE Summative Learning Project 2	3	CIVE 595A
	Engineering Technical Elective IV	3	Per course requirements
	Engineering Technical Elective V	3	Per course requirements

	Social Science/ Humanities/Fine Arts Elective	3	

* BS bound students are required to take CIVE400 - Summative Learning (BS) Project instead.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
6 credit restricted Departmental Electives
9 credits from any Engineering discipline at the 400 level or above.

Course Coding

Code numbers designate the courses offered in the Civil Engineering program in the form of (CIVE abc), where:

a	Year (level): 2 = first year, 3 = second Year, 4 = Third year, 5 = Fourth Year
b	Concentration Areas (as follows) 0: Common Courses and Construction Management; 1: Structural Analysis and Design; 2: Construction Materials and Geotechnics; 4: Public Works; 7: Water Resources and Treatment;
c	Course sequence in area: 0, 1, …, 9

Courses Description

I. Mandatory Courses

Non-Major Courses

A description of the non-major mandatory courses follows.

CHEM 211	**Environmental Chemistry and Lab**	**3(2,2)**
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality.		

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

ELEC 210	**Electrical Circuits**	**3(3,0)**
Circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; Laplace Transform; sinusoids, phasors,		

and an introduction to network theory. Sinusoidal (AC) steady-state, mutual inductance, transformers, and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	**Engineering Economics and Management**	**3(3,0)**
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements. **Prerequisite**: None. Annually.		

ENGR 510	**Engineering Project Management**	**3(3,0)**
The course covers key components of engineering project management, including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing the impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	**Calculus III**	**3(3,0)**
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

MECH 220	Dynamics	3(3,0)
Brief review of statics; proper use of different coordinate systems: Cartesian and normal/tangential axes; kinematics of a particle: rectilinear, curvilinear, and relative motion; kinetics of a particle: force and acceleration, work and energy, impulse and momentum; planar kinematics of a rigid body: translating and rotating axes, absolute and relative motion; planar kinetics of a rigid body: force and acceleration, work and energy, impulse and momentum. **Prerequisite**: CIVE 211. Annually.		

MECH 320	Mechanics of Materials	3(3,0)
Mechanical stress, strain, and stress/strain relations under pure tension, compression, pure bending, and pure torsion of circular bars; 1-D thermal strain & stress; combined		

stresses; principal stresses; plane stress transformation & Mohr's circle; basic 3-D elastic stress/strain relations; Euler-elastic buckling of columns. **Prerequisite**: CIVE 211. Annually.

MECH 333	**Thermal Fluid Sciences**	**3(3,0)**
This course seeks to impart fundamental concepts of thermal-fluid (including thermodynamics, fluid mechanics, and heat transfer) to non-mechanical engineering students. These cover: thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law) and simple flow devices; fluid statics; Bernoulli's equation; Conservation of mass, momentum & energy; viscous flow in pipes; introduction to conduction, convection, and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

PHYS 210	**Fundamentals of Physics and Lab**	**3(3,1)**
Welcome to the Introductory Physics course. This course consists of: Review of Classical Mechanics; Fluid Mechanics; General Properties of Waves; Electric Forces; Electric Fields for Discrete and Continuous Charge Distributions; Gauss's Law; Electric Potential; Kirchhoff's Rules; Magnetic Fields and Forces; and Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

Major Courses

A description of the civil engineering mandatory courses follows.

CIVE 200	**Introduction to Civil Engineering**	**2(2,0)**
A broad introductory course that exposes students to the history and heritage of civil engineering; specialized sub-disciplines (Structural, Transportation, Construction Materials, Geotechnical, Environmental); challenges of the civil engineering profession; professionalism, professional registration, and societies; and Codes and Standards. This course also introduces students to the business aspects of the civil engineering profession, including construction management and engineering economics.		

CIVE 202	**Civil and Architectural Drawings**	**3(2,2)**
Essential engineering drawing concepts for civil and architectural applications, along with computer drafting skills in 2D and 3D, are taught and applied to produce floor plans, sections, elevations, and reinforcement detailing for reinforced concrete elements. Emphasis is directed at teaching students proper placement of symbols, annotation, and scheduling information required for site construction. **Prerequisite**: None. Annually.		

CIVE 211	**Statics**	**3(3,0)**
Static laws; force vectors and operations; force systems and moments; free-body diagrams; equilibrium of particles and rigid bodies in 2D and 3D; plane trusses and frames; cables; internal forces: shear and moment diagrams; centroid and center of mass; moment of inertia; and friction. **Prerequisite**: None. Annually.		

CIVE 240	Surveying	3(2,2)
Surveying fundamentals; theory of errors in observations; leveling; distance measurement; angles, azimuths, and bearings; traversing; topographic surveys and maps. The course also has a field component in which students work in groups to conduct surveys on the topics covered and write technical reports. **Prerequisite**: None. Annually.		

CIVE 312	Structural Analysis I	3(3,0)
Shear and bending moment diagrams for frames, equilibrium, stability, and determinacy; influence lines for determinate structures; Elastic deformation of beams, frames, and trusses; introduction to indeterminate structures. **Prerequisite**: MECH 320. Annually.		

CIVE 314	Concrete I	3(3,0)
Behavior of reinforced concrete structural elements; concepts of design and proportioning sections for strength and serviceability; theory of flexure and shear; analysis and design of beams for flexure and shear; analysis for crack width and deflection; design philosophies and process; design of one way solid and joist slabs; bond theory; development, anchorage and splicing of reinforcement; bars cut-off; continuous beams and one-way slabs. **Prerequisite**: MECH 320. Annually.		

CIVE 321	Construction Materials	3(3,0)
Properties and tests of materials used in the construction of civil structures, such as aggregate, cement, mixing water, admixtures, steel, masonry, Portland cement concrete, asphalt concrete, and timber; laboratory and field measurement techniques to assess material properties and performance; emphasis on fresh and hardened concrete, types of concrete and applications, and concrete mix design. Annually.		

CIVE 321L	Construction Materials Laboratory	1(0,2)
Los Angeles abrasion test; specific gravity and absorption of coarse and fine aggregates and soils; moisture content; bulk unit weight and voids in aggregates and soils; sieve analysis; concrete mix design; mixing concrete and slump test; compressive strength of concrete; absorption, moisture content and density of concrete masonry units. Report writing and data analysis. **Co-req**: CIVE 321. Annually.		

CIVE 341	Highway Engineering	3(3,0)
Introduction to highway engineering standards; types and classifications of roads; characteristics of the driver, the pedestrian, the vehicle, and the road; highway location and survey methods; earthwork, geometric design of highway components: vertical and horizontal alignment, transition curves, superelevations and intersections, highway materials and evaluations, construction techniques and plants, quality control and testing, visual assessment, and maintenance. **Prerequisite**: CIVE 240. Annually.		

CIVE 342	Traffic Engineering	3(3,0)
Social and economic impact of traffic engineering on society; basis of traffic engineering; transportation planning; traffic flow theory; traffic studies and analysis; travel demand modeling and forecasting; queuing theory; capacity analysis; level of service; traffic control and analysis at signalized intersections. **Prerequisite**: CIVE 341. Annually.		

CIVE 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience in which students work on a meaningful problem of concern to industry or the community at large, or to develop a promising idea. The project involves whatever is needed to achieve the intended outcome: literature review, design work, data collection, experimentation, data analysis, report writing, and presentation. **Prerequisite**: ENGL 217 and Senior Standing. Annually.		

CIVE 413	Computer Modeling of Structures	3(3,0)
Principles of structural modeling of different types of structural elements: trusses, beams, columns, and frames; modeling the structural behavior of buildings under the effect of vertical and lateral loads; computational applications using structural analysis software; case studies and projects. **Prerequisite**: CIVE 312. Annually.		

CIVE 415	Concrete II	3(3,0)
Short columns, slender columns, uniaxially and biaxially bent columns; two-way slab behavior, analysis, and design; retaining walls; wall footings, concentrically and eccentrically loaded single column footings, and combined footings; staircases. **Prerequisite**: CIVE 312 and CIVE 314. Annually.		

CIVE 417	Steel Structures	3(3,0)
Structural steel properties, loads, design methods, analysis, and design of members and connections in steel structures: tension, compression, and flexure members. **Prerequisite**: CIVE 312. Annually.		

CIVE 423	Soil Mechanics	3(3,0)
Soil classification; origin of soil and grain size; soil structure, weight-volume relationships; plasticity and structure of soil; soil compaction; permeability; seepage, stresses in a soil mass; compressibility of soil and consolidation; shear strength of soil. Prerequisites: MECH 320 and GEOL 221. Annually.		

CIVE 423L	Soil Mechanics Laboratory	1(0,2)
Experimental testing of soil: Sieve Analysis, Atterberg Limits, Proctor test; sand cone field density measurement, consolidation, and direct shear test; report writing, and data analysis. Co-requisite: CIVE 423. Annually.		

CIVE 424	Foundation Engineering	3(3,0)
Geotechnical site investigation; field and laboratory tests; characterization of strata; foundation systems; ultimate bearing capacity and settlement of shallow foundations (spread footings, strip footings, combined footings, and mat); effect of water table level		

on bearing capacity; structural design of mat foundations; ultimate bearing capacity and settlement of deep foundations (single pile and group of piles); pile load test. **Prerequisite**: CIVE 423. Annually.

CIVE 432	**Environmental Engineering**	**3(3,0)**
Chemical and biological processes related to environmental science and engineering, materials and energy balances; environmental systems analysis with application to air, water, and soil; hydrology; water treatment; water quality management; wastewater treatment; air pollution; noise pollution; solid waste management; analysis of environmental problems and development of engineering solutions. **Prerequisite**: CHEM 211. Annually.		

CIVE 435	**Hydraulics**	**3(3,0)**
Engineering approaches to the measurement, control, and conveyance of water flows, with particular emphasis on the analysis, design, characteristics, and selection of hydraulic models, as well as the design of water and wastewater systems. **Prerequisite**: MECH 333. Annually.		

CIVE 499	**Co-op Work Experience**	**3(3, 0)**
This Co-op work experience is designed to provide students with an eight-week short-term opportunity in Civil and Environmental Engineering. Students will be aligned with discipline-related industries, institutions of higher learning, or government agencies. **Prerequisite**s: ENGL 217; Senior Standing.		

CIVE 501	**Construction Planning and Management**	**3(3,0)**
Fundamental elements of management of civil engineering projects; roles of all participants in the process: owners, designers, contractors, and suppliers; emphasis on contractual aspect, project estimate, planning and control of construction projects; management functions, network techniques (CPM), resource scheduling, construction financing, and cost/schedule relationship. **Prerequisite**: Senior standing. Annually.		

CIVE 507	**Structural Analysis II**	**3(3,0)**
Statically indeterminate structures, force method (beams, trusses, and frames). Cable-supported indeterminate structures. Temperature and settlement effects. Displacement methods: slope-deflection equations and moment distribution (beams and rigid frames). Qualitative influence lines for statically indeterminate beams. **Prerequisite**: CIVE 312. Annually.		

CIVE 508	AI Applications in Civil Engineering	**3(3,0)**
This course introduces students to the fundamentals of Artificial Intelligence (AI) and its practical applications in civil engineering. The course includes hands-on programming, case studies from real-world civil projects, and AI-enhanced tools to empower students with the skills to lead data- and AI-driven innovation in infrastructure and construction. **Prerequisite**: CCEE 214. Annually.		

CIVE 536	Hydrology	3(3,0)
Measurement and analysis of precipitation data; flood analysis; methods for estimating evaporation and transpiration; streamflow measurement; groundwater hydrology; groundwater reservoirs; groundwater discharge; rainfall-runoff relationship; statistical analysis in hydrology. **Prerequisite**: CIVE 435. Annually.		

CIVE 538	Water and Wastewater Treatment	3(3,0)
Quality and control tests for water and wastewater, design principles of water and wastewater treatment plants, and physical, chemical, and biological operational processes. **Prerequisite**: CIVE 432, CIVE 435. Annually.		

CIVE 595A	BE Summative Learning Project 1	1(1,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes a literature review, a scope of work, and a project proposal. **Prerequisite**s: Senior Standing, ENGL 217. Annually.		

CIVE 595B	BE Summative Learning Project 2	3(3,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes literature review, design work, data collection, experimentation, data analysis, and technical writing. **Prerequisite**: CIVE 595A. Annually.		

II. Elective Courses

Students may satisfy their technical elective requirements by selecting from the following courses.

1. Common Courses and Construction Management Sequence

CIVE 502	Engineering Risk Analysis	3(3,0)
Elements of set theory, basic elements of probability theory, random variables, multiple random variables, functions of random variables, probabilistic models, statistical inference, and formulation of the reliability problem. **Prerequisite:** MATH 351.		

CIVE 503	Computer Methods in Civil Engineering	3(3,0)
Using the computer for analysis, design, and decision making in civil engineering. Conversion of CAD or REVIT elements to structural elements. Shop drawings. Value engineering. Applications. **Prerequisite:** Senior Standing.		

CIVE 504	Infrastructure Asset Management	3(3,0)
Problems of infrastructure, asset management principles, methods of inspection and monitoring of civil infrastructures, risk assessment, asset condition information and data, and the replace/maintain decision. **Prerequisite:** Senior Standing.		

CIVE 506	LEED and Green Construction Principles in Construction Management	3(3,0)
Green construction methods and benefits of applying the Leadership in Energy and Environmental Design (LEED) principles. **Prerequisite:** Senior Standing.		

CIVE 597	Topics in Civil Engineering	3(3,0)
This course covers topics of special interest in Civil Engineering. These topics may vary each time the course is offered, and the course content will be available with each course offering. May be repeated for credit as topics vary. **Prerequisite:** Consent of Advisor.		

2. Structural Analysis & Design Sequence

CIVE 510	Finite Elements Analysis	3(3,0)
Theory of the finite element method; spring, truss, and beam elements; constant-strain elements; plane-stress or plane-strain for axisymmetric problems; plates and shells; introduction to nonlinear material analysis; introduction to structural dynamic analysis; applications. **Prerequisite**: CIVE 312. On demand.		

CIVE 511	Structural Dynamics	3(3,0)
Dynamic response of single- and multi-degree-of-freedom systems; modal analysis of lumped and continuous mass systems; natural frequencies and mode shapes; deterministic time-domain approach and design response spectra; seismic response of structures; peak response; seismic design principles. **Prerequisite**: CIVE 312. On demand.		

CIVE 512	Advanced Structural Analysis	3(3,0)
Flexibility and Stiffness Matrix Methods for Analysis of Indeterminate Trusses, Beams, and Frames. Grid beams and structures on elastic supports. Composite Structural Systems. Influence lines for indeterminate structures using the concepts of virtual work and moment distribution methods – Qualitative and Quantitative approaches. **Prerequisite**: CIVE 507. On demand.		

CIVE 514	Earthquake Engineering	3(3,0)
Earthquake causes and measures; earthquake faults and waves; plate tectonics; structural dynamics of single- and multi-degree-of-freedom systems; seismic hazard evaluation in engineering practice; response analysis of structures. Equivalent static lateral force method, lateral load-resisting systems, architectural considerations for earthquakes, and mitigation of earthquake forces. **Prerequisite**: CIVE 312. On demand.		

CIVE 515	Prestressed Concrete	3(3,0)
Definitions and Methods of prestressing. Materials and their properties. Discussion of losses in prestressing. Elastic behavior of stress distribution under different load stages—analysis and design of homogeneous sections. Shear cracking load. Behavior		

of beam. Load Balancing Procedures. Deflection. Stresses resulting from shrinkage and creep. Partial prestressing. **Prerequisite**: CIVE 314. On demand.

CIVE 519	**Tall Building Structures**	**3(3,0)**
Loadings; structural systems and analysis modeling; braced frames; rigid frame structures; stability of high-rise buildings; dynamic response; shear wall structures; coupled shear wall structures; wall-frames; dual structural systems. **Prerequisite**: CIVE 413. On demand.		

3. Construction Material and Geotechnics Sequence

CIVE 521	**Special Concretes**	**3(3,0)**
Design and characteristics of lightweight concrete, heavyweight concrete, self-consolidating concrete, fiber-reinforced concrete, mass concrete; concrete in hot weather, concrete in cold weather, high-strength concrete, high-performance concrete, roller-compacted concrete, and shrinkage-compensating concrete. **Prerequisite**: CIVE 321. On demand.		

CIVE 522	**Durability of Construction Materials**	**3(3,0)**
Definition and importance of durability; properties of main building materials (concrete, ferrous and non-ferrous metals, wood, building stones, clay bricks, gypsum, lime, plastics); factors affecting durability; mechanisms of decay; precautions to increase durability; selection of durable material; durability tests; economic aspects and impact of durability. **Prerequisite**: CIVE 321. On demand.		

CIVE 523	**Repair of Concrete Structures**	**3(3,0)**
Conventional methods of investigation; nondestructive methods; structural health monitoring; corrosion risk evaluation; protection of reinforcing steel against corrosion; instrumentation techniques; repair materials and techniques; removal and preparation of concrete for repair; crack repair; concrete patching; structural strengthening; case studies. **Prerequisite**: CIVE 321. On demand.		

CIVE 524	**Nondestructive Evaluation & Instrumentation of Infrastructure**	**3(3,0)**
Overview of the different types of civil infrastructure, their problems and durability, use of nondestructive evaluation methods to assess their actual conditions, basic NDE methods, sonic and ultrasonic methods, ground penetrating radar (GPR), infrared thermography, electrical resistivity, radioactive and nuclear methods, sensing technologies, different instruments/sensors and their use, wireless technology, and structural health monitoring. Case studies and laboratory demonstrations. **Prerequisite**: CIVE 321. On demand.		

CIVE 525	**Concrete Technology**	**3(3,0)**
Cement and supplementary cementitious materials, mechanical behavior of concrete, properties and applications of various specialized concretes, hot weather concreting, cold weather concreting, chemical admixtures, concrete mix proportioning, deterioration mechanisms, environmental and physical causes of degradation, alkali-		

aggregate reaction, carbonation, chloride attack, sulfate attack, acid and seawater attack, freeze-thaw cycle. **Prerequisite**: CIVE 321. Annually.

CIVE 526	**Admixtures for Concrete**	**3(3,0)**
Definition and classification of concrete admixtures; precautions in their use; chemical, mineral, and miscellaneous admixtures; types and properties of admixtures; standard specifications; classification and purpose of use; beneficial and detrimental effects on the properties of fresh and hardened concrete; mechanisms of action; advantages and disadvantages**. Prerequisite**: CIVE 321. On demand.		

CIVE 527	**Seepage and Dams**	**3(3,0)**
Measurement of hydraulic conductivity (permeability), analysis of seepage using analytical solutions, graphical procedures, and numerical methods; control of seepage; wells, filters, and drains; types of dams; performance of earth and earth-rock dams. **Prerequisite**: CIVE 424; Co-requisite: CIVE 536. On demand.		

CIVE 528	**Earth Retaining Structures**	**3(3,0)**
Types of earth-retaining systems; earth pressure theory; design of externally stabilized walls; design of internally stabilized walls; reinforcing elements; fundamentals of soil-reinforcement interaction. **Prerequisite**: CIVE 424, CIVE 314. On demand.		

CIVE 529	**Advanced Foundation Engineering**	**3(3,0)**
Lateral earth pressure, retaining walls, sheet-pile walls, anchors, braced cuts, drilled-shaft foundations, foundations on collapsible soils, foundations on expansive soils, foundations on rock, soil improvement methods. **Prerequisite**: CIVE 424. On demand.		

4. Water and Environmental Engineering Sequence

CIVE 571	**Water Quality**	**3(3,0)**
Water quality parameters; sources of drinking water; drinking water problems and their impact on human life; drinking water treatment methods; common operations in water treatment plants; design of distribution systems; assessment and monitoring of water quality. **Prerequisite**: CIVE 432. On demand.		

CIVE 573	**Solid Waste Management**	**3(3,0)**
Study of the types and sources of non-hazardous municipal solid waste; estimation of waste quantities and storage volumes; identification of collection, transfer, treatment, and disposal alternatives; preliminary design of landfills; solid waste management principles and processes; and pollution control management. **Prerequisite**: CIVE 432. On demand.		

CIVE 575	**Water Supply & Wastewater Systems**	**3(3,0)**
Identification and evaluation of water supply solutions; design of water distribution networks; pumping; population forecasting and requirements; collecting and		

disposing of stormwater and sewage; design of storm and sewage systems; excess stormwater management; environmental risk and protection measures. **Prerequisite**: CIVE 435. On demand.

CIVE 576	**Hydraulic Structures**	**3(3,0)**
Hydraulic aspects of the theory and design of hydraulic structures; types of dams and their functions; stability analysis of gravity dams; culverts, navigation canals, reservoirs, and port facilities; and control structures such as gates and weirs. **Prerequisite**: CIVE 435 and CIVE 536. On demand.		

5. Public Works Sequence

CIVE 540	**GIS for Civil Engineering**	**3(3,0)**
Review of basic cartographic principles and the use of geographic information systems for thematic mapping, spatial analysis, and application in the water resources sector. The laboratory emphasizes experience with GIS software. **Prerequisite**: Senior Standing. On demand.		

CIVE 541	**Pavement Design**	**3(3,0)**
Analysis and design of flexible and rigid pavements; pavement type selection; loading; failure criteria and reliability; mechanistic pavement design; design exercises using existing methods. **Prerequisite**: CIVE 423. On demand.		

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT (ECE)

Faculty Members

Chairperson:	Mohamad Diab
Professor:	Mohamad Diab, Mohamad Taha, Rached Zantout, Toufic Hijazi
Associate Professor:	Bassam Moslem, Maher Sabbah, Ousama Mustafa
Assistant Professor:	Roaa El Chammaa
Lecturer:	Amani Fawaz, Amal Arid, Milana Kassab

Vision

The Electrical and Computer Engineering Department (ECE) envisages being a leader in shaping intellectual and human capital in electrical, computer, communication, and biomedical engineering. The department seeks to produce application-oriented graduates in all fields of electrical, computer, communication, and biomedical engineering. Our graduates are trained to enrich Lebanon, the Middle East, and the world in general, technologically, economically, and socially.

Mission

The mission of the department is to be recognized as one of the best departments of electrical and computer engineering. Such excellence is not limited to education but should also encompass research, especially in translating research outcomes to the community at large.

Programs Offered

The ECE Department offers three programs – Electrical Engineering, Computer and Communications Engineering, and Biomedical Engineering. Each program grants three degrees: Bachelor of Science, Bachelor of Engineering, and Master of Science.

ELECTRICAL ENGINEERING PROGRAM

Mission

The Electrical Engineering program at RHU aims to help students explore their innate creativity and potential and endow them with the knowledge, skills and abilities to: pursue successful careers in electrical engineering and related fields; think critically in solving complex problems using modern tools and technologies; communicate and work effectively with diverse groups; learn more every day, and succeed in graduate studies in renowned institutions if they so choose.

Program Educational Objectives

The program's educational objectives are to enable graduates to:

1. Foster an environment that encourages excellence in endeavor.
2. Provide quality and professional education that prepares leaders.
3. Solve problems facing society and industry in Lebanon, the Middle East, and the world.
4. Engage in multidisciplinary research.

Student Outcomes

The Electrical Engineering program adopts the learning outcomes of ABET, so our graduates will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Electrical Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Electrical and Similarly Named Engineering Programs.

Career Opportunities

An Electrical engineer could develop components for some of the most fun things in our lives (MP3 players, digital cameras, or roller coasters) as well as the most essential (medical tests or communications systems). This largest field of engineering encompasses the macro (such as large power grids that light up cities) as well as the micro (including devices smaller than a millimeter that trigger a car's airbags to inflate). An electrical engineer may work in robotics, computer networks, electric transportation systems, medical imaging, or renewable energy systems—at the very forefront of technological innovation—as well as in the low-power industry and power plants.

Program Overview

The Electrical (ELEC) Engineering Program at Rafik Hariri University puts students' best interests first. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Electrical Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: General Education requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and the BE programs is shown in the following tables:

BS in Electrical Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education Requirement	0	0	20	7	27	23
College Requirement	3	0	18	3	24	20
Program Requirement	46	0	21	0	67	57
Credits	49	0	59	10	118	100

BE in Electrical Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	10	30	20
College Requirement	3	0	21	3	27	18
Program Requirement	56	15	24	0	95	62
Credits	59	15	65	13	152	100

I. General Education Requirements

The list of the GE required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented on the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Electrical Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of engineering technical electives for BE only.

A. Mandatory requirements

The mandatory courses for the major and non-major ELEC programs are listed in the table below.

Course #	Title	Credits	Prerequisites
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	Pre-req.: CCEE 221

CCEE 331	Signals and Systems	3	MATH 216
CCEE 341	Communication Systems	3	CCEE 331
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
CCEE 534	Signal Processing	2	CCEE 331
ELEC 200	Introduction to Electrical and Computer Engineering	3	
ELEC 301	Sensors and Sensor Circuit Design	3	ELEC 210
ELEC 301L	Sensors and Sensor Circuit Design Lab	1	Co-req.: ELEC 301
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 323	Electronics Circuit Design	3	ELEC 301, ELEC 320
ELEC 330	Power Systems	3	ELEC 210
ELEC 331	Electric Machines	3	ELEC 210
ELEC 331L	Electric Machines Lab	1	Co-req.: ELEC 331
ELEC 340	Engineering Electromagnetics	3	PHYS 211
ELEC 400	BS Summative Learning Project*	3	ENGL 217
ELEC 405	Electrical System Design	3	ELEC 210
ELEC 406	Modeling and Simulation	3	ELEC 210
ELEC 431	Introduction to Renewable Energy Systems	3	ELEC 210
ELEC 432	Power Electronics	3	ELEC 320
ELEC 432L	Power Electronics Lab	1	Co-req.: ELEC 432
ELEC 451	Control Systems	3	CCEE 331
ELEC 499	Co-op Work Experience	3	ENGL 217
ELEC 533	Advanced Machines and Drives	3	ELEC 331, 432
ELEC 534	Advanced Power Systems	3	ELEC 330
ELEC 563	Smart Grid Technology	3	ELEC 432
ELEC 595A	BE Summative Learning Project 1	1	ENGL 217
ELEC 595B	BE Summative Learning Project 2	3	ELEC 595A
MECH 333	Thermal Fluid Sciences	3	MATH 216

* Required for BS-bound students. BE students take a technical elective instead.

B. Technical Electives

As part of the Bachelor of Engineering in ELEC Engineering program, the student is required to take 15 credit hours of engineering technical electives. Students must be of

senior standing to take engineering technical electives. Technical electives allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain broader exposure to the various ELEC Engineering disciplines. The student should, in cooperation with their academic advisor, select a list of electives that best meet their needs and aspirations. The listed engineering technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas: (1) Power, (2) Renewable Energy, (3) Intelligent Systems.

It is highly recommended that students take elective courses from the list below after completing the mandatory requirements in their respective areas. Students may select technical electives from other programs, subject to approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
ELEC 433	High Voltage Engineering	3	ELEC 330
ELEC 434	Electrical Power Distribution	3	ELEC 330
ELEC 444	Electromagnetic Compatibility	3	ELEC 210
ELEC 498	Special Topics in Electrical Engineering	3	Instructor's consent
ELEC 503	Artificial Neural Systems	3	MATH 311
ELEC 504	Lasers and Laser Applications in Engineering	3	
ELEC 531	Electrical Power Systems for Transportation Applications	3	ELEC 432
ELEC 532	Application of Power Electronics in Power Systems	3	ELEC 330, ELEC 432
ELEC 535	Renewable Energy Systems for the Built Environment	3	
ELEC 536	Renewable Energy Systems	3	ELEC 431
ELEC 542	Advanced Power Electronics	3	ELEC 432
ELEC 551	Nonlinear Adaptive Control	3	ELEC 451
ELEC 560	Sustainable Energy	3	None
ELEC 561	Wind Energy Technology	3	ELEC 431 and ELEC 432
ELEC 562	Solar Energy Technology	3	ELEC 210
ELEC 597	Advanced Topics in Electrical Engineering	3	Instructor's Consent Senior Standing

C. Summative Learning Project SLP

Students must complete a 3-credit-hour course for BS (4 credits for BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to innovate a promising idea. The SLP could be an extension of the projects students had worked on in previous courses, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on students' progress throughout the Co-op period by conducting field visits and ensuring that students' performance is aligned with their aspirations and employers' needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

The courses offered in the ELEC Engineering program are designated code numbers in the form of (ELEC abc), where:

a:	Designates course year/level
b:	Designates the focus area as follows: 0: General; 1: Electrical; 2: Electronics; 3: Power; 4: Magnetism; 5: Control
c:	Designates the course sequence in the area

For example, ELEC 320 is the *first Electronics course* offered by the *electrical engineering program* in the *second year*.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in ELEC Engineering. The study plan serves as a roadmap to facilitate students' smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (14 Credits)			
ARAB 212	Arabic Language and Communication	2	
CCEE 214	Introduction to Programming	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 211	Physics: Electricity and Magnetism & Lab	3	
Year 1, Spring Semester (15 Credits)			
CCEE 221	Logic Design	3	
ELEC 200	Introduction to Electrical & Computer Engineering	3	
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
PHYS 312	Modern Physics	3	

Year 1, Summer Semester (10 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
MATH 311	Linear Algebra and Applications	3	
	Humanities/ Fine Arts Elective I	3	
	Science Elective	3	
Year 2, Fall Semester (17 Credits)			
CCEE 331	Signals and Systems	3	MATH 216
ELEC 301	Sensors and Sensor Circuit Design	3	ELEC 210
ELEC 301L	Sensors and Sensor Circuit Design Lab	1	Co-req.: ELEC 301
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ENGL 217	Professional English	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 216
Year 2, Spring Semester (17 Credits)			
ELEC 323	Electronics Circuit Design	3	ELEC 301, ELEC 320
ELEC 330	Power Systems	3	ELEC 210
ELEC 331	Electric Machines	3	ELEC 210
ELEC 331L	Electric Machines Lab	1	Co-req.: ELEC 331
ELEC 340	Engineering Electromagnetics	3	PHYS 211
MATH 314	Ordinary Differential Equations	3	MATH 216
	Community and Sustainability Elective	1	
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 421	Numerical Analysis	3	MATH 311
	Social Science Elective I	3	
Year 3, Fall Semester (16 Credits)			
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
ELEC 405	Electrical System Design	3	ELEC 210
ELEC 406	Modeling and Simulation	3	ELEC 210
ELEC 431	Introduction to Renewable Energy Systems	3	ELEC 210
ELEC 451	Control Systems	3	CCEE 331
Year 3, Spring Semester (17 Credits)			
CCEE 341	Communication Systems	3	CCEE 331
CEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
ELEC 432	Power Electronics	3	ELEC 320
ELEC 432L	Power Electronics Lab	1	Co-req.: ELEC 432
ELEC	Engineering Technical Elective 1* or BS Project	3	
MECH 333	Thermal Fluid Sciences	3	MATH 216

Year 3, Summer Semester (3 Credits)			
ELEC 499	Co-op Work Experience	3	ENGL 217
The above 118 credits complete the requirements for a Bachelor of Science degree in Electrical Engineering.			
Year 4, Fall Semester (16 Credits)			
CCEE 534	Signal Processing	3	CCEE 331
ELEC 534	Advanced Power Systems	3	ELEC 330
ELEC 595A	BE Summative Learning Project 1	1	ENGL 217, Senior standing
ENGR 510	Engineering Project Management	3	ENGR 300
	Engineering Technical Elective 2	3	
	Engineering Technical Elective 3	3	
Year 4, Spring Semester (18 Credits)			
ELEC 533	Advanced Machines and Drives	3	ELEC 331, 432
ELEC 563	Smart Grid Technology	3	ELEC 432
ELEC 595B	BE Summative Learning Project 2	3	ELEC 595A
	Engineering Technical Elective 4	3	
	Engineering Technical Elective 5	3	
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take ELEC 400 – Summative Learning (BS) Project instead.

Note: Engineering Technical Electives (levels 400 or above) are selected as such:
6 credits restricted Departmental Electives
9 credits from any Engineering discipline at the 400 level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

Description of the mandatory, non-major courses follows.

CCEE 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to COSC 214.		

CCEE 221	**Logic Design**	**3(3,0)**
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices, including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices. Equivalent to COSC 351.		

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: CCEE 221. **Prerequisite**: CCEE 221. Equivalent to COSC 351L.		

CCEE 331	Signals and Systems	3(3,0)
This course covers the mathematical description and classification of continuous and discrete signals and systems. Topics include types of signals and systems, Fourier series, Fourier transforms, Discrete-Time Fourier transforms (DTFT), Discrete and Fast Fourier Transforms (DFT and FFT), Laplace transforms, z-transforms, and transfer functions. **Prerequisite:** MATH 216		

CCEE 341	Communication Systems	3(3,0)
This course introduces the fundamentals of transmission and reception in communication systems, as well as the effects of noise. Topics include power spectral density; amplitude modulation and demodulation; angle modulation and demodulation; analog communication system performance in the presence of noise; sampling and analog-to-digital conversion; introduction to digital modulation; and channel capacity. **Prerequisite**: CCEE 331.		

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

CCEE 426L	Design of Embedded Systems Lab	1(0,2)
This lab complements topics covered in the CCEE 426 course. It involves the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Experiments include design using finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, scheduling, system-level simulation, and interface-based design. **Co-requisite**: CCEE 426. Equivalent to COSC 356L.		

CCEE 460	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning methods in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 460L	Artificial Intelligence Lab	1(0,2)
This lab will help prepare students to handle the huge amount of data produced by both humans and machines today. The student will learn to use state-of-the-art AI tools to analyze data and make complex decisions based on that data using real-world practical examples and case studies. At the end of the lab, the student will be able to apply AI knowledge in their field at work or in day-to-day life to make better decisions. **Co-req.:** CCEE 460.		

CCEE 534	Signal Processing	3(3,0)
This course covers time series analysis, frequency analysis, time-frequency, and time-scale analysis. It also covers the design of digital filters and signal modeling. **Prerequisite**: CCEE 331.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

MECH 333	Thermal Fluid Sciences	3(3,0)
This course seeks to impart thermal-fluid fundamental concepts (including Thermodynamics, Fluid Mechanics, and Heat Transfer) to non-mechanical engineering students. These cover: thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law) and simple flow devices; fluid statics; Bernoulli's equation; Conservation of mass, momentum & energy; viscous flow in pipes; introduction to conduction, convection and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and to particle physics.		

Major Courses

Mandatory electrical engineering courses are described below.

ELEC 200	Introduction to Electrical and Computer Engineering	3(3,0)
In this course, the student will be trained in core skills to excel in electrical, biomedical, computer, and communications engineering. The student will be trained in application programming using an appropriate programming environment. Students will build on the basics of Programming learned in the introductory programming course to implement Digital Signal Processing applications and perform data analysis. Students will also be introduced to basic lab equipment such as Oscilloscopes, Function Generators, and Multi-Meters. Students will be introduced to microcontroller-based projects to explore more of the Electrical and Computer Engineering majors (CCE, BIOM, and ELEC majors). Throughout the course, students will work in multidisciplinary teams to study engineering cases for the community and implement a project to address a community need.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers;		

sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.

ELEC 301	**Sensors and Sensor Circuit Design**	**3(3,0)**
This course provides an introductory overview of the multidisciplinary field of sensors. It includes interpreting relevant signals; measuring noise, pressure, flow, motion, and force; and electrode theory. **Prerequisite**: ELEC 210		

ELEC 301L	**Sensors and Sensor Circuit Design Lab**	**1(0,2)**
The lab teaches students how to measure and interpret signals. Experiments include signal, noise, pressure (strain gauge), and temperature measurements; fluid and gas flow, motion, and force measurements; and applications of electrodes in medical equipment and in research and simulation kits—co-requisite: ELEC 301.		

ELEC 320	**Electronics**	**3(3,0)**
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	**Electronics Lab**	**1(0,2)**
This is a Lab course with experiments in Electrical and Electronic Circuits. It covers passive electrical elements and sources; lab instruments; voltage divider circuits; Thevenin's theorem; RC circuits; diode circuits; Op-Amp circuits; BJT and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ELEC 323	**Electronics Circuit Design**	**3(3,0)**
This course offers students a methodology for the design process with an emphasis on the feasibility and preliminary design phases, including conditioning circuits, Op-amps and feedback systems, filters, oscillators, sample/hold, DAC/ADC, and data acquisition Boards. Students will learn how to write proposals and reports and increase their creativity through industry-based group projects on interdisciplinary topics. **Prerequisite**: ELEC 301; ELEC 320.		

ELEC 330	**Power Systems**	**3(3,0)**
This course covers three-phase power systems, symmetrical components, transmission lines, power transformers, power system modeling, per-unit analysis, power flow studies, and fault analysis. **Prerequisite:** ELEC 210.		

ELEC 331	**Electric Machines**	**3(3,0)**
This course covers magnetic circuits; fundamentals of electromechanical energy conversion; and the construction, testing, steady-state analysis, and operational characteristics of DC machines, transformers, induction machines, and synchronous machines. **Prerequisite**: ELEC 210.		

ELEC 331L	Electric Machines Lab	1(0,2)
This is a Lab course with experiments on electric machines. It covers the following topics: the basics of DC motors and generators; DC series, shunt, and separately excited motors and generators; single-phase AC induction machines; and 3-phase machines. **Co-requisite:** ELEC 331.		

ELEC 340	Engineering Electromagnetics	3(3,0)
This course covers vector analysis, static electric fields, static magnetic fields, time-varying fields, Maxwell's equations, and electromagnetic waves. **Prerequisite:** PHYS 211.		

ELEC 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program. Annually.		

ELEC 405	Electrical System Design	3(3,0)
This course aims to teach students how to design various electrical systems, including equipment selection, protection systems, EMC and safety requirements, international standards, drawing and interpreting schematics and wiring diagrams, cost considerations, and energy efficiency. **Prerequisite:** ELEC 210.		

ELEC 406	Modeling and Simulation	3(3,0)
This course aims to teach students how to represent, characterize, understand, analyze, and solve real-world problems through modeling and simulation. In this course, students are introduced to the tools and techniques for modeling and simulating systems, ranging from basic circuits to more complex technical systems across various engineering domains. **Prerequisite:** PHYS 211 and ELEC 210.		

ELEC 431	Introduction to Renewable Energy Systems	3(3,0)
This course aims to provide students with information on renewable energy sources, including solar, wind, and wave energy, as well as biomass energy. This course will also illustrate how to link these sources to the grid and synchronize them to deliver electricity reliably and efficiently. **Prerequisite:** ELEC 210.		

ELEC 432	Power Electronics	3(3,0)
This course familiarizes the students with basic power switch technology and associated electronic circuits. In this course, power electronic circuits and switching devices are studied. Their applications in AC/DC, DC/DC, DC/AC, and AC/AC converters, as well as switching power supplies, are studied. **Prerequisite:** ELEC 320.		

ELEC 432L	Power Electronics Lab	1(0,2)
This Lab course is designed to teach students the fundamentals of power electronics. It covers the following topics: uncontrolled single- and 3-phase rectifiers using diodes;		

controlled single-phase and 3-phase rectifiers using thyristors; DC-to-AC converters; DC-to-DC buck and boost converters; and applications of DIAC, TRIAC, and IGBT. **Co-requisite:** ELEC 432.

ELEC 451	Control Systems	3(3,0)
This course is to explore the modeling of linear dynamic systems via differential equations and transfer functions utilizing state-space and input-output representations; analysis of control systems in the time and frequency domains, study of the classical stability tests, such as the Routh-Hurwitz and Nyquist criteria, and design methods using root-locus plots and Bode plots; and the development of control techniques based on PID, lead and lag networks. **Prerequisite**: CCEE 331.		

ELEC 499	Co-op Work Experience	3(3,0)
This Co-op work experience is designed to provide students with 5 weeks of short-term work in electrical engineering. Students are encouraged to network in the discipline-related industries. A report, a poster, and a PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

ELEC 533	Advanced Electric Machines and Drives	3(3,0)
This course focuses on the basic structure and principles of DC and AC drive systems. Constant flux and field weakening. Constant torque and constant hp operation. Effect of non-ideal waveform on motor performance. Application of Space Vector to AC machine analysis. Pulse Width Modulation techniques. Multi-level converters and multi-phase motor drives. Modern methods of motor control: Field Oriented Control, Direct Torque Control, etc. Methods of Sensorless Control. **Prerequisite:** ELEC 331 and ELEC 432.		

ELEC 534	Advanced Power Systems	3(3,0)
The course is designed to give students a deeper understanding of power system analysis by covering the following topics: Circuit models, Per-unit representation, Network methods, Load-flow studies, Load-flow control, and Economic dispatch. It also covers the Symmetrical three-phase faults, Unsymmetrical faults, Power system stability, and computer methods. **Prerequisite**: ELEC 330.		

ELEC 595A	BE Summative Learning Project 1	1(1,0)
The first phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite**: ENGL 217.		

ELEC 595B	BE Summative Learning Project 2	3(3,0)
The second phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite**: ELEC 595A.		

II. Elective Courses

Students may meet their engineering technical electives requirements from the following selection of elective courses.

ELEC 433	High Voltage Engineering	3(3,0)
This course introduces high-voltage engineering, withstand levels, S curves, insulation coordination, and breakdown mechanisms. Non-destructive testing of apparatus: insulation resistance, partial discharge, and HV production measurements for test objects. System over-voltages, switching, lightning, and over-voltage protection devices. **Prerequisite:** ELEC 330.		

ELEC 434	Electrical Power Distribution	3(3,0)
This course covers general considerations; load characteristics; sub-transmission and distribution substations; primary and secondary distribution; secondary network systems; distribution transformers; voltage regulation and the application of capacitors; voltage fluctuations; and protective device coordination. **Prerequisite:** ELEC 330.		

ELEC 444	Electromagnetic Compatibility	3(3,0)
This course introduces Electromagnetic Interference, Conducted and Radiated Emission, Conducted and Radiated Susceptibility, and Product Slippage. EMC standards and Regulations. EMC Theory and Materials Relating to EMC Design. EMC Measurement and test facilities. **Prerequisite:** ELEC 210.		

ELEC 498	Special Topics in Electrical Engineering	3(3,0)
This course is designed to enable students to study current special topics of interest, carefully selected from ELEC-related areas. The contents of such a course are to be determined by the instructor and the department.		

ELEC 503	Artificial Neural Systems	3(3,0)
This course focuses on the foundations of neural network theory and its application in engineering, cognitive science, and artificial intelligence. The course also covers the foundations of machine learning and neural processing algorithms: supervised and unsupervised learning of feed-forward and recurrent neural networks, perception layers, associative memories, and feature maps. **Prerequisite**: MATH 311		

ELEC 504	Lasers and Laser Applications in Engineering	3(3,0)
This course is designed to give students a functional knowledge of lasers. The course covers light, atoms, absorption processes, and spontaneous and stimulated emission of radiation. Moreover, the types and characteristics of lasers are discussed. Laser sources, resonators, and amplifiers are discussed. In addition, applications in engineering, technology, and the biomedical field will be discussed.		

ELEC 531	Electrical Power Systems for Transportation Applications	3(3,0)
Electrical power and energy systems are at the forefront of application development in, for example, electric and hybrid road vehicles, more-electric and all-electric aircraft,		

and traction applications. The associated hardware, technologies, and control methods are crucial to achieving critical global targets in energy efficiency, low-carbon, and low-emissions operations. The greatest challenges arise when we combine new technologies at a large scale and at complex system levels. This course will focus on power systems and control equipment for electric transportation. The course will focus on the equipment needed to support new forms of electric transportation, such as Electric cars, aircraft, and trains. **Prerequisite**: ELEC 432.

ELEC 532	**Application of Power Electronics in Power Systems**	**3(3,0)**
This course focuses on power electronics and its applications in power systems, covering the following topics: Flexible AC transmission systems (FACTS), conventional thyristor-controlled reactors and phase shifters, voltage source inverter (VSI)- based static condenser (STATCON), and unified power flow controller (UPFC). **Prerequisites**: ELEC 330; ELEC 432.		

ELEC 535	**Renewable Energy Systems for the Built Environment**	**3(3,0)**
This course covers Energy and carbon emissions in the built environment, including data on energy consumption in Lebanon. Overview of building services – HVAC, Lighting, HWS, Lifts, and Small Power; data and benchmarks for energy consumption by end use in buildings; methods for estimating predicted energy consumption in buildings; time-based energy demand schedules; the importance of energy efficiency and the Lean-Clean-Green principle. **Prerequisite**: ELEC 210.		

ELEC 536	**Renewable Energy Systems**	**3(3,0)**
This course aims to provide knowledge about energy production from renewable sources, the structure and nature of the interconnected electric power system, and the critical need for environmentally sensitive solutions. In addition, the economic and regulatory policy aspects of electricity and electricity markets are discussed. The course covers the basics of Wind energy conversion systems and microgrids with hybrid power sources. Modeling and control of renewable energy sources such as wind turbine generation, solar panels, fuel cells, and power electronics interfaces will be presented. **Prerequisite**: ELEC 431.		

ELEC 542	**Advanced Power Electronics**	**3(3,0)**
This course is designed to provide students with a functional knowledge of modeling switching power converters, advanced power converter topologies, design constraints, and control methods. It also covers the operation of multi-level DC/AC inverters and matrix converters. **Prerequisite**: ELEC 432.		

ELEC 550	**Advanced Control Systems**	**3(3,0)**
This course focuses on the formulation of the linear control problem using state-space methods, frequency-response and time-response analysis; it introduces students to advanced concepts such as controllability, observability, canonical forms, state transition matrices, stability, nonlinear system linearization, and discrete and time-invariant systems. **Prerequisite**: ELEC 451.		

ELEC 551	Nonlinear Adaptive Control	3(3,0)
This course presents a comprehensive exposition of the theory of nonlinear dynamical systems and their control. It will focus on methods for characterizing and understanding the behavior of systems described by nonlinear ordinary differential equations, as well as methods for designing controllers for such systems. In this course, both classical and modern concepts from nonlinear system theory will be introduced. **Prerequisite**: ELEC 451.		

ELEC 560	Sustainable Energy	3(3,0)
This course is designed to provide you with the methods, tools, and perspectives to understand, critique, and ultimately influence the management of technical, economic, and policy choices regarding energy generation and use. We will focus equally on the technical, economic, political, and environmental impacts of energy. **Prerequisites**: ELEC 431		

ELEC 561	Wind Energy Technology	3(3,0)
This course aims to provide the students with an in-depth understanding of the theoretical, technological, and economic aspects of wind energy systems. Upon successful completion of this course, students will be able to identify, assess, and select the types of wind turbines, estimate the power output of specific wind energy devices and systems, assess the structural suitability of wind towers, and evaluate the key aspects of onshore (urban) and offshore wind energy systems. **Prerequisite**: ELEC 210.		

ELEC 562	Solar Energy Technology	3(3,0)
This course aims to provide students with a systematic understanding of current knowledge, problems, and insights in solar photovoltaic technologies; enable students to evaluate current research and advances in the field; and assess solar PV technologies, develop critiques, and propose solutions. **Prerequisite**: ELEC 210.		

ELEC 597	Advanced Topics in Electrical Engineering	3(3,0)
This course is designed to enable students to study a specific advanced topic of interest in Electrical Engineering. The content outline for such a topic is to be determined by the instructor and approved by the department Chair. **Prerequisite**: Instructor's consent. On demand.		

MINOR IN RENEWABLE ENERGY SYSTEMS

Rationale

The College of Engineering/ECE Department offers a minor in Renewable Energy Systems (RES). The minor aims to address the expected national, regional, and international need for personal experts in renewable energy systems for the generation, transmission, and use of energy.

Program Objectives

The aims of the minor are to:

- Provide students with advanced expertise in renewable energy sources, systems, and policies;
- Provide students with the necessary tools to contribute to the fast-growing renewable energy systems sector;
- Prepare undergraduate students for graduate studies in renewable energy.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:

- Use current techniques, architectures, and tools to generate, store, and distribute energy from renewable sources;
- Design systems and tools to generate, store, and distribute renewable energy.

Program Requirements

While most relevant to electrical engineering students, this minor is offered to undergraduate RHU students in the BE or MS program, from different engineering disciplines. It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their major, interested RHU engineering students need to complete the appropriate form declaring their intent to minor in RES while completing their regular major.

To complete the RES minor, a student must

1) Declare a minor in Renewable Energy Systems by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the student's college major and the dean of the College of Engineering
3) Earn a Minor Cumulative Grade Point Average of no less than 70%.
4) Complete 18 credits of coursework as specified below.

It should be noted that overlapping courses between the student's major requirements and the renewable energy systems minor requirements are counted toward fulfilling the minor requirements. The student must complete a minimum of 3 credit hours of coursework that is not counted toward the requirement for their major or any other minor.

Career Options

The RES minor consists of six courses (18 credits), of which three are mandatory and three are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Curriculum and Program

The RES minor consists of six courses (18 credits), of which three are mandatory and three are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. **Mandatory Courses**

The three mandatory courses (9 credits) are:

ELEC 320	Electronics	3(3,0)
ELEC 431	Introduction to Renewable Energy Systems	3(3,0)
ELEC 432	Power Electronics	3(3,0)

II. **Elective Courses**

Students may choose three elective courses (9 credits) from the following list.

ELEC 532	Application of Power Electronics in Power Systems	3(3,0)
ELEC 535	Renewable Energy Systems for the Built Environment	3(3,0)
ELEC 536	Renewable Energy Systems	3(3,0)
ELEC 542	Advanced Power Electronics	3(3,0)
ELEC 560	Sustainable Energy	3(3,0)
ELEC 561	Wind Energy Technology	3(3,0)
ELEC 562	Solar Energy Technology	3(3,0)
ELEC 563	Smart Grid Technology	3(3,0)
Approved Electives from other engineering programs		
Civil Engineering		
CIVE 506	LEED Principles in Construction Management	3(3,0)
Computer and Communication Engineering		
CCEE 426	Design of Embedded Systems	3(3,0)
CCEE 564	Machine Learning and Data Mining	3(3,0)
Mechanical and Mechatronics Engineering		
MECH 453	Buildings and Energy	3(3,0)
MECH 471	Optimization in Engineering Design	3(3,0)

MECH 541	Renewable Energy	3(3,0)
MECA 581	Systems Engineering	3(3,0)

III. Courses Description

ELEC 320	**Electronics**	**3(3,0)**
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite**: ELEC 210.		

ELEC 431	**Introduction to Renewable Energy Systems**	**3(3,0)**
This course aims to provide students with information on renewable energy sources, including solar, wind, and wave energy, as well as biomass energy. This course will also illustrate how to link these sources to the grid and synchronize them to deliver electricity reliably and efficiently. **Prerequisite**: ELE210		

ELEC 432	**Power Electronics**	**3(3,0)**
This course familiarizes the students with basic power switch technology and associated electronic circuits. In this course, power electronic circuits and switching devices are studied. Their applications in AC/DC, DC/DC, DC/AC, and AC/AC converters, as well as switching power supplies, are studied. **Prerequisite:** ELEC 320.		

ELEC 560	**Sustainable Energy**	**3(3,0)**
This course is designed to provide you with the methods, tools, and perspectives to understand, critique, and ultimately influence the management of technical, economic, and policy choices regarding energy generation and use. We will focus equally on the technical, economic, political, and environmental impacts of energy. **Prerequisites**: **ELEC 431**		

ELEC 532	**Application of Power Electronics in Power Systems**	**3(3,0)**
This course focuses on power electronics and its applications in power systems, covering the following topics: Flexible AC transmission systems (FACTS), conventional thyristor-controlled reactors and phase shifters, voltage source inverter (VSI)- based static condenser (STATCON), and unified power flow controller (UPFC). **Prerequisites**: ELEC 330; ELEC 432.		

ELEC 535	**Renewable Energy Systems for the Built Environment**	**3(3,0)**
This course covers Energy and carbon emissions in the built environment, including data on energy consumption in Lebanon. Overview of building services – HVAC, Lighting, HWS, Lifts, and Small Power; data and benchmarks for energy consumption by end use in buildings; methods for estimating predicted energy consumption in buildings; time-based energy demand schedules; the importance of energy efficiency and the Lean-Clean-Green principle. **Prerequisite:** ELEC 210.		

ELEC 536	Renewable Energy Systems	3(3,0)
This course aims to provide knowledge about energy production from renewable sources, the structure and nature of the interconnected electric power system, and the critical need for environmentally sensitive solutions. In addition, the economic and regulatory policy aspects of electricity and electricity markets are discussed. The course covers the basics of Wind energy conversion systems and microgrids with hybrid power sources. Modeling and control of renewable energy sources such as wind turbine generation, solar panels, fuel cells, and power electronics interfaces will be presented. **Prerequisite**: ELEC 431.		

ELEC 542	Advanced Power Electronics	3(3,0)
This course is designed to provide students with a functional knowledge of modeling switching power converters, advanced power converter topologies, design constraints, and control methods. It also covers the operation of multi-level DC/AC inverters and matrix converters. **Prerequisite:** ELEC 432.		

ELEC 561	Wind Energy Technology	3(3,0)
This course aims to provide the students with an in-depth understanding of the theoretical, technological, and economic aspects of wind energy systems. Upon successful completion of this course, students will be able to identify, assess, and select the types of wind turbines, estimate the power output of specific wind energy devices and systems, assess the structural suitability of wind towers, and evaluate the key aspects of onshore (urban) and offshore wind energy systems. **Prerequisite:** ELEC 210.		

ELEC 562	Solar Energy Technology	3(3,0)
This course aims to provide students with a systematic understanding of current knowledge, problems, and insights in solar photovoltaic technologies; enable students to evaluate current research and advances in the field; and assess solar PV technologies, develop critiques, and propose solutions. **Prerequisite:** ELEC 210.		

ELEC 563	Smart Grid Technology	3(3,0)
This course will examine smart grid technologies and the transformational impacts of the smart grid on the industry. Students in this course will learn the fundamentals of the smart grid: its purpose and objectives, technologies, architectures, and management. Students will also learn about many of the challenges facing the smart grid as it evolves. **Prerequisite:** ELEC 432		

COMPUTER AND COMMUNICATIONS ENGINEERING PROGRAM

Mission

The CCE program aims to help students explore their innate creativity and potential and endow them with the knowledge, skills, and abilities to pursue successful careers in computer and communication engineering and related fields; think critically in solving complex problems using modern tools and technologies; communicate and work effectively with diverse groups; learn more every day, and succeed in graduate studies in renowned institutions if they so choose.

Program Educational Objectives

The program's educational objectives are to enable graduates to:

1. Foster an environment that encourages excellence in endeavor.
2. Provide quality and professional education that prepares leaders.
3. Solve problems facing society and industry in Lebanon, the Middle East, and the world.
4. Engage in multidisciplinary research.

Student Outcomes

The Computer and Communication Engineering program adopts the learning outcomes of ABET, so our graduates will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze, interpret data, and use engineering judgment to conclude.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Computer and Communications Engineering is accredited by the Engineering Accreditation Commission of ABET, https://www.abet.org, under the commission's General Criteria and Program Criteria for Computer and Communications and Similarly Named Engineering Programs.

Program Overviews

The Computer and Communications Engineering Program at Rafik Hariri University puts students' best interests first and foremost. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum tailored to market needs, facilities that meet the highest standards, top-notch faculty members who graduated from premier higher education institutions and are committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Computer and Communications Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 116 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or BE degree is conferred only if a student's earned Cumulative GPA, Major GPA, and Summative Learning Project grade are all at least 70.

The required credit hours are divided into three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS in Computer and Communication Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	3	0	14	10	27	23
College Requirement	0	0	24	0	24	20
Program Requirement	46	0	21	0	67	57
Credits	49	0	59	10	118	100

BE in Computer and Communication Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	3	0	14	13	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	53	18	24	0	95	62
Credits	56	18	65	13	152	100

I. University Requirements (General Education)

The list of GE-required courses and their descriptions is presented on the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Computer and Communications Engineering (CCE) encompass 95 credits (67 credits for the BS degree) distributed as follows: 77 credits of mandatory courses (67 credits for the BS degree) and 18 credits (0 credits for the BS degree) of engineering technical elective courses for BE only.

A. Mandatory Requirements

The mandatory courses for the major and non-major CCE programs are listed in the table below.

Course #	Title	Credits	Pre-requisites
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	CCEE 221
CCEE 216	Advanced Programming and Data Structure	3	CCEE 214
CCEE 310	Software Engineering	3	CCEE 214
CCEE 315	Database Management Systems	3	CCEE 214
CCEE 324	Computer Organization	3	CCEE 221
CCEE 331	Signals and Systems	3	MATH 216
CCEE 341	Communication Systems	3	CCEE 331
CCEE 354	Computer Networks	3	CCEE 214
CCEE 354L	Computer Networks Lab 1	1	Co-req.: CCEE 354
CCEE 400	BS Summative Learning Project	3	ENGL 217
CCEE 411	Web Programming	3	CCEE 214

CCEE 426	Design of Embedded Systems	3	Pre-req.: CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE 444	Antennas and Propagation	3	Pre-req.: ELEC 340
CCEE 444L	Antennas and Propagation Lab	1	Co-req.: CCEE 444
CCEE 447	Digital Communications	3	Pre-req.: CCEE 341
CCEE 447L	Communications Lab	1	Co-req.: CCEE 447
CCEE 449	Wireless Communications	3	Pre-req.: CCEE 447
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
CCEE 499	Co-op Experience	3	Pre-req.: ENGL 217
CCEE 528	Computer Architecture	3	Pre-req.: CCEE 324
CCEE 534	Signal Processing	3	Pre-req.: CCEE 331
CCEE 543	Mobile Communications	3	Pre-req.: CCEE 449
CCEE 564	Machine Learning and Data Mining	3	Pre-req.: CCEE 214; MATH 351 or BADM 250
CCEE 595A	BE Summative Learning Project 1	1	Pre-req.: ENGL 217
CCEE 595B	BE Summative Learning Project 2	3	Pre-req.: CCEE 595A
CCEE	CCEE Selected Elective	3	
ELEC 200	Introduction to Electrical and Computer Engineering	3	
ELEC 320	Electronics	3	Pre-req.: ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 340	Engineering Electromagnetics	3	Pre-req.: PHYS 211

* Required for BS-bound students. BE students take technical electives instead.

B. Technical Electives

As part of the Bachelor of Engineering degree program in CCE, the student is required to take 18 credit hours of engineering technical elective courses. Students must be of senior standing to take engineering technical electives. Technical electives allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match engineering technical elective courses from different areas to gain broader exposure to the various CCE disciplines. Students should, in cooperation with their academic advisor, select a list of electives that best meet their needs and aspirations. The listed engineering technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

1. Computer Software
2. Computer Hardware
3. Signal Processing
4. Communication Systems
5. Computer Networks
6. Artificial Intelligence

The list of additional available CCE electives is given in the table below:

Course #	Title	Credits	Pre-/Co-requisites
CCEE 498	Special Topics in Computer and Communication Engineering	3	
CCEE 597	Advanced Topics in Computer and Communication Engineering	3	Senior Standing
Computer Software			
CCEE 511	Mobile Application Development	3	Pre-req.: CCEE 214
CCEE 513	Operating Systems	3	Pre-req.: CCEE 214
CCEE 514	Advanced Web Programming	3	Pre-req.: CCEE 411
CCEE 515	Distributed Computing	3	CCEE 528
CCEE 516	Advanced Programming and Database Management Systems	3	Pre-req.: CCEE 315 or equivalent
Computer Hardware			
CCEE 520	Parallel Processing	3	Pre-req.: CCEE 221
CCEE 521	Hardware-Software Co-Design	3	Pre-req.: CCEE 221
Signal Processing			
CCEE 535	Digital Signal Processing	3	Pre-req.: CCEE 331
CCEE 536	Digital Image Processing	3	Pre-req.: CCEE 214
Communication Systems			
CCEE 540	Satellite Communication Systems	3	Pre-req.: CCEE 447
CCEE 542	Microwave Communications	3	Pre-req.: ELEC 340
CCEE 545	Advanced Mobile Communications	3	Co-requisite: CCEE 449
CCEE 546	Array Processing	3	Pre-req.: CCEE 544
CCEE 547	Optical Communications	3	Pre-req.: CCEE 447
Computer Networks			
CCEE 457	Cybersecurity Fundamentals	3	Pre-req.: CCEE 354
CCEE 552	Network Security	3	Pre-req.: CCEE 354
CCEE 553	Advanced Communication Networks	3	Pre-req.: CCEE 354
CCEE 554L	Computer Networks Lab 2	1	Co-req.: CCEE 354L
CCEE 555	Advanced Computer Networks	3	Pre-req.: CCEE 354
CCEE 555L	Computer Networks Lab 3	1	Pre-req.: CCEE 354L
CCEE 556	Network and System Administration	3	Pre-req.: CCEE 354
CCEE 556L	Computer Networks Lab 4	1	Pre-req.: CCEE 555L
Artificial Intelligence			
CCEE 561	Computer Vision	3	Pre-req.: CCEE 214
CCEE 563	Robotics	3	
CCEE 566	Natural Language Processing	3	Pre-req.: CCEE 214
CCEE 567	Optimization	3	Pre-req.: MATH 215 or equivalent
CCEE 568	Big Data and Analytics/Big Data System	3	Pre-req.: CCEE 564 or equivalent

RHU is a CISCO Academy. As such, students will be awarded CISCO certificates after completing the required Computer Networks Lab 1. In addition, students can take any three of the following labs, which will count as one technical elective.

Course #	Title	Credits	Pre-/Co-requisites
Computer Networks			
CCEE 554L	Computer Networks Lab 2	1	Co-req.: CCEE 354L
CCEE 555L	Computer Networks Lab 3	1	Pre-req.: CCEE 354L
CCEE 556L	Computer Networks Lab 4	1	Pre-req.: CCEE 555L

Undergraduate students can also take 600-level courses, provided that their cumulative GPA is above 80 and the instructor approves.

Students may up to 9 credits of (400-level or above) engineering technical electives from other programs,.

C. Summative Learning Project (SLP)

Students must complete a 3-credit-hour course for BS (4 credits for BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to innovate a promising idea.

D. Co-op Experience

Students must complete 8 weeks of practical training working in an area related to their field of interest. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on the student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

Code numbers designate the courses offered in the CCE program in the form of (CCEE abc), where:

a	Designates the year (level)
b	Designates the focus area as follows: 0: General; 1: Computer Software; 2: Computer Hardware; 3: Signal Processing; 4: Communication Systems; 5: Computer Networks; 6: Artificial Intelligence
c	Designates the course sequence in a focus area

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in CCE. The study plan serves as a roadmap to facilitate students' smooth progression toward graduation.

Course #	Title	Credits	Pre-/Co-requisites
Year 1, Fall Semester (14 Credits)			
CCEE 214	Introduction to Programming	3	
ARAB 212	Arabic Language & Communication	2	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 211	Physics: Electricity and Magnetism and Lab	3	
Year 1, Spring Semester (15 Credits)			
CCEE 216	Advanced Programming and Data Structure	3	CCEE 214
CCEE 221	Logic Design	3	
ELEC 200	Introduction to Electrical and Computer Engineering	3	
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
Year 1, Summer Semester (10 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
MATH 311	Linear Algebra and Applications	3	
	Science Elective	3	
	Humanities/ Fine Arts Elective I	3	
Year 2, Fall Semester (17 Credits)			
CCEE 324	Computer Organization	3	CCEE 221
CCEE 331	Signals and Systems	3	MATH 216
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ENGL 217	Professional English Communication	3	ENGL 210
CCEE 354	Computer Networks	3	Co-req.: CCEE 221
CCEE 354L	Computer Networks Lab 1	1	Co-req.: CCEE 354
Year 2, Spring Semester (16 Credits)			
CCEE 310	Software Engineering	3	CCEE 214
CCEE 315	Database Management Systems	3	CCEE 214
CCEE 341	Communication Systems	3	CCEE 331
ELEC 340	Engineering Electromagnetics	3	PHYS 211
MATH 314	Ordinary Differential Equations	3	MATH 216
	Community and Sustainability Elective	1	
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economy and Management	3	

MATH 421	Numerical Analysis	3	MATH 311
	Social Science Elective I	3	
Year 3, Fall Semester (17 Credits)			
CCEE 411	Web Programming	3	Co-req.: CCEE 214
CCEE 447	Digital Communications	3	CCEE 341
CCEE 447L	Communications Lab	1	Co-req.: CCEE 447
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE	Selected CCEE Elective	3	
MATH 351	Probability and Statistics	3	MATH 216
Year 3, Spring Semester (17 Credits)			
CCEE 444	Antennas and Propagation	3	ELEC 340
CCEE 444L	Antennas and Propagation Lab	1	Co-req.: CCEE 444
CCEE 449	Wireless Communications	3	Co-req.: CCEE 447
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
MATH 210	Discrete Mathematics	3	MATH 215
	Engineering Technical Elective 1 or BS Project*	3	
Year 3, Summer Semester (3 Credits)			
CCEE 499	Co-op Work Experience	3	ENGL 217
The above 118 credits complete the requirements for a Bachelor of Science degree in Computer and Communication Engineering.			
Year 4, Fall Semester (16 Credits)			
CCEE 534	Signal Processing	3	CCEE 331
CCEE 528	Computer Architecture	3	CCEE 324
CCEE 595A	BE Summative Learning Project 1	1	ENGL 217; Senior Standing
ENGR 510	Engineering Project Management	3	ENGR 300
	Engineering Technical Elective 2	3	
	Engineering Technical Elective 3	3	
Year 4, Spring Semester (18 Credits)			
CCEE 543	Mobile Communications	3	Co-req.: CCEE 449
CCEE 564	Machine Learning and Data Mining	3	CCEE 214; MATH 351 or BADM 250
CCEE 595B	BE Summative Learning Project 2	3	CCEE 595A
	Engineering Technical Elective 4	3	
	Engineering Technical Elective 5	3	
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take CCEE 400 – Summative Learning (BS) Project instead.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
9 credits restricted Departmental Electives
9 credits in any Engineering discipline at the 400 level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

A description of the non-major mandatory courses follows.

ELEC 200	Introduction to Electrical and Computer Engineering	3(3,0)
In this course, the students will be trained in core skills to equip them to excel in electrical, biomedical, and computer and communications engineering. The student will be trained in application programming using an appropriate programming environment. Students will build on the basics of Programming learned in the introductory programming course to implement Digital Signal Processing applications and perform data analysis. Students will also be introduced to basic lab equipment such as Oscilloscopes, Function Generators, and Multi-Meters. Students will be introduced to microcontroller-based projects to further explore the Electrical and Computer Engineering majors (CCE, BIOM, and ELEC). Throughout the course, students will work in multidisciplinary teams to study engineering cases for the community and implement a project to address a community need.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 320	Electronics	3(3,0)
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	Electronics Lab	1(0,2)
This is a Lab course with experiments in Electrical and Electronic Circuits. It covers passive electrical elements and sources; lab instruments; voltage divider circuits; Thevenin's theorem; RC circuits; diode circuits; Op-Amp circuits; BJT and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ELEC 340	Engineering Electromagnetics	3(3,0)
Introduction to vector analysis; Electrostatic fields in vacuum and dielectrics; Conductors; Capacitance; Electrostatic energy and forces; Poisson's equation. Magnetic fields, Maxwell's equations, static electric and magnetic fields, boundary-value problems, Laplace's and Poisson's equations, and static electromagnetic fields. Time-dependent Maxwell's equations and Plane-wave propagation. **Prerequisite**: PHYS 211.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting and financial statements.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, and structural induction, sets and set operations, functions, growth of functions, relations, and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar		

curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

Major Courses

Mandatory computer and communications engineering courses are described below.

CCEE 214	Introduction to Programming	3(2,2)
This course presents the fundamentals of structured programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to COSC 214.		

CCEE 216	Advanced Programming and Data Structures	3(2,2)
This course covers advanced object-oriented programming concepts, including overloading, inheritance, and polymorphism. In addition, the course covers data structure concepts, including analysis, sorting, and searching algorithms, as well as stacks, queues, trees, and graphs. **Prerequisite**: CCEE 214. Equivalent to COSC 215.		

CCEE 221	Logic Design	3(3,0)
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers; the design of sequential logic devices, including flip-flops, registers, and counters, as well as the analysis of devices used to build logic networks, including programmable logic devices. **Equivalent to** COSC 351.		

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Co-requisite**: CCEE 221. **Equivalent to** COSC 351L.		

CCEE 310	Software Engineering	3(3,0)
Analysis, design, construction, maintenance, and evolution of large software systems are covered. Students are introduced to the system life cycle, project management techniques, and database systems. Analysis, design, and implementation of software systems are also included. **Prerequisite**: CCEE 214. **Equivalent to** COSC 341.		

CCEE 315	Database Management Systems	3(3,0)
The course covers the nature and purpose of database systems; an introduction to data modeling: the Entity-Relationship Model and the Relational Model, including relational algebra, relational calculus, and SQL; integrity constraints; file organization and index files; and normalization. It also covers an introduction to object databases, data mining, schema evolution, distributed databases, web-enabled databases, and databases for e-commerce applications. **Prerequisite**: CCEE 214. **Equivalent to** COSC 231.		
CCEE 324	**Computer Organization**	**3(3,0)**

Computer system organization and design; implementation of CPU data path and control; instruction set design; memory hierarchy (caches, main memory, virtual memory) organization and management; input/output subsystems (bus structures, interrupts, DMA); performance evaluation; and pipelined processors. **Prerequisite**: CCEE 221. **Equivalent to** COSC 353.

CCEE 331	**Signals and Systems**	**3(3,0)**
This course covers the mathematical description and classification of continuous and discrete signals and systems. Topics include types of signals and systems, Fourier series, Fourier transforms, Discrete-Time Fourier Transforms (DTFT), Discrete and Fast Fourier Transforms (DFT and FFT), Laplace Transforms, Z-Transforms, and transfer functions. **Prerequisite:** MATH 216		

CCEE 341	**Communication Systems**	**3(3,0)**
This course introduces the fundamentals of transmission and reception in communication systems, as well as the effects of noise. Topics include power spectral density; amplitude modulation and demodulation; angle modulation and demodulation; analog communication system performance in the presence of noise; sampling and analog-to-digital conversion; and an introduction to digital modulation. **Prerequisite**: CCEE 331.		

CCEE 354	**Computer Networks**	**3(3,0)**
This course enables students to gain fundamental knowledge of computer networks, appreciate the trade-offs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field. **Co-requisite**: CCEE 221. Equivalent to COSC 360.		

CCEE 354L	**Computer Networks Lab I**	**1(0,2)**
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. **Co-requisite**: CCEE 354.		

CCEE 400	**Summative Learning (BS) Project**	**3(3,0)**
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program.		

CCEE 411	**Web Programming**	**3(3,0)**
This course teaches students how to develop and implement web-based programs with an emphasis on interface programming. It introduces students to web development and the client-side and server-side languages and technologies needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, PHP, and responsive design. **Co-requisite:** CCEE 214. Equivalent to COSC 333.		

CCEE 411L	Internet Engineering and Web Programming Lab	1(0,2)
This lab provides students with hands-on skills in Web programming. Experiments mainly cover the following topics: HTML and CSS, JavaScript and JQuery, Responsive web design, Bootstrap, Web content management systems, and server-side development. **Co-requisite:** CCEE 411.		

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

CCEE 426L	Design of Embedded Systems Lab	1(0,2)
This lab complements topics covered in the CCEE 426 course. It involves the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Experiments include Design using finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, scheduling, system-level simulation, and interface-based design. **Co-requisite**: CCEE 426. Equivalent to COSC 356L.		

CCEE 444	Antennas and Propagation	3(3,0)
This course introduces the fundamental principles of antennas and propagation for communications-oriented electrical engineers. Topics include: fundamental antenna parameters, wire antennas, analysis and design of linear antenna arrays, radio wave propagation, free-space path loss, and link budget. **Prerequisite**: ELEC 340.		

CCEE 444L	Antennas and Propagation Lab	1(0,2)
This lab provides students with hands-on experience in simulating and designing basic antennas using a microwave computational tool. It also includes a demonstration of basic antenna performance measurements. **Co-req.**: CCEE 444.		

CCEE 447	Digital Communications	3(3,0)
This course introduces the fundamentals of modern digital communication systems. Topics include: deterministic and random signal analysis; Channels; signal space representation; Maximum Likelihood Detection; Coherent and Non-Coherent Digital Modulation Schemes (ASK, PSK and M-PSK, FSK and M-FSK, M-QAM); error probabilities; bandwidth efficiency and energy efficiency tradeoffs; inter-symbol interference (ISI) and pulse shaping; matched filter and optimal receiver; source coding. **Prerequisite**: CCEE 341.		

CCEE 447L	Communications Lab	1(0,2)
This lab taken with CCEE 447. The student will acquire knowledge on implementing and analyzing analog and digital communication and systems using the NI EMONA Add-in Module and the NI USRP. analog and digital modulation, sampling and reconstruction, coding, and **Co-requisite:** CCEE 447.		

CCEE 449	Wireless Communications	3(3,0)
Wireless communications are omnipresent in today's world. It does not transport only conversational information; it is used for a plethora of tasks, from short-range exchange of photos over Bluetooth to satellite communication and deep-space information transfer, from a few bits used to control robots and for sensory information to big data collection. This course introduces current wireless systems and standards, shadowing and multipath fading effects in wireless communications, radio channel capacity, multiple access techniques, and channel coding methods. **Co-requisite:** CCEE 447.		

CCEE 460	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 460L	Artificial Intelligence Lab	1(0,2)
This lab will help prepare students to handle the huge amount of data produced by both humans and machines today. The student will learn to use state-of-the-art AI tools to analyze data and make complex decisions based on that data using real-world practical examples and case studies. At the end of the lab, the student will be able to apply AI knowledge in their field at work or in day-to-day life to make better decisions. **Co-req.:** CCEE 460.		

CCEE 499	Co-op Work Experience	3(3,0)
This Co-op work experience is designed to provide students with an eight-week short-term work experience in the field of computer and communication engineering. Students are encouraged to network with industry professionals in their discipline. A report, poster, and PowerPoint presentation are required. **Prerequisites**: Senior Standing. ENGL 217.		

CCEE 528	Computer Architecture	3(3,0)
Computer system organization and design; implementation of CPU data path and control; instruction set design; memory hierarchy (caches, main memory, virtual memory) organization and management; input/output subsystems (bus structures, interrupts, DMA); performance evaluation; pipelined processors. **Prerequisite**: CCEE 324. **Equivalent to** COSC 353.		

CCEE 534	Signal Processing	3(3,0)
This course covers time series analysis, frequency analysis, time-frequency analysis, and time-scale analysis. It also covers the design of digital filters and signal modeling. **Prerequisite**: CCEE 331.		

CCEE 543	Mobile Communications	3(3,0)
This course deals with the evolution of cellular technologies from 2G to LTE: cellular systems; medium access techniques; wireless standardization; GSM networks (history, architecture overview, access network, GSM air interface, dimensioning aspects, network design and planning, handover, call handling, network operation); 3G mobile systems architecture, 3G air interface, protocols and procedures used in 3G, 3G functionalities such as: idle mode camping, power control and capacity management, handover in 3G; introduction to 4G/LTE. **Co-requisite**: CCEE 449.		

CCEE 562	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 564	Machine Learning and Data Mining	3(3,0)
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process and applying the corresponding approaches to solve practical problems and develop intelligent software applications. The course covers several topics that lie within classification, prediction, and clustering. **Prerequisite**: CCEE 214. Equivalent to COSC 480.		

CCEE 595A	Summative Learning Project 1	1(1,0)
The first phase of a team-oriented, project-based experience culminates in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisites**: ENGL 217, Senior Standing.		

CCEE 595B	BE Summative Learning Project 2	3(3,0)
The second phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite:** CCEE 595A.		

II. Elective Courses

Students may meet their engineering technical electives requirements from the following selection of elective courses.

CCEE 457	Cybersecurity Fundamentals	3(3,0)
This course equips learners with workforce preparedness skills and establishes a foundation for success in cybersecurity. Participants acquire, apply, and refine cybersecurity knowledge and skills through a sequence of comprehensive, practical experiences that solidify their learning. This course imparts fundamental cybersecurity		

concepts and abilities at the introductory level, encompassing threat reduction, defense strategies, and post-incident forensics. **Prerequisite: CCEE 354**

CCEE 498	Special Topics in Computer & Communication Engineering	3(3,0)
This course is designed to enable students to study current special topics of interest, carefully selected from CCE-related topics. The contents of such a course are to be determined by the instructor and the department.		

CCEE 511	Mobile Application Development	3(3,0)
Today's applications are increasingly mobile. Computers are no longer confined to desks and laps; they live in our pockets and hands. This course teaches students how to build mobile apps for Android and iOS, two of today's most popular platforms, and how to deploy them in the Android Market and the App Store. Students learn how to write native apps for Android using Eclipse and the Android SDK, for iPhones, iPod touches, and iPads using Xcode and the iOS SDK, and for web apps on both platforms. **Prerequisite:** CCEE 214. Equivalent to COSC 435.		

CCEE 513	Operating Systems	3(3,0)
This course covers operating systems principles; scheduling and resource management; virtual memory; file systems; concurrent processing and synchronization; security and protection; the Internet; network structures; distributed operating systems; Web technologies; and operating systems (URL, HTML, HTTP, applets). A case study of a UNIX operating system is covered during the course. **Prerequisite**: CCEE 214. Equivalent to COSC 451.		

CCEE 514	Advanced Web Programming	3(3,0)
This course focuses on server-side programming. It teaches students how to connect their website or web application to a database and save and retrieve data from it. The course exposes students to web controls, validation controls, data source controls, data binding controls, state management, and working with third-party media such as XML and web services. **Prerequisite**: CCEE 411/COSC 333. Equivalent to COSC 434.		

CCEE 515	Distributed Computing	3(3,0)
This course will introduce students to the challenges faced in constructing parallel and distributed applications, including testing, debugging, and performance evaluation. The student will be trained in various implementation techniques, paradigms, architectures, and programming languages. **Prerequisite**: CCEE 528 or equivalent.		

CCEE 516	Advanced Programming and Database Management Systems	3(2,2)
This course will introduce students to managing their own databases and querying them, as well as to managing Data Warehousing. Students will also learn advanced programming tools, including Bigtable, NoSQL, R, Python, SCALA, MapReduce, Elasticsearch, and apply these tools to address big data issues. **Prerequisite**: CCEE 315 or equivalent.		

CCEE 520	Parallel Processing	3(3,0)
This course introduces students to various aspects of parallel processing, including parallel architectures, algorithms, systems, programming languages, and implementations. Students will be expected to work with recent existing parallel machines and design and implement parallel processing projects. **Prerequisite**: CCEE 221.		

CCEE 521	Hardware-Software Co-Design	3(3,0)
This course introduces the student to the design of complex embedded systems. Several models and methods are presented to help the student design software and hardware components that work together. Models to describe hardware and software components are introduced (specifications). Then, hardware-software partitioning and design space exploration are introduced as part of the system design. Performance analysis and estimation techniques are then explained. Finally, a current hardware-software co-design topic is chosen and introduced. Students will be expected to design and implement a project using the information in this course. **Prerequisite**: CCEE 221.		

CCEE 535	Digital Signal Processing	3(3,0)
This course focuses on digital signal processing (DSP) and its applications. It helps students understand how digital signal processing can be applied in operational systems. Students are required to develop simulations of the learned concepts using MATLAB. Specific topics covered include a review of mathematical tools used in DSP, digital filter structure, digital filter design, simple DSP algorithm implementation, and spectral analysis of signals. **Prerequisite**: CCEE 331.		

CCEE 536	Digital Image Processing	3(3,0)
Two-dimensional signals and systems. Image formation and perception. Representation, coding, filtering, restoration, and enhancements. Feature extraction and scene analysis. Introduction to computer vision. Introduction to Medical Imaging. **Prerequisite:** CCEE 331 or Equivalent.		

CCEE 540	Satellite Communication Systems	3(3,0)
This course focuses on the fundamentals of satellite technology. Topics include the history of satellites; satellite missions and applications; the study of satellite orbits and trajectories; satellite launch and in-orbit operations; satellite link fundamentals; satellite hardware; Communication Satellites; GPS; and Direct Broadcast Satellites. **Prerequisite**: CCEE 447.		

CCEE 542	Microwave Communications	3(3,0)
This course introduces the fundamental principles of microwave circuit analysis and design. Topics include review of electromagnetic theory, Transmission Line Theory, Impedance Matching and Tuning, Microwave Network Analysis, Power Dividers, and Microwave Propagation. This course also includes some laboratory experiments on microwave circuits—prerequisite: ELEC 340.		

CCEE 545	Advanced Mobile Communications	3(3,0)
This course seeks to provide insight into the development of the latest generations of mobile networks. In this course, we will focus mainly on LTE and LTE-Advanced (LTE-A): Motivation, architecture, air interface, radio procedures, quality of service, and planning. In addition, we will briefly discuss 5G networks and the Internet of Things. **Co-requisite**: CCEE 449.		

CCEE 546	Array Processing	3(3,0)
This course includes an introduction to Mono- and Multi-Antenna Concepts: multi-antenna benefits; beamforming techniques; adaptive antennas; MIMO systems: diversity/multiplexing tradeoff; multi-antenna transmission techniques (Alamouti and spatio-temporal techniques); multi-antenna reception techniques; transmission strategies; applications (MIMO in 3G and LTE). **Prerequisite**: CCEE 444.		

CCEE 547	Optical Communications	3(3,0)
This course is designed to progress from the description of the components in a fiber link to the interconnections within a link or a network. The components of fiber links will be discussed (fiber, connections, sources, and receivers). These optical components will be integrated into a complete optical link. **Prerequisite**: CCEE 331.		

CCEE 552	Network Security	3(3,0)
Network security is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft. It involves creating a secure infrastructure for devices, applications, and users to operate securely. **Pre-requisite**: CCEE 354 or COSC 360.		

CCEE 553	Advanced Communication Networks	3(3,0)
In this course, we will explore current network architectures and topologies, such as telephone systems (PDH/SDH, SONET), Frame Relay, ATM, and MPLS, as well as technologies under development. We will also learn how to assess network performance and which technologies are used to enhance service quality. At the end of the course, the student will be able to identify the technologies to be used in a given environment and study its performance. **Prerequisite**: CCEE 354.		

CCEE 554L	Computer Networks Lab 2	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: CCEE 354L.		

CCEE 555	Advanced Computer Networks	3(3,0)
This is an advanced networking course that requires active student participation. In this course, we will explore TCP; Internet inter-domain routing; Internet quality of service and traffic engineering; Internet traffic measurement and analysis; data		

centers; the analysis and performance of content distribution networks; and Software-defined networking. In addition, the student will learn how to build client/server computer network applications using sockets. **Prerequisite**: CCEE 354.

CCEE 555L	Computer Networks Lab 3	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 2) Certification. The primary focus of this Lab is routers and switches in small networks. The student will learn the architecture, components, and operations of routers and switches, as well as their basic configuration and functionality. They will also learn how to configure and troubleshoot static and dynamic routing protocols, access-control lists, VLANs, DHCP for IPv4 and IPv6 networks, and NAT operations. **Prerequisite**: CCEE 354L.		

CCEE 556	Network and System Administration	3(3,0)
This course will help students become system administrators and install, configure, maintain, and troubleshoot a Microsoft Windows Server Operating system. It provides them with the knowledge and skills to work with Active Directory technologies in environments including complex network services. This course also provides students with basic knowledge of installing, configuring, and maintaining Linux operating systems. By the end of this course, students will have all the required materials to submit and pass the MCSA: Windows Server certification exam. **Prerequisite**: CCEE 354.		

CCEE 556L	Computer Networks Lab 4	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn how to manage CISCO IOS software licensing and configuration files. **Prerequisite**: CCEE 555L.		

CCEE 561	Computer Vision	3(3,0)
This course introduces the principles, models, and applications of computer vision. The course will cover image structure, projection, stereo vision, and the interpretation of visual motion. Case studies of industrial (robotic) applications of computer vision, including visual navigation for autonomous robots, robot hand-eye coordination, and novel man-machine interfaces. **Prerequisite**: CCEE 214 or COSC 214.		
CCEE 563	**Robotics**	**3(3,0)**
Introductory historical development of robotics; rigid-object kinematics; robot arm kinematics; inverse kinematics; dynamics; introduction to trajectory planning and manipulator control. **Prerequisite**: MATH 311 and MATH 314. Equivalent to MECA 524.		

CCEE 566	Natural Language Processing	3(3,0)
This course introduces students to the field of natural language processing (NLP). The student is first introduced to word and sentence tokenization. The student then uses the learned skills to implement systems for text classification and sentiment analysis, spelling correction, information extraction, parsing, meaning extraction, and question answering, Machine learning algorithms as well as algorithms like n-gram language modeling, naive bayes and maxent classifiers, sequence models like Hidden Markov Models, probabilistic dependency and constituent parsing, and vector-space models of meaning will be introduced as needed for the above NLP applications. **Prerequisite**: CCEE 214, COSC 214, or Equivalent.		

CCEE 567	Optimization	3(3,0)
This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. It includes a team project in which students select and solve a problem in practice. **Prerequisite**: MATH 215 or equivalent. Equivalent to: BADM 420 and MECH 571.		

CCEE 568	Big Data and Analytics/Big Data System	3(3,0)
This course enables students to understand why the Big Data Era has emerged. Students will become conversant in the terminology and core concepts of big data problems, applications, and systems. Students will learn how to make Big Data useful in their business or career. Students will be introduced to one of the most common frameworks, Hadoop, that has made big data analysis easier and more accessible -- increasing the potential for data to transform our world. **Prerequisite**: CCEE 564 or equivalent.		

CCEE 597	Advanced Topics in Computer & Communications Engineering	3(3,0)
This course is designed to enable students to study a selected advanced topic of interest, carefully chosen from the CCE engineering-related topics. The content outline of such a topic is to be determined by the instructor and approved by the department. **Prerequisite**: Instructor's consent.		

MINOR IN ARTIFICIAL INTELLIGENCE

Rationale

The College of Engineering/ECE Department offers a minor in Artificial Intelligence (AI). The minor aims to meet the anticipated national, regional, and international demand for experts in Artificial Intelligence, Knowledge Representation, Data Mining, Machine Learning, and Automated Reasoning.

Minor Objectives

The aims of the minor are to:
1. Provide students with advanced expertise in AI.
2. Prepare students to apply AI in industry.
3. Prepare students for graduate studies in AI.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:
1. Use current techniques, architectures, and tools to build intelligent systems.
2. Design systems and tools to evaluate the performance of intelligent systems.
3. Apply AI tools and techniques to improve the performance of existing non-AI-enabled systems.

Minor Requirements

While most relevant to students in Computer and Communications Engineering and Computer Science, this minor is also offered to undergraduate RHU students in the BE or MS program from other disciplines (Engineering and Computer Science). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their major, interested RHU engineering students need to complete the appropriate form declaring their intent to minor in AI while completing their regular major.

To complete the AI minor, a student must:
1. Declare a minor in Artificial Intelligence by completing the Minor Declaration Form.
2. Obtain the approval of the dean of the student's college and the dean of the College of Engineering.
3. Earn a Minor Cumulative Grade Point Average of no less than 70 %.
4. Complete 18 credits of coursework as specified below.

It should be noted that overlap among student major requirements, other minor requirements, and AI minor requirements is counted toward fulfilling the minor

requirements. The student must complete a minimum of 3 credit hours of coursework that is not counted toward the requirement for their major or any other minor.

Career Options

This minor allows its holders to pursue careers in a variety of sectors, regardless of their major. These sectors include, but are not limited to, communications, computers, computer networks, environmental engineering, energy, smart cities, autonomous vehicles, biomedical engineering, health, security, biology, and physics.

Curriculum and Program

The Artificial Intelligence minor consists of six courses (18 credits), two of which are mandatory and four electives selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Mandatory Courses

The three mandatory courses (9 credits) are:

CCEE 460	Artificial Intelligence	3(3,0)
CCEE 564	Machine Learning and Data Mining	3(3,0)

Elective Courses

Students may choose three elective courses (9 credits) from the following list.

CCEE 516	Advanced Programming and Database Management Systems	3(2,2)
CCEE 536	Digital Image Processing	3(3,0)
CCEE 561	Computer Vision	3(3,0)
CCEE 566	Natural Language Processing	3(3,0)
CCEE 567	Optimization	3(3,0)
CCEE 568	Big Data and Analytics/Big Data Systems	3(3,0)
CCEE 612	Advanced Data Mining	3(3,0)

Courses Description

CCEE 460	**Artificial Intelligence**	**3(3,0)**
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search.		

The Python language libraries will also be introduced. **Prerequisite**: CCEE 214 or equivalent.

CCEE 516	**Advanced Programming and Database Management Systems**	**3(2,2)**
This course will introduce students to database management and querying, as well as to data warehousing. Students will also learn advanced programming tools, including Bigtable, NoSQL, R, Python, Scala, MapReduce, and Elasticsearch, and apply them to address big data challenges. **Prerequisite**: CCEE 315 or equivalent.		

CCEE 536	**Digital Image Processing**	**3(2,2)**
Two-dimensional signals and systems. Image formation and perception. Representation, coding, filtering, restoration, and enhancements. Feature extraction and scene analysis. Introduction to computer vision. Introduction to Medical Imaging. **Prerequisite**: CCEE 214		

CCEE 561	**Computer Vision**	**3(3,0)**
This course introduces the principles, models, and applications of computer vision. The course will cover image structure, projection, stereo vision, and the interpretation of visual motion. Case studies of industrial (robotic) applications of computer vision, including visual navigation for autonomous robots, robot hand-eye coordination, and novel man-machine interfaces. **Prerequisite**: CCEE 214 or COSC 214.		

CCEE 563	**Robotics**	**3(3,0)**
Introductory historical development of robotics; rigid-object kinematics; robot arm kinematics; inverse kinematics; dynamics; introduction to trajectory planning and manipulator control. **Equivalent to**: MECA 544. **Prerequisite**: None		

CCEE 564	**Machine Learning and Data Mining**	**3(3,0)**
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process and applying the corresponding approaches to solve practical problems and develop intelligent software applications. The course covers several topics related to classification, prediction, and clustering. **Prerequisite:** CCEE 214		

CCEE 566	**Natural Language Processing**	**3(3,0)**
This course introduces students to the field of natural language processing (NLP). The student is first introduced to word and sentence tokenization. The student then uses the learned skills to implement systems for text classification and sentiment analysis, spelling correction, information extraction, parsing, meaning extraction, and question answering, Machine learning algorithms as well as algorithms like n-gram language modeling, naive bayes and maxent classifiers, sequence models like Hidden Markov Models, probabilistic dependency and constituent parsing, and vector-space models of meaning will be introduced as needed for the above NLP applications. **Prerequisite**: CCEE 214		

CCEE 567	Optimization	3(3,0)
This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. It includes a team project in which students select and solve a problem in practice. **Equivalent to**: BADM 420 and MECH 571. **Prerequisite**: MATH 215 or equivalent.		

CCEE 568	Big Data and Analytics/Big Data System	3(3,0)
This course enables students to understand why the Big Data Era has emerged. Students will become conversant in the terminology and core concepts of big data problems, applications, and systems. Students will learn how to make Big Data useful in their business or career. Students will be introduced to one of the most common frameworks, Hadoop, which has made big data analysis easier and more accessible -- increasing the potential for data to transform our world. **Prerequisite**: CCEE 564 or equivalent.		

Minor in Computer Networks

Rationale

In today's world, we cannot imagine a company without computer networks. Therefore, networking skills are becoming a valuable asset for employees across domains.

In response to national, regional, and international needs and demand for experts in computer networks, the College of Engineering/ECE Department and the College of Arts and Sciences/CS Department offer a minor in Computer Networking.

Program's Purpose

The main aim of the minor is to expose students from different backgrounds to the concepts of computer networks and the protocols that allow devices and systems to communicate. It will equip them with the knowledge and skills needed to install and operate networks at various scales. Furthermore, the minor will enable students to earn the Cisco CCNA Routing and Switching certification. The minor shall also serve as good preparation for further and more advanced graduate and/or professional studies in computer networks.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:

- Compare different computer network protocols;
- Design simple computer networks;
- Perform simple computer network troubleshooting;
- Investigate simple computer network attacks.

Minor Requirements

This minor is structured to accommodate undergraduate RHU students in the BE or BS program from different disciplines (engineering, computer science, or business). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their studies, interested RHU students need to declare their intent to pursue a minor in computer networks by completing the relevant minor declaration form and informing their advisor.

To complete the computer networking minor, a student must

1. Declare a minor in computer networks by completing the Minor Declaration Form;
2. Obtain the approval of the Dean of the student's college major and the Dean of the College of Engineering;

3. Earn a Minor Cumulative Grade Point Average (MCGPA) of no less than 70%;
4. Complete 19 credits of coursework as specified below.

It should be noted that common courses between the student's major requirements and the computer networks minor requirements count toward both the major and the minor. The student must complete a minimum of 3 credit hours of coursework that is not counted toward the requirement for their major or any other minor.

Career Options

Computer network experts are well-suited to work in a variety of sectors, including but not limited to telecommunications, hospitals, banks, the police, and the army.

Curriculum and Program

The computer networking minor consists of 5 courses and 4 labs (19 credits), of which 7 credits are mandatory, and the remaining credits are elective, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Mandatory Courses

The mandatory courses (7 credits) are:

CCEE 354	Computer Networks	3(3,0)
CCEE 354L	Computer Networks Lab	1(0,2)
CCEE 554L	Computer Networks Lab 2	1(0,2)
CCEE 555L	Computer Networks Lab 3	1(0,2)
CCEE 556L	Computer Networks Lab 4	1(0,2)

Elective Courses

Students may choose elective courses (12 credits) from the following list.

CCEE 411	Web Programming	3(3,0)
CCEE 514	Advanced Web Programming	3(3,0)
CCEE 547	Optical Communications	3(3,0)
CCEE 552	Network Security	3(3,0)
CCEE 553	Advanced Communication Networks	3(3,0)
CCEE 555	Advanced Computer Networks	3(3,0)
CCEE 556	Network and System Administration	3(3,0)

Courses Description

CCEE 411	Web Programming	3(3,0)
This course teaches students how to develop and implement web-based programs with an emphasis on interface programming. It introduces students to web development and the client-side and server-side languages and technologies needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, PHP, and responsive design. **Co-requisite:** CCEE 214. Equivalent to COSC 333.		

CCEE 354	Computer Networks	3(3,0)
This course enables students to gain fundamental knowledge of computer networks, appreciate the trade-offs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field. **Co-requisite**: CCEE 221. Equivalent to COSC 360.		

CCEE 354L	Computer Networks Lab	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. Accompanying Lab for CCEE 354. **Co-requisite**: CCEE 354.		

CCEE 514	Advanced Web Programming	3(3,0)
This course focuses on server-side programming. It teaches students how to connect their website or web application to a database and save and retrieve data from it. The course exposes students to web controls, validation controls, data source controls, data binding controls, state management, and working with third-party media such as XML and web services. **Prerequisite**: CCEE 411/COSC 333. Equivalent to COSC 434.		

CCEE 547	Optical Communications	3(3,0)
This course is designed to progress from the description of the components in a fiber link to the interconnections within a link or a network. The components of fiber links will be discussed (fiber, connections, sources, and receivers). These optical components will be integrated into a complete optical link. **Prerequisite**: CCEE 331.		

CCEE 552	Network Security	3(3,0)
Network security is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft. It involves creating a secure infrastructure for devices, applications, and users to operate securely. **Pre-requisite**: CCEE 354 or COSC 360.		

CCEE 553	Advanced Communication Networks	3(3,0)
In this course, we will explore current network architectures and topologies, such as telephone systems (PDH/SDH, SONET), Frame Relay, ATM, and MPLS, as well as technologies under development. We will also learn how to assess network performance and which technologies are used to enhance service quality. At the end of the course, the student will be able to identify the technologies to be used in a given environment and study its performance. **Prerequisite**: CCEE 354.		

CCEE 554L	Computer Networks Lab 2	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: CCEE 354L.		

CCEE 555	Advanced Computer Networks	1(0,2)
This is an advanced computer networks course that requires active student participation. In this course, we will explore TCP; Internet inter-domain routing; Internet quality of service and traffic engineering; Internet traffic measurement and analysis; data centers; the analysis and performance of content distribution networks; and Software-defined networking. In addition, the students will learn how to build client/server computer network applications using sockets. **Prerequisite**: CCEE 354.		

CCEE 555L	Computer Networks Lab 3	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 2) Certification. The primary focus of this Lab is routers and switches in small networks. The student will learn the architecture, components, and operations of routers and switches, as well as their basic configuration and functionality. They will also learn how to configure and troubleshoot static and dynamic routing protocols, access-control lists, VLANs, DHCP for IPv4 and IPv6 networks, and NAT operations. **Prerequisite**: CCEE 354L.		

CCEE 556	Network and System Administration	3(3,0)
This course will help students become system administrators and install, configure, maintain, and troubleshoot a Microsoft Windows Server Operating system. It provides them with the knowledge and skills to work with Active Directory technologies in environments including complex network services. This course also provides students with basic knowledge of installing, configuring, and maintaining Linux operating systems. By the end of this course, students will have all the required materials to submit and pass the MCSA: Windows Server certification exam. **Prerequisite**: CCEE 354.		

CCEE 556L	Computer Networks Lab 4	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn how to manage CISCO IOS software licensing and configuration files. **Prerequisite**: CCEE 555L.		

BIOMEDICAL ENGINEERING PROGRAM

Mission

The program aims to help students explore their innate creativity and potential and endow them with the knowledge, skills, and abilities to: pursue successful careers in biomedical/clinical engineering and subsequent fields; think critically in solving health-care complex problems using modern tools and technologies; communicate and work effectively with diverse groups; learn more every day and succeed in graduate studies in renowned institutions if they choose to.

Program Educational Objectives

The program's educational objectives are to enable graduates to:

1. Foster an environment that encourages excellence in endeavor.
2. Provide quality and professional education that prepares leaders.
3. Solve problems facing society and industry in Lebanon, the Middle East, and the world.
4. Engage in multidisciplinary research.

Student Outcomes

The Electrical Engineering program adopts the learning outcomes of ABET, so our graduates will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze, interpret data, and use engineering judgment to conclude.

7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Biomedical Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Biomedical and Similarly Named Engineering Programs.

Career Opportunities

Biomedical engineering is a multidisciplinary career that draws on knowledge from different scientific fields, instrumentation, and design. Our program graduates will act as technology entrepreneurs capable of interacting with healthcare professionals.

Biomedical engineers develop tools and machines to improve human health and ensure proper diagnostic procedures. Biomedical courses and laboratories teach how to perform well in advanced therapeutic and surgical devices, medical imaging, artificial organs, physiological modeling, rehabilitation engineering, biorobotics, and management of medical equipment.

Biomedical engineering is the fastest-growing career, a trend that is expected to continue over the next decade.

Program Overview

The Biomedical (BIOM) Engineering Program at Rafik Hariri University puts students' best interests first. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) degree in the BIOM Engineering program comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) degree curriculum comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: General Education requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS in BIOM Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education Requirement	0	0	20	7	27	23
College Requirement	0	0	24	0	24	20
Program Requirement	46	0	21	0	67	57
Credits	46	0	65	7	118	100
BE in BIOM Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	10	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	56	15	24	0	95	62
Credits	56	15	71	10	152	100

I. General Education Requirements

The list of GE-required courses and their descriptions is presented on the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in BIOM Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of technical engineering electives for BE only.

A. Mandatory Requirements

The BIOM mandatory core courses (major and non-major) are listed in the table below.

Course #	Title	Credits	Prerequisites
BIOM 301	Biomedical Sensors	3	ELEC 210
BIOM 301L	Biomedical Sensors Lab	1	Co-req.: BIOM 301
BIOM 311	Biomedical Instrumentation	3	BIOM 301 ELEC 210

BIOM 312	Biomedical Engineering Design	3	BIOM 301 ELEC 210
BIOM 312L	Biomedical Engineering Design Lab	1	Co-req.: BIOM 312
BIOM 400*	BS Summative Learning Project	3	
BIOM 313	Bio-electronics Circuit Design	3	BIOM 301, ELEC 320
BIOM 414	Advanced Biomedical Instrumentation	3	BIOM 311
BIOM 415L	Advanced Biomedical Lab	1	Co-req.: BIOM 414
BIOM 417	Critical Equipment	3	BIOM 311
BIOM 418	Clinical Laboratory Instruments	3	BIOM 311
BIOM 421	Medical Imaging	3	PHYS 312
BIOM 440	Computational Neuroscience	3	BIOM 312, MATH 311
BIOM 499	Co-op Work Experience	3	ENGL 217; Senior Standing
BIOM 519	Advanced Therapeutic Devices	3	BIOM 414
BIOM 522	Advanced Medical Imaging	3	BIOM 421
BIOM 531	Clinical Engineering	3	BIOM 417 and BIOM 421
BIOM 595A	BE Summative Learning Project 1	1	
BIOM 595B	BE Summative Learning Project 2	3	
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	CCEE 221
CCEE 331	Signals and Systems	3	MATH 216
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
CCEE 534	Signal Processing	3	CCEE 331
ELEC 200	Introduction to Electric and Computer Engineering	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 340	Engineering Electromagnetics	3	PHYS 211
ELEC 342	Power Electronics	3	ELEC 320
ELEC 451	Control Systems	3	CCEE 331

* Required for BS-bound students. BE students take engineering technical electives instead.

B. Technical Electives

As part of the Bachelor of Engineering in BIOM Engineering program, the student is required to take 12 credit hours of technical electives. Students must be of senior standing to take engineering technical electives. Technical elective courses allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match engineering technical elective courses from different areas to gain a broader exposure to the various BIOM Engineering disciplines. The student should, in cooperation with the academic advisor, select a list of electives that best meet their needs and aspirations. The listed engineering technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

1. Medical Instrumentation
2. Medical Imaging
3. Clinical Engineering
4. Artificial organs

It is highly recommended that the student take engineering technical electives from the list below after completing the mandatory requirements in the relevant area. Students may select engineering technical electives from other programs, subject to approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
BIOM 402	Radio Frequency Applications in Biomedical Engineering	3	CCEE 331
BIOM 498	Special Topics in Biomedical Engineering	3	
BIOM 501	Emerging Healthcare Technologies	3	
BIOM 502	AI Applications in Healthcare	3	
BIOM 504	Bio-Image Processing	3	CCEE 331
BIOM 505	Computer Simulations for Life Sciences	3	
BIOM 506	Biomaterials	3	BIOM 311
BIOM 507	Bio-fluids	3	
BIOM 508	Microcontroller and Embedded Systems for Biomedicine	3	CCEE 426
BIOM 512	Biomechanics and Rehabilitation Engineering	3	CIVE 211 BIOM 311
BIOM 513	Hemodynamics Measurement	3	
BIOM 521	Introduction to E-Healthcare	3	BIOM 421
BIOM 532	Biomedical Servicing	3	
BIOM 533	Health-care Facility Management	3	
BIOM 597	Advanced Topics in Biomedical Engineering	3	Senior Standing

C. Summative Learning Project (SLP)

Students must complete a 3-credit-hour course for BS (4 credits for BE; taken 3 credits in the first regular semester and 1 credit in the following regular semester) in which they

work, preferably in groups, on a problem of concern to industry or the community at large, or to innovate a promising idea. The SLP could be an extension of the projects students had worked on in previous courses, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training working in an area related to their field of interest. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on the student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

The courses offered in the BIOM Engineering program are designated code numbers in the form of (BIOM abc), where:

a	Designates year (level)
b	Designates focus areas (as follows) 0: General Biomedical courses; 1: Medical Instrumentation; 2: Medical Imaging; 3: Clinical Engineering
c	Designates course sequence in a focus area

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in BIOM Engineering. The study plan serves as a roadmap to facilitate students' smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (14 Credits)			
ARAB 212	Arabic Language & Communication	2	
CCEE 214	Introduction to Programming	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 211	Physics: Electricity and Magnetism & Lab	3	
Year 1, Spring Semester (15 Credits)			
CCEE 221	Logic Design	3	
ELEC 200	Introduction to Electrical & Computer Engineering	3	

ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
PHYS 312	Modern Physics	3	
Year 1, Summer Semester (10 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
MATH 311	Linear Algebra and Applications	3	
BIOL 210	Human Anatomy and Physiology and Lab	3	
	Humanities/ Fine Arts Elective I	3	
Year 2, Fall Semester (17 Credits)			
MATH 351	Probability and Statistics	3	MATH 215
BIOM 301	Biomedical Sensors	3	ELEC 210
BIOM 301L	Biomedical Sensors Lab	1	Co-req.: BIOM 301
CCEE 331	Signals and Systems	3	MATH 216
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ENGL 217	Professional English	3	ENGL 210
Year 2, Spring Semester (17 Credits)			
BIOM 311	Biomedical Instrumentation	3	BIOM 301 ELEC 210
BIOM 312	Biomedical Engineering Design	3	BIOM 301
BIOM 312L	Biomedical Engineering Design Lab	1	Co-req.: BIOM 312
BIOM 313	Bioelectronics Circuits Design	3	BIOM 301, ELEC 320
ELEC 340	Engineering Electromagnetics	3	PHYS 211
	Community and Sustainability	1	
MATH 314	Ordinary Differential Equations	3	MATH 216
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 421	Numerical Analysis	3	MATH 311
	Social Science Elective I	3	
Year 3, Fall Semester (17 Credits)			
BIOM 414	Advanced Biomedical Instrumentation	3	BIOM 311
BIOM 415L	Advanced Biomedical Lab	1	Co-req.: BIOM 414
BIOM 417	Critical Care Equipment	3	BIOM 311
BIOM 421	Medical Imaging	3	PHYS 312
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426

ELEC 451	Control Systems	3	CCEE 331
Year 3, Spring Semester (16 Credits)			
BIOM 418	Clinical Laboratory Instruments	3	BIOM 311
BIOM 440	Computational Neuroscience	3	BIOM 312 and MATH 311
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
ELEC 432	Power Electronics	3	ELEC 320
	Engineering Technical Elective 1	3	
Year 3, Summer Semester (3 Credits)			
BIOM 499	Co-op Work Experience	3	ENGL 217
The above 118 credits complete the requirements for a Bachelor of Science degree in Biomedical Engineering.			
Year 4, Fall Semester (16 Credits)			
BIOM 519	Advanced Therapeutic Devices	3	BIOM 414
BIOM 522	Advanced Medical Imaging	3	BIOM 421
BIOM 595A	BE Summative Learning Project 1	1	ENGL 217
CCEE 534	Signal Processing	3	CCEE 331
ENGR 510	Engineering Project Management	3	
	Engineering Technical Elective 2	3	
Year 4, Spring Semester (18 Credits)			
BIOM 531	Clinical Engineering	3	BIOM 417 BIOM 421
BIOM 595B	BE Summative Learning Project 2	3	BIOM 595A
	Engineering Technical Elective 3	3	
	Engineering Technical Elective 4	3	
	Engineering Technical Elective 5	3	
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take BIOM 400 – Summative Learning (BS) Project instead.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
6 credits restricted Departmental Electives
9 credits from any Engineering discipline at the 400 level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

A description of the non-major mandatory courses follows.

BIOL 210	**Human Anatomy and Physiology and Lab**	**3(2,2)**
A general overview of cell structure and function, and the anatomy and physiology of human body systems. These include cardiovascular, central nervous, respiratory,		

urinary, digestive, immune, and musculoskeletal systems. The course also offers a set of experiments that deal with basic biological processes and advanced biological assessments. **Prerequisite**: None.

CCEE 214	Programming	3(2,2)
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures.		

CCEE 221	Logic Design	3(3,0)
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices, including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices.		

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: CCEE 221. **Prerequisite**: CCEE 221. Equivalent to COSC 351L.		

CCEE 331	Signals and Systems	3(3,0)
This course covers the mathematical description and classification of continuous and discrete signals and systems. Topics include signal types, Fourier series, Fourier transforms, Discrete-Time Fourier Transform (DTFT), Fast Fourier Transform (FFT), Laplace transforms, z-transforms, linear time-invariant systems, transfer functions, and state-space representation. **Prerequisite:** MATH 216.		

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

CCEE 426L	Design of Embedded Systems Lab	1(0,2)
This lab complements topics covered in the CCEE 426 course. It involves the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Experiments include Design using finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, scheduling, system-level simulation, and interface-based design. **Co-requisite**: CCEE 426. Equivalent to COSC 356L.		

CCEE 460	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 460L	Artificial Intelligence Lab	1(0,2)
This lab will help prepare students to handle the huge amount of data produced by both humans and machines today. The student will learn to use state-of-the-art AI tools to analyze data and make complex decisions based on that data using real-world practical examples and case studies. At the end of the lab, the student will be able to apply AI knowledge in their field at work or in day-to-day life to make better decisions. **Co-req.:** CCEE 460.		

CCEE 534	Signal Processing	3(3,0)
This course covers time series analysis, frequency analysis, time-frequency, and time-scale analysis. It also covers the design of digital filters and signal modeling. **Prerequisite**: CCEE 331.		

ELEC 200	Introduction to Electrical and Computer Engineering	3(3,0)
In this course, the student will be trained in core skills to excel in electrical, biomedical, computer, and communications engineering. The student will be trained in application programming using an appropriate programming environment. Students will build on the basics of Programming learned in the introductory programming course to implement Digital Signal Processing applications and perform data analysis. Students will also be introduced to basic lab equipment such as Oscilloscopes, Function Generators, and Multi-Meters. Students will be introduced to microcontroller-based projects to further explore the Electrical and Computer Engineering majors (CCE, BIOM, and ELEC). Throughout the course, students will work in multidisciplinary teams to study engineering cases for the community and implement a project to address a community need.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 320	Electronics	3(3,0)
A course on semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, JFETs, MOSFETs, and operational amplifiers. **Prerequisite**: ELEC 210		

ELEC 320L	Electronics Lab	1(0,2)
This is a Lab course with experiments in Electrical and Electronic Circuits. It covers passive electrical elements and sources; lab instruments; voltage divider circuits; Thevenin's theorem; RC circuits; diode circuits; Op-Amp circuits; BJT and MOSFET characteristics. **Co-requisite**: ELEC 320.		

ELEC 340	Engineering Electromagnetics	3(3,0)
Introduction to vector analysis; Electrostatic fields in vacuum and dielectrics; Conductors; Capacitance; Electrostatic energy and forces; Poisson's equation. Magnetic fields, Maxwell's equations, static electric and magnetic fields, boundary-value problems, Laplace's and Poisson's equations, and static electromagnetic fields. Time dependence in Maxwell's equations and plane-wave propagation. **Prerequisite**: PHYS 211.		

ELEC 451	Control Systems	3(3,0)
Introduction to control systems; Modeling of systems in various energy domains; transfer functions and block diagrams; time-domain analyses; Laplace-domain analyses; frequency-domain analyses; stability; precision; rapidity; root locus; Bode, Nyquist, and Nichols diagrams; design of PID controllers and dynamic compensators. **Prerequisite**: CCEE 331.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements. **Prerequisite**: None.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

Major Courses

Mandatory biomedical engineering courses are described below.

BIOM 301	Biomedical Sensors	3(3,0)
This course provides an introductory overview of the multidisciplinary field of sensors. The student is first introduced to the fundamental concepts of signals and noise in measurement, the characteristics of the measurement system, and the determination of the absolute quantity. Then the student learns about the different types of sensors to measure pressure, force, flow, motion, and temperature. **Prerequisite**: ELEC 210.		

BIOM 301L	Biomedical Sensors Lab	1(0,2)
The lab teaches students how to measure and interpret different types of signals. Experiments include measurements of signal, noise, pressure (strain gauge), and temperature; flow, motion, and force; and applications using research and simulation kits. **Co-requisite**: BIOM 301.		

BIOM 311	Biomedical Instrumentation	3(3,0)
The course describes the major medical electronic instruments used in diagnostic medicine. It explains the theoretical background, block diagrams, and circuitry of prosthetic devices, including different types of cardiac pacemakers, muscle and		

bladder prostheses, cochlear prostheses, and cardiac defibrillators and cardioverters. **Prerequisite**: BIOM 301.

BIOM 312	Biomedical Engineering Design	3(3,0)
This course examines the origins of various bioelectric phenomena and signals routinely recorded in modern clinical practice. Given adequate monitoring equipment, the engineer of today can record many forms of bioelectric phenomena with relative ease. This course describes the theoretical, block-diagram, and circuitry aspects of electrophysiological devices such as electrocardiogram, electroencephalogram, electroneurogram, electromyogram, and the like. **Prerequisite**: BIOM 301.		

BIOM 312L	Biomedical Engineering Design Lab	1(0,2)
The accompanying lab for BIOM 301 teaches students to measure and interpret biological and medical signals. Experiments cover the following topics: signal, noise, pressure (strain gauge), and temperature measurements; blood and gas flow, motion, and force measurements; and applications using electrodes based on medical equipment and research and simulation kits. **Co-requisite**: BIOM 312.		

BIOM 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program. Annually.		

BIOM 313	Bio-electronics Circuit Design	3(3,0)
This course focuses on design and measurement processes for medical systems and associated constraints. Students are required to apply the learned concepts in projects to design healthcare technology. **Prerequisites**: BIOM 301, ELEC 320.		

BIOM 414	Advanced Biomedical Instrumentation	3(3,0)
The course describes the advanced medical electronic instruments used in therapeutic and diagnostic medicine. It includes extracorporeal shock wave lithotripsy, an infant incubator, a medical laser, a ventilator, and various medical devices used in the operating room, such as electrosurgical units and anesthesia units. **Prerequisite**: BIOM 311.		

BIOM 415L	Advanced Biomedical Lab	1(0,2)
The accompanying lab for BIOM 414 teaches students about advanced medical electronic instruments used in therapeutic and diagnostic medicine. Experiments cover the following topics: extracorporeal shock wave lithotripsy, infant incubator, medical laser applications, ventilator, and various medical devices used in the operating room, such as electrosurgical units and anesthesia units. **Co-requisite**: BIOM 414.		

BIOM 417	Critical Care Equipment	3(3,0)
This course focuses on the interpretation of clinical data outputs, the description of modules integration, and the manipulation of features relating to medical equipment (ME) found in surgical theaters, intensive care units, and emergency departments. Students are required to simulate ME's system function, intervene in case of ME's failure, upgrade available ME(s), and, in a team-oriented project, propose and implement a new design for a high-risk medical technology. **Prerequisite:** BIOM 311.		

BIOM 418	Clinical Laboratory Instruments	3(3,0)
This course covers the clinical laboratory responsible for analyzing patient specimens to provide information that aids in the diagnosis of disease and evaluates the effectiveness of therapy. The major topics covered in this course are spectrophotometry, flame photometers, spectrofluorometry, chemistry analyzers, hematology analyzers, and chromatography columns. **Prerequisite**: BIOM 311.		

BIOM 421	Medical Imaging	3(3,0)
This course focuses on the physical and technical features, as well as the clinical applications, of imaging modalities used in healthcare facilities, including diagnostic tools. Specific topics covered include Radiation; X-ray generators; Interaction; Attenuation; Photoelectric effect; Compton scattering; Filters; Collimators; Grids; Intensifying Screen; Fluoroscopic Image Intensifier; Resolution; Ultrasound Generation; US Scan; Transducers; Modes; Doppler. **Prerequisites**: PHYS 312		

BIOM 440	Computational Neuroscience	3(3,0)
This course introduces basic computational methods for understanding what nervous systems do and for determining how they function. It will explore the computational principles governing various aspects of neural coding, dynamics, learning, and memory. Specific topics cover the representation of information by spiking neurons, the processing of information in neural networks, graph theory, and algorithms for adaptation and learning. MATLAB/Python demonstrations and exercises will be used to deepen understanding of various concepts and methods. **Prerequisites**: BIOM 312; MATH 311.		

BIOM 499	CO-OP Work Experience	3(3,0)
This CO-OP work experience is designed to provide students with an eight-week short-term opportunity in biomedical engineering. Students are encouraged to network with industry professionals in their discipline. A report, poster, and PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

BIOM 519	Advanced Therapeutic Devices	3(3,0)
This course describes the latest biomedical concepts and technologies used for therapeutic purposes. The major topics covered in this course are hemodialysis, hemofiltration, and hemodiafiltration; ventilators and their modes; respiratory modeling and spirometry parameters; and various advanced therapeutic devices. **Prerequisite**: BIOM 414		

BIOM 522	Advanced Medical Imaging	3(3,0)
This course seeks to provide the student with a basic understanding of the concepts of Advanced Biomedical Imaging systems. It involves working with tomographic modalities that produce 3D slice images. The course focuses on the nature of waves, the physical principles, and the instrumentation of each of the three imaging technologies. It applies to radiation interactions with human tissues and to the fundamentals of slice reconstruction systems. Major applications are in Magnetic Resonance, Computed Tomography, Nuclear Medicine, PET, and SPECT scans. Specific topics covered include Radon, 2DFT, translations, rotations, generations, magnetic moment, spin, relaxations, gradient, superconductivity, sequence, contrast, k-space, resolution, artifacts, scintillation, positron, and single photon emission. **Prerequisite**: BIOM 421		

BIOM 531	Clinical Engineering	3(3,0)
This course presents the mission, objectives, and responsibilities of biomedical engineering in health care delivery institutions. It focuses on possessing in-depth knowledge regarding the effective and efficient usage of hospital technological resources. **Prerequisites**: BIOM 417/ BIOM421.		

BIOM 595A	BE Summative Learning Project 1	1(1,0)
The first phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisites**: ENGL 217.		

BIOM 595B	BE Summative Learning Project 2	3(3,0)
The second phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite**: BIOM 595A.		

II. Elective Courses

Students may meet their engineering technical electives requirements from the following selection of elective courses.

BIOM 402	Radio Frequency Applications in Biomedical Engineering	3(3,0)
This course addresses radio-frequency design and applications in biomedical engineering and medicine, covering electromagnetic radiation, circuit elements, coupling devices, modulation and demodulation, transmitters, and receivers. Applications include telemetry, transcutaneous power transfer, hyperthermia, RF ablation, tagging, and identification. **Prerequisite**: CCEE 331.		

BIOM 498	Special Topics in Biomedical Engineering	3(3,0)
This course is designed to enable students to study current special topics of interest, carefully selected from BIOM-related areas. The contents of such a course are to be determined by the instructor and the department.		

BIOM 501	Emerging Healthcare Technologies	3(3,0)
This course aims to enable students to review the latest innovations in research areas of interest within the Bio-Engineering field. The course will mainly focus on introducing students to the evolving domain of biomedical technology by examining the interaction and integration of skills among physicians, scientists, engineers, firms, and users. For instance, it will include explicit discussions of fast-growing health care practices that exhibit radical novelty. Students will learn to analyze how such evolutionary technologies can impact a wide range of sectors and associate with several stakeholders. The studies will emphasize topics such as endoscopy, brain mapping, heat therapy, and advanced vital-signs measurements.		

BIOM 502	AI Applications in Healthcare	3(3,0)
This course introduces artificial intelligence (AI) and its applications in healthcare, including machine learning and precision medicine. Students learn the core skills needed to assess clinical information datasets and apply them to enhance evidence-based healthcare outcomes. In this course, students apply AI knowledge and skills to promote effective disease management and patient engagement models.		

BIOM 504	Bio-Image Processing	3(3,0)
This course introduces the basic theories and methodologies of digital image processing. Topics include intensity transformations for image enhancement, the two-dimensional discrete Fourier transform, spatial- and frequency-domain linear image filtering, nonlinear image filtering, binary image processing, edge detection, image segmentation, and the basics of digital video processing. This course makes extensive use of MATLAB as an analysis, design, and visualization tool. **Prerequisite**: CCEE 321.		

BIOM 505	Computer Simulations for Life Sciences	3(3,0)
This course introduces mathematical tools and algorithms implemented to simulate dynamic biological and medical processes. The course examines modeling methodologies and simulation techniques for existing software such as MATLAB/Simulink, LabVIEW, and others. Specific topics covered include System's mode, Compartments, Physiological mechanisms, monitoring medical parameters, Simulation tools, Parameter estimation, Identification and prediction.		

BIOM 506	Biomaterials	3(3,0)
The course studies cell-surface interactions mediated by adhesive proteins in the physiological medium, the analysis of signal transduction, and cellular functions induced by surface control. Emphasis is placed on biomaterial polymers, ceramics, and bone substitutes. Research applications include aiding in the diagnosis of		

malignant tumors and the use of nanobiotechnology to coat medical devices. **Prerequisite**: BIOM 311.

BIOM 507	**Bio-fluids**	**3(3,0)**
This course is designed to enable students to review the main fluids of the human body, their behavior, and the latest related clinical and surgical applications. The course will mainly provide an overview of bodily fluids, basic fluid mechanics concepts, the functional properties of blood, and gas exchange and transport. Also, it includes in-depth discussions of recent health care practices based on the control aspects of the vascular system and the hemodynamics of large arteries.		

BIOM 508	**Microcontroller and Embedded Systems for Biomedicine**	**3(3,0)**
This course explains the architecture of microcontroller-based systems: registers, timers, interrupts, ports, memory, serial communication, and design. It also covers parallel communication and design, analog-to-digital conversion, and the use of PIC 16F84 and PIC 16F877 microcontrollers. MPLAB, PROTEUS, and mikroC software are used. **Prerequisite**: CCEE 426.		

BIOM 512	**Biomechanics and Rehabilitation Engineering**	**3(3,0)**
This course will focus on the human musculoskeletal system. It emphasizes various engineering principles governing biological materials and applies these to the study of bone, cartilage, ligaments, tendons, and muscles. It covers: Bone, Joint, Ligament, Tendon, Cartilage, Viscosity, Elasticity, Plasticity, Stress, Strain, Muscle, Length, Velocity, Moment Arm, Contractile Element, EMG, Lagrangian, and Rehabilitation Mechanics. **Prerequisites**: CIVE 211 and BIOM 311.		

BIOM 513	Hemodynamics Measurement	**3(3,0)**
This course enables the students to study the basic principles of hemodynamics measurement. The students will learn different techniques for measuring blood flow, pressure, compliance, and resistance using invasive and non-invasive methods. In addition, there will be coverage of the qualitative and quantitative measures of cardiopulmonary functions.		

BIOM 521	**Introduction to E-Health Care**	**3(3,0)**
This course emphasizes current applications of telecommunications in medical diagnosis and patient care. It examines collaborative arrangements for real-time management of patients at a distance using wireless, mobile, and robotic technologies. **Prerequisite**: BIOM 421.		

BIOM 532	**Biomedical Servicing**	**3(3,0)**
This course provides students with the technical knowledge and skills necessary to become a broad-based electro-medical specialist. Students will learn to install, set up, troubleshoot, integrate, program, test, operate, and repair systems and components. The topics covered in this course include inspection, corrective and		

preventive maintenance, calibration, repair, supply management, best practices, and accreditation of medical equipment.

BIOM533	**Health-care Facility Management**	**3(3,0)**
This course presents procedures followed for medical and health services management. It covers the duties of the administrator and his assistants, as well as workflow between staff of different services and departments. This course emphasizes topics such as credentials, quality control, human resources, finances, recovery, insurance, sterilization, infection control, and health safety procedures.		

BIOM 597	**Advanced Topics in Biomedical Engineering**	**3(3,0)**
This course is designed to enable students to study a specific advanced topic of interest, carefully selected from biomedical engineering-related topics. The content outline for such a topic is to be determined by the instructor and approved by the department Chair. **Prerequisite**: Senior Standing		

MECHANICAL AND MECHATRONICS ENGINEERING DEPARTMENT (MME)

Chairperson:	Nadim Diab
Professor:	Ahmad Chaaban, Nadim Diab,
Associate Professor:	Ahmad Kobeissi
Assistant Professor:	Manar Younis
Lecturer:	Mohamad Al Kaderi
Adjunct Faculty:	Omar Berjaoui, Maya Maaotuk

Programs Offered

The Mechanical and Mechatronics Engineering (MME) Department offers two programs – Mechanical Engineering (MECH) and Mechatronics Engineering (MECA). Each program grants three degrees: Bachelor of Science, Bachelor of Engineering, and Master of Science. The details of each program follow.

Vision

To be recognized for the ability, agility, and tenacity of its graduates in confronting the challenges of a changing workplace.

MECHANICAL ENGINEERING PROGRAM

Mission

The mission of the Mechanical Engineering program at RHU is to provide a solid education that enables students to start a professional career and/or continue graduate studies in mechanical engineering. The program seeks to instill in students a sense of and understanding of moral and professional responsibilities, the ability to learn more, and the leadership required in their calling, primarily in Lebanon and the Middle East.

Objectives

In a few years after graduation, Mechanical Engineering program graduates will:

1. Apply their multidisciplinary knowledge in Mechanical engineering to meet employers' expectations.
2. Possess self-learning motivation and qualifications to pursue advanced studies.
3. Demonstrate professional skills and embrace leadership roles in their profession and/or communities.

Program Outcomes

Upon successful completion of the BS/BE program in Mechanical Engineering, graduates will have:

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. Ability to communicate effectively with a range of audiences
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Mechanical and Similarly Named Engineering Programs.

Career Opportunities

Mechanical engineers attain a broad spectrum of skills sought after by a wide range of professions. Industrial sectors, enterprises, and services in which a mechanical engineer can pursue a career include power generation and distribution, building and construction, medicine and pharmacology, aerospace, automotive, chemicals, computers and electronics, renewable energy, entertainment, water resources, sports, environmental institutions, and government.

Mechanical engineering also serves as an excellent foundation for careers in business management and business consulting.

Program Overviews

The Mechanical Engineering Program at Rafik Hariri University puts students' best interests first and foremost. Every little experience the student gains represents a block in the building of a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Mechanical Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS in Mechanical Engineering (118 Credits)						
Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	6	27	23
College Requirements	0	0	24	0	24	20

Program Requirements	49	0	18	0	67	57
Credits	49	0	63	6	118	100

BE in Mechanical Engineering (152 Credits)						
Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	9	30	20
College Requirements	0	0	27	0	27	18
Program Requirements	59	15	21	0	95	62
Credits	59	15	69	9	152	100

I. University Requirements (General Education)

The list of the University-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of college-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Mechanical Engineering

Engineering encompasses 95 credits (67 credits for the BS degree) distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits for elective courses.

A. Mandatory Requirements

The mandatory courses for the major and non-major Mechanical Engineering programs are listed in the table below.

Course #	Title	Credits	Prerequisites
CIVE 211	Statics	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
MECA 340	Dynamic Systems Modeling	3	MATH 314
MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req.: MECA 340

MECH 400*	Summative Learning (BS) Project	3	ENGL 217
MECA 440	Control Systems Design	3	MECA 340
MECA 440L	Control Systems Design Lab	1	MECA 440
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
MECH 223	Material Science	3	
MECH 224	Mechanical Engineering Graphics and CAD	3	
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
MECH 330	Thermodynamics I	3	MATH 215
MECH 334	Thermodynamics II	3	MECH 330
MECH 334L	Thermodynamics II Lab	1	Co-req.: MECH 334
MECH 335	Fluid Mechanics	3	MECH 330
MECH 335L	Fluid Mechanics Lab	1	Co-req.: MECH 335
MECH 421	Mechanical Design I	3	MECH 320
MECH 422	Manufacturing processes	3	MECH 320
MECH 423	Mechanical Design II	3	MECH 223, MECH 321, and MECH 421
MECH 431	Heat Transfer	3	MECH 333 or MECH 335 and MATH 317
MECH 431L	Heat Transfer Lab	1	Co-req.: MECH 341
MECH 435	HVAC	3	MECH 431
MECH 499	Co-op Work Experience	3	ENGL 217; Senior Standing
MECH 520	Finite Element Method for Engineers	3	MECH 320 MATH 421 MECH 431
MECH 521	Vibrations and Acoustics	3	MECA 340
MECA 541	CAD/CAM/CAE	3	MECH 224, MECH 421
MECH 595A	BE Summative Learning Project - I	1	ENGL 217 Senior Standing
MECH 595B	BE Summative Learning Project - II	3	MECH 595A
MECH 4xx/5xx	Technical Electives	15	Course requirement

* Required for BS-bound students. BE students take a Technical Elective instead.

B. Technical Electives

As part of the Bachelor of Engineering in Mechanical Engineering program, the student is required to take 15 credit hours of technical electives. These courses allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to the various Mechanical Engineering disciplines. The student should, in cooperation with the academic advisor, select a list of electives that best meet their needs and aspirations. The listed technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

- Mechanical Design and Manufacturing
- Energy and Thermo-fluid Systems

It is highly recommended that the student take elective courses from the list below after completing the mandatory requirements in the related area. Students may select technical electives from other programs, subject to approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
Mechanical Design and Manufacturing			
MECH 445	Automotive Engineering	3	MECH 330 or 333 and MECH 321
MECH 470	Product Design and Development	3	MECH 421
MECH 498	Special Topics in Mechanical Engineering	3	
MECH 522	Mechanism Synthesis: Classical Kinematics to AI-Driven Optimization	3	MECH 321
MECH 523	Mechanics of Composite Materials	3	MECH 320
MECH 524	Elasticity and Plasticity	3	MECH 320
MECH 525	Advanced Dynamics	3	MECH 220
MECH 527	Modern Material Science	3	MECH 223
MECH 570	Digital Manufacturing	3	MECH 422 and MECA 541
MECH 571	Optimization in Engineering Design	3	MATH 421
Energy and Thermo-fluid Systems			
MECH 436	Petroleum Refining Technology for Mechanical Engineers	3	CHEM 211, MECH320, MECH 330
MECH 437	Heat Exchanger Design, Performance, and Inspection	3	MECH 320, MECH 431
MECH 445	Automotive Engineering	3	MECH 330 or 333 and MECH 321
MECH 448	Plumbing Engineering	3	MECH 335 or MECH 333
MECH 451	Fluid Mechanics II	3	MECH 335

MECH 453	Buildings and Energy	3	MECH 330 or MECH 333
MECH 455	Gas Turbines	3	MECH 330 or MECH 333
MECH 461	Thermal Systems Design	3	MECH 431
MECH 468	Internal Combustion Engines	3	MECH 334
MECH 471	Refrigeration	3	MECH 334
MECH 498	Special Topics in Mechanical Engineering	3	
MECH 530	Computational Fluid Dynamics	3	MECH 431
MECH 541	Renewable Energy	3	MECH 330 or MECH 333

C. Summative Learning Project

Students must complete a 3-credit-hour course for the BS (4 credits for the BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to develop a promising idea. The SLP could be an extension of the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in Mechanical Engineering. The following study plan serves as a roadmap for smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
CCEE 214	Introduction to Programming	3	
CIVE 211	Statics	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 210	Fundamentals of Physics and Lab	3	

Year 1, Spring Semester (15 Credits)			
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
MECH 223	Material Science	3	
Year 1, Summer Semester (9 Credits)			
CHEM 211	Environmental Chemistry and Lab	3	
MECH 224	Mechanical Engineering Graphics and CAD	3	
MATH 311	Linear Algebra with Applications	3	
Year 2, Fall Semester (17 Credits)			
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronic Lab	1	Co-req: ELEC 320
MATH 314	Ordinary Differential Equations	3	MATH 216
MECH 330	Thermodynamics I	3	MATH 215
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
Year 2, Spring Semester (17 Credits)			
ENGL 217	Professional English	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 215
MECA 340	Dynamic Systems Modeling	3	MATH 314
MECH 334	Thermodynamics II	3	MECH 330
MECH 334L	Thermodynamics II Lab	1	Co-req: MECH 334
MECH 335	Fluid Mechanics	3	MECH 330
MECH 335L	Fluid Mechanics Lab	1	Co-req.: MECH 335
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 317	Partial Differential Equations	3	MATH 314
	Social Sciences/Cultural Elective I	3	
Year 3, Fall Semester (16 Credits)			
MATH 421	Numerical Analysis	3	MATH 311
MECA 440	Control Systems Design	3	MECA 340
MECA 541	CAD/CAM/CAE	3	MECH 224, MECH 421
MECH 421	Mechanical Design I	3	MECH 320
MECH 431	Heat Transfer	3	MECH 333 or MECH 335 and MATH 317

MECH 431L	Heat Transfer Lab	1	Co-req.: MECH 431
Year 3, Spring Semester (17 Credits)			
ARAB 212	Arabic Language & Communication	2	
MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req: MECA 341
MECH 423	Mechanical Design II	3	MECH 223, MECH 321, and MECH 421
MECA 440L	Control Systems Design Lab	1	MECA 440
	Community & Sustainability	1	Co-req.: ENGL 210
	Humanities/ Fine Arts Elective I	3	
	Engineering Technical Elective I*	3	Per course requirements
Year 3, Summer Semester (3 Credits)			
MECH 499	Co-op Work Experience	3	ENGL 217 Senior standing
The 118 credits above complete the requirements for a Bachelor of Science Degree in Mechanical Engineering.			
Year 4, Fall Semester (16 Credits)			
ENGR 510	Engineering Project Management	3	ENGR 300
MECH 422	Manufacturing Processes	3	MECH 320
MECH 435	HVAC	3	MECH 431
MECH 595A	BE Summative Learning Project I	1	ENGL 217 Senior Standing
	Engineering Technical Elective II	3	Per course requirements
	Engineering Technical Elective III	3	Per course requirements
Year 4, Spring Semester (18 Credits)			
MECH 520	Finite Element Method for Engineers	3	MECH 320; MATH 421, MECH 431
MECH 521	Vibrations and Acoustics	3	MECA 340
MECH 595B	BE Summative Learning Project II	3	MECH 595A
	Engineering Technical Elective IV	3	Per course requirements
	Engineering Technical Elective V (AI-Restricted**)	3	Per course requirements
	Social Sciences/Humanities II	3	
Total BE credits: 152			

* BS-bound students are required to take MECH 400 – Summative Learning (BS) Project I instead.
** Students are required to choose at least one technical elective (i.e., 3 credits) from the list of AI-Restricted elective courses.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
9 credits restricted Departmental Electives
6 credits from any engineering discipline at the 400-level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

Description of the mandatory, non-major courses follows.

CCEE 214	Introduction to Programming	2(2,0)
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures. **Prerequisite**: None.		

CCEE 214L	Introduction to Programming Lab	1(0,2)
Laboratory experiments in programming for concepts covered in CCEE 212. **Co-req**: CCEE 214.		

CHEM 211	Environmental Chemistry and Lab	3(2,2)
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.		

CIVE 211	Statics	3(3,0)
Static laws; force vectors and operations; force systems and moments; free-body diagrams; equilibrium of particles and rigid bodies in 2D and 3D; plane trusses and frames; cables; internal forces: shear and moment diagrams; centroid and center of mass; moment of inertia; and friction. **Prerequisite**: None. Annually.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 320	Electronics	3(3,0)
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	Electronics Lab	1(0,2)
This is a lab course that experiments with electrical and electronic circuits.l It covers passive electrical elements and sources, lab instruments, voltage divider circuits, Thevenin's theorem, RC circuits, diode circuits, Op-Amp circuits, BJT, and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements. **Prerequisite**: None. Annually.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management, including project selection and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics, and professional conduct, and realizing the impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers		

infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.

MATH 216	**Calculus IV**	**3(3,0)**
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	**Linear Algebra with Applications**	**3(3,0)**
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	**Ordinary Differential Equations**	**3(3,0)**
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	**Partial Differential Equations**	**3(3,0)**
Lagrange's theorem, boundary conditions for first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables, introduction to Fourier series and integrals, boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 351	**Probability and Statistics**	**3(3,0)**
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	**Numerical Analysis**	**3(3,0)**
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using		

Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.

MECA 340	**Dynamic Systems Modeling**	**3(3,0)**
Introduction to dynamic modeling of mechanical, electrical, electromechanical, thermal, fluid, and mixed-discipline systems; Block diagrams and computer simulation (MATLAB); State-variable solution; Laplace Transform solution; Laplace transform properties; inverse Laplace transform; transfer function, frequency response, first- and second-order systems. **Prerequisite**: MATH 314. Annually.		

MECA 341	**Measurements and Instrumentation**	**3(3,0)**
This course covers the fundamentals of instrumentation and measurement of various physical quantities. Topics include: sensor types, technologies, characteristics and calibration; design of a measurement system; statistical analysis of measured data; measurement noise and uncertainties; data acquisition, data storage, and display devices; signal conditioning and interface electronics concepts including filtering, A/D and D/A conversion, amplification, modulation, compensation; applications; **Prerequisite**: ELEC 320. Annually.		

MECA 341L	**Measurements and Instrumentation Lab**	**1(0,2)**
The lab experience complements the topics covered in MECA 341. Students learn to use the NI ELVIS platform, LabVIEW programming, and data acquisition system characteristics to build a measurement system, perform data analysis and sensor calibration, and design and build signal conditioning circuits for various types of measurements. The lab involves a team project to integrate all instrumentation and measurement tools and techniques to build a measurement system and submit a report. **Co-requisite**: MECA 341. Annually.		

MECA 440	**Control Systems Design**	**3(3,0)**
This course focuses on the analysis and design of control systems. Measure control system performance; Transient and steady-state response analyses; Topics include analytical solutions in time and S-domain, root locus, Bode, Nyquist, Lead-Lag controller design, and PID compensator design; Analysis and design of control systems in state space. Control system analysis and design using MATLAB and SIMULINK; set up a control system, understand it, and optimize its performance; **Prerequisites**: MECA 340. Annually.		

MECA 440L	**Control Systems Design Lab**	**1(0,2)**
This 1-credit laboratory course provides students with practical experience in the design, analysis, and implementation of linear control systems. Students will gain hands-on proficiency in using MATLAB and Simulink as powerful tools for modeling, simulating, and analyzing control system behavior in both the time and frequency domains. A significant component of the lab involves the implementation of fundamental automatic controllers, including ON/OFF, Proportional-Derivative (PD), Proportional-Integral (PI), and Proportional-Integral-Derivative (PID) controllers, using Arduino microcontrollers for various real-world applications such as temperature		

control and speed regulation. Students will also be introduced to the diverse range of control systems training kits available in the laboratory, providing them with opportunities to experiment with different hardware and system dynamics. This lab-intensive course bridges the gap between theoretical concepts and practical implementation, equipping students with essential skills to design and deploy control solutions across a wide array of engineering applications. **Prerequisite**: MECA 440. Annually.

MECA 541	**CAD/CAM/CAE**	**3(3,0)**
Fundamental principles of computer-aided design (CAD), computer-aided manufacturing (CAM), and computer-aided engineering (CAE); prototyping and distributed systems; specific rapid prototyping technologies; design for assembly; small batch manufacturing; continuous manufacturing; biotechnology and manufacturing. **Prerequisite**: MECH 224 and MECH 421. Annually.		

PHYS 210	**Fundamentals of Physics and Lab**	**3(3,1)**
Welcome to the Introductory Physics course. This course consists of a review of Classical Mechanics, Fluid Mechanics, General Properties of Waves, Electric Forces, Electric Fields for Discrete and Continuous Charge Distributions, Gauss's Law, Electric Potential, Kirchhoff's Rules, Magnetic Fields and Forces, and a Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

Major Courses

Mandatory mechanical engineering courses are described below.

MECH 200	**Introduction to Mechanical Engineering**	**3(3,0)**
This course will give students a comprehensive overview of Mechanical Engineering concepts and applications. Engineering topics will include an introduction to learn-by-practice concepts; microcontrollers (Arduino); sensors and actuators; materials, mechanisms, codes, and standards; and robotics (LEGO Mindstorms). In addition, students will be introduced to engineering ethics and learn how to apply the ASME/ABET code of ethics in practical case studies. The course also integrates the development of hands-on skills, teamwork, and communication; exercises and projects engage students in building, implementing, and testing their designs. **Prerequisite**: None. Annually.		

MECH 220	**Dynamics**	**3(3,0)**
Brief review of statics; proper use of different coordinate systems: Cartesian and normal/tangential axes; kinematics of a particle: rectilinear, curvilinear, and relative motion; kinetics of a particle: force and acceleration, work and energy, impulse, and momentum; planar kinematics of a rigid body: translating and rotating axes, absolute and relative motion; planar kinetics of a rigid body: force and acceleration, work and energy, impulse and momentum. **Prerequisite**: CIVE 211. Annually.		

MECH 223	**Material Science**	**3(3,0)**
The course introduces the fundamentals of materials science. The following topics will be discussed: Classification of materials; atomic structure and interatomic bonding;		

crystalline structures & amorphous materials; and imperfections in solids. Overview of Alloying, Analysis of Phase Diagrams & Heat Treatment of Metals. Discussion of electrical, thermal, magnetic, and optical properties. **Prerequisite**: None. Annually.

MECH 224	Mechanical Engineering Graphics and CAD	3(3,0)
Drafting principles; part and assembly drawings; generating 2D and 3D drawings and solid models using AutoCAD; customizing AutoCAD for specific applications; generating 2D drawings using CAD/CAM software; generating parametric, feature-based solids using Autodesk Mechanical Desktop. Introducing mechanical Revit, a Building Information Modeling (BIM) software tailored for the mechanical, electrical, and plumbing (MEP) disciplines. **Prerequisite**: None. Annually.		

MECH 320	Mechanics of Materials	3(3,0)
Mechanical stress, strain, and stress/strain relations under pure tension, compression, pure bending, and pure torsion of circular bars; 1-D thermal strain & stress; combined stresses; basic thin-walled cylindrical and spherical shells under internal pressure; principal stresses; plane stress transformation & Mohr's circles; basic 3-D elastic stress/strain relations; strain measurement using strain gauge technique. **Prerequisite**: CIVE 211. Annually.		

MECH 320L	Mechanics of Materials Lab	1(0,2)
This hands-on laboratory course complements theoretical concepts from Mechanics of Materials (MECH 320) by providing practical experience in the experimental analysis of material behavior under various loading conditions. Students will conduct tests involving axial tension/compression, bending, shear, and torsion to investigate stress-strain relationships, deformation, and failure modes in structural materials. Emphasis is placed on accurate measurement techniques using strain gauges, extensometers, and deflection indicators, as well as data analysis, interpretation, and comparison with theoretical predictions. Through structured experiments and lab reports, students will develop skills in experimental design, data acquisition, critical analysis, and technical communication. **Co-requisite**: MECH 320.		

MECH 321	Mechanisms and Dynamics of Machinery	3(3,0)
Mechanization of motion; kinematic analysis of linkage mechanisms; graphical synthesis of planar mechanisms; synthesis of cam-follower mechanisms (graphical and analytical designs); gear terminology and types of gears used in machines; analysis and synthesis of gear trains; velocity, acceleration, and force analysis of mechanisms. **Prerequisite**: MECH 200 and MECH 220. Annually.		

MECH 330	Thermodynamics I	3(3,0)
The course introduces the interconversion of heat and work. It covers concepts of a given system and its surroundings; thermodynamic state and properties of pure substances; work and heat; energy analysis (First Law of thermodynamics) of closed systems and control volumes; applications to thermal conversion systems; the Second Law of thermodynamics and Entropy. **Prerequisite**: MATH 215. Annually.		

MECH 333	Thermal Fluid Sciences	3(3,0)
This course imparts fundamental concepts of thermal fluid (including thermodynamics, fluid mechanics, and heat transfer) to non-mechanical engineering students. This course covers the following topics: thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law of Thermodynamics) and simple flow devices; fluid statics; Bernoulli's equation; conservation of mass and momentum; viscous flow in pipes; introduction to conduction, convection and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

MECH 334	Thermodynamics II	3(3,0)
This course focuses on real-world engineering examples & applications to give students a feel for how thermodynamics is applied in engineering practice. These cover Exergy analysis, vapor- and gas-power cycles, refrigeration and heat pump systems, psychrometric basics, and air conditioning. **Prerequisite**: MECH 330. Annually.		

MECH 334L	Thermodynamics II Lab	1(0,2)
This laboratory takes an experimental approach to demonstrate the fundamentals of thermodynamics. The lab includes a bomb and gas calorimeter, a miniature steam power plant, an internal combustion engine bench, and an air conditioning unit. One or two design-oriented experiments following the Design-Build-Test methodology. **Co-requisite**: MECH 334. Annually.		

MECH 335	Fluid Mechanics	3(3,0)
This course introduces the fundamentals of fluid mechanics. The topics covered are fluid statics; Bernoulli's equation; conservation of mass, momentum, and energy; an overview of fluid kinematics; an overview of finite-control-volume analysis; the Navier-Stokes equations; and viscous flow in pipes. **Prerequisite**: MECH 330. Annually.		

MECH 335L	Fluid Mechanics Lab	1(0,2)
This laboratory course offers a practical exploration of key fluid mechanics concepts through a variety of experimental setups. Students will use a pressure-measurement bench and a hydraulic bench to analyze head losses in pumps, piping systems, and flow over weirs. A wind tunnel is used for boundary layer analysis and flow visualization, including experiments on flow over a NACA 2412 airfoil, a sphere, and a cylinder. The lab also features a smoke generator for enhanced visualization of flow patterns and separation phenomena. **Co-requisite**: MECH 335. Annually.		

MECH 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program. Annually.		

MECH 421	Mechanical Design I	3(3,0)
Engineering materials properties; tools of machine design; stress, strain, and deformation of machine parts; buckling of columns; statistical methods and safety consideration; static and fatigue failure theories; design of machine parts for strength under steady loads; design of machine parts for rigidity, deflection of beams and rotation of shafts; design of machine parts for strength under cyclic/dynamic loads; codes and standards. **Prerequisite**: MECH 320. Annually.		

MECH 422	Manufacturing Processes	3(3,0)
General overview of manufacturing; properties of engineering materials; steelmaking; dimensions and tolerances; manufacturing processes: rolling, forging, extrusion, powder metallurgy; mechanical assembly: welding, brazing, soldering, bonding; welding QA/QC; non-destructive testing (NDT); machining; casting; sheet metal forming; plastic injection molding; polymers and composites. **Prerequisite**: MECH 320. Annually.		

MECH 423	Mechanical Design II	3(3,0)
Static & fatigue design of shafts and axles; design of permanent and non-permanent joints (screws, fasteners, bolting, welding); rolling-contact bearings; lubrication and journal bearings; Gears-general, spur, and helical gears; flexible mechanical elements; power transmission case study; material selection; codes and standards; computer modeling; static and fatigue design using the principles of applied fracture mechanics combined with finite elements analysis results; failure analysis; team project. **Prerequisites**: MECH 223, MECH 321, and MECH 421. Annually.		

MECH 431	Heat Transfer	3(3,0)
The course introduces the relationship between heat transfer and thermodynamics; steady and transient heat conduction in one and two dimensions; analytical and numerical solutions; extended surfaces; internal and external flow; forced and natural convection; and an introduction to heat radiation. **Prerequisite**: MECH 333 or MECH 335 and MATH 317. Annually.		

MECH 431L	Heat Transfer Lab	1(0,2)
This laboratory course provides hands-on experience with fundamental heat transfer principles. Students will work with a variety of experimental setups, including a temperature-measurement bench for calibrating and recording thermal data and heat-exchanger units configured for cross-flow, parallel-flow, and mixed-flow operations. Additionally, a thermal conductivity apparatus is used to investigate temperature gradients and calculate conductivity across various solid specimens. **Co-requisite**: MECH 431. Annually.		

MECH 435	HVAC	3(3,0)
Air conditioning basics; occupant comfort and indoor air quality; heating and cooling load calculations; HVAC controls; air conditioning systems and air delivery systems; refrigeration basics, methods, and applications. **Prerequisite**: MECH 431. Annually.		

MECH 499	Co-op Work Experience	3(3,0)
This Co-op work experience is designed to provide students with a five-week short-term work experience in mechanical engineering. Students are encouraged to network with industry professionals in their discipline. A report, a poster, and a PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

MECH 520	Finite Element Method for Engineers	3(3,0)
This course focuses on the basic concepts of finite element methods; formulating shape functions and basic structural element equations; modeling and analysis of 1-D, 2-D, and 3-D structural analysis and heat transfer problems; finite element modeling and analysis using MATLAB and ANSYS. **Prerequisites**: MECH 320, MATH 421, and MECH 431. Annually.		

MECH 521	Vibrations and Acoustics	3(3,0)
This course covers Free Vibrations, Vibrations with Damping, Harmonically Excited Vibrations, Complex N-Degree-of-Freedom Systems, Continuous Vibrations, Fundamentals of Sound Waves and Acoustics, and Basic Machinery Vibration. Prerequisite: MECA 340. Annually.		

MECH 595A	BE Summative Learning Project I	1(1,0)
A team-oriented project includes conducting and formulating experimental laboratory work and design problems on a specific, approved specialty topic. The project includes a literature review, a scope of work, and a project proposal. **Prerequisite**: Senior Standing, ENGL 217. Annually.		

MECH 595B	BE Summative Learning Project II	3(3,0)
A team-oriented project includes conducting and formulating experimental laboratory work and design problems on a specific, approved specialty topic. The project includes a literature review, design work, data collection, experimentation, data analysis, and technical writing. **Prerequisite**: MECH 595A. Annually.		

II. Elective Courses

Students may meet their technical electives requirements from the following selection of elective courses.

MECH 436	Petroleum Refining Technology for Mechanical Engineers	3(3,0)
Crude oil; petroleum products; refinery configuration; major process equipment; crude distillation; catalytic reforming; hydrotreating; gasoline; gas treating; product blending; risk-based inspection; environmental control; economics. **Prerequisite**s: CHEM 211, MECH 330, and MECH 320. On demand.		

MECH 437	Heat Exchangers Design, Performance, and Inspection	3(3,0)
Quick review of “mechanics of materials” and “heat transfer”; Codes & Standards; Heat exchanger types and applications; Thermal design; Mechanical design;		

Inspection and testing; Efficiency; TEMA standard use; Design project; Laboratory testing. **Prerequisites**: MECH 320 and MECH 431. On demand

MECH 445	**Automotive Engineering**	**3(3,0)**
Studies of automotive components: engine parts (crankshaft, camshaft, inlet/exhaust valves, piston/cylinder mechanism); power boosting (turbocharging/supercharging); transmission; steering mechanism design; engine combustion and emissions; major project on simulation of overall engine performance. **Prerequisites**: MECH 330 or 333 and MECH 321. On demand.		

MECH 448	**Plumbing Engineering**	**3(3,0)**
Soil, waste, and vent piping; drainage system; piping systems for domestic and industrial hot and cold water, fuel oil, de-ionized and de-mineralized water; laboratory compressed air and vacuum; laboratory outlets; laboratory drains; vacuum pumps; general piping installation. **Prerequisites**: MECH 335 or MECH 333. On demand.		

MECH 451	**Fluid Mechanics II**	**3(3,0)**
Review of incompressible flow; potential flow; similitude, dimensional analysis, and modeling; external and open channel flow; pumps and turbines. **Prerequisite**: MECH 335. On demand		

MECH 453	**Buildings and Energy**	**3(3,0)**
Selected topics in low-energy building design: building history and energy use; building physics; architectural and material considerations; building energy systems and the use of renewable energy systems; hands-on project. **Prerequisite**: MECH 330 or MECH 333. On demand.		

MECH 455	**Gas Turbines**	**3(3,0)**
Thermodynamics of gas turbines; Gas turbine design; comparison of designs; shaft power and aircraft propulsion cycles; component efficiencies; basic components; general considerations of turbomachinery design; axial compressors; combustors; clean combustion systems; fuel control; design-point performance calculations; off-design behavior and matching procedure; project. **Prerequisite**: MECH 330 or MECH 333. On demand.		

MECH 461	**Thermal Systems Design**	**3(3,0)**
Modeling of thermal systems; system solution; design applications of optimization methods: Lagrange multipliers, search methods, and dynamic, geometric, and linear programming**. Prerequisite**: MECH 431. On demand.		

MECH 468	**Internal Combustion Engines**	**3(3,0)**
Fuels and their properties; review of basic thermodynamics and gaseous mixtures; combustion thermodynamics; engine types; combustion chambers; valve mechanisms; dynamic balance of ICE; flame temperature; dissociation and detonation; turbocharging and supercharging; testing and measurements; engine		

characteristics and performance; laboratory demonstration. **Prerequisite**: MECH 334. On demand.

MECH 470	Product Design and Development	3(3,0)
Idea generation; concept generation; concept selection; functional analysis; engineering design process for systems and components; economic consideration; reliability analysis; product safety; design project. **Co-requisite**: MECH 421. On demand.		

MECH 471	Refrigeration	3(3,0)
Methods of producing cold: thermodynamic basis, first and second law analyses – vapor compression systems; ideal vs. actual cycles; single-stage, multistage and cascaded systems; environmental considerations and alternate refrigerants; vapor absorption systems. **Prerequisite**: MECH 334. On demand.		

MECH 498	Special Topics in Mechanical Engineering	3(3,0)
Guided study and research on major-related topics. **Prerequisite**: Instructor's Consent. On demand.		

MECH 523	Mechanics of Composite Materials	3(3,0)
Analysis of a thin sheet of aligned-fiber reinforced, elastic matrix (lamina) to layered structures comprised of these sheets (laminates); mechanics of the thick-sectioned components (of many microstructural compositions) desirable in many engineering applications; processing of composite materials; mechanical and thermal properties and their degradation with time; practical use; fatigue versus static strength; finite element application. **Prerequisite**: MECH 320. On demand.		

MECH 524	Elasticity and Plasticity	3(3,0)
Analysis of Cartesian tensors using indicial notation; 3-D stress and strain concepts; point stress and strain about tensor concepts; deviatoric and spherical stress state; Octahedral shear stress; equations of equilibrium and compatibility; constitutive laws for 3-D elastic and elastic-plastic stress state; kinematic & isotropic hardening models in plasticity; finite element applications. **Prerequisite**: MECH 320. On demand.		

MECH 525	Advanced Dynamics	3(3,0)
General introduction; rigid body kinematics: translation (3-D), rotation (3-D), compositions of rotations, Euler-angles presentation of rotation, Roll-pitch-yaw presentation of rotation, axis-angle presentation of rotation, homogenous transformations, rotation matrix derivative, angular velocity, linear velocity, acceleration; rigid body kinetics: review of Newtonian dynamics, generalized coordinates, holonomic and non-holonomic constraints, virtual displacements, virtual work, D'Alembert's principle, Lagrange equations, Hamilton's principle, canonical transformations; applications of Hamiltonian dynamics; multi-body dynamics. **Prerequisite**: MECH 220. On demand.		

MECH 527	Modern Material Science	3(3,0)
Advanced coverage of metals and alloys; applications and processing of ceramics; characteristics, applications, and processing of polymers; surface property enhancement; materials simulation; economic, environmental, and societal issues in materials science; biomaterials and nanoscale materials. Prerequisite: MECH 223. On demand		

MECH 530	Computational Fluid Dynamics	3(3,0)
This course introduces computational techniques to solve thermal-fluid problems. It integrates applications using ANSYS and CFD theory where it covers the following: Structured and unstructured grids; an overview of various discretization methods with emphasis on the finite volume method; Diffusion term discretization; Convection term discretization; Source term discretization; Transient term discretization, implicit and explicit schemes; Direct and Iterative solvers, tridiagonal matrix algorithm (TDMA), Gauss-Seidel; CFD analysis process (domain generation, domain meshing, domain setup or mathematical modeling, and post-processing); customizing Ansys fluent solver through user-defined functions; application of ANSYS code to practical engineering problems. Prerequisite: MECH 431.		

MECH 541	Renewable Energy	3(3,0)
Energy situation; energy conversion; solar power; Photovoltaic and thermoelectric conversion, ocean, tidal, and wave power; wind power; biomass energy; the case of nuclear power; project. **Prerequisite**: MECH 330 or MECH 333. On demand.		

MECH 570	Digital Manufacturing	3(3,0)
In this course, the student will get an insight into diagnosing and correcting operational flaws; modeling and fabrication based on additive manufacturing technologies; automation technologies; production planning, process quality control; use of modern four-axis Computer Numerical Control (CNC) machines, 3D printers, and up-to-date software programs, including AutoCAD, Inventor, SolidWorks, and VisualCam. **Prerequisite**: MECH 422 and MECA 541. On demand.		

MECH 571	Optimization in Engineering Design	3(3,0)
Problem definition, cost function and constraints; local vs. global methods; deterministic vs. stochastic methods; linear vs. non-linear programming; constrained vs. unconstrained; continuous vs. discrete; gradient-based methods; combinatorial optimization techniques; applications to various engineering problems (single/multi-objective); project. **Prerequisite**: MATH 421. On demand.		

III. AI-Restricted Elective Courses

Students must choose at least one elective course (3 credits) from the following selection of AI-Restricted courses.

MECA 445	Advanced Programming for AI & Robotics	3(3,0)
This advanced course, built upon the prerequisite "Mechatronics Systems Programming," delves into the practical application of Python for developing intelligent robotic systems. A significant portion of the course will be dedicated to the Robot Operating System (ROS), providing students with the skills to design, implement, and test complex robotic software. Students will learn core ROS concepts, including nodes, topics, services, and parameters, as well as essential ROS tools for simulation and real-world robot interaction. Furthermore, the course will explore techniques for integrating Artificial Intelligence (AI) into mechatronics applications using Python. This will include an introduction to relevant Python libraries for machine learning, computer vision, and data processing, along with practical examples of how AI can be applied to tasks such as perception, decision-making, and control in robotic systems. The course will emphasize hands-on projects and assignments using platforms such as Raspberry Pi and NVIDIA Jetson Nano to provide students with practical experience in developing and deploying advanced robotic and AI-driven mechatronic solutions. **Prerequisite**: MECA 444L. Annually.		

MECA 535	Expert Systems and Applied Intelligence	3(3,0)
This course explores the core principles of intelligent systems and their application in solving complex engineering and decision-making problems. Students will learn perception methods, reasoning techniques, and search strategies, as well as key AI tools such as neural networks, evolutionary computing, fuzzy logic, and intuitionistic fuzzy logic. The course also introduces intelligent programming using Prolog/python to model knowledge and build rule-based expert systems. Through practical examples, students will gain hands-on experience in developing AI-driven solutions for real-world applications. **Prerequisite**: MECA 444L.		

MECA 536	Applied Deep Learning for Intelligent Systems	3(3,0)
This course introduces the basics of computer vision and machine learning, focusing on their use in intelligent systems. Students will learn techniques for object detection, image understanding, and sensor data interpretation, using deep learning tools to build decision-making models. Hands-on practice with simulation platforms will allow students to design and test vision-based solutions in realistic scenarios. At the end of the course, participants will gain practical skills in combining perception, planning, and control, preparing them for applications in autonomous systems and other technologies. **Prerequisite**: MECA 444L.		

MECA 546	Smart Energy Management Systems	3(3,0)
This course delves into smart energy management systems (SEMS), focusing on versatile operating modes for solar photovoltaic (PV) and battery energy storage (BESS) systems. It equips mechatronics engineering students with the fundamental knowledge and practical skills required to design, implement, and optimize these crucial technologies for a sustainable future. Furthermore, the course discusses the potential of using generative AI for the design and development of smart energy management systems. A capstone case study is presented and analyzed. **Prerequisite**: MECA 442.		

MECH 522	Mechanism Synthesis: Classical Kinematics to AI-Driven Optimization	3(3,0)
This course introduces the theory and computational practice of mechanism synthesis, with emphasis on the design of four-bar linkages for path, motion, and function generation. Classical kinematic synthesis methods are integrated with numerical optimization techniques, including Newton–Raphson, least-squares, and gradient-based methods, as well as intelligent metaheuristic algorithms such as genetic algorithms, tabu search, and ant colony optimization. The course also exposes students to data-driven and AI-assisted approaches, including deep neural networks for inverse mechanism design, as advanced computational tools for automation, design-space exploration, and decision support in mechanism synthesis. Computational projects provide students with hands-on experience in mechanism modeling, simulation, error analysis, optimization, and the synthesis of mechanical systems under practical design constraints. **Prerequisite:** MECH 321.		

MINOR IN APPLIED PETROLEUM ENGINEERING TECHNOLOGY

Rationale

The Applied Petroleum Engineering Technology (APET) minor addresses the expected national need for engineering personnel knowledgeable in the Oil and gas industries' transportation, processing, and storage. It also prepares undergraduate students for possible graduate studies in petroleum engineering and related fields.

It is primarily geared towards mechanical engineering students, but it should also accommodate students from other engineering backgrounds, provided all prerequisite courses are completed.

The program is primarily concerned with the technical, environmental, and economic aspects of oil and gas extraction as well as downstream processing and handling. The goal is to apply fundamental knowledge of upstream and downstream operations to enhance the graduate's ability to perform effectively as a practitioner in a short period of time.

Program participants are required to complete at least 18 credits of related coursework. Although not required, with proper and timely advising, they may also complete their Co-op work experience or their BS project in Oil and Gas, thereby enhancing their capabilities.

To effectively cover the added coursework in the fall and spring semesters, one full-time faculty member with a specialty in Petroleum/Chemical Engineering is required.

Program Objectives

The primary objectives of the APET minor program are to:

- Give engineering students basic knowledge of the Oil and Gas industry;
- Endow students with the ability to deal with applied Oil and Gas downstream engineering;
- Prepare qualified Lebanese graduates to become technically productive contributors to the future Lebanese petroleum sector;
- Provide an impetus for undergraduate students to pursue graduate studies in the Oil and Gas domain.

Learning Outcomes

After completing the minor program, students should have the ability to:

- Apply relevant knowledge of mathematics, science, and engineering to identify, formulate, and solve practical problems related to downstream petroleum engineering processes;
- Use modern engineering tools relevant to core petroleum engineering practices;
- Deal with the uncertainty and risk involved in petroleum transportation, refining, and storage problems.

Career Options

This minor enables APET holders to begin careers as field technologists in the exploration, transportation, refining, and storage sectors of the conventional oil and gas production industry. Typical employers include oil and gas Companies; oil service companies, including the QA/QC sector; government-related agencies; and petrochemical plants and refineries.

Program Requirements

While most relevant to mechanical engineering students, this minor is offered to undergraduate RHU students in the BS program, from different disciplines (mechanical, mechatronics, civil, electrical, computer, communications, and biomedical). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements. Proper advising is pertinent to the timely completion of the minor.

To complete the APET minor, a student must:

1) Declare an APET minor by completing the Minor Declaration Form.
2) Obtain the approval of the dean of the college major and the dean of the College of Business Administration;
3) Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
4) Complete 18 credits of coursework as specified below.

APET Curriculum

The program strikes a proper balance between fundamentals and practice while preparing students to become productive contributors in the oil and gas industry. Accordingly, this minor shall include:

- Oil and gas chemistry and properties
- Design and analysis of systems and procedures for producing and handling fluids;
- Decision-making under conditions of risk and uncertainty;
- Safety hazards and environmental impact consideration and protection;
- Pipeline design and installation;
- Above-ground storage and tanks;
- Gas and oil processing and storage (gas, crude oil, and refined oil products);

- Refinery units and equipment inspection using codes and standards of international organizations such as API and ASME;
- Quality assurance and quality control aspects;
- Non-Destructive Testing (NDT)
- National and regional macroeconomic impact.

The APET curriculum comprises 18 credits: 15 mandatory and 3 elective, selected to satisfy the requirements of the proposed program objectives and learning outcomes. The courses that encompass the minor are given below.

I. Mandatory Courses

Description of the five mandatory courses (15 credits) follows.

APET 200	**Petroleum Engineering**	**3(3,0)**
Overview of the petroleum engineering and petroleum industry: Nature of oil and gas reservoirs, exploration and drilling techniques, formation evaluation, well completions and production, surface facilities, reservoir mechanics, offshore and onshore structures, environmental considerations; new technologies, prospects; petroleum industry activities in Lebanon and the Middle East; available industry and challenges; Oil & gas safety issues.		

APET 201	**Crude Oil & Natural Gas Transportation & Storage**	**3(3,0)**
This course will introduce the petroleum student to natural gas and crude oil transportation and storage equipment design and operation; following the flow of gas from the wellhead to the gas & oil pipelines, to the storage tanks; calculations are performed on pipeline sizing, inspection, and installation; storage tank design, installation, and inspection; use of commercial software. **Prerequisite**: APET 200.		

APET 203	**Modern Refining Technology**	**3(3,0)**
This course presents an introduction to petroleum refining technology, an overview of crude oil supply and petroleum product demand, and a description of refinery process technology. Some refining technologies are described, such as crude oil distillation, heavy oil conversion options, hydrotreating, and catalytic reforming. Some gas processing is also discussed. **Prerequisite**: APET 200.		

CHEM 213	**Physical and Organic Chemistry**	**3(3,0)**
This course will allow students to learn chemical principles related to the general, Physical, and Organic fields of chemistry. Topics include study of different states of matter, homogeneous and heterogeneous mixtures, study of endothermic and exothermic processes, discussion on surface phenomena and industrial applications of adsorption and absorption processes; Structure and properties of organic compounds, synthesis and reactions of various compounds, mechanism of radical substitution, electrophilic addition and electrophilic aromatic substitution, Liquids and liquid-gas solutions, viscosity and surface tension of liquids, name and write formulas		

of selected organic compounds generally found in oil and gas industries; Catalyst types and use. **Prerequisite**: general or environmental chemistry.

GEOL 210	**Petroleum Geology**	**3(3,0)**
This course introduces the geological concepts important to petroleum exploration and exploitation. Topics include the composition of the Earth, plate tectonics, rocks and minerals, the formation of sediments and sedimentary rocks, stratigraphy, geologic structures, oil and gas sources, reservoir engineering properties, exploration techniques, well evaluation, and reservoir development.		

II. Elective Course

Students may choose one three-credit course from the following list.

APET 202	**Materials Selection – Corrosion Monitoring and Control**	**3(3,0)**
Topics will provide the student with an understanding of the factors involved in materials utilization in petroleum equipment design and construction, such as pressure vessels & piping, materials properties, basic welding principles, and the applicable codes, standards, and engineering specifications used for materials specification in the hydrocarbon industries. Common non-destructive testing procedures used on equipment include identifying defect types, such as corrosion and cracks. Equipment strategy: monitoring and control. **Prerequisite**: APET 200.		

APET 300	**Applied Fluid Mechanics**	**3(3,0)**
Review of incompressible flow, turbulence, and dimensional analysis; pipe flows; boundary layers, lift, drag, and flow separation; compressible flow; turbomachinery and rotating machinery; oil and gas issues.		

III. Program Prerequisites

Students who wish to study the APET minor must complete the following courses.

Course Code	Title	Credits
CHEM 211	Environmental Chemistry and Lab	3
ENGL 217	Professional English Communication	3
MATH 215	Calculus III	3
MATH 311	Linear Algebra with Applications	3
MATH 314	Ordinary Differential Equations	3
MATH 351	Probability and Statistics	3
MECH 320	Mechanics of Materials	3
MECH 330	Thermodynamics I	3
MECH 335	Fluid Mechanics	3
PHYS 210	Fundamentals of Physics and Lab	3

MINOR IN INDUSTRIAL ENGINEERING AND MANAGEMENT SYSTEMS

Rationale

The Industrial Engineering and Management Systems (IEMS) minor aims to expand engineering students' expertise in system-wide optimization, decision science, and digital integration. It prepares students to architect the "intelligence layer" of the enterprise and prepares undergraduate students for graduate studies in Industrial Engineering, Engineering Management, and Operations Research.

It primarily targets mechanical and mechatronics engineering students, but it also accommodates students from various engineering backgrounds, provided all prerequisite courses are completed. The program focuses on mathematical optimization, financial engineering, facility logistics, quality assurance, and digital twin simulation. By the end of this minor, a student is expected to be proficient in designing, simulating, and managing complex industrial ecosystems that are cost-effective, data-driven, and industry-ready.

Program participants are required to complete at least 18 credits of related coursework. Although not required, with proper and timely advising, they may also complete their Co-op work experience or their BE project in Industrial Engineering or Systems Management, thereby enhancing their capabilities.

Program Objectives

The primary educational objectives of the IEMS minor program are to:

- Give engineering students a sophisticated understanding of Industrial Engineering principles and management systems.
- Equip students with the knowledge to pursue graduate studies in Industrial Engineering, Engineering Management, Systems Engineering, Data Analytics, and other related disciplines.

Learning Outcomes

After completing the minor program, students should have the ability to:

- Apply advanced mathematical modeling and operations research to optimize resource allocation and industrial scheduling.
- Analyze the financial viability of engineering projects using rigorous economic tools and ROI assessments.

- Design and validate complex systems using CAD-integrated Digital Twin and discrete-event simulation software.
- Manage global supply chains and logistics networks while ensuring high standards of quality through Six Sigma and Lean methodologies.

Career Options

This minor allows IEMS holders to start careers in engineering management, systems analytics, supply chain architecture, industrial consultancy, project management, operations strategy, and digital manufacturing integration in both regional and global markets.

Program Requirements

While most relevant to mechanical and mechatronics engineering students, this minor is offered to undergraduate RHU students in the BS/BE program from different disciplines. It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts. Proper advising is pertinent to the timely completion of the minor. To complete the IEMS minor, a student must:

1. Declare an IEMS minor by completing the Minor Declaration Form;
2. Obtain the approval of the dean of the college major;
3. Obtain a Minor Cumulative Grade Point Average of no less than 70%;
4. Complete 18 credits of coursework as specified below.

IEMS Curriculum

The program offers a rich blend of mathematical optimization and virtual system modeling, equipping students with the tools to navigate the modern industrial world. Accordingly, this minor shall include:

- Engineering financial management and project economics.
- Advanced quality engineering and lean operations.
- Mathematical optimization and deterministic operations research.
- Physical facility planning and systematic layout design.
- Digital Twin architecture and virtual system simulation.
- Global supply chain engineering and logistics optimization.

Mandatory Courses

The student must complete a minimum of 6 credit hours of coursework exclusive to the IEMS minor that will not count toward requirements for their major or any other minor. A description of the 18 credits from which a student must select is given below:

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

BMGT 410	Quality Management	3(3,0)
This course highlights that TQM has become crucial for business excellence in worldwide markets. It allows students to learn techniques that establish sustainable quality improvement in both product and service industries. Topics include seven quality control tools; SPC; quality function deployment and the house of quality; product reliability; Six Sigma; Benchmarking; quality certification and awards; and quality costing. **Prerequisite:** BADM 250		

ENGR 400	Operations Research	3(3,0)
This course introduces mathematical modeling and algorithmic techniques for optimizing decision-making in complex engineering systems. Students will master the application of Linear Programming and the Simplex Method to solve problems involving constrained resources. Key topics include duality and sensitivity analysis (determining how changes in the system affect the optimal solution), network flow optimization (shortest-path and PERT/CPM), and integer programming. The course emphasizes translating real-world industrial challenges—such as production scheduling, trajectory planning for robots, and resource allocation—into rigorous mathematical frameworks. **Prerequisite:** MATH 311		

ENGR 410	Supply Chain Engineering & Logistics	3(3,0)
This course applies engineering and mathematical principles to the design and optimization of global supply chain networks. Moving beyond general procurement, students will focus on the quantitative modeling of inventory control (including deterministic and stochastic models), warehousing operations, and transportation logistics. Key topics include the "Bullwhip Effect" in multi-echelon systems, facility location modeling using center-of-gravity methods, and the strategic integration of Just-in-Time (JIT) principles. The course specifically addresses Design for Logistics (DFL), teaching students how mechanical design decisions (packaging, weight, and modularity) impact the total landed cost and supply chain resilience in a global market. **Prerequisite:** MATH 351 and ENGR 300		

MECH 424	Manufacturing Systems & Facilities Planning	3(3,0)
This course focuses on the strategic design and organization of integrated manufacturing environments. Moving from individual machine operations to system-wide flow, students will study Systematic Layout Planning (SLP), group technology, and the design of Cellular Manufacturing units. The curriculum emphasizes quantitative methods for line balancing, machine-requirement calculations, and material-handling systems. Students will analyze the physical constraints of the shop floor to minimize "waste" in movement and optimize the integration of automated workstations and robotic cells. **Prerequisite:** MECH 224 and MECH 422		

MECA 425	Digital Twin & System Simulation	3(3,0)
This course explores the creation and use of Digital Twins to model, simulate, and predict the behavior of complex manufacturing and mechatronic systems. Utilizing discrete-event simulation (DES) software, students will develop virtual replicas of physical systems to analyze the impact of stochastic variables, such as machine downtimes and fluctuating arrival rates. The course covers data distribution fitting, model verification, and the statistical validation of simulation outputs. By performing "what-if" experimental analyses, students will learn to identify system bottlenecks and optimize throughput in a risk-free virtual environment before physical deployment. **Prerequisite:** MATH 351 and MECA 541		

MECHATRONICS ENGINEERING PROGRAM

Mission

The mission of the Mechatronics Engineering program at RHU is to provide a solid education that enables students to start a professional career and/or continue graduate studies in mechatronics engineering. The program seeks to instill in students a sense of and understanding of moral and professional responsibilities, the ability to learn more, and the leadership required in their calling, primarily in Lebanon and the Middle East.

Objectives

In a few years after graduation, Mechatronics Engineering program graduates will:

1. Apply their multidisciplinary knowledge in Mechatronics engineering to meet employers' expectations.
2. Possess self-learning motivation and qualifications to pursue advanced studies.
3. Demonstrate professional skills and embrace leadership roles in their profession and/or communities.

Program Outcomes

Upon successful completion of the BS/BE program in Mechatronics Engineering, graduates will have:

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. Ability to communicate effectively with a range of audiences
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Mechatronics Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria, with no applicable program criteria.

Career Opportunities

Mechatronics engineers attain a broad spectrum of skills that are central to the practice of almost any professional field. Mechatronics engineers can find career opportunities in industries and enterprises such as food processing, building and construction, medical and pharmaceutical industries, aerospace, automotive, chemical, oil and gas, computer and electronics, renewable energy, and far more. Most importantly, you will be prepared to imagine something that never was and drive it forward to make it a reality! There is so much to be gained in ensuring the future of humanity, and mechatronics engineers can tap into these possibilities through the spirit of innovation and entrepreneurship.

Program Overview

At RHU, every experience the student has represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. Mechatronics engineering, being a holistic approach to designing and developing multifaceted smart systems, has remarkable strength. Student transformation is accomplished through a curriculum that orbits the edges of knowledge, facilities equipped with the latest technology and conforming to the highest standards, faculty members who are passionate and committed to students' academic success and personal growth, and an environment that inspires learning and drives discovery.

The Bachelor of Engineering (BE) program in Mechatronics Engineering comprises 152 credit hours across eight regular semesters and three Summer Semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three Summer Semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned cumulative GPA, major GPA, and the Summative Learning Project grades are all 70 or above.

The required credit hours span three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of credit hours for the BS and BE programs is shown in the following tables, which reveal the apparent multidisciplinary nature of Mechatronics. This demonstrates the value of a degree in Mechatronics as a starting point for several higher degree paths.

BS in Mechatronics Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	6	27	23

College Requirement	0	0	24	0	24	20
Program Requirement	28	0	39	0	67	57
Credits	28	0	84	6	118	100

BE in Mechatronics Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	9	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	41	15	39	0	95	62
Credits	41	15	87	9	152	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of required courses and descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Mechatronics Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of electives.

A. Mandatory Requirements

The mandatory core courses in Mechatronics Engineering (major and non-major) are listed in the table below.

Course #	Title	Credits	Prerequisites
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	CCEE 221
CIVE 211	Statics	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 331	Electric Machines	3	ELEC 210
MECA 340	Dynamic Systems Modeling	3	MATH 314

MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req.: MECA 341
MECA 400*	Summative Learning (BS) Project	3	ENGL 217
MECA 440	Control Systems Design	3	MECA 340
MECA 440L	Control Systems Design Lab	1	MECA 440
MECA 441	Sensors and Actuators	3	MECA 341
MECA 442	Microcontrollers for Mechatronics	3(2+2)	CCEE 221 and MECH 200
MECA 443	Mechatronic System Design	3	MECA 440, 441, and 442
MECA 443L	Mechatronic System Design Lab	1	Co-req.: MECA 443
MECA 444L	Mechatronic Systems Programming	1	CCEE 214 and MECH 200
MECA 445	Advanced Programming for AI and Robotics	3	MECA 444L
MECA 444	Mechatronic Systems Programming	3	CCEE 214 and MECH 200
MECA 499	Co-op Work Experience	3	ENGL 217 Senior Standing
MECA 541	CAD/CAM/CAE	3	MECH 224 and MECH 421
MECA 542	Industrial and Manufacturing Control	3	ELEC 320
MECA 544	Robotics	3	MECH 321
MECA 595A	BE Summative Learning Project I	1	ENGL 217 Senior Standing
MECA 595B	BE Summative Learning Project II	3	MECA 595A
MECA 4xx/5xx	Technical Electives	15	Per course requirement
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
MECH 224	Mechanical Engineering Graphics & CAD	3	
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
MECH 333	Thermal Fluid Sciences	3	MATH 216
MECH 421	Mechanical Design I	3	MECH 320
MECH 422	Manufacturing Processes	3	MECH 320

* Required for BS-bound students. BE students take a Technical Elective instead.

B. Technical Electives

As part of the Bachelor of Engineering in Mechatronics Engineering program, the student is required to take 15 credit hours of technical electives. These courses allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to interrelated Engineering disciplines. The student should, in cooperation with the academic advisor, select a list of electives that best meet their needs and aspirations. The listed technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

- Mechanisms and Robotics
- Automation

The student is highly recommended to take elective courses after completing the mandatory requirements in the related area. Students may select technical electives from other programs, subject to the approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
Mechanisms and Robotics			
MECA 485	Autotronics	3	MECA 440
MECA 498	Special Topics in Mechatronics Engineering	3	
MECA 535	Expert Systems and Applied Intelligence	3	MECA 444L
MECA 536	Applied Deep Learning for Intelligent Systems	3	MECA 444L
MECA 543	Fluid Power Systems	3	MECH 333 or MECH 335
MECA 581	Systems Engineering	3	
MECH 522	Mechanism Synthesis Classical Kinematics to AI-Driven Optimization	3	MECH 321
Automation			
MECA 484	Civionics	3	
MECA 485	Autotronics	3	MECA 440
MECA 498	Special Topics in Mechatronics Engineering	3	
MECA 515	Modern Control Systems	3	MECA 440
MECA 545	Integrated Automation Lab	1	Co-req: MECA 542
MECA 546	Smart Energy Management Systems	3	MECA 442
MECA 586	Mechatronics Drives	3	ELEC 320

C. Summative Learning Project

Students must complete a 3-credit-hour course for the BS (4 credits for the BE; take 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to develop a promising idea. The SLP could be an extension of

the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in MECA Engineering. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
CCEE 214	Introduction to Programming	3	
CIVE 211	Statics	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 210	Fundamentals of Physics and Lab	3	
Year 1, Spring Semester (15 Credits)			
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
	Humanities/Fine Arts Elective I	3	
Year 1, Summer Semester (9 Credits)			
CHEM 211	Environmental Chemistry and Lab	3	
MECH 224	Mechanical Engineering Graphics & CAD	3	
MATH 311	Linear Algebra with Applications	3	
Year 2, Fall Semester (17 Credits)			
CCEE 221	Logic Design	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req: ELEC 320
MATH 314	Ordinary Differential Equations	3	MATH 216
MECH 320	Mechanics of Materials	3	CIVE 211

MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
Year 2, Spring Semester (17 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
ENGL 217	Professional English Communication	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 215
MECA 340	Dynamic Systems Modeling	3	MATH 314
MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req: MECA 341
MECH 333	Thermal Fluid Sciences	3	MATH 216
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 317	Partial Differential Equations	3	MATH 314
	Social Sciences/ Cultural Elective I	3	
Year 3, Fall Semester (16 Credits)			
MECA 440	Control Systems Design	3	MECA 340
MECA 441	Sensors & Actuators	3	MECA 341
MECA 442	Microcontrollers for Mechatronics	3 (2+2)	CCEE 221 and MECH 200
MECA 444L	Mechatronic Systems Programming	1	CCEE 214 and MECH 200
MECA 541	CAD/CAM/CAE	3	MECH 224 and MECH 421
MECH 421	Mechanical Design I	3	MECH 320
Year 3, Spring Semester (17 Credits)			
ARAB 212	Arabic Language & Communication	2	
ELEC 331	Electric Machines	3	ELEC 210
MATH 421	Numerical Analysis	3	MATH 311
MECA 440L	Control Systems Design Lab	1	MECA 440
MECA 443	Mechatronic System Design	3	MECA 440, 441, and 442
MECA 443L	Mechatronic System Design Lab	1	Co-req.: MECA 443
	Community & Sustainability	1	Co-req.: ENGL 210
	Engineering Technical Elective I*	3	Per course requirements
Year 3, Summer Semester (3 Credits)			
MECA 499	Co-op Work Experience	3	ENGL 217 Senior Standing
The 118 credits above complete the requirements for a Bachelor of Science Degree in Mechatronics Engineering.			
Year 4, Fall Semester (16 Credits)			
ENGR 510	Engineering Project Management	3	ENGR 300

MECA 445	Advanced Programming for AI and Robotics	3	MECA 444L
MECA 595A	BE Summative Learning Project I	1	ENGL 217; Senior Standing
MECH 422	Manufacturing Processes	3	MECH 320
	Engineering Technical Elective II	3	Per course requirements
	Engineering Technical Elective III	3	Per course requirements
Year 4, Spring Semester (18 Credits)			
MECA 542	Industrial and Manufacturing Control	3 (2+2)	ELEC 320
MECA 544	Robotics	3	MECH 321
MECA 595B	BE Summative Learning Project II	3	MECA 595A
	Engineering Technical Elective IV	3	Per course requirements
	Engineering Technical Elective V (AI-Restricted**)	3	Per course requirements
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take the MECA 400 – Summative Learning (BS) Project instead.

** Students are required to choose at least one technical elective (i.e., 3 credits) from the list of AI-Restricted Elective courses.

Note: Engineering Technical Electives (levels 400 or above) are selected as such:
9 credit restricted Departmental Electives
6 credits from any engineering discipline at the 400-level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

Description of the mandatory, non-major courses follows.

CCEE 214	**Programming & Lab**	**3(2,2)**
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures. Laboratory experiments. **Prerequisite**: None.		

CCEE 221	**Logic Design**	**3(3,0)**
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices,		

including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices. **Prerequisite**: None.

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: CCEE 221. Equivalent to COSC 351L.		

CHEM 211	Environmental Chemistry and Lab	3(2,2)
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.		

CIVE 211	Statics	3(3,0)
Static laws; force vectors and operations; force systems and moments; free-body diagrams; equilibrium of particles and rigid bodies in 2D and 3D; plane trusses and frames; cables; internal forces: shear and moment diagrams; centroid and center of mass; moment of inertia; and friction. **Prerequisite**: None. Annually.		

ELEC 320	Electronics	3(3,0)
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	Electronics Lab	1(0,2)
This is a lab course that experiments with electrical and electronic circuits. It covers passive electrical elements and sources, lab instruments, voltage divider circuits, Thevenin's theorem, RC circuits, diode circuits, Op-Amp circuits, and BJT and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, R, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 331	Electric Machines	3(3,0)
This course covers magnetic circuits; fundamentals of electromechanical energy conversion; and the construction, testing, steady-state analysis, and operational characteristics of DC machines, transformers, induction machines, and synchronous machines. **Prerequisite**: ELEC 210.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting and financial statements. **Prerequisite**: None. Annually.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including project selection and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics, and professional conduct, realizing the impact of engineering solutions in various contexts (global, economic, environmental, societal, etc.), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar		

curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	Partial Differential Equations	3(3,0)
Lagrange theorem, boundary conditions of first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables**,** introduction to Fourier series and integrals**,** boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

MECH 200	Introduction to Mechanical Engineering	3(3,0)
This course aims to provide students with a comprehensive overview of the fundamentals of Mechanical Engineering concepts and applications. Engineering topics to be covered shall include an introduction to learn-by-practice concepts; microcontrollers (Arduino); sensors and actuators; materials, mechanisms, codes, and standards; and robotics (LEGO Mindstorms). In addition, students will be introduced to engineering ethics and will learn how to apply the ASME/ABET code of ethics in practical case studies. The course also integrates the development of hands-on skills, teamwork, and communication; exercises and projects engage students in building, implementing, and testing their designs. **Prerequisite**: None. Annually.		

MECH 220	Dynamics	3(3,0)
Brief review of statics; proper use of different coordinate systems: Cartesian and normal/tangential axes; kinematics of a particle: rectilinear, curvilinear, and relative motion; kinetics of a particle: force and acceleration, work and energy, impulse and momentum; planar kinematics of a rigid body: translating and rotating axes, absolute and relative motion; planar kinetics of a rigid body: force and acceleration, work and energy, impulse and momentum. **Prerequisite**: CIVE 211. Annually.		

MECH 224	Mechanical Engineering Graphics and CAD	3(3,0)
Drafting principles; part and assembly drawings; generating 2D and 3D drawings and solid models using AutoCAD; customizing AutoCAD for specific applications; generating 2D drawings using CAD/CAM software; generating parametric, feature-based solids using Autodesk Mechanical Desktop. Introducing mechanical Revit, a Building Information Modeling (BIM) software tailored for the mechanical, electrical, and plumbing (MEP) disciplines. **Prerequisite**: None. Annually.		

MECH 320	Mechanics of Materials	3(3,0)
Mechanical stress, strain, and stress/strain relations under pure tension, compression, pure bending, and pure torsion of circular bars; 1-D thermal strain & stress; combined stresses; basic thin-walled cylindrical and spherical shells under internal pressure; principal stresses; plane stress transformation & Mohr's circles; basic 3-D elastic stress/strain relations; strain measurement using strain gauge technique. **Prerequisite**: CIVE 211. Annually.		

MECH 320L	Mechanics of Materials Lab	1(0,2)
This hands-on laboratory course complements theoretical concepts from Mechanics of Materials (MECH 320) by providing practical experience in the experimental analysis of material behavior under various loading conditions. Students will conduct tests involving axial tension/compression, bending, shear, and torsion to investigate stress-strain relationships, deformation, and failure modes in structural materials. Emphasis is placed on accurate measurement techniques using strain gauges, extensometers, and deflection indicators, as well as data analysis, interpretation, and comparison with theoretical predictions. Through structured experiments and lab reports, students will develop skills in experimental design, data acquisition, critical analysis, and technical communication. **Co-requisite**: MECH 320.		

MECH 321	Mechanisms and Dynamics of Machinery	3(3,0)
Mechanization of motion; kinematic analysis of linkage mechanisms; graphical synthesis of planar mechanisms; synthesis of cam-follower mechanisms (graphical and analytical designs); gear terminology and types of gears used in machines; analysis and synthesis of gear trains; velocity, acceleration, and force analysis of mechanisms. **Prerequisite**: MECH 200 and MECH 220. Annually.		

MECH 333	Thermal Fluid Sciences	3(3,0)
This course imparts thermal-fluid fundamental concepts (including Thermodynamics, Fluid Mechanics, and Heat Transfer) to non-mechanical engineering students. These cover thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law) and simple flow devices; fluid statics; Bernoulli's equation; Conservation of mass, momentum & energy; viscous flow in pipes; introduction to conduction, convection and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

MECH 421	Mechanical Design I	3(3,0)
Engineering materials properties; tools of machine design; stress, strain, and deformation of machine parts; buckling of columns; statistical methods and safety consideration; static and fatigue failure theories; design of machine parts for strength under steady loads; design of machine parts for rigidity, deflection of beams and rotation of shafts; design of machine parts for strength under cyclic/dynamic loads; codes and standards. **Prerequisite**: MECH 320. Annually.		

MECH 422	Manufacturing Processes	3(3,0)
General overview of manufacturing; properties of engineering materials; steelmaking; dimensions and tolerances; manufacturing processes: rolling, forging, extrusion, powder metallurgy; mechanical assembly: welding, brazing, soldering, bonding; welding QA/QC; non-destructive testing (NDT); machining; casting; sheet metal forming; plastic injection molding; polymers and composites. **Prerequisite**: MECH 320. Annually.		

PHYS 210	Fundamentals of Physics and Lab	3(3,1)
Welcome to the Introductory Physics course. This course consists of a review of Classical Mechanics, Fluid Mechanics, General Properties of Waves, Electric Forces, Electric Fields for Discrete and Continuous Charge Distributions, Gauss's Law, Electric Potential, Kirchhoff's Rules, Magnetic Fields and Forces, and a Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

Major Courses

Mandatory mechatronics courses are described below.

MECA 340	Dynamic Systems Modeling	3(3,0)
Introduction to dynamic modeling of mechanical, electrical, electromechanical, thermal, fluid, and mixed-discipline systems; Block diagrams and computer simulation (MATLAB); State-variable solution; Laplace Transform solution; Laplace transform		

properties; inverse Laplace transform; transfer function, frequency response, first- and second-order systems. **Prerequisite**: MATH 314. Annually.

MECA 341	Measurements and Instrumentation	3(3,0)
This course covers the fundamentals of instrumentation and measurement of various physical quantities. Topics include sensor types, technologies, characteristics, and calibration; design of a measurement system; statistical analysis of measured data; measurement noise and uncertainties; data acquisition, data storage and display devices; signal conditioning and interface electronics concepts including filtering, A/D and D/A conversion, amplification, modulation, compensation; applications; **Prerequisite**: ELEC 320. Annually.		

MECA 341L	Measurements and Instrumentation Lab	1(0,2)
The lab experience complements the topics covered in MECA 341. Students learn to use the NI ELVIS platform, LabVIEW programming, and the characteristics of data acquisition systems to build a measurement system, perform data analysis and sensor calibration, and design and build signal-conditioning circuits for various types of measurements. The lab involves a team project to integrate all instrumentation and measurement tools and techniques to build a measurement system and submit a report. **Co-requisite**: MECA 341. Annually.		

MECA 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 for the last semester of the BS Program. Annually.		

MECA 440	Control Systems Design	3(3,0)
This course focuses on the analysis and design of control systems. Measure control system performance; Transient and steady-state response analyses; Topics include analytical solutions in the time and S domains, root locus, Bode, Nyquist, Lead-Lag controller design, and PID compensator design; Analysis and design of control systems in state space. Control system analysis and design using MATLAB and SIMULINK; set up a control system, understand it, and optimize its performance; **Prerequisites**: MECA 340. Annually.		

MECA 440L	Control Systems Design Lab	1(0,2)
This 1-credit laboratory course provides students with practical experience in the design, analysis, and implementation of linear control systems. Students will gain hands-on proficiency in using MATLAB and Simulink as powerful tools for modeling, simulating, and analyzing control system behavior in both the time and frequency domains. A significant component of the lab involves the implementation of fundamental automatic controllers, including ON/OFF, Proportional-Derivative (PD), Proportional-Integral (PI), and Proportional-Integral-Derivative (PID) controllers, using Arduino microcontrollers for various real-world applications such as temperature control and speed regulation. Students will also be introduced to the diverse range of		

control systems training kits available in the laboratory, providing them with opportunities to experiment with different hardware and system dynamics. This lab-intensive course bridges the gap between theoretical concepts and practical implementation, equipping students with essential skills to design and deploy control solutions across a wide array of engineering applications. **Prerequisite**: MECA 440. Annually.

MECA 441	**Sensors and Actuators**	**3(3,0)**
Sensor and actuator technologies; classification and terminology of sensors and actuators; sensors characteristics; physical principles of sensing; measurement applications (position and speed, stress and strain, temperature, vibration and acceleration, pressure and flow…); Electromagnetic principles; solenoids and relays; electric motors (DC motors, Stepper motors…); Hydraulics actuators; Pneumatic actuators; interface methodologies and circuits; integration aspects; manufacturing techniques and material properties; team projects. **Prerequisite**: MECA 341. Annually.		

MECA 442	**Microcontrollers for Mechatronics**	**3(2,2)**
This course aims to provide a deep understanding of the integration of microcontrollers in mechatronic systems, starting from the technologies used to develop embedded systems to the internal features and peripherals of microcontrollers/microprocessors (timers, interrupts, analog-to-digital converters…). In this course, students develop various microcontroller programs for mechatronic applications, including the usage of I/O and communication peripherals. **Prerequisite**: CCEE 221 and MECH 200.		

MECA 443	**Mechatronic System Design**	**3(3,0)**
This course integrates all modules in mechatronics systems design. It consists of an overview of mechatronics; mechatronics systems design strategies; advanced modeling; use of control strategies; design of electric actuators; design of sensors; power electronics; hardware and software integration; and team projects on the design and implementation of integration between mechanical, electrical, and control systems. **Prerequisite**: MECA 440, 441, and 442.		

MECA 443L	**Mechatronic System Design Lab**	**1(0,2)**
Overview of mechatronic systems; integration of mechanical components, sensors, and actuators into mechatronics applications; use of switchgear components; integration of microcontrollers and FPGAs; LabVIEW programming; Design of power supplies. **Co-requisite**: MECA 443. Annually.		

MECA 444L	**Mechatronic Systems Programming**	**1(0,2)**
This introductory 1-credit course provides mechatronics students with a foundational understanding of Python. The curriculum is specifically designed to equip students with the essential Python skills necessary for subsequent coursework and applications in robotics, automation, and embedded systems. Emphasis will be placed on developing a strong grasp of core Python concepts, data structures, and control flow, laying the groundwork for using Python in the Robot Operating System (ROS), for artificial intelligence (AI) applications on platforms like Raspberry Pi and NVIDIA Jetson Nano,		

and for general scripting in mechatronic projects. Through hands-on exercises and practical examples relevant to mechatronics, students will gain the confidence to apply Python for tasks such as data acquisition, sensor interfacing, basic control algorithms, and interaction with robotic software frameworks. **Prerequisite**: CCEE 214 and MECH 200. Annually.

MECA 445	**Advanced Programming for AI & Robotics**	**3(3,0)**
This advanced course, built upon the prerequisite "Mechatronics Systems Programming," delves into the practical application of Python for developing intelligent robotic systems. A significant portion of the course will be dedicated to the Robot Operating System (ROS), providing students with the skills to design, implement, and test complex robotic software. Students will learn core ROS concepts, including nodes, topics, services, and parameters, as well as essential ROS tools for simulation and real-world robot interaction. Furthermore, the course will explore techniques for integrating Artificial Intelligence (AI) into mechatronics applications using Python. This will include an introduction to relevant Python libraries for machine learning, computer vision, and data processing, along with practical examples of how AI can be applied to tasks such as perception, decision-making, and control in robotic systems. The course will emphasize hands-on projects and assignments using platforms such as Raspberry Pi and NVIDIA Jetson Nano to provide students with practical experience in developing and deploying advanced robotic and AI-driven mechatronic solutions. **Prerequisite**: MECA 444L. Annually.		

MECA 499	**Co-op Work Experience**	**3(3,0)**
This Co-op work experience is designed to provide students with a five-week short-term opportunity in mechanical engineering. Students are encouraged to network in the discipline-related industries. A report, a poster, and a PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

MECA 541	**CAD/CAM/CAE**	**3(3,0)**
Fundamental principles of computer-aided design (CAD), computer-aided manufacturing (CAM), and computer-aided engineering (CAE); prototyping and distributed systems; specific rapid prototyping technologies; design for assembly; small batch manufacturing; continuous manufacturing; biotechnology and manufacturing. **Prerequisite**: MECH 224 and MECH 421. Annually.		

MECA 542	**Industrial and Manufacturing Control**	**3(2,2)**
This course will give an overview of programmable logic controllers (PLCs) and explores different PLC hardware components; number systems and codes; fundamentals of Logic; generation, loading and testing of PLC programs required to operate electrical motors and mechanical actuators used in industrial processes; developing fundamental PLC Wiring Diagrams and Ladder Logic Programs with an emphasis on programming timers and counters; integrated automation solutions through various programming architecture such as function-block and ladder diagrams; configuration of process visualization of a panel. Prerequisite: ELEC 320. On demand.		

MECA 544	Robotics	3(3,0)
Introductory historical development of robotics; static grasp analysis; rigid-object kinematics; degrees of freedom; robot arm kinematics; inverse kinematics; dynamics; kinematic singularities; introduction to trajectory planning and control of manipulators; screw motions. **Prerequisite**: MECH 321. Annually.		

MECA 595A	BE Summative Learning Project I	1(1,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes a literature review, a scope of work, and a project proposal. **Prerequisites**: Senior Standing, ENGL 217. Annually.		

MECA 595B	BE Summative Learning Project II	3(3,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes literature review, design work, data collection, experimentation, data analysis, and technical writing. **Prerequisite**: MECA 595A. Annually.		

II. Elective Courses

Students may meet their technical electives requirements from the following selection of elective courses.

MECA 484	Civionics	3(3,0)
Applications of mechatronic techniques to civil construction. This course will provide engineers with the background necessary to optimize design techniques and understand infrastructure performance, behavior, and condition. The successful integration of intelligent sensing of innovative structures will allow civil structural engineers to expand the design envelope and introduce new design concepts, materials, and innovation in civil engineering. **Prerequisite**: Instructor's consent. On demand.		

MECA 485	Autotronics	3(3,0)
This course explores topics such as sensors and actuators in automotive systems, communication systems, the CAN protocol, electronic circuits, automotive standards, electronic suspension and steering, engine management, and on-board diagnostics. **Prerequisite**: MECA 440.		

MECA 498	Special Topics in Mechatronics Engineering	3(3,0)
Guided study and research on major-related topics. **Prerequisite**: Instructor's consent. On demand.		

MECA 515	Modern Control Systems	3(3,0)
In this course, the student learns how to use data and correlate it with physical theory to build state-space and control models; design, build, and test controllability,		

feedback, observability, and stability; and perform black-box modeling in both the time and frequency domains. **Prerequisite**: MECA 440. On demand.

MECA 543	**Fluid Power Systems**	**3(3,0)**
Introduction to hydraulic and pneumatic systems: study of basic hydraulic/pneumatic circuit components: pipes, valves, pumps, motors, filters, accumulators, sensors, and silencers; analysis/design of various hydraulic/pneumatic circuits such as high-low, regenerative, and synchronization circuits; use of Matlab/Simulink for simulation of fluid power system circuits. **Prerequisites**: MECH 333 or MECH 335. On demand.		

MECA 545	**Integrated Automation Lab**	**1(0,2)**
This lab explores PID control algorithms; digital controllers and their components; tuning methods and process control; diagnosis and error search; analog value processing; communication between two or more PLCs. Co-requisite: MECA 542. On demand.		

MECA 581	**Systems Engineering**	**3(3,0)**
A course with an interdisciplinary approach intended to enable the realization of successful systems. It focuses on defining customer needs and required functionality early in the development cycle, documenting requirements, then proceeding with design synthesis and system validation while considering the complete problem, including operations, performance, test, manufacturing, cost, and schedule. This course emphasizes the links between systems engineering and the fundamentals of decision theory, statistics, and optimization. Introduction to the latest commercially successful techniques in systems engineering. **Prerequisite**: Instructor's consent. On demand.		

MECA 586	**Mechatronics Drives**	**3(3,0)**
Switched-mode converter; steady-state voltages and currents of step-down, step-up, inverting, and other power converters; power semiconductor devices in a switching converter; discontinuous conduction mode; solve converters operating in DCM; implement transformer isolation in a dc-dc converter; design and analyze the feedback systems of switching regulators; Design Project. **Prerequisite**: ELEC 320. On demand. **Prerequisite**: ELEC 320		

MECA 597	**Advanced Topics in Mechatronics Engineering**	**3(3,0)**
This course is designed to enable students to study a specific advanced topic of interest, carefully selected from the field of mechatronics engineering. The content outline for such a topic is to be determined by the instructor and approved by the department Chair. **Prerequisite**: Instructor's consent. On demand.		

III. AI-Restricted Elective Courses

Students must choose at least one elective course (3 credits) from the following selection of AI-Restricted elective courses.

MECA 535	Expert Systems and Applied Intelligence	3(3,0)
This course explores the core principles of intelligent systems and their application in solving complex engineering and decision-making problems. Students will learn perception methods, reasoning techniques, and search strategies, as well as key AI tools such as neural networks, evolutionary computing, fuzzy logic, and intuitionistic fuzzy logic. The course also introduces intelligent programming using Prolog/python to model knowledge and build rule-based expert systems. Through practical examples, students will gain hands-on experience in developing AI-driven solutions for real-world applications. **Prerequisite**: MECA 444L.		

MECA 536	Applied Deep Learning for Intelligent Systems	3(3,0)
This course introduces the basics of computer vision and machine learning, focusing on their use in intelligent systems. Students will learn techniques for object detection, image understanding, and sensor data interpretation, using deep learning tools to build decision-making models. Hands-on practice with simulation platforms will allow students to design and test vision-based solutions in realistic scenarios. At the end of the course, participants will gain practical skills in combining perception, planning, and control, preparing them for applications in autonomous systems and other technologies. **Prerequisite**: MECA 444L.		

MECA 546	Smart Energy Management Systems	3(3,0)
This course delves into smart energy management systems (SEMS), focusing on versatile operating modes in solar photovoltaic (PV) and battery energy storage (BESS) systems. It equips mechatronics engineering students with the fundamental knowledge and practical skills required to design, implement, and optimize these crucial technologies for a sustainable future. Furthermore, the course discusses the potential of using generative AI for the design and development of smart energy management systems. A capstone case study is presented and analyzed. **Prerequisite**: MECA 442.		

MECH 522	Mechanism Synthesis: Classical Kinematics to AI-Driven Optimization	3(3,0)
This course introduces the theory and computational practice of mechanism synthesis, with emphasis on the design of four-bar linkages for path, motion, and function generation. Classical kinematic synthesis methods are integrated with numerical optimization techniques, including Newton–Raphson, least-squares, and gradient-based methods, as well as intelligent metaheuristic algorithms such as genetic algorithms, tabu search, and ant colony optimization. The course also exposes students to data-driven and AI-assisted approaches, including deep neural networks for inverse mechanism design, as advanced computational tools for automation, design-space exploration, and decision support in mechanism synthesis. Computational projects provide students with hands-on experience in mechanism modeling, simulation, error analysis, optimization, and the synthesis of mechanical systems under practical design constraints. **Prerequisite:** MECH 321.		

MINOR IN MANUFACTURING AND AUTOMATION ENGINEERING

Rationale

The Manufacturing and Automation Engineering (MAE) minor aims to broaden engineering students' knowledge of up-to-date automation techniques and manufacturing methods. It also prepares undergraduate students for potential graduate studies in Manufacturing and Automation Engineering and related fields.

It primarily targets mechanical and mechatronics engineering students, but it should also accommodate students from a variety of engineering backgrounds, provided all prerequisite courses are completed.

The program mainly focuses on decision-making, materials science, manufacturing processes, quality control, system integration, automation, and many other related topics. Therefore, by the end of this minor, a student is expected to be familiar with the economic study, design, material selection, manufacturing, assembly, control, and automation of the entire engineering production process.

Program participants are required to complete at least 18 credits of related coursework. Although not required, with proper and timely advising, they may also complete their Co-op work experience or their BE project in Manufacturing and Automation Engineering, thereby enhancing their capabilities.

Program Objectives

The primary educational objectives of the MAE minor program are to:

- Give engineering students basic knowledge of the Manufacturing and Automation Engineering principles
- Equip students with the knowledge to pursue graduate studies in Industrial Engineering, Manufacturing Engineering, Automation Engineering, Materials Science, Engineering Management, and other related disciplines

Learning Outcomes

After completing the minor program, students should have the ability to:

- Apply relevant knowledge of mathematics, science, and engineering to identify, formulate, and solve practical problems related to Automation Engineering and modern Manufacturing processes
- Use modern CAD/CAM/CAE engineering tools
- Automate and control complex industrial equipment and processes
- Make apt decisions based on their comprehension of Professional and Ethical Behavior

Career Options

This minor allows MAE holders to start careers in supply chain management, product design and development, quality control and assurance, production lines, cost engineering departments, design, automation, and operation of integrated systems, and many other fields.

Program Requirements

While most relevant to mechanical and mechatronics engineering students, this minor is offered to undergraduate RHU students in the BS/BE program from different disciplines (mechanical, mechatronics, civil, electrical, computer and communications, and biomedical). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements. Proper advising is pertinent to the timely completion of the minor.

To complete the MAE minor, a student must

1) Declare an MAE minor by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the college major;
3) Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
4) Complete 18 credits of coursework as specified below

MAE Curriculum

The program offers a rich blend of fundamental sciences and practical, hands-on experience, equipping students with the essential knowledge to delve into Manufacturing and Automation Engineering. Accordingly, this minor shall include:

- Materials science and metallurgy
- Advanced/modern manufacturing processes
- Product design, optimization, and development
- Manufacturing process quality control
- Economic, environmental, and societal aspects of the industry
- Industrial equipment automation and production process control

The MAE curriculum encompasses 18 mandatory credits selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Mandatory Courses

The student must complete a minimum of 3 credit hours of coursework exclusive to the MAE minor that will not count toward requirements for their major or any other minor. A description of the courses from which a student should select 18 credits is given below:

MECA 515	Modern Control Systems	3(3,0)
In this course, the student learns how to use data and correlate it with physical theory to build state-space and control models; design, build, and test controllability, feedback, observability, and stability; and perform black-box modeling in both the time and frequency domains. **Prerequisite**: MECA 440. On demand.		

MECA 541	CAD/CAM/CAE	3(3,0)
Fundamental principles of computer-aided design (CAD), computer-aided manufacturing (CAM), and computer-aided engineering (CAE); prototyping and distributed systems; specific rapid prototyping technologies; design for assembly; small batch manufacturing; continuous manufacturing; biotechnology and manufacturing. **Prerequisite:** MECH 224 and MECH 421. Annually.		

MECA 542	Industrial and Manufacturing Control	3(2,2)
This course will give an overview of programmable logic controllers (PLCs) and explores different PLC hardware components; number systems and codes; fundamentals of Logic; generation, loading and testing of PLC programs required to operate electrical motors and mechanical actuators used in industrial processes; developing fundamental PLC Wiring Diagrams and Ladder Logic Programs with an emphasis on programming timers and counters; integrated automation solutions through various programming architecture such as function-block and ladder diagrams; configuration of process visualization of a panel. **Prerequisite**: ELEC 320. On demand.		

MECA 545	Integrated Automation Lab	1(0,2)
This lab explores PID control algorithms; digital controllers and their components; tuning methods and process control; diagnosis and error search; analog value processing; communication between two or more PLCs. **Co-requisite**: MECA 542. On demand.		

MECA 586	Mechatronics Drives	3(3,0)
Switched-mode converter; steady-state voltages and currents of step-down, step-up, inverting, and other power converters; power semiconductor devices in a switching converter; discontinuous conduction mode; solve converters operating in DCM; implement transformer isolation in a dc-dc converter; design and analyze the feedback systems of switching regulators; Design Project. **Prerequisite**: ELEC 320. On demand.		

MECH 422	Manufacturing Processes	3(3,0)
General overview of manufacturing; properties of materials; dimensions and tolerances; machining; casting; sheet metal forming; injection molding; thermoforming; plastic processing; fundamentals of process performance (rate, quality, cost, flexibility); variation and quality; sustainability; overview of additive manufacturing. Project. **Prerequisite**: MECH 320. Annually.		

MECH 470	Product Design and Development	3(3,0)
Idea generation; concept generation; concept selection; functional analysis; engineering design process for systems and components; economic consideration; reliability analysis; product safety; design project. **Co-requisite**: MECH 421. On demand.		

MECH 527	Modern Material Science	3(3,0)
Advanced coverage of metals and alloys; applications and processing of ceramics; characteristics, applications, and processing of polymers; surface property enhancement; materials simulation; economic, environmental, and societal issues in materials science; biomaterials and nanoscale materials. **Prerequisite**: MECH 223. On demand.		

MECH 570	Digital Manufacturing	3(3,0)
In this course, the student will get an insight into diagnosing and correcting operational flaws; modeling and fabrication based on additive manufacturing technologies; automation technologies; production planning, process quality control; use of modern four-axis Computer Numerical Control (CNC) machines, 3D printers, and up-to-date software programs, including AutoCAD, Inventor, SolidWorks, and VisualCam. **Prerequisite**: MECH 422 and MECA 541. On demand.		

MECH 571	Optimization in Engineering Design	3(3,0)
Problem definition, cost function and constraints; local vs. global methods; deterministic vs. stochastic methods; linear vs. non-linear programming; constrained vs. unconstrained; continuous vs. discrete; gradient-based methods; combinatorial optimization techniques; applications to various engineering problems (single/multi-objective); project. **Prerequisite**: MATH 421. On demand.		

Faculty List

Al Wardany, Riad; Associate Professor, Ph.D. in Civil Engineering, University of Sherbrooke, 2005

Arid, Amal; Lecturer, Master's in Computer and Electrical Engineering, AUB, 2010.

Chaaban, Ahmad; Professor, Ph.D. in Mechanical Engineering, University of Waterloo, 1985.

Rouaa, El Chammaa; Assistant Professor, Ph.D. in Computer and Communication Engineering, USJ/ Lyon University, 2020

Berjaoui, Omar; Instructor, Master in Mechatronics Engineering, RHU,2011.

Diab, Mohammad; Professor, Ph.D. in Biomedical Engineering, University of Compiègne, 2007.

Diab, Nadim; Professor, Ph.D. in Mechanical Engineering, American University of Beirut, 2013.

Fawaz, Amani; Lecturer, Ph.D. in Electrical and Energy Engineering, ESIB-USJ, 2025.

Hajj Chehade, Rana; Assistant Professor, Master 2 in Civil Engineering, Lille University, 2015.

Hijazi, Toufic; Professor, Ph.D. in Electrical Engineering, Clarkson University, 1988.

Kaderi, Mohamad; Lecturer, MS in Mechatronics Engineering, Rafik Hariri University, 2024

Kasab, Milana; Lecturer, M.Sc. in Biomedical Engineering, Rafik Hariri University, 2021.

Koubeissi, Ahmad; Associate Professor, Ph.D. in Robotics and Automation Engineering, university of Lille, 2015.

Machaka, Muheiddein; Associate Professor, Ph.D. in Civil Engineering, Beirut Arab University, 2015.

Maatouk, Maya: Instructor, Master in Mechatronics Engineering, Politecnico di Torino, Turin, Italy, 2021.

Moslem, Bassam; Associate Professor, Ph.D. in Signal Processing and Bioengineering, UTC, 2011.

Mrad, May; Instructor, M.Sc. in Civil & Environmental Engineering, Rafik Hariri University, 2018.

Moustafa, Ousama; Associate Professor, Ph.D. in Engineering, University Le Havre,2008.

Ramadan, Rawan; Lecturer, Ph.D. Structural and Geotechnical Engineering, BAU, 2024.

Sabbah, Maher; Associate Professor, Ph.D. in Biomedical Engineering, University of Technology of Compiègne (UTC), France, and Université Claude Bernard, Lyon, 2016.

Taha, Mohamad; Professor, Ph.D. in Electrical Engineering, Aston University, 1992.

Zantout, Rached; Professor, Ph.D. in Communications Engineering, Ohio State University, 1994.

Younis, Manar; Assistant Professor, Ph.D. in Mechanical Engineering, American University of Beirut, 2021



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